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DIFFERENTIATION OF THE SPECIES OF THE GENUS *LEGIONELLA*

Selected characteristics useful for differentiating legionellae from other organisms and from one another are presented in Tables BXII.γ.56 and BXII.γ.57. Cellular fatty acids of *Legionella* species are shown in Table BXII.γ.58, ubiquinone content in Table BXII.γ.59, and DNA relatedness data in Table BXII.γ.62.

Immunologic cross-reactions are detailed in Table BXII.γ.60, and the serogroups of *Legionella* species are described in Table BXII.γ.61. Morphology is shown in Figs. BXII.γ.94, BXII.γ.95, and BXII.γ.96.

List of species of the genus *Legionella*

1. *Legionella pneumophila* Brenner, Steigerwalt and McDade 1979, 658.^{AL}

pneu.mo'phi.la. Gr. n. *pneumo* lung; Gr. adj. *philos* loving; M.L. adj. *pneumophila* lung-loving.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Hydrolysis of hippurate is not positive for all strains of *L. pneumophila* (Hébert, 1981). Serologic cross-reactions are detailed in Table BXII.γ.60. The species contains 15 serogroups (Table BXII.γ.61). Causative agent of legionellosis, presenting either as a pneumonia with a high fatality rate in untreated patients (Legionnaires' disease) or as an acute, self-limited, febrile illness (Pontiac fever). Isolated from human lung, bronchoalveolar lavage fluid, sputum, blood, intravascular devices, peritoneal fluid, abscess fluids, and potable and surface water. DNA relatedness between strains of *L. pneumophila* is 75–100%. *L. pneumophila* is 4–21% related to other *Legionella* species (Table BXII.γ.62). The fatty acid composition of *L. pneumophila* closely resembles that of *L. longbeachae*, and the two species cannot be distinguished by this technique (Bibb et al., 1981). The phenotype of strains, derived by serogrouping, does not always match their genotype. Strains within the same serogroup may be genetically disparate, whereas strains in different serogroups may be closely related genetically. *L. pneumophila* was divided into three DNA groups, based on a study of 60 strains (Brenner et al., 1988c). Subspecies designations were proposed for these nucleic acid groups.

The mol% G + C of the DNA is: 39 (T_m , Bd).

Type strain: Philadelphia 1, ATCC 33152, DSM 7513.

GenBank accession number (16S rRNA): M59157, M36023.

Additional Remarks: The type strain was isolated from human lung tissue by McDade and Weaver in 1977. The oldest isolate is strain OLDA isolated by E. Jackson in 1947 (Jackson et al., 1952).

- a. *Legionella pneumophila* subsp. *pneumophila* (Brenner, Steigerwalt and McDade 1979) Brenner, Steigerwalt, Epple, Bibb, McKinney, Starns, Colville, Selander, Edelstein and Moss 1989, 205^{VP} (Effective publication: Brenner, Steigerwalt, Epple, Bibb, McKinney, Starns, Colville, Selander, Edelstein and Moss 1988c, 1701) (*Legionella pneumophila* Brenner, Steigerwalt and McDade 1979, 658.) *pneu.mo'phi.la*. Gr. n. *pneumo* lung; Gr. adj. *philos* loving; M.L. adj. *pneumophila* lung-loving.

This subspecies represents DNA group I and contains strains in serogroups 1–10. All strains hydrolyze sodium

hippurate. Strains of *L. pneumophila* subsp. *pneumophila* are >70% related to each other under optimal conditions (60°C) with ≤3.0% divergence in related sequences; they are >60% related under stringent conditions (75°C). DNA relatedness of these strains to other subspecies is below the species level in at least one and usually two of these hybridization parameters. Isolated from human lung tissue.

The mol% G + C of the DNA is: 39–41 (T_m); type strain: 39 (T_m).

Type strain: Philadelphia 1, ATCC 33152, DSM 7513.

GenBank accession number (16S rRNA): M59157, M36023.

- b. *Legionella pneumophila* subsp. *fraseri* Brenner, Steigerwalt, Epple, Bibb, McKinney, Starns, Colville, Selander, Edelstein and Moss 1989, 205^{VP} (Effective publication: Brenner, Steigerwalt, Epple, Bibb, McKinney, Starns, Colville, Selander, Edelstein and Moss 1988c, 1702.) *fra'seri*. M.L. gen. n. *fraseri* in honor of D.W. Fraser, who headed the investigation of the Legionnaires' disease outbreak in Philadelphia that led to the recognition of legionellosis and isolation of *L. pneumophila*.

This subspecies represents DNA group II and contains strains from serogroups 1, 4, 5, and Lansing 3. Strains from serogroups 1 and 5 hydrolyze sodium hippurate; strains from serogroups 4 and Lansing 3 do not hydrolyze hippurate. Strains of *L. pneumophila* subsp. *fraseri* are >65% related to each other under optimal conditions with <3% divergence and >60% related under stringent conditions. DNA relatedness of these strains to other subspecies is below the species level in at least one and usually two of these hybridization parameters. Isolated from human lung tissue.

The mol% G + C of the DNA is: 38–39 (T_m); type strain: 39 (T_m).

Type strain: Los Angeles 2, ATCC 33156, DSM 7514.

GenBank accession number (16S rRNA): M36025.

- c. *Legionella pneumophila* subsp. *pascallei* Brenner, Steigerwalt, Epple, Bibb, McKinney, Starns, Colville, Selander, Edelstein and Moss 1989, 205^{VP} (Effective publication: Brenner, Steigerwalt, Epple, Bibb, McKinney, Starns, Colville, Selander, Edelstein and Moss 1988c, 1702.)

pas.cul'le.i. M.L. gen. n. *pascallei* in honor of A.W. Pascalle for his pioneering studies on media, isolation, and nosocomial spread of *L. pneumophila* and *L. micdadei*.

This subspecies represents DNA group III that hydrolyze sodium hippurate. Strains of subsp. *pascallei* are

TABLE BXII.γ.62. Base composition and DNA relatedness within and among species under optimal conditions

<i>Legionella</i> species	Mol% G + C Content	% Relatedness within species under optimal conditions	% Relatedness with other species under optimal conditions	Closest relative(s)	16S-23S rDNA Group ^a
<i>L. adelaidensis</i>	40	NA ^b	≤31	None	Distinctive
<i>L. anisa</i>	42	84–99	5–57	<i>L. bozemanii</i>	A
<i>L. birminghamensis</i>	42.9	NA	1–3	None	Distinctive
<i>L. bozemanii</i>	41–43	56–73	3–65	<i>L. parisiensis</i>	A
<i>L. brunensis</i>	41.5	NA	1–8	None	Distinctive
<i>L. cherrii</i>	39–40	94–100	6–67	<i>L. steigerwaltii</i> , <i>L. bozemanii</i> , <i>L. dumoffii</i> , <i>L. gormanii</i> , <i>L. anisa</i> , <i>L. parisiensis</i>	A
<i>L. cincinnatiensis</i>	36.3	NA	<44	<i>L. santicrocus</i> , <i>L. longbeachae</i> , <i>L. sainthelensi</i>	Distinctive
<i>L. dumoffii</i>	41–42	90	3–57	<i>L. steigerwaltii</i> , <i>L. cherrii</i> , <i>L. parisiensis</i>	A
<i>L. erythra</i>	51	NA	5–59	<i>L. rubrilucens</i>	B
<i>L. fairfieldensis</i>	42	87–94	1–17	None	Distinctive
<i>L. feeleei</i>	46	78–100 (stringent conditions)	≤15	None	Distinctive
<i>L. geestiana</i>	52	NA	≤11	None	Distinctive
<i>L. gormanii</i>	41	>90	6–47	<i>L. cherrii</i> , <i>L. parisiensis</i> , <i>L. steigerwaltii</i>	A
<i>L. gratiana</i>	38	NA	3–46	<i>L. longbeachae</i> , <i>L. cincinnatiensis</i> , <i>L. sainthelensi</i> , <i>L. parisiensis</i>	Distinctive
<i>L. hackeliae</i>	40	NA	3–22	<i>L. cherrii</i> , <i>L. jamestowniensis</i>	Distinctive
<i>L. israelensis</i>	40.9	95–100	≤12	None	Distinctive
<i>L. jamestowniensis</i>	42	NA	≤17	None	Distinctive
<i>L. jordanis</i>	45	93–100	2–20	<i>L. spiritensis</i>	Distinctive
<i>L. lansingensis</i>	42.6	NA	1–12	None	Distinctive
<i>L. londiniensis</i>	43	94–100	0–36	None	Distinctive
<i>L. longbeachae</i>	40	97–100	2–22	<i>L. cherrii</i>	Distinctive
<i>L. lytica</i> ; amoeba- associated strains	NA	NA	NA	NA	NA
<i>L. maceachernii</i>	41	100	3–23	<i>L. micdadei</i>	Distinctive
<i>L. micdadei</i>	41–44	>90	1–23	<i>L. maceachernii</i>	Distinctive
<i>L. moravica</i>	41.5	NA	1–9 (named species); 24–36 (unnamed species)	None	D
<i>L. nautarum</i>	41	70–100	2–24	<i>L. micdadei</i>	Distinctive
<i>L. oakridgensis</i>	43–45	96–100	2–22	<i>L. spiritensis</i>	Distinctive
<i>L. parisiensis</i>	42	NA	NA	<i>L. bozemanii</i> , <i>L. dumoffii</i> , <i>L. gormanii</i> , <i>L. cherrii</i> , <i>L. steigerwaltii</i> , <i>L. wadsworthii</i> (37)	A
<i>L. pneumophila</i>	38–41	>70	4–21	<i>L. cherrii</i>	C
<i>L. pneumophila</i> subsp. subsp.	39–41	72–100		—	C
<i>L. pneumophila</i> ^c					
<i>L. pneumophila</i> subsp. subsp. <i>fraseri</i>	38–39	>70		—	C
<i>L. pneumophila</i> subsp. subsp. <i>pascallei</i>	39	96–100		—	C
<i>L. quateirensis</i>	39	NA	2–36	<i>L. moravica</i>	D
<i>L. quintivanii</i>	43	92–100	1–30	<i>L. birminghamensis</i>	Distinctive
<i>L. rubrilucens</i>	52	NA	3–59	<i>L. erythra</i>	B
<i>L. sainthelensi</i>	41	91–100	3–58	<i>L. longbeachae</i> , <i>L. santicrocus</i>	Distinctive
<i>L. santicrocus</i>	38	NA	7–64	<i>L. sainthelensi</i>	Distinctive
<i>L. shakespearei</i>	45.5	NA	1–25	<i>L. jordanis</i>	Distinctive
<i>L. spiritensis</i>	46	NA	4–34	<i>L. rubrilucens</i>	Distinctive
<i>L. steigerwaltii</i>	40	NA	6–54	<i>L. cherrii</i> , <i>L. bozemanii</i> , <i>L. dumoffii</i> , <i>L. gormanii</i> , <i>L. parisiensis</i>	A

(continued)

TABLE BXII.γ.62. (cont.)

<i>Legionella</i> species	Mol% G + C Content	% Relatedness within species under optimal conditions	% Relatedness with other species under optimal conditions	Closest relative(s)	16S-23S rDNA Group ^a
<i>L. tucsonensis</i>	44	NA	1–66	<i>L. bozemanii</i> , <i>L. anisa</i> , <i>L. parisiensis</i> , <i>L. gormanii</i> , <i>L. wadsworthii</i> , <i>L. longbeachae</i> , <i>L. micdadei</i> , <i>L. dumoffii</i>	A
<i>L. taurinensis</i>	46	78–100	<3–64	<i>L. rubrilucens</i> , <i>L. erythra</i> , <i>L. spiritensis</i>	NA
<i>L. wadsworthii</i>	42	NA	3–36	<i>L. steigerwaltii</i> , <i>L. tucsonensis</i>	Distinctive
<i>L. waltersii</i>	NA	NA	1–8	None	Distinctive
<i>L. worleiensis</i>	41	NA	1–24	<i>L. moravica</i>	D
Genomospecies 1	NA	NA	1–69	<i>L. quintivanii</i>	Distinctive

^aRiffard et al., 1998

^bNA, not available.

^cThe DNA relatedness among the three subspecies of *L. pneumophila* is below the species level in at least one and usually two hybridization parameters.



FIGURE BXII.γ.94. Electron micrograph of *Legionella pneumophila* grown in broth (× 90,000).

>95% related to each other at optimal conditions with a divergence of 1% and >95% related at stringent conditions. DNA relatedness of these strains to other subspecies is below the species level in at least one and usually two of these hybridization parameters.

The type strain was isolated from a hospital water supply, but has not been isolated from humans.

The mol% G + C of the DNA is: 39 (*T_m*).

Type strain: U8W, ATCC 33737, DSM 7515.

2. **Legionella adelaidensis** Benson, Thacker, Lanser, Sangster, Mayberry and Brenner 1991b, 580^{VP} (Effective publication: Benson, Thacker, Lanser, Sangster, Mayberry and Brenner 1991a, 1005.)

a.del.aid.en' sis. M.L. fem. adj. *adelaidensis* coming from Adelaide, Australia.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Not demonstrated to cause human disease. Isolated from surface water. *L. adelaidensis* was >31% related to 28 other *Legionella* species tested.

The mol% G + C of the DNA is: 40 (*T_m*).

Type strain: 1762-AUSE, ATCC 49625.

Additional Remarks: The type strain was isolated from cooling tower water in Adelaide, Australia in 1987.

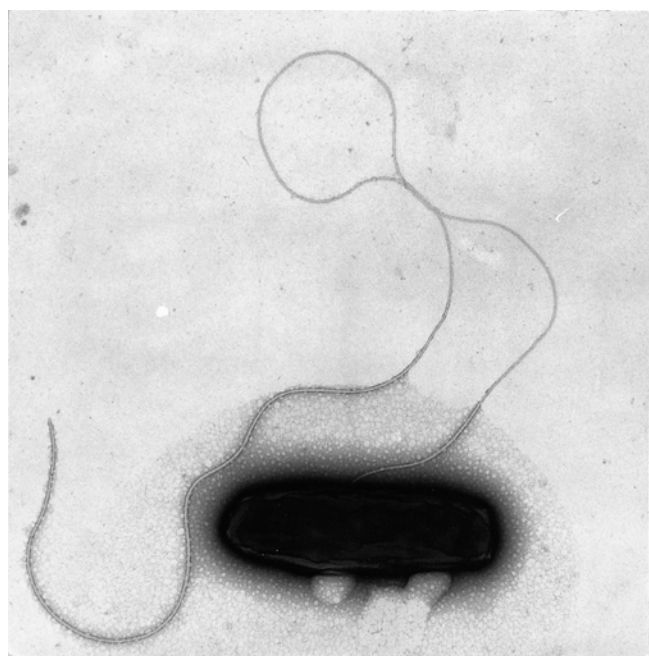


FIGURE BXII.γ.95. Transmission electron micrograph of *Legionella pneumophila* grown on CYE agar, showing a single, probably lateral, flagellum ($\times 61,700$).

3. ***Legionella anisa*** Gorman, Feeley, Steigerwalt, Edelstein, Moss and Brenner 1985b, 375^{VP} (Effective publication: Gorman, Feeley, Steigerwalt, Edelstein, Moss and Brenner 1985a, 308.)

a'ni.sa. M.L. gen. n. *anisa* from Gr. adj. *anisos* unequal, referring to the fact that blue-white autofluorescence is exhibited by most, but not all, strains.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. One of the original five strains did not exhibit autofluorescence (Gorman et al., 1985a). Serologic cross-reactions are detailed in Table BXII.γ.60. Causative agent of human pneumonia and empyema and of acute, non-pneumonic, febrile illness (Pontiac fever) (Gorman et al., 1985a; Bornstein et al., 1989b; Fenstersheib et al., 1990; Thacker et al., 1990a). Isolated from human bronchial lavage and pleural fluid and from surface and potable water. The strain that was associated with acute, nonpneumonic, febrile illness was isolated from a public fountain (Fenstersheib et al., 1990). The DNA relatedness among strains of *L. anisa* is 84–99% with divergence of 0.5–1.0% at 60°C; relatedness at 75°C is 84–92%. *L. anisa* is 5–57% related to other *Legionella* species with divergence of 7.0–14% at 60°C and 1–27% at 75°C. It is most closely related to other species that exhibit blue-white autofluorescence (Table BXII.γ.62).

The mol% G + C of the DNA is: 42 (T_m).

Type strain: WA-316-C3, ATCC 35292.

GenBank accession number (16S rRNA): X73394, Z32635.

Additional Remarks: The type strain was isolated in Los Angeles, California, from hot water in a sink by George W. Gorman in 1981.

4. ***Legionella birminghamensis*** Wilkinson, Thacker, Benson, Polt, Brookings, Mayberry, Brenner, Gilley and Kirklin

1988b, 220^{VP} (Effective publication: Wilkinson, Thacker, Benson, Polt, Brookings, Mayberry, Brenner, Gilley and Kirklin 1987, 2121.)

bir.ming.ham.en'sis. N.L. fem. adj. *birminghamensis* from Birmingham, Alabama, where the original isolate was made.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Causative agent of human pneumonia (Wilkinson et al., 1987). Isolated from human lung biopsy. *L. birminghamensis* was 1–3% related to 18 other *Legionella* species tested (Table BXII.γ.62).

The mol% G + C of the DNA is: 42.9 (T_m).

Type strain: 1407-AL-H, ATCC 43702.

Additional Remarks: The type strain was isolated from a human lung biopsy in Birmingham, Alabama in 1986.

5. ***Legionella bozemanii*** (sic) Brenner, Steigerwalt, Gorman, Weaver, Feeley, Cordes, Wilkinson, Patton, Thomason and Lewallen Sasserville 1980d, 676^{VP} (Effective publication: Brenner, Steigerwalt, Gorman, Weaver, Feeley, Cordes, Wilkinson, Patton, Thomason and Lewallen Sasserville 1980c, 114.)

boze.ma'ni.i. M.L. gen. n. *bozemanii* named after F. Marilyn Bozeman, the microbiologist who isolated and first studied the organism.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. *L. bozemanii* cannot be distinguished reliably from *L. micdadei*, *L. dumoffii*, or *L. gormanii* by analysis of cellular fatty acids (Karr et al., 1982). There are two serogroups of *L. bozemanii* (Table BXII.γ.61) (Tang et al., 1984). Causative agent of human pneumonia and empyema. Isolated from human lung, pleural fluid, and potable water. DNA relatedness between strains of *L. bozemanii* is 56–77%. *L. bozemanii* is 3–65% related to other *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 43 (T_m).

Type strain: WIGA, ATCC 33217.

GenBank accession number (16S rRNA): M36031.

Additional Remarks: The type strain was isolated in 1959 by Bozeman from human lung tissue.

6. ***Legionella brunensis*** Wilkinson, Drasar, Thacker, Benson, Schindler, Potuznkiova, Mayberry and Brenner 1989, 205^{VP} (Effective publication: Wilkinson, Drasar, Thacker, Benson, Schindler, Potuznkiova, Mayberry and Brenner 1988a, 399.)

bru.nen'sis. N.L. gen. n. *brunensis* coming from Brno, Czechoslovakia.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Not demonstrated to cause human disease. Isolated from surface water. *L. brunensis* is 1–8% related to 32 other described *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 41.5 (T_m).

Type strain: 441-1, ATCC 43878, CDC 1635-CZK-E.

GenBank accession number (16S rRNA): Z32636.

Additional Remarks: The type strain was isolated from cooling tower water in Brno, Czechoslovakia; first reported in 1986.

7. ***Legionella cherrii*** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Ska-



FIGURE BXII.γ.96. Carbol fuchsin-stained *Legionella pneumophila* grown on CYE agar, showing both short rods and filaments ($\times 1000$).

liy, Martin, Brake, Fields, McEachern and Corcoran 1985, 57^{VP}

cher'ri.i. N.L. gen. n. *cherrii* in honor of William B. Cherry for his many pioneering contributions to our knowledge of legionellae.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Not demonstrated to cause human disease. Isolated from surface water. *L. cherrii* strains are $\geq 94\%$ related to each other at both 60°C and 75°C with 0.0–0.5% divergence. It is highly related to the following species that also exhibit blue-white autofluorescence: *L. steigerwaltii* (67%), *L. dumoffii* (57%), *L. bozemanii* (51%), and *L. gormanii* (47%). It is 24% related to another species that produces blue-white autofluorescence, *L. parisiensis*, and is 6–35% related to other *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 39–40 (T_m).

Type strain: ORW, ATCC 35252.

GenBank accession number (16S rRNA): X73404.

Additional Remarks: The type strain was isolated from thermally polluted water in Minnesota by R.L. Tyndall and C.B. Duncan in 1982.

8. ***Legionella cincinnatiensis*** Thacker, Benson, Staneck, Vincent, Mayberry, Brenner and Wilkinson 1989b, 205^{VP} (Effective publication: Thacker, Benson, Staneck, Vincent, Mayberry, Brenner and Wilkinson 1988a, 419.)
cin.cin.na.ti.en'sis. N.L. fem. adj. *cincinnatiensis* from Cincinnati, Ohio, where the species was first isolated.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Isolates from two clinical cases of *L. cincinnatiensis* cross-reacted with *L. longbeachae*, serogroup 1, and *L. du-*

moffii in the direct immunofluorescence test and were initially misidentified (Thacker et al., 1988a; Jernigan et al., 1994). Causative agent of human pneumonia. Isolated from an open lung biopsy. At 60°C *L. cincinnatiensis* was 44% related to *L. longbeachae*, 39% related to *L. santacrucis*, and 36% related to *L. sainthelensi*. At 75°C it is 5–10% related to these species (Table BXII.γ.62).

The mol% G + C of the DNA is: 36 (T_m).

Type strain: 72-OH-H, ATCC 43753.

GenBank accession number (16S rRNA): X73407.

Additional Remarks: The type strain was isolated from a human open lung biopsy in Cincinnati, Ohio in 1982.

9. ***Legionella dumoffii*** Brenner, Steigerwalt, Gorman, Weaver, Feeley, Cordes, Wilkinson, Patton, Thomason and Lewallen Sasseville 1980d, 676^{VP} (Effective publication: Brenner, Steigerwalt, Gorman, Weaver, Feeley, Cordes, Wilkinson, Patton, Thomason and Lewallen Sasseville 1980c, 115.)
du.mof'fi.i. M.L. gen. n. *dumoffii* of Dumoff, named after Morris Dumoff, who first isolated *L. pneumophila* directly on bacteriologic media.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. One strain of *L. dumoffii* failed to produce brown pigment on tyrosine-containing media (Edelstein and Pryor, 1985). Serologic cross-reactions are detailed in Table BXII.γ.60. *L. dumoffii* cannot be distinguished reliably from *L. bozemanii*, *L. micdadei*, or *L. gormanii* by analysis of cellular fatty acids (Karr et al., 1982). Causative agent of human pneumonia (Dumoff, 1979). Isolated from human lung tissue, prosthetic heart valves (Tompkins et al., 1988), and from water. DNA relatedness between strains of *L. dumoffii* is 90%. *L. dumoffii* is 3–57% related to other *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 41–42 (T_m).

Type strain: NY-23, ATCC 33279.

GenBank accession number (16S rRNA): Z32637, X73405.

Additional Remarks: The type strain was isolated from water by G. W. Gorman in 1978.

10. **Legionella erythra** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Skaliy, Martin, Brake, Fields, McEachern and Corcoran 1985, 55.^{VP}

erythra. Gr. fem. adj. *erythra* red after the red autofluorescence exhibited by the species.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. A second serogroup of *L. erythra* cannot be distinguished serologically from *L. rubrilucens* (Table BXII.γ.60) (Saunders et al., 1992). The species contains two serogroups (Table BXII.γ.61) (Saunders et al., 1992). Not yet demonstrated to cause human disease. Isolated from surface water. *L. rubrilucens* and *L. erythra* are 59–62% interrelated, with 9.0–9.5% divergence in related sequences. Relatedness to other *Legionella* species ranges from 5–59%. The closest other species was *L. spiritensis* with 20% relatedness. At 75°C *L. rubrilucens* and *L. erythra* are 27–34% related (Table BXII.γ.62).

The mol% G + C of the DNA is: 51 (T_m).

Type strain: SE-32A-C8, ATCC 35303.

GenBank accession number (16S rRNA): Z32638.

Additional Remarks: The type strain was isolated from cooling tower water in Seattle, Washington, by George W. Gorman in 1981.

11. **Legionella fairfieldensis** Thacker, Benson, Hawes, Gidding, Dwyer, Mayberry and Brenner 1991b, 580^{VP} (Effective publication: Thacker, Benson, Hawes, Gidding, Dwyer, Mayberry and Brenner 1991a, 477.)

fairfieldensis. N.L. fem. adj. *fairfieldensis* coming from Fairfield, Victoria, Australia.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Not demonstrated to cause human disease. Isolated from surface water. Strains of *L. fairfieldensis* are 87–94% related at 60°C, with 0.0–0.5% divergence. *L. fairfieldensis* was ≤17% related to 30 other *Legionella* species tested (Table BXII.γ.62).

The mol% G + C of the DNA is: 42 (T_m).

Type strain: 1725-AUS-E, ATCC 49588.

Additional Remarks: The type strain was isolated from cooling tower water in Fairfield, Victoria, Australia, in 1987.

12. **Legionella feeleeii** Herwaldt, Gorman, McGrath, Toma, Brake, Hightower, Jones, Reingold, Boxer, Tang, Moss, Wilkinson, Brenner, Steigerwalt and Broome 1984b, 355^{VP} (Effective publication: Herwaldt, Gorman, McGrath, Toma, Brake, Hightower, Jones, Reingold, Boxer, Tang, Moss, Wilkinson, Brenner, Steigerwalt and Broome 1984a, 338.)

feeleeii. M.L. gen. n. *feeleeii* of the late James Feeley, who contributed greatly to early work on *Legionella* and developed effective agar media for primary isolation.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Hydrolysis of hippurate by *L. feeleeii* has been re-

ported by some investigators, but not others (Brenner et al., 1985). Serologic cross-reactions are detailed in Table BXII.γ.60. The species contains two serogroups (Thacker et al., 1985). Causative agent of human pneumonia (Thacker et al., 1985) and acute, nonpneumonic, febrile illness (Pontiac fever) (Herwaldt et al., 1984a). Isolated from human sputum and lung tissue and from surface water. The DNA relatedness among strains of *L. feeleeii* is 78–100% at 75°C; *L. feeleeii* is 1–15% related to other *Legionella* species (Table BXII.γ.60).

The mol% G + C of the DNA is: 46 (T_m).

Type strain: WO-44C, ATCC 35072.

GenBank accession number (16S rRNA): X73395.

Additional Remarks: The type strain was isolated from industrial coolant by George W. Gorman in 1981.

13. **Legionella geestiana** Dennis, Brenner, Thacker, Wait, Vesey, Steigerwalt and Benson 1993, 335.^{VP}

geestiana. M.L. gen. n. *geestiana* pertaining to the Geest office building.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Not demonstrated to cause human disease. Isolated from potable water. *L. geestiana* is 0–10% related to other *Legionella* species tested (Table BXII.γ.62).

The mol% G + C of the DNA is: 52 (T_m).

Type strain: 1308ATCC 49504.

Additional Remarks: The type strain was isolated from a domestic hot water tap in the Geest Office Building in London, United Kingdom, between 1982 and 1984.

14. **Legionella gormanii** Morris, Steigerwalt, Feeley, Wong, Martin, Patton and Brenner 1980b, 676^{VP} (Effective publication: Morris, Steigerwalt, Feeley, Wong, Martin, Patton and Brenner 1980a, 720.)

gormanii. M.L. gen. n. *gormanii* of Gorman; named after G.W. Gorman, who isolated and first studied the organism and pioneered in the isolation of legionellae from environmental and clinical sources.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. *L. gormanii* cannot be distinguished reliably from *L. bozemanii*, *L. micdadei*, or *L. dumoffii* by analysis of cellular fatty acids (Karr et al., 1982). Causative agent of human pneumonia (Griffith et al., 1988). Isolated from bronchial lavage fluid and from water. *L. gormanii* is 1–47% related to other *Legionella* species. The relationship of strains within the species is >90%.

The mol% G + C of the DNA is: 41 (T_m).

Type strain: LS-13, ATCC 33297.

GenBank accession number (16S rRNA): Z32639.

Additional Remarks: The type strain was isolated by G.W. Gorman from creek bank soil in 1978.

15. **Legionella gratiana** Bornstein, Marmet, Surgot, Nowicki, Meugnier, Fleurette, Ageron, Grimont, Grimont, Thacker, Benson and Brenner 1991, 580^{VP} (Effective publication: Bornstein, Marmet, Surgot, Nowicki, Meugnier, Fleurette, Ageron, Grimont, Grimont, Thacker, Benson and Brenner 1989a, 550.)

gratiana. M.L. fem. adj. *gratiana* of Gratianus, the Roman emperor who bathed in the hot springs where the species was first isolated.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Not demonstrated to cause human disease. Isolated from surface and potable water. Using the hydroxyapatite method at 60°C, *L. gratiana* was most closely related to *L. longbeachae* (46%), *L. cincinnatiensis* (41%), *L. saintheleusi* (38%), and *L. parisiensis* (32%). The relationship to 23 other species was <16% (Table BXII.γ.62).

The mol% G + C of the DNA is: 38 (T_m).

Type strain: Lyon 8420412, ATCC 49413, CDC 1242.

Additional Remarks: The type species was isolated from spring water at a spa in the Savoy region of France, first reported in 1989.

16. **Legionella hackeliae** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Skaliy, Martin, Brake, Fields, McEachern and Corcoran 1985, 55.^{VP}

hack.el' i. ae. M.L. gen. n. *hackeliae* after Meredith Hackel, who first isolated the organism.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Causative agent of human pneumonia (Brenner et al., 1985; Wilkinson et al., 1985b). Isolated from human bronchial biopsy tissue (Brenner et al., 1985; Wilkinson et al., 1985b). *L. hackeliae* is 3–22% related to other *Legionella* species under optimal conditions; it is 14–17% related to *L. jamestowniensis* in reciprocal studies and is 22% related to *L. cherrii* (Table BXII.γ.62).

The mol% G + C of the DNA is: 40 (T_m).

Type strain: Lansing 2, ATCC 35250.

GenBank accession number (16S rRNA): M36028.

Additional Remarks: The type strain was isolated from a bronchial biopsy specimen by Meredith Hackel in 1981.

17. **Legionella israelensis** Bercovier, Steigerwalt, Derhi-Cochin, Moss, Wilkinson, Benson and Brenner 1986, 369.^{VP}
is.ra.el.en' sis. M.L. fem. adj. *israelensis* coming from Israel.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Not demonstrated to cause human disease. Isolated from surface water. Strains of *L. israelensis* are 95–100% related to each other under optimal conditions. The relationship to other species tested was 3–12% (Table BXII.γ.62).

The mol% G + C of the DNA is: 40 (T_m).

Type strain: Bercovier 4, ATCC 43119.

GenBank accession number (16S rRNA): X73408, Z32640.

Additional Remarks: The type strain was isolated from an oxidation pond in Gaash, Israel, by Hervé Bercovier.

18. **Legionella jamestowniensis** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Skaliy, Martin, Brake, Fields, McEachern and Corcoran 1985, 55.^{VP}

james' town. i. en. sis. M.L. gen. n. *jamestowniensis* named after Jamestown, New York, where the organism was first isolated.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Not yet demonstrated to cause human disease.

Isolated from wet soil. When tested under stringent conditions at 75°C *L. jamestowniensis* is 1–17% related to other *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 42 (T_m).

Type strain: JA-26-G1-E2, ATCC 35298.

GenBank accession number (16S rRNA): X73409.

Additional Remarks: The type strain was isolated by G.W. Gorman from wet soil in 1979.

19. **Legionella jordanis** Cherry, Gorman, Orrison, Moss, Steigerwalt, Wilkinson, Johnson, McKinnery and Brenner 1982b, 384^{VP} (Effective publication: Cherry, Gorman, Orrison, Moss, Steigerwalt, Wilkinson, Johnson, McKinnery and Brenner 1982a, 294.)

jor.da' nis. M.L. gen. n. *jordanis* named after the Jordan River in Bloomington, Indiana, source of the first isolate.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Causative agent of human pneumonia (Thacker et al., 1988b). Isolated from human lung tissue and from water and sewage. DNA relatedness among strains of *L. jordanis* is >90%. *L. jordanis* is 2–20% related to other *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 45 (T_m).

Type strain: BL-540, ATCC 33623.

GenBank accession number (16S rRNA): Z32667, X73396.

Additional Remarks: The type strain was isolated from surface water by G.W. Gorman in 1978.

20. **Legionella lansingensis** Thacker, Dyke, Benson, Havlichek, Robinson-Dunn, Stiefel, Schneider, Moss, Mayberry and Brenner 1994, 595^{VP} (Effective publication: Thacker, Dyke, Benson, Havlichek, Robinson-Dunn, Stiefel, Schneider, Moss, Mayberry and Brenner 1992, 2400.)

lan.sing.en' sis. M.L. fem. adj. *lansingensis* from Lansing, Michigan, where the species was first isolated.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Causative agent of human pneumonia (Thacker et al., 1992). Isolated from human bronchoscopic washings. *L. lansingensis* was 1–12% related to the 28 other *Legionella* species tested (Table BXII.γ.62).

The mol% G + C of the DNA is: 42.6 (T_m).

Type strain: 1677-MI-H, ATCC 49751.

Additional Remarks: The type strain was isolated from human bronchoscopic washings in Lansing, Michigan, in 1986.

21. **Legionella londiniensis** Dennis, Brenner, Thacker, Wait, Vese, Steigerwalt and Benson 1993, 335.^{VP}
lon.din.i.en' sis. M.L. gen. n. *londiniensis* coming from London where the first isolate was made.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Weak hydrolysis of hippurate may be observed. Not demonstrated to cause human disease. Isolated from surface water. Strains of *L. londiniensis* are >90% related to each other. They are 0–36% related to other *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 43 (T_m).

Type strain: 1477, ATCC 49505.

Additional Remarks: The type strain was isolated between 1982 and 1984 from cooling tower pond water in London, England.

22. ***Legionella longbeachae*** McKinney, Porschen, Edelstein, Bissett, Harris, Bondell, Steigerwalt, Weaver, Ein, Lindquist, Kops and Brenner 1982, 266^{VP} (Effective publication: McKinney, Porschen, Edelstein, Bissett, Harris, Bondell, Steigerwalt, Weaver, Ein, Lindquist, Kops and Brenner 1981, 743.)

long.beach' ae. M.L. gen. n. *longbeachae* of Long Beach (California) where the organism was first isolated.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. *L. longbeachae* cannot be distinguished from *L. pneumophila* by analysis of fatty acid composition (Table BXII.γ.58) (Bibb et al., 1981). The species contains two serogroups (Table BXII.γ.61) (Bibb et al., 1981). Causative agent of human pneumonia. Isolated from human lung tissue and transtracheal aspirates and from surface water and moist potting soils. The DNA relatedness among strains of *L. longbeachae* is 97–100% *L. longbeachae* is 2–22% related to other *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 40 (T_m).

Type strain: Long Beach 4, ATCC 33462, DSM 10572, NCTC 11477.

GenBank accession number (16S rRNA): M36029.

Additional Remarks: The type strain was isolated from a transtracheal aspirate by R. Porschen in 1980.

23. ***Legionella lytica*** (Drozanski 1991) Hookey, Saunders, Fry, Birtles and Harrison 1996, 529^{VP} (*Sarcobium lyticum* Drozanski 1991, 86.)

ly'ti.ca. M.L. neut. adj. *lyticum* that which causes lysis.

Identity was established between an amoebal pathogen, *Legionella lytica*, and a previously described amoebal pathogen, *Sarcobium lyticum* (Drozanski, 1991). The new nomenclature is proposed, because the genus *Legionella* has priority over *Sarcobium*. This species is an obligate intracellular parasite of amoebae with limited ability to grow on agar. The few known characteristics are listed in Table BXII.γ.56. Etiologic agent of human disease (Fry et al., 1991a). Isolated from human sputum and lung with amoebae. Isolated bacteria are not available for DNA hybridization experiments. The genetic sequence of *L. lytica* is <97% related to other *Legionella* spp. and thus likely to represent a new species (Fry et al., 1991aM).

The mol% G + C of the DNA is: 43 (T_m).

Type strain: L2, PCM 2298.

GenBank accession number (16S rRNA): X97364.

24. ***Legionella maceachernii*** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Skaliy, Martin, Brake, Fields, McEachern and Corcoran 1985, 55.^{VP}

mac.kath'erni.i. M.L. gen. n. *maceachernii* named after Harold V. McEachern, who first isolated the organism.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Causative agent of human pneumonia (Wilkinson et al., 1985a).

Isolated from human bronchial biopsy tissue, water from a home evaporative condenser, and a potable water cistern. When tested under stringent conditions at 75°C *L. maceachernii* is 3–23% related to other *Legionella* species; it is 14–17% related to *L. jamestowniensis* in reciprocal studies and is 22% related to *L. cherrii* (Table BXII.γ.62).

The mol% G + C of the DNA is: 41 (T_m).

Type strain: PX-1-G2-E2, ATCC 35300.

GenBank accession number (16S rRNA): Z32641.

Additional Remarks: The type species was isolated by George W. Gorman from water from a home evaporative condenser in 1979.

25. ***Legionella micdadei*** Hébert, Steigerwalt and Brenner 1980c, 676^{VP} (Effective publication: Hébert, Steigerwalt and Brenner 1980b, 257) (*Legionella pittsburghensis* Pasculle, Feeley, Gibson, Cordes, Myerowitz, Patton, Gorman, Carmack, Ezzell and Dowling 1980, 731.)

mic.da'de.i. M.L. gen. n. *micdadei* of McDade, named after Joseph E. McDade, who isolated the etiologic agent of the 1976 Legionnaires' disease outbreak in Philadelphia.

Selected characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Causative agent of pneumonia. Originally referred to as the Pittsburgh pneumonia agent. Not acid-fast by the Ziehl-Neelsen method (Hébert et al., 1980d), but may appear acid-fast in tissue preparations stained by other methods (Pasculle et al., 1979). Browning was demonstrated on F-G agar only after some strains were well adapted to and growing well on this medium. This took 12 or more subcultures (Hébert et al., 1980d). Without adaptation or in the absence of heavy growth, no browning occurs on F-G agar (Morris et al., 1980a). Isolated from human lung, blood, soft tissue, and from potable water. DNA relatedness between strains of *L. micdadei* is 90–100%. *L. micdadei* is 1–23% related to other *Legionella* species (Table BXII.γ.62). *L. micdadei* cannot be distinguished reliably from *L. bozemanii*, *L. dumoffii*, or *L. gormanii* by analysis of cellular fatty acids (Karr et al., 1982).

The mol% G + C of the DNA is: 41–44 (T_m).

Type strain: TATLOCK, ATCC 33218.

GenBank accession number (16S rRNA): M36032.

Additional Remarks: The type strain was isolated by H. Tatlock from a guinea pig inoculated with human blood in 1943.

26. ***Legionella moravica*** Wilkinson, Drasar, Thacker, Benson, Schindler, Potuznkiova, Mayberry and Brenner 1989, 205^{VP} (Effective publication: Wilkinson, Drasar, Thacker, Benson, Schindler, Potuznkiova, Mayberry and Brenner 1988a, 399.) *mo.ra'vi.ca.* M.L. gen. n. *moravica* coming from Moravia, Czechoslovakia.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Not demonstrated to cause human disease. Isolated from surface water. *L. moravica* was 1–9% related to 31 other named or described *Legionella* species and 24–36% related to two undescribed species (Table BXII.γ.62).

The mol% G + C of the DNA is: 41.5 (T_m).

Type strain: 316-36, ATCC 43877, CDC 1634-CZK-E.

Additional Remarks: The type strain was isolated from cooling tower water in Moravia, where the species was first isolated; first reported in 1987.

27. **Legionella nautarum** Dennis, Brenner, Thacker, Wait, Vesey, Steigerwalt and Benson 1993, 335.^{VP}
nau.ta'rum. M.L. gen. n. *nautarum* of sailors.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Not demonstrated to cause human disease. Isolated from potable water. Two strains of *L. nautarum* were 70% related to each other at 60°C and 78% at 75°C. Relatedness with other *Legionella* species ranges from 2–24% at 60°C (Table BXII.γ.62).

The mol% G + C of the DNA is: 41 (T_m).

Type strain: 1224, ATCC 49506.

Additional Remarks: The type strain was isolated from a domestic water system in London, England.

28. **Legionella oakridgensis** Orrison, Cherry, Tyndall, Fliermans, Gough, Lambert, McDougal, Bibb and Brenner 1983b, 672^{VP} (Effective publication: Orrison, Cherry, Tyndall, Fliermans, Gough, Lambert, McDougal, Bibb and Brenner 1983a, 544.)
oak.ridge.en'sis. M.L. gen. n. *oakridgensis* named after Oak Ridge, Tennessee, where the species was first isolated.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Causative agent of human pneumonia (Tang et al., 1985). Isolated from human lung (Tang et al., 1985) and from cooling tower water (Orrison et al., 1983a). The DNA relatedness among strains of *L. oakridgensis* is 90–100%; *L. oakridgensis* is 2–22% related to other *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 43–45 (T_m).

Type strain: Oak Ridge 10, ATCC 33761.

GenBank accession number (16S rRNA): X73397, Z32642.

Additional Remarks: The type strain was isolated from cooling tower water by R.L. Tyndall, C.B. Duncan, and E.L. Domingue in 1981.

29. **Legionella parisiensis** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Skaliy, Martin, Brake, Fields, McEachern and Corcoran 1985, 55.^{VP}
pa.ri.si.en'sis. M.L. fem. adj. *parisiensis* coming from Paris, France.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Causative agent of human pneumonia (Lo Presti et al., 1997). Isolated from surface water and human tracheal aspirate. *L. parisiensis* is substantially related to the other *Legionella* species that exhibit blue-white autofluorescence: *L. bozemanii*, *L. dumoffii*, *L. gormanii*, *L. cherrii*, and *L. steigerwaltii*. Of the species that do not exhibit blue-white autofluorescence, *L. parisiensis* is most closely related to *L. wadsworthii* (37%) (Table BXII.γ.62).

The mol% G + C of the DNA is: 42 (T_m).

Type strain: PF-209C-C2, ATCC 35299.

Additional Remarks: The type strain was isolated from cooling tower water by George W. Gorman in 1981.

30. **Legionella quateirensis** Dennis, Brenner, Thacker, Wait, Vesey, Steigerwalt and Benson 1993, 336.^{VP}
qua.tei.ren'sis. M.L. gen. n. *quateirensis* coming from Quateira.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. *L. quateirensis* cannot be distinguished serologically from *L. worsleiensis* or *L. pneumophila* serogroup 4. Not demonstrated to cause human disease. Isolated from potable water. *L. quateirensis* is 2–36% related at 60°C to other *Legionella* species tested. It is most closely related to *L. moravica* (36%) (Table BXII.γ.62).

The mol% G + C of the DNA is: 39 (T_m).

Type strain: 1335, ATCC 49507.

Additional Remarks: The type strain was isolated from a shower in a bathroom of a hotel in Quateira, Portugal, between 1982 and 1984.

31. **Legionella quinlivanii** Benson, Thacker, Waters, Quinlivan, Mayberry, Brenner and Wilkinson 1990b, 105^{VP} (Effective publication: Benson, Thacker, Waters, Quinlivan, Mayberry, Brenner and Wilkinson 1989, 196.)
quin.li.va'd ni.i. N.L. gen. n. *quinlivanii* named after P.A. Quinlivan.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Not demonstrated to cause human disease. There are two serogroups of *L. quinlivanii* (Birtles et al., 1991). Isolated from surface water. The relationship among strains of *L. quinlivanii* is 92–100%. *L. quinlivanii* is 1–30% related to other *Legionella* species; under optimum conditions the most closely related *Legionella* sp. was *L. birming-hamensis* (23–30%) (Table BXII.γ.62).

The mol% G + C of the DNA is: 43 (T_m).

Type strain: 14442-AUSE, ATCC 43830, NCTC 12433.

Additional Remarks: The type strain was isolated from water in Australia in 1986.

32. **Legionella rubrilucens** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Skaliy, Martin, Brake, Fields, McEachern and Corcoran 1985, 55.^{VP}
ru.bri.lu'sens. M.L. gen. n. *rubrilucens* “red shining” after the red autofluorescence exhibited by the species.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. A second serogroup of *L. erythra* cannot be distinguished serologically from *L. rubrilucens* (Table BXII.γ.60) (Saunders et al., 1992). Not yet demonstrated to cause human disease. Isolated from tap water. *L. rubrilucens* and *L. erythra* are 59–62% interrelated, with 9.0–9.5% divergence in related sequences. Relatedness to other *Legionella* species ranges from 3–59%. The most closely related other species was *L. spiritensis* at 34%. At 75°C *L. rubrilucens* and *L. erythra* are 27–34% related (Table BXII.γ.62).

The mol% G + C of the DNA is: 52 (T_m).

Type strain: WA-270A-C2, ATCC 35304, DSMZ 11884.

GenBank accession number (16S rRNA): X73398, Z32643.

Additional Remarks: The type strain was isolated from tap water in Los Angeles, California, by George W. Gorman in 1980.

33. ***Legionella sainthelensi*** Campbell, Bibb, Lambert, Eng, Steigerwalt, Allard, Moss and Brenner 1984b, 355^{VP} (Effective publication: Campbell, Bibb, Lambert, Eng, Steigerwalt, Allard, Moss and Brenner 1984a, 372.)

sainthelensi. M.L. gen. n. *sainthelensi* from Mount St. Helens, where the original isolate was made from surface water in the devastated area after a volcanic eruption.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. The species contains two serogroups (Benson et al., 1990a). Causative agent of human pneumonia (Benson et al., 1990a). Isolated from human respiratory secretions (Benson et al., 1990a) and from surface water (Campbell et al., 1984a). The DNA relatedness among strains of *L. sainthelensi* is 91–100%; the two species most closely related to *L. sainthelensi* are *L. santicrucis* (58% related with a divergence of 9% at 60°C and 11% related at 75°C) and *L. longbeachae* (37% related with 8.5% divergence at 60°C and 4% related at 75°C). *L. spiritensis* was 3–14% related to the other *Legionella* species tested (Table BXII.γ.62).

The mol% G + C of the DNA is: 41 (T_m).

Type strain: Mt. St. Helens 4, ATCC 35248.

GenBank accession number (16S rRNA): X73399.

Additional Remarks: The type strain was isolated from spring water by J. Campbell and S. Eng in 1981.

34. ***Legionella santicrucis*** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Skaliy, Martin, Brake, Fields, McEachern and Corcoran 1985, 58.^{VP}

santicrucis. M.L. gen. n. *santicrucis* named after Santa Cruz, the Latin name for St. Croix, Virgin Islands.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Not demonstrated to cause human disease. Isolated from potable water. *L. santicrucis* is 7–64% related to other *Legionella* species. It is closely related to *L. sainthelensi* (64% related at 60°C and 23% at 75°C) and is 38% related to *L. longbeachae* (Table BXII.γ.62).

The mol% G + C of the DNA is: 38 (T_m).

Type strain: SC-63-C7, ATCC 35301.

Additional Remarks: The type strain was isolated from tap water in St. Croix, Virgin Islands, by George W. Gorman in 1982.

35. ***Legionella shakespearei*** Verma, Brenner, Thacker, Benson, Vesey, Kurtz, Dennis, Steigerwalt, Robinson and Moss 1992, 406.^{VP}

shakespearei. N.L. gen. n. *shakespearei* named after the playwright William Shakespeare, because the organism was isolated in Stratford-upon-Avon.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. The species is distinctive in production of uniformly pink colonies on BCYE agar. Not demonstrated to cause human disease. Isolated from surface water. *L. shakespearei* was 25% related to *L. jordanis* at 60°C. It was <10%

related to 26 other *Legionella* species tested (Table BXII.γ.62).

The mol% G + C of the DNA is: 45.5 (T_m).

Type strain: 214, ATCC 49655.

Additional Remarks: The type strain was isolated from cooling tower water in Stratford-upon-Avon, United Kingdom, reported in 1992.

36. ***Legionella spiritensis*** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Skaliy, Martin, Brake, Fields, McEachern and Corcoran 1985, 55.^{VP}

spiritensis. M.L. fem. adj. *spiritensis* coming from Spirit Lake near Mount St. Helens in Oregon.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Not yet associated with human disease. There are two serogroups of *L. spiritensis* (Harrison et al., 1988). Isolated from surface water. *L. spiritensis* is most closely related (34%) to *L. rubrilucens*, a red fluorescing species. It is 4–34% related to a number of species, including *L. erythra*, another red fluorescing species (Table BXII.γ.62).

The mol% G + C of the DNA is: 46 (T_m).

Type strain: Mt. St. Helens 9, ATCC 35249.

GenBank accession number (16S rRNA): M36030.

Additional Remarks: The type strain was isolated from lake water by J. Campbell in 1981.

37. ***Legionella steigerwaltii*** Brenner, Steigerwalt, Gorman, Wilkinson, Bibb, Hackel, Tyndall, Campbell, Feeley, Thacker, Skaliy, Martin, Brake, Fields, McEachern and Corcoran 1985, 57.^{VP}

steigerwaltii. M.L. gen. n. *steigerwaltii* named in honor of Arnold G. Steigerwalt, who did the definitive experiments to classify the first 22 *Legionella* species.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Not demonstrated to cause human disease. Isolated from potable water. *L. steigerwaltii* is 32–54% related to other species that exhibit blue-white autofluorescence. It is 31–34% related to three species that do not autofluoresce: *L. santicrucis*, *L. wadsworthii*, and *L. sainthelensi*. The relationship with other species is 6–33% (Table BXII.γ.62).

The mol% G + C of the DNA is: 40 (T_m).

Type strain: SC-18-C9, ATCC 35302.

GenBank accession number (16S rRNA): X73400.

Additional Remarks: The type strain was isolated from tap water in St. Croix, Virgin Islands, by George W. Gorman in 1982.

38. ***Legionella taurinensis*** Lo Presti, Riffard, Meugnier, Reyrolle, Lasne, Grimont, Grimont, Vandenesch, Etienne, Fleurette and Freney 1999, 401.^{VP}

taurinensis. M.L. adj. *taurinensis* pertaining to Turin, Italy, whose classical Latin name was Augusta Taurinorum.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Not demonstrated to cause human disease. Isolated from potable and surface water. Strains of *L. taurinensis* were 78–100% related to each other. *L. taurinensis* was >20% related to *L. rubrilucens*, *L. erythra*, and *L. spiritensis*. Its closest relative was *L. rubrilucens*, which was 54% related. In reciprocal reactions *L. taurinensis* was 64% re-

lated to *L. rubrilucens* with 7.5% divergence under optimal conditions tested (Table BXII.γ.62).

The mol% G + C of the DNA is: 46 (T_m , HPLC).

Type strain: Turin 1, ATCC 700508.

GenBank accession number (16S rRNA): AF037597.

Additional Remarks: The type strain was isolated from a humidifier in Turin, Italy.

39. ***Legionella tucsonensis*** Thacker, Benson, Schiffman, Pugh, Steigerwalt, Mayberry, Brenner and Wilkinson 1990b, 105^{VP} (Effective publication: Thacker, Benson, Schiffman, Pugh, Steigerwalt, Mayberry, Brenner and Wilkinson 1989a, 1833.)

tuc.so.nen'sis. M.L. fem. adj. *tucsonensis* from Tucson, Arizona, where the species was first isolated.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Causative agent of human pneumonia (Thacker et al., 1989a). Isolated from pleural fluid. Of the species tested, *L. tucsonensis* was most closely related to the following species that show blue-white autofluorescence: *L. bozemanii*, *L. gormanii*, and *L. parisiensis*. It is only 1–11% related to *L. birminghamensis*, *L. cincinnatiensis*, *L. brunensis*, *L. moravica*, and *L. quinlivanii* (Table BXII.γ.62).

The mol% G + C of the DNA is: 44 (T_m).

Type strain: 1087-AZ-H, ATCC 49180.

GenBank accession number (16S rRNA): Z32644.

Additional Remarks: The type strain was isolated from human pleural fluid in Tucson, Arizona in 1984.

40. ***Legionella wadsworthii*** Edelstein, Brenner, Moss, Steigerwalt, Francis and George 1983, 672^{VP} (Effective publication: Edelstein, Brenner, Moss, Steigerwalt, Francis and George 1982a, 812.)

wads' worthii.i. M.L. gen. n. *wadsworthii* of V.A. Wadsworth Medical Center after James Wadsworth, and the location where the patient was hospitalized and the bacterium isolated.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Causative agent of human pneumonia (Edelstein

et al., 1982a). Isolated from human sputum. *L. wadsworthii* is 3–36% related to other *Legionella* species with a divergence of 9.5–16% at 60°C. When tested under stringent conditions at 75°C *L. wadsworthii* is 2–15% related to other *Legionella* species (Table BXII.γ.62).

The mol% G + C of the DNA is: 42 (T_m).

Type strain: Wadsworth 81-716A, ATCC 33877.

GenBank accession number (16S rRNA): X73401.

Additional Remarks: The type strain was isolated from sputum by Paul H. Edelstein in 1981.

41. ***Legionella waltersii*** Benson, Thacker, Daneshvar and Brenner 1996, 633.^{VP}

wal'ters.i.i. M.L. gen. n. *waltersii* of Walters, to honor R.P. Walters, who isolated the organism.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Not demonstrated to cause human disease. Isolated from potable and surface water. It was <10% related to other *Legionella* spp. tested.

The mol% G + C of the DNA is: unknown.

Type strain: 2074-AUS-E, ATCC 51914.

Additional Remarks: The type strain was isolated from a drinking water distribution system in South Australia by R.P. Waters in 1987–88.

42. ***Legionella worsleiensis*** Dennis, Brenner, Thacker, Wait, Vese, Steigerwalt and Benson 1993, 336.^{VP}

wors.lei.en'sis. M.L. gen. n. *worsleiensis* coming from Worsley.

The characteristics are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. It cannot be distinguished serologically from *L. pneumophila* serogroup 4 or from *L. quateirensis*. Not demonstrated to cause human disease. Isolated from surface water. *L. worsleiensis* is 1–24% related to other *Legionella* species tested at 60°C (Table BXII.γ.62).

The mol% G + C of the DNA is: 41 (T_m).

Type strain: 1347, ATCC 49508.

Additional Remarks: The type strain was isolated from cooling tower return flow at an industrial site in Worsley, United Kingdom, between 1982 and 1984.

Other Organisms

1. *Legionella* genomospecies 1 Benson, Thacker, Daneshvar, and Brenner 1996.

This genetically distinct organism was reported as a genomospecies, because it could not be distinguished biochemically or serologically from *L. quinlivanii*, serogroup 2. The characteristics of the genomospecies are as described for the genus and as listed in Tables BXII.γ.56, BXII.γ.57, BXII.γ.58, and BXII.γ.59. Serologic cross-reactions are detailed in Table BXII.γ.60. Not demonstrated to cause human disease. Isolated from surface water. The relationship between *Legionella* genomospecies 1 and *L. quinlivanii* in reciprocal reactions was 53–69% with 4.5–6.5% under optimal conditions and 31–51% under stringent conditions. In the same laboratory the relationship between strains of *L. quinlivanii* was 79–99% with 0.5–3.0% under optimal con-

ditions and 85–100% under stringent conditions is unknown. It was <10% related to other *Legionella* spp. tested (Table BXII.γ.62).

The mol% G + C of the DNA is: unknown.

Deposited strain: 2055-AUS-E, ATCC 51913.

Additional Remarks: The type strain was isolated from cooling tower water in South Australia, Australia.

Bacterial isolates recovered by conventional techniques may represent additional, as yet unnamed *Legionella* species (Wilkinson et al., 1990; Benson et al., 1996). Strains that have been described as probably new species include LC878, "*L. donaldsonii*" (Riffard et al., 1998i) and strain "glasgow" (Hookey et al., 1996). At least six genomospecies distinct from all named species except *L. taurinensis* have been characterized (D.J. Brenner, personal communication).

Family II. **Coxiellaceae** *fam. nov.*

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Co.xi.el.la'ce.ae. M.L. fem. n. *Coxiella* type genus of the family; *-aceae* ending to denote family;
M.L. fem. pl. n. *Coxiellaceae* the *Coxiella* family.

The family *Coxiellaceae* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rRNA sequences; the family contains the genera *Coxiella* (type genus) and *Rickettsiella*.

Found only in vacuoles in cells of invertebrate or vertebrate hosts. Serial passage in cell-free medium has not been achieved.

Type genus: **Coxiella** (Philip 1943) Philip 1948, 58^{AL} (**Rickettsia** (**Coxiella**) **burnetii** Philip 1943, 306.)

Genus I. *Coxiella* (Philip 1943) Philip 1948, 58^{AL} (*Rickettsia* (*Coxiella*) *burnetii* Philip 1943, 306)

MICHEL DRANCOURT AND DIDIER RAOULT

Co.xi.el'la. M.L. fem. dim. *-ella* ending; M.L. fem. n. *Coxiella* named after Harold R. Cox, who in collaboration with G.E. Davis, first isolated this organism in the United States shortly after its discovery in Australia and who introduced the technique of yolk sac inoculation of the chick embryo, which greatly facilitated the study of this and other genera.

Strictly intracellular bacteria, usually 0.2–0.4 μm $\times$ 0.4–1.0 μm . Best stained by **Gimenez staining**. They have no flagella or capsule. They live in close natural association with **arthropod and vertebrate hosts**. The genus includes *Coxiella burnetii*—the **agent of Q fever**—and endosymbionts of ticks and aquatic invertebrates. *C. burnetii* grows well in cultured cell lines and in the yolk sac of chicken embryos, where it undergoes a cycle of development with formation of an endospore-like body. *Coxiella burnetii* possesses **two antigenic phases**: natural virulent phase I and attenuated phase II, which is obtained after passage in cultured cell lines and yolk sac. The organism **grows in vacuoles of the host cell** rather than in the cytoplasm or the nucleus. *C. burnetii* phase I vacuoles share characteristics of late-stage, immature phagosomes whereas *C. burnetii* phase II vacuoles are bactericidal phagolysosomes. Highly resistant to chemical agents and elevated temperature. Coxiellae occur worldwide in ticks and various vertebrates, including humans. Infection is particularly prevalent in cattle, sheep and goats. Extremely infectious; a single aerosol-borne organism can cause the human disease Q fever. *C. burnetii* is one of the ten most feared potential **bioterrorism agents**.

The mol% G + C of the DNA is: 42.7.

Type species: **Coxiella burnetii** (Derrick 1939) Philip 1948, 58^{AL} (*Rickettsia burnetii* (sic) Derrick 1939, 14.)

FURTHER DESCRIPTIVE INFORMATION

Coxiellae are generally regarded as Gram-negative rods, but they are best stained by Gimenez staining (Gimenez, 1964) (Fig.BXII.γ.97). They have no flagella or capsule and are not always retained by ordinary bacteriological filters. Coxiellae have never been cultured axenically in laboratory media, although outside the host, the organisms metabolize glutamate, glucose and other substrates if the pH is low. Electron microscopy of bacteria separated from host cell components reveals great variation in morphology.

Cell morphology Two forms of *C. burnetii* cells are found in infected eucaryotic cells: a metabolically inactive sporelike “small-cell variant” (SCV; 204–450 nm) and a “large-cell variant” (LCV;

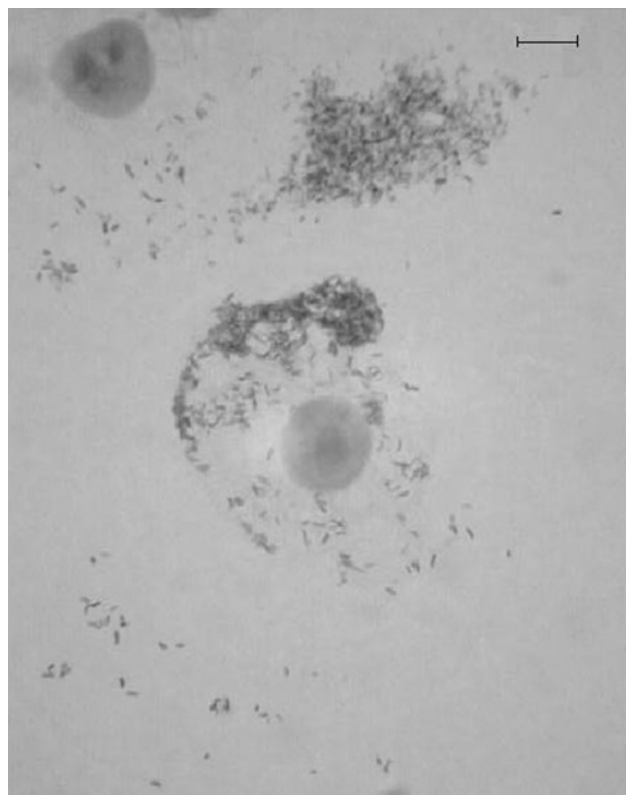


FIGURE BXII.γ.97. Gimenez-stained *Coxiella burnetii* inside a large vacuole of a L 929 cell. Bar = 5 μm .

up to 2 μm in length). SCVs develop from LCVs following an asymmetric cell division (McCaul, 1991) and are distinguished from LCVs by having a regular rod shape, a dense layer of peptidoglycan and protein between the two layers of the cell envelope, and condensed nucleoids. The more pleomorphic LCVs possess dispersed nucleoids and granules or sometimes fibrils in their cytoplasm. Condensation or dispersal of the nucleoid may be associated with the presence in SCVs of a 20-kDa protein—Hq1—that binds DNA; Hq1 is a homolog of histone H1 (Heinzen and Hackstadt, 1996).

SCVs also possess greater resistance than LCVs to drying, UV irradiation, acid or alkaline pH, disinfectants, and other chemicals (Ranson and Huebner, 1951; Babudieri, 1959; Maurin and Raoult, 1999). At 4°C, viability of SCVs is retained for 1 year or more in dried fomites such as tick feces and wool, as well as in sterile skim milk or unchlorinated water. Meats remain infected for at least 1 month. Complete inactivation is not always accomplished by exposure to 63°C for 30 min or 85–90°C for a few seconds. *C. burnetii* is rapidly inactivated by diethyl ether but not by ethanol (Ranson and Huebner, 1951; Babudieri, 1959).

Thus, the morphogenesis and properties of SCVs are comparable, although not identical, to cycles of cell differentiation and formation of resistant, inactive cell forms in other bacteria (McCaul and Williams, 1991). SCVs are released from host cells and bind to the membranes of phagocytic cells; hence, SCVs appear to be a resistant stage adapted to survive transmission between individual hosts. In addition, *C. burnetii* is highly infectious: a single organism can cause aerosol-borne disease. It has therefore been considered a potential weapon for biological warfare (Lederberg, 2000) and has been classified as a category B biological weapon by the Centers for Disease Control and Prevention, Atlanta, GA.

Antigenic structure Repeated serial passage (10–100 times) in cultured eucaryotic cells or in embryonated eggs produces attenuated “rough” or Phase II variants of naturally-occurring virulent “smooth” or Phase I *C. burnetii*. This phenomenon resembles the production of rough variants of smooth strains of enterobacteria (Stoker and Fiset, 1956; Brezina, 1978) and involves mutation in the genes encoding enzymes that synthesize the lipopolysaccharide (LPS) (Hackstadt et al., 1985; Hackstadt, 1988). Phase II *C. burnetii* strains exhibit changes in both the sugar composition and the length of the LPS (Amano and Williams, 1984). The sugars dihydrohydroxytreptose, galactosamine uronyl- α -(1,6)-glucosamine and L-virenose are present in Phase I but not Phase II LPS (Schramek and Mayer, 1982; Schramek et al., 1985; Amano et al., 1987; Hackstadt, 1990). The protein complement of the outer membrane also changes during the transition from Phase I to Phase II (Amano and Williams, 1984). Vodkin and Williams (1986) demonstrated large deletions in the genome of the Phase II Nine Mile strain.

Genetics The genome of *C. burnetii* consists of a chromosome plus a facultative 36–42-Kb plasmid whose function remains undetermined. The genome of the Nine Mile strain of *C. burnetii* is 1.99 Mb based on the complete genome sequence (GenBank accession number NC002971), but the genome size varies from 1.5 to 2.4 Mb among different *C. burnetii* strains (Willems et al., 1998). In addition, complete genome sequencing confirmed that the chromosome is circular, contrary to previous speculations of a linear geometry (Willems et al., 1996; 1998; Dr. J.E. Samuel,

personal communication). The *C. burnetii* Nine Mile strain genome comprises 6653 open reading frames (ORFs) that code for peptides of ≥ 50 residues. A physical macrorestriction map of *C. burnetii* Nine Mile phase I was constructed by Willems et al. (1996; 1998); 25 DNA fragments were distinguished by pulsed field gel electrophoresis (PFGE) after digestion of total DNA with *NotI*. Vodkin and Williams (1988) and Hoover and Williams (1990) have elucidated operon structures in *C. burnetii*.

Fifteen *C. burnetii* genes have been cloned to date in *Escherichia coli*, including genes involved in metabolism (*gltA* (Heinzen and Mallavia, 1987), *pyrB* (Hoover and Williams, 1990), and *icd* (Nguyen and Hirai, 1999)), transcription and translation (*rpoD* and *rpoS* (Seshadri and Samuel, 2001) and *serS* (Willems et al., 1996)); stress responses, (*sodB* (Heinzen et al., 1992), *htpA* and *htpB* (Vodkin and Williams, 1988), and *dnaJ* (Zuber et al., 1995a)), production of surface antigens (*omp* (Hendrix et al., 1990), *mucZ* (Zuber et al., 1995b), and *algC* (Willems et al., 1996)), and environmental sensing (*qrsA* (Mo and Mallavia, 1994)). In addition, a possible chromosomal origin of replication (*oriC*) has been cloned (Chen et al., 1990a), but Suhan et al. (1996) were unable to demonstrate the ability of this DNA fragment to initiate DNA replication in *C. burnetii*.

Plasmid DNA can be introduced into *C. burnetii* by electroporation; plasmids bearing both antibiotic resistance determinants (Suhan et al., 1994; 1996) and a green fluorescent protein gene (Lukacova et al., 1999) have been introduced and the genes expressed.

Four plasmids native to *C. burnetii* have been characterized: QpH1 (Samuel et al., 1983), QpRS (Samuel et al., 1985), QpDG (Mallavia, 1991), and QpDV (Valkova and Kazar, 1995). These plasmids are 33–42 Kb in size and share 30 kb of homology with each other (Minnick et al., 1990; Mallavia, 1991). The chromosomal DNAs of some plasmid-free genomic group V *C. burnetii* strains contain a region of homology to the DNA of plasmid QpRS (Savinelli and Mallavia, 1990); this region is derived from plasmid sequences (Lautenschlager et al., 2000). The entire DNA sequences of plasmids QpH1 and QpDV have been obtained (Thiele et al., 1994, GenBank accession number NC002118; and Valkova and Kazar (1995), GenBank accession number NC002131; respectively).

DNA–DNA hybridization studies showed that *C. burnetii* strains are homogenous (Vodkin and Williams, 1986). Early studies of genetic differences among strains revealed by RFLP analysis (Hendrix et al., 1991), plasmid analysis (Samuel et al., 1983, 1985; Mallavia, 1991), and immunoblotting of LPS (Hackstadt, 1986) suggested that six RFLP-defined groups differed in their association with “acute” or “chronic” cases of Q fever. However, recent studies on larger numbers of strains from a variety of locations and sources using improved techniques—genomic RFLP (Thiele and Willems, 1994; Jäger et al., 1998), PCR-based plasmid analysis (Stein and Raoult, 1993), immunoblotting of LPS with specific monoclonal antibodies (Yu and Raoult, 1994), and gene sequence analysis (Nguyen and Hirai, 1999; Sekeyova et al., 1999)—have shown that genetic variation in this pathogen is geographic in origin. (See also Thiele et al. (1993) and Willems et al. (1996).) Moreover, variation in host factors is more likely to explain whether individual cases of Q fever are “acute” or “chronic” than is genetic variation in the pathogen (La Scola et al., 1998b; Maurin and Raoult, 1999; Raoult et al., 2000; Fenollar et al., 2001).

Epidemiology In humans, contact between *C. burnetii* and a nonimmune individual results in a primary infection which can be either asymptomatic or symptomatic. The symptomatic primary infection is acute Q fever, which is usually followed by complete recovery of a normal host. In some hosts, *C. burnetii* can multiply despite an antibody response following primary infection, symptomatic or not. In these cases, chronic infection develops because the immune system is unable to control the infection. This distinction between acute and chronic infection is based on clinical expression, temporal course, and the serological profile of the patient, and is mainly determined by host factors. This hypothesis is supported by all available data from humans as well as from animal models.

The aerosol route is the principal mode of acquisition of *C. burnetii* infection (Babudieri, 1959; Dupuis et al., 1987; Tissot-Dupont et al., 1999). *C. burnetii* is highly infectious, and a single organism can cause the aerosol-borne Q fever. This fact and the great resistance of *C. burnetii* to environmental stresses have led to the classification of *C. burnetii* as a group B bioterrorism agent. Indeed, due to its sporelike form, *C. burnetii* may remain viable in soil and milk for several months following contamination. Cattle, sheep, and goats are the main reservoirs of *C. burnetii* infection for humans. High densities of *C. burnetii* are found in the placenta of infected parturient animals and are shed in the environment following labor. In Europe, most cases occur between January and June because of the heavily contaminated environment following lambing season. Cases among students following visits to farms as part of school activities and outbreaks following exposure to birth products of cats, dogs, and rabbits, have been reported. *C. burnetii* can also be transmitted by ingestion of unpasteurized dairy products (Fishbein and Raoult, 1992). Recently, cases were linked to ingestion of pasteurized milk in Canada (Hatchette et al., 2001). Human cases have been reported following contact with parturient products of an infected woman (in an obstetrician) and through transplacental, blood, and sexual transmission (Maurin and Raoult, 1999). Based on the results of animal experimentation (La Scola et al., 1997), it has been hypothesized that the route of acquisition of *C. burnetii* infection may influence the clinical presentation of the disease; i.e., infection via an aerosol results in pneumonia, whereas infection via intraperitoneal injection results in hepatitis.

Q fever has been described worldwide except in New Zealand. Estimates of the incidence of Q fever are complicated by the lack of systematic investigation of this illness worldwide. In regions of France where Q fever is endemic, the annual incidence of acute Q fever is estimated at 50/100,000 persons (Maurin and Raoult, 1999). The incidence of hospital admission for acute and chronic Q fever is estimated at 1.7 and 0.5/1,000,000 persons, respectively. Comparable figures have been reported from England.

Pathogenicity in humans In Q fever outbreaks, there is a high ratio of males to females in infected patients and a low ratio of children to adults (Maltezou and Raoult, 2002). Because the levels of exposure and seroprevalence are not different in these groups, it is suspected that host factors determine different clinical expression of acute disease. The incubation period for acute Q fever is 1–3 weeks. Data from the National Reference Center for Rickettsial Diseases, Marseille, France, and from large outbreaks show that up to 60% of infected patients remain asymptomatic. Among symptomatic patients, most (95%) develop a mild, non-specific flu-like illness, whereas the remaining 5% re-

quire hospitalization. Chronic Q fever accounts for about 10% of hospital cases and 0.2% of all cases (Dupuis et al., 1987; Maurin and Raoult, 1999). Acute Q fever has two primary clinical presentations: atypical pneumonia and hepatitis. Most infected patients experience a transient bacteremia with *C. burnetii*, usually late in the incubation period. The spread of the organisms to other organs can result in severe complications; endocarditis is the primary manifestation of chronic Q fever. Other aspects of the pathogenicity of *C. burnetii* in humans, the clinical aspects of Q fever, and current recommendations for treatment of Q fever have been reviewed in detail by Maurin and Raoult, 1999.

Animal infection and reservoirs Livestock—including cattle, sheep and goats—and pets are major reservoirs of *C. burnetii* (Langley et al., 1988; Laughlin et al., 1991; Buhariwalla et al., 1996). The pathogen can be found in urine, feces, milk, and birth products of these animals and transmitted to humans by these substances as well as by aerosols. Animals are usually not harmed by *C. burnetii*, although association of the organisms with infertility (Ho et al., 1995), abortion (Waldhalm et al., 1978), low birth weight (Ho et al., 1995), and death of newborns (Buhariwalla et al., 1996) in animal hosts has been reported. Serological studies of cattle herds have shown that large proportions of seronegative animals brought into a region where *C. burnetii* is endemic are exposed to *C. burnetii* (Huebner and Bell, 1951) and that the incidence of exposure may be increasing (Krauss, 1989). Some animals, including cattle and goats, may become chronically infected (Babudieri, 1959; Baca and Paretsky, 1983; Lang, 1990). There is a lack of current data on *C. burnetii* infection in livestock.

The pathogenicity of *C. burnetii* in various laboratory animals has been reviewed by Maurin and Raoult, 1999. Rats, mice, guinea pigs, rabbits, and monkeys have been used in laboratory studies; infection can produce no symptoms, can produce fever and granulomas, or can lead to death.

Interactions with host cells *C. burnetii* infects mammalian monocytes and macrophages *in vivo* (Dellacasagrande et al., 2000); the organism can be grown in embryonated eggs and in Vero, fibroblast, and mouse macrophagelike cells *in vitro*. *In vitro* cultures of *C. burnetii*-infected cells can be maintained for long periods (Burton et al., 1978; Baca et al., 1981, 1985; Akporiaye et al., 1983; Roman et al., 1986). Both the mechanisms of entry of Phase I and Phase II *C. burnetii* cells into mammalian cells and the adaptations that promote long-term survival and reproduction of *C. burnetii* cells inside mammalian cells specialized to kill them have been investigated extensively (Hackstadt and Williams, 1981; Akporiaye et al., 1983; Zuerner and Thompson, 1983; Hendrix and Mallavia, 1984; Roman et al., 1986; Chen et al., 1990b; Raoult et al., 1990; Maurin et al., 1992; Ghigo et al., 2002).

Antibiotic susceptibility Antibiotic susceptibility testing of *C. burnetii* has been carried out in cell cultures, in animals, and in embryonated eggs. These methods and the results obtained have been reviewed in detail by Maurin and Raoult (1999).

Vaccination Vaccination is a logical strategy for the prevention of Q fever both among humans at high risk for this infection and among animals. A killed Q fever vaccine was approved for use in Australia (O-Vax; Commonwealth Serum Laboratories). Further information about the effectiveness of this vaccine can be found in Ormsbee and Marmion (1990). In areas where Q fever is endemic, the following groups should be offered vacci-

nation: dairy workers, abattoir workers; and others who work with cattle, sheep, or goats. Vaccination of cattle, sheep, and goats in these areas is also indicated, although few data are available to indicate the effectiveness and cost benefit of such a strategy in eliminating Q fever.

ENRICHMENT AND ISOLATION PROCEDURES

Isolation of *C. burnetii* requires a Biosafety Level 3 facility and poses a substantial risk to laboratory personnel. In the past, isolation was achieved by injecting potentially infected specimens into guinea pigs, mice, or embryonated eggs (Ormsbee, 1952; Perrin and Bengtson, 1942; Williams et al., 1986a); organisms are most readily recovered from the spleens of guinea pigs 5–8 days after infection. Tissue culture cells can be substituted for animals (Maurin and Raoult, 1999).

MAINTENANCE PROCEDURES

C. burnetii is maintained in liquid nitrogen but can be more conveniently maintained at -80°C . Phase I organisms are obtained by experimental passage in laboratory animals (guinea pig). Phase II organisms are obtained by passage in cultured cell systems or embryonated eggs.

DIFFERENTIATION OF THE GENUS *COXIELLA* FROM OTHER GENERA

Coxiella can be differentiated from *Rickettsiella* by its relatively high heat resistance and by its formation of an endospore-like structure. Although electron-dense forms have been observed in the developmental cycles of both *Rickettsiella* and *Coxiella burnetii* (Avakyan et al., 1983; Popov et al., 1991), *Rickettsiella* species do not develop intracellularly into larger particles that multiply and then condense to reform the smaller forms in a cycle that resembles that of *Chlamydia*.

TAXONOMIC COMMENTS

Phylogenetic analysis, chiefly based on 16S rRNA gene sequences, has shown that *Coxiella* belongs to the *Gamma*proteobacteria (Weisburg et al., 1985, 1989; Wilson et al., 1989b; Stein et al., 1993), with *Legionella*, *Francisella*, and *Rickettsiella* as its closest relatives (Fig. BXII.γ.98). Bacteria of the genus *Rickettsia* and of the family *Bartonellaceae* belong to the *Alphaproteobacteria* (Fig. BXII.γ.99). In addition, uncultured endosymbionts of ticks as diverse as *Rhipicephalus sanguineus*, *Haemaphysalis longicornis*, *Haemaphysalis con-*

cinnae, and *Ornithodoros moubata* are closely related to *C. burnetii* by 16S rRNA gene sequence analysis (Noda et al., 1997). The 16S rDNA sequence of one rickettsia-like organism isolated in the yolk sac of chicken embryos using material from farm-reared *Cherax quadricarinatus* crayfish possessed 95.6% similarity to the *C. burnetii* 16S rRNA gene sequence; this organism has been proposed as *Coxiella cheraxi* sp. nov. (Tan and Owens, 2000) (Fig. BXII.γ.100), but the new name has not yet been validated.



FIGURE BXII.γ.98. Transmission electron micrograph of *Coxiella burnetii* inside a vacuole of a Vero cell.

List of species of the genus *Coxiella*

1. ***Coxiella burnetii*** (Derrick 1939) Philip 1948, 58^{AL} (*Rickettsia burneti* (sic) Derrick 1939, 14.)
burne' ti. i. M.L. gen. n. *burnetii* of Burnet, named after Frank MacFarlane Burnet, who first studied the properties of this organism.

The characteristics are as described for the genus. The complete genome sequence of strain RSA 493 is available (GenBank accession number NC002971).

The mol% G + C of the DNA is: 42.6 (by sequencing).

Type strain: ATCC VR 615 strain (Nine Mile Phase I).

Reference strains include: (1) ATCC V5-145 [strain Henzerling; essentially in phase II]; (2) VR 616 [strain Nine Mile Q (Davis and Cox, 1938), essentially Phase II]; (3) VR 147 Strain Dyer, from blood from patient, USPHS Rocky Mountain Laboratory, 1938; (4) VR 542 Strain Ohio 314, from milk from cow; collected by Ohio Dept. Hlth., 1955; (5) VR 614 Strain California 76, from a dairy cow in California; and (6) VR 730 Strain Bangui, from a human in Central Africa.

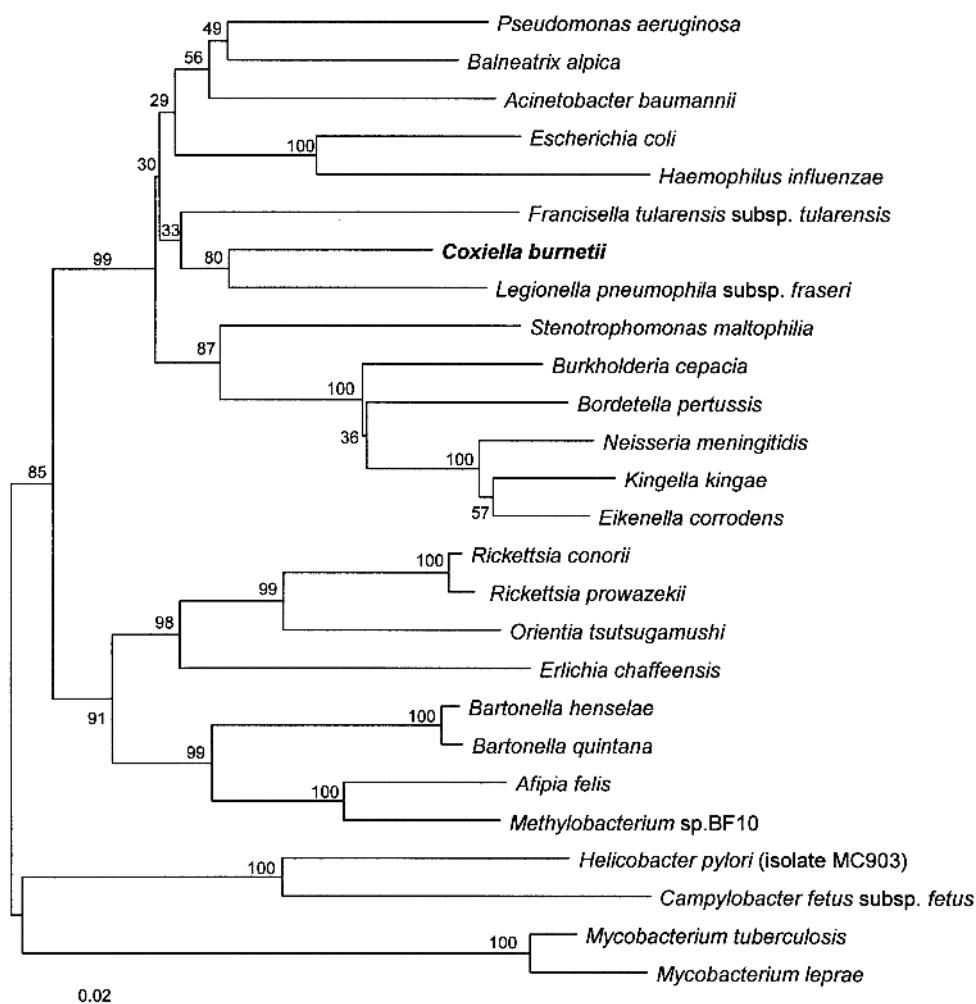


FIGURE BXII-γ.99. Dendrogram based on 16S rRNA gene sequence comparison of *Coxiella burnetii* with other *Proteobacteria*. This dendrogram includes *Coxiella* sequences derived from: *Coxiella burnetii* (strain 607, taxon 777), GenBank accession number D89792; *Francisella tularensis* (strain SCHU, isolate FSC 043, taxon:263), GenBank accession number Z21932; *Rickettsia conorii* (taxon 781), GenBank accession number L36107; *Rickettsia prowazekii* (taxon 782), GenBank accession number M21789; *Bartonella henselae* (strain BA-TE, taxon:38323, clone PL 84-1), GenBank accession number Z11684; *Bartonella quintana* (taxon 803), GenBank accession number AJ250247; *Mycobacterium tuberculosis* (taxon 1773), GenBank accession number X52917; *Mycobacterium leprae* (taxon 1769), GenBank accession number X55022; *Pseudomonas aeruginosa* (taxon 287), GenBank accession number M34133; *Haemophilus influenzae* (strain NCTC 11146, taxon:67859), GenBank accession number AF024530; *Orientia tsutsugamushi* (taxon 784), GenBank accession number AF062074; *Ehrlichia chaffeensis* (taxon 945), GenBank accession number AF147752; *Afipia felis* (taxon 1035), GenBank accession number M65248; *Methylobacterium* sp. BF10 (taxon 194432), GenBank accession number Z23156; *Bordetella pertussis* (ATCC 9797, taxon 520), GenBank accession number U094950; *Burkholderia cepacia* (taxon 292), GenBank accession number AB091761; *Kingella kingae* (ATCC 23330, taxon 504), GenBank accession number M22517; *Neisseria meningitidis* (strain NCTC 10025^T, taxon 487), GenBank accession number X74900; *Helicobacter pylori* (isolate MC903, taxon 210), GenBank accession number U01332; *Campylobacter fetus* (taxon 196), GenBank accession number M65012; *Stenotrophomonas maltophilia* (strain N4-15, taxon:40324), GenBank accession number AF017749; *Legionella pneumophila* (strain Los Angeles 1, taxon:44), GenBank accession number M36025; *Balneatrix alpica* (taxon:75684), GenBank accession number Y17112; *Acinetobacter baumannii* (isolate DSM 30008, taxon 470), GenBank accession number X81667; *Escherichia coli* (taxon 562), GenBank accession number J01859; *Eikenella corrodens* (taxon 539), GenBank accession number M22513.

Genus II. *Rickettsiella* Philip 1956, 267^{AL}

PIERRE-EDOUARD FOURNIER AND DIDIER RAOULT

Rick.ett.si.el'la. M.L. dim. -ella ending; M.L. fem. n. *Rickettsia* genus of parasitic bacteria; M.L. fem. n. *Rickettsiella* small *Rickettsia*.

Rod- or disk-shaped organisms, developing intracellularly into larger particles that multiply and then condense to reform the smaller forms in a cycle that resembles that of *Chlamydia*. Sometimes produce or induce the formation of large crystalline bodies. Gram negative. Grow in cell vacuoles of the fat body, he-

patopancreas, and other organs of insects and other invertebrate hosts. In some instances the organisms have been cultivated in invertebrates other than the host of origin, but such cultivation has occurred rarely and only for a few passages in vertebrate and invertebrate cell cultures and in a cell-free medium. Pathogenic

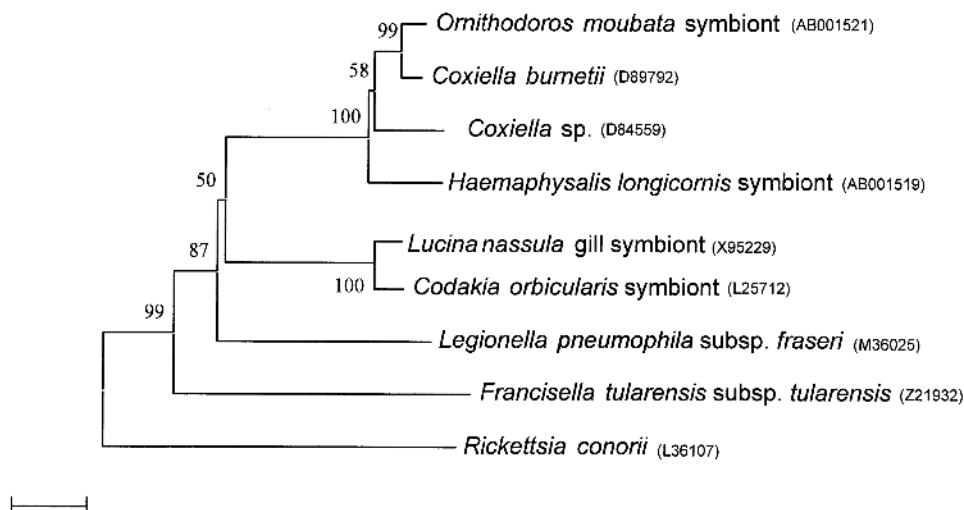


FIGURE BXII.γ.100. Dendrogram based on 16S rRNA gene sequence comparison of *Coxiella burnetii* to its close relatives. *Rickettsia conorii* (part of the *Alphaproteobacteria*) was used as an outgroup to root the tree. Bootstrap values are indicated for each node. Bar = 2 inferred nucleotide substitutions per 100 nucleotides.

for the larvae of the host species and for young and mature stages of other invertebrate hosts. Pathogenicity for vertebrate hosts demonstrated only in a mouse experimental model. Natural hosts include insects, crustaceans, and arachnids.

The mol% G + C of the DNA is: 36–41.

Type species: *Rickettsiella popilliae* (Dutky and Gooden 1952) Philip 1956, 267 (*Coxiella popilliae* Dutky and Gooden 1952, 749)

FURTHER DESCRIPTIVE INFORMATION

Most of the knowledge of this genus is based on light and electron microscopic observations. Bacteria were classified in the genus *Rickettsiella* mainly based on morphologic criteria, including their intracellular location, their oval, rod-like, or pleomorphic forms, the occurrence of a complex intravacuolar cycle, and the occurrence of crystals. Studies of host specificity, attempts at cultivation in cells or axenic media, or antigenic analyses have not been extensive and only minimally successful. Table BXII.γ.63 lists the species that, to the best of our limited knowledge, appear to belong to this genus.

Phylogenetic classification In the previous edition of the *Manual*, the genus *Rickettsiella* was classified in the order *Rickettsiales* based on a very few morphological data, including a close association with arthropods and intravacuolar location during multiplication. Recently, analysis of the 16S rDNA sequence of *Rickettsiella grylli* has revealed that this bacterium is most closely related to *Coxiella burnetii* (Philip, 1943) in the *Gamma*proteobacteria; therefore, it has been provisionally removed from the order *Rickettsiales* (Roux et al., 1997). An analysis of the 16S rDNA sequences from another two rickettsiellae—isolates from the psyllid *Cecidotrioza sozanica* (Insecta, Psylloidea) (Spaulding and Hennessy, 1960) and from the tick *Ixodes woodii* (Arachnida, Acarina) (Kurti et al., 2002)—has demonstrated that these organisms cluster with *R. grylli* with a high statistical probability (Fig. BXII.γ.101). The 16S rDNA sequence of *R. grylli* has 98.2% and 94.3% similarity to that of the psyllid and the tick isolates, respectively, and 88.3% similarity to that of *Coxiella burnetii*. However, the 16S rDNA sequences of *Rickettsiella popilliae* and *Rick-*

ettsiella chironomi are not available, and thus their phylogenetic position is not clear.

Cell structure All strains studied in some detail have been shown to undergo a cycle of development. The infectious form is a small, dense particle (elementary body) that gains entrance into a vacuole of its host cell by phagocytosis. There it enlarges into intermediate and large forms of much lower electron density and which are capable of multiplication (reticulate body). As the host-cell vacuole becomes crowded with multiplying bacteria, the large forms condense to reform the small dense particles, which eventually escape from the vacuole to start another cycle.

There are striking differences between the cycles of *R. popilliae* and *R. chironomi*. The cycle of *R. grylli* resembles that of *R. popilliae*. *R. popilliae* cells, although displaying some plasticity, retain typical outer membranes and a rod-shaped appearance through most of their cycle (Devauchelle et al., 1972). Some large forms become giant round cells from which bipyramidal crystalline bodies ($0.8\text{--}1.8 \times 3.8 \mu\text{m}$ in size) arise. The roles of the bacteria and of the host cell in the production of the crystalline bodies are not entirely clear. Except for the absence of tyrosine, these bodies have approximately the same amino acid composition as the albuminoid spheres that are found in greatest numbers in late larval and pupal stages in normal insects. It has been postulated that the crystalline bodies are derived from the albuminoid reserve as the result of a disturbance of host cell metabolism (Huger, 1959; Krieg, 1959). Other evidence suggests a close association of the crystalline bodies with the bacterial cycle (Huger, 1962; Devauchelle et al., 1972) and host dependence. A *Cetonia* strain (Table BXII.γ.63) produces crystalline bodies in *Coleoptera* but not in *Orthoptera* or *Lepidoptera* (Meynadier and Monsarrat, 1969), and a *Melolontha* strain does not produce crystalline bodies in mammalian cell cultures (Pourquier et al., 1963).

In *R. chironomi*, the infectious particle is a disk-shaped elementary body $0.06 \times 0.6 \mu\text{m}$ in size. This body enlarges in a newly infected cell into a spherical initial body $1 \mu\text{m}$ in diameter,

TABLE BXII-7.63. Ecology of the species of the genus *Rickettsiella* and proposed nomenclature

Nomenclature used in the present edition of the <i>Manual</i>	Host zoological class	Host zoological order	Host common name (scientific name)	Location	Subjective synonyms	References
<i>R. popilliae</i>	<i>Insecta</i>	<i>Coleoptera</i>	Japanese beetle (<i>Popillia japonica</i>)	USA	" <i>Rickettsiella popilliae</i> "	Dutky and Gooden, 1952; Philip, 1956
			Other scarabid and carabid beetles	USA, UK	Unnamed	Sutter and Kirk, 1968; Carter and Luff, 1977
			Cockchafer (<i>Melolontha melolontha</i>) ^a	Germany	" <i>Rickettsiella melolonthae</i> " ("Rickettsia melolonthae")	Krieg, 1955a; Philip, 1956
		<i>Diptera</i> <i>Dicolyptera</i> <i>Isopoda</i>	Mealworm (<i>Tenebrio molitor</i>)	Germany	" <i>Rickettsiella tenebrionis</i> "	Krieg, 1965
			Cetonid beetle (<i>Cetonia</i> sp.)	Madagascar	" <i>Rickettsiella cetonidarum</i> "	Meynadier and Monsarrat, 1969
<i>R. grylli</i>	<i>Crustacea</i> <i>Insecta</i>	<i>Dicolyptera</i> <i>Isopoda</i> <i>Orthoptera</i>	Crane fly (<i>Tipula paludosa</i>)	Germany	" <i>Rickettsiella tipulae</i> "	Müller-Kogler, 1958
			Cockroach (<i>Blattia orientalis</i>)	Germany	" <i>Rickettsiella blattae</i> "	Huger, 1964
			Isopode (<i>Armadillidium vulgare</i>) ^a	France	" <i>Rickettsiella armadillidii</i> "	Vago et al., 1970
	<i>Crustacea</i> <i>Insecta</i>	<i>Orthoptera</i> <i>Amphipoda</i>	Cricket (<i>Gryllus binaculatus</i>)	France	<i>Rickettsiella grylli</i>	Vago and Martoja, 1963
			Desert locust (<i>Schistocerca gregaria</i>)	Jordan	" <i>Rickettsiella schistocerae</i> "	Vago and Meynadier, 1965
<i>R. chironomi</i>	<i>Crustacea</i> <i>Insecta</i> <i>Arachnida</i>	<i>Amphipoda</i> <i>Diptera</i> <i>Aranea</i>	Amphipod (<i>Crangonyx floridanus</i>)	USA	Unnamed	Federici et al., 1974
			Midge (<i>Chironomus tentans</i>) ^a	Germany	<i>Rickettsiella chironomi</i>	Weiser, 1963
			Spider (<i>Argrodes gibbosus</i>)	Spain (Canary islands)	Unnamed	Meynadier et al., 1974
	<i>Insecta</i>	<i>Coleoptera</i>	Spider (<i>Pisaura mirabilis</i>)	France	Unnamed	Morel, 1977
			Stethorus beetle (<i>Stethorus punctum</i>)	Morocco	<i>Rickettsiella stethorae</i>	Hall and Badgley, 1957
Organisms for which insufficient data exist to support assignment to a species	<i>Lepidoptera</i>	<i>Lepidoptera</i>	Sturnid moth (<i>Samia cynthia</i>)	UK	Unnamed	Entwistle et al., 1968
			Navel orange worm (<i>Paratyelois transitella</i>)	USA	Unnamed	Kellen et al., 1972
			Psyllid (<i>Cecidotrioza sozanica</i>)	USA	Unnamed	Spaulding and von Dohlen, 2001
			Cockroach (<i>Blattia orientalis</i>)	Germany	" <i>Rickettsiella crassifrons</i> "	Radek, 2000
			Mite (<i>Phytoseiulus persimilis</i>)	USSR	" <i>Rickettsiella phytoseiuli</i> "	Sutakova and Rutigen, 1978
	<i>Arachnida</i>	<i>Acarina</i>	Tick (<i>Ixodes woodii</i>)	USA	<i>Rickettsiella</i> sp. Strain GSU	Kurti et al., 2002
			Scorpion (<i>Buthus occitanus</i>)	France	" <i>Rickettsiella buthi</i> " ("Porochlamydia buthi")	Morel, 1976
			Crab (<i>Carcinus mediterraneus</i>)	France (Mediterranean sea)	Unnamed	Bonami and Pappalardo, 1980
	<i>Crustacea</i>	<i>Decapoda</i>	Redclaw crayfish (<i>Cherax quadricarinatus</i>)	UK	Unnamed	Romero et al., 2000
			Isopode (<i>Porcellio scaber</i>)	Slovenia	Unnamed	Drobne et al., 1999
	<i>Crustacea</i>	<i>Decapoda</i>				
	<i>Crustacea</i>	<i>Decapoda</i>				

^aAnd related species.

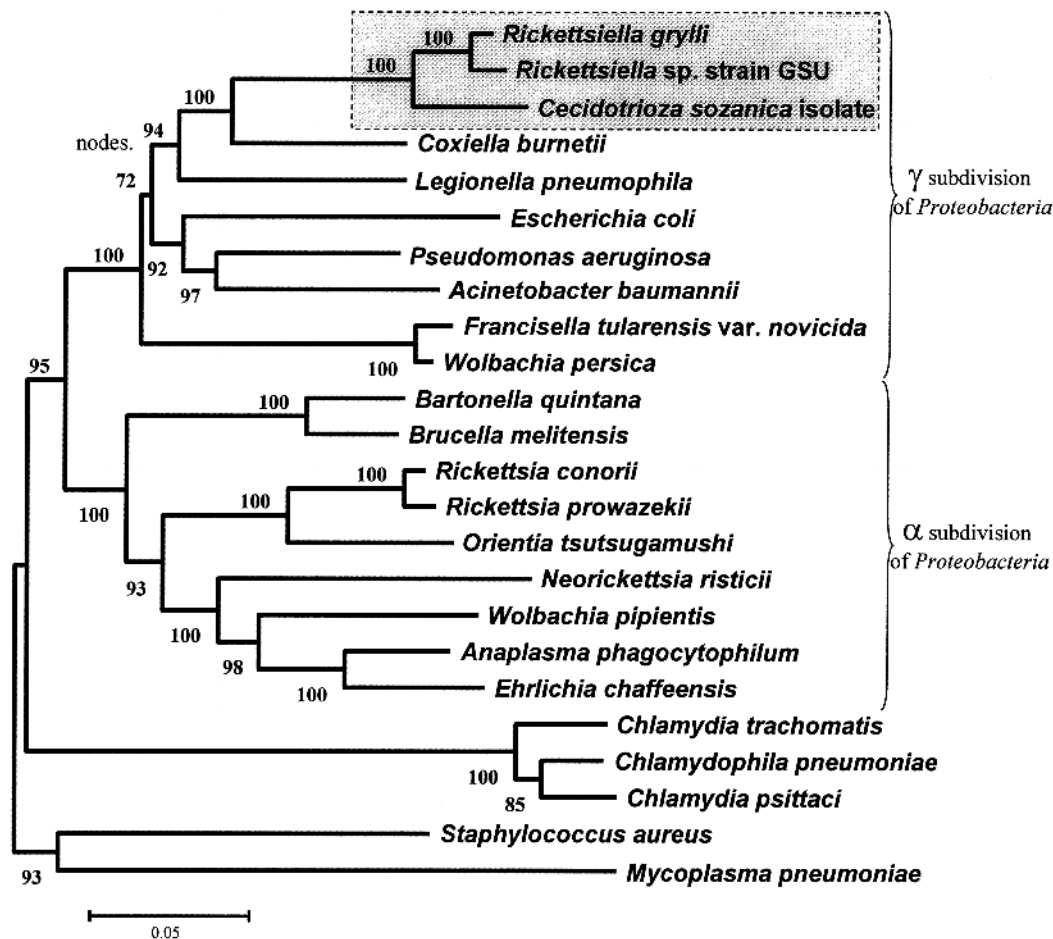


FIGURE BXII.γ.101. Unrooted 16S rDNA-based phylogenetic tree showing the position of *Rickettsiella grylli*. The tree was constructed using the neighbor-joining method, and the bootstrap values are indicated at the nodes. This tree includes *Rickettsiella* sequences derived from: *Rickettsiella grylli*, GenBank accession number U97547; *Rickettsiella* sp. (strain GSU), GenBank accession number AF383621; isolate from *Cecidotrioza sozanica*, GenBank accession number AF286124; *Coxiella burnetii* (strain Q177), GenBank accession number M21291; *Legionella pneumophila* (strain ATCC 33152), GenBank accession number M59157; *Escherichia coli*, GenBank accession number Z83204; *Pseudomonas aeruginosa*, GenBank accession number AB073312; *Acinetobacter baumannii* (ATCC 19606T), GenBank accession number Z93435; *Francisella tularensis* subsp. *novicida*,* GenBank accession number L26084; "*Francisella persica*" (= *Wolbachia persica*), GenBank accession number M21292; *Bartonella quintana* (strain Fuller), GenBank accession number M11927; *Brucella melitensis* biovar abortus, GenBank accession number X13695; *Rickettsia conorii* (strain Moroccan), GenBank accession number L36105; *Rickettsia prowazekii* (strain Breinl), GenBank accession number M21789; *Orientia tsutsugamushi* (strain Karp), GenBank accession number D38623; *Neorickettsia risticii* (strain SHSN-1), GenBank accession number AF037210; *Wolbachia pipientis*, GenBank accession number U23709; *Anaplasma phagocytophilum*, GenBank accession number U02521; *Ehrlichia chaffeensis*, GenBank accession number AF147752; *Chlamydia trachomatis* (strain:B/TW-5/OT), GenBank accession number D85719; *Chlamydia pneumoniae* (strain TW183), GenBank accession number L06108; *Chlamydia psittaci* (strain 6BC, ATCC VR-125T), GenBank accession number AB001778; *Staphylococcus aureus* (strain M2), GenBank accession number Y1585; *Mycoplasma pneumoniae* (strain FH, ATCC 15531), GenBank accession number M29061.

*Editorial Note: In this volume Sjöstedt proposes that *Francisella tularensis* var. *novicida* be regarded as a subspecies of *Francisella tularensis* (see page 208).

which divides by binary fission. The initial bodies are eventually reduced to intermediate forms that subsequently condense to reform the elementary bodies. Occasionally the initial bodies continue to grow to become particles $1.5 \times 2.0 \mu\text{m}$ in diameter, which divide equally or unequally (Federici, 1980), or which may undergo multiple divisions that result in the formation of as many as 30 elementary bodies (Götz, 1972). Unlike those of *R. popilliae*, the giant cells in *R. chironomi* do not evolve into typical crystalline bodies (Weiser and Zizka, 1968).

Cultivation Most strains of *R. popilliae* have been grown in the larvae of *Melolontha*, the common cockchafer. The *R. popilliae* and *R. grylli* strains from the cetonid beetle have been grown in *Orthoptera*, *Coleoptera*, and *Lepidoptera* (Meynadier and Monsarrat, 1969). Intracoeleomically inoculated female *Dermacentor reticulatus* ticks (*Arachnida*, *Acarina*) have been used to isolate and maintain "*Rickettsiella phytoseiuli*", a *Rickettsiella* sp. isolated from the mite *Phytoseiulus persimilis* (*Arachnida*, *Acarina*) (Sutakova and Ruttgen, 1978).

Limited growth of *R. popilliae* has been obtained in chicken embryo endodermal cell cultures and in McCoy cells incubated at 28 or 32°C. Typical organisms were first detected after 1 week and their numbers increased during the following 2 weeks. Infectivity was reduced progressively during serial passage. No growth occurred after 4–5 passages (Suitor, 1964). *R. popilliae* has also been passed three times in other mammalian cell lines (Pourquier et al., 1963). *R. grylli* was grown in cricket cardiac cell cultures (Meynadier et al., 1967).

The only cultivation of a *Rickettsiella* in a cell-free medium has been reported for "*R. phytoseiuli*" in SM IMV-72 medium, which is used for cultivation of phytopathogenic mycoplasmas (Sutakova et al., 1991). The bacteria were maintained until day 66 in primary culture and until day 16 after the first passage. Further passages failed.

Drugs and antibiotic susceptibility Currently, there are no data on the antimicrobial susceptibility of rickettsiellae available.

Pathogenicity The chief interest in rickettsiellae stems from the effect of these organisms on laboratory insectaries and other animal collections as well as from the possible use of these organisms as biological control agents for agricultural pests. Rickettsiellae are not as virulent and persistent as some of the viral and mycotic pathogens of agricultural pest insects; however, rickettsiellae stay alive in the soil for years. Infection of offspring is effected through contamination of soil, rather than by transovarian passage (Hurpin and Robert, 1972, 1976, 1977). The wide distribution of rickettsiellae geographically and in arthropod taxa suggests an early appearance in the course of evolution.

R. popilliae causes blue disease of insect larvae, a deadly disease characterized by the discoloration of the infected larvae. In the cricket and locust, *R. grylli* causes swelling of the abdomen and, at later stages, turgidity of the intersegmentary membranes and ventral inclination of the head, leading to death. *R. grylli* induces a "behavioral fever" in experimentally infected crickets *Acheta domesticus* and *Gryllus bimaculatus* (Louis et al., 1986; Adamo, 1998). These infected invertebrates, when exposed to a temperature gradient, are more likely to survive at 32–33°C than at 26°C—their optimal temperature when uninfected. Infected insects move to areas that are at the higher temperature. This behavior is thought to be used by the insects to create a fever and protect them from infection.

Rickettsiella species have never been reported to be human pathogens. Virulence for vertebrate hosts has not extensively been studied. *R. popilliae* (Melolontha strain) is virulent for mice only when massive doses are inoculated intraperitoneally (Krieg, 1955b). Successful infection by the intranasal route has also been demonstrated (Giroud et al., 1958). When inoculated intraperitoneally or by inhalation, *R. grylli* multiplies in the mouse, but the severity of the infection depends on the inoculum size. With a low inoculum, the bacterium has either no effect or causes local inflammation; the mouse overcomes the infection and the microorganism disappears within 45 d (Croizier and Meynadier, 1972; Delmas and Timon-David, 1985). When injected with a high inoculum, mice develop a cirrhotic hepatitis and malignant tumors, including hematosarcomas.

ENRICHMENT AND ISOLATION PROCEDURES

Successful propagation of *Rickettsiella* has been achieved by the inoculation of healthy larvae of the same or compatible insect

species with infected blood or other infectious material. Japanese beetles or cockchafer inoculated with *R. popilliae* are incubated under appropriate conditions of temperature, food and moisture for about 35 d. Following surface sterilization, the organisms are obtained from the surviving larvae by grinding. The bacteria can be separated from tissue components by cycles of adsorption and dilution from Celite and filtration through diatomaceous filters of medium porosity (Dutky, 1959) or by comparable more recently developed methods such as differential centrifugation (Morel, 1980).

Isolation in cell culture is useful for the study of antigenic cross-reactions by fluorescent antibody techniques or for other comparative investigations. However, unlimited growth in cell culture has not been demonstrated.

MAINTENANCE PROCEDURES

If *R. popilliae* suspensions are not contaminated with other bacteria, viability can be retained for 1 year at 4°C but not for 3 years. Viability is retained for least 3 years at –70°C.

DIFFERENTIATION OF THE GENUS *RICKETTSIELLA* FROM OTHER GENERA

These intracellular bacteria are pathogenic for, or symbiotic with, invertebrates and exhibit a developmental cycle similar to that of *Chlamydia* species. Rickettsiellae are phylogenetically related to *Coxiella burnetii* within the *Gammaproteobacteria*.

TAXONOMIC COMMENTS

Little of the sort of information that is generally used for bacterial classification is available for the classification of *Rickettsiella*.

Historically, based on the similarity in the ultrastructure and developmental cycles, many investigators suggested that rickettsiellae more closely resembled *Chlamydia* than *Rickettsia* and that they should be transferred to the order *Chlamydiales* (Yousfi et al., 1979; Morel, 1980; Avakyan and Popov, 1984). Points of similarity with *Chlamydia* are: (1) both *Chlamydia* and rickettsiellae multiply in the vacuoles of their host cells, and (2) both undergo a developmental cycle consisting of up to six stages that involve the formation of a small dense particle whose function is to infect and a much larger particle of lower density whose function is to multiply. Points of difference between *Chlamydia* and rickettsiellae are (1) the rod or disk shape of the infectious particles of rickettsiellae, (2) the formation of large crystalline bodies, and (3) the absence of relatedness between *Rickettsiella* and *Chlamydia* based on serological analyses (Croizier et al., 1975). Resemblances to other members of the *Rickettsiales* include a close association with arthropods and the intravacuolar location during multiplication—characters that are found in *Ehrlichia* and other genera in the order *Rickettsiales*. By pulsed field gel electrophoresis, the genome sizes of members of the genus *Rickettsiella* range from 1.55 Mb to 2.65 Mb (Frutos et al., 1989) (Table BXII.γ.64). On the basis of DNA–DNA hybridization, DNA restriction site polymorphism, DNA melting profiles, and mol% G + C data (Table BXII.γ.64), Frutos et al. (1994) concluded that the classification of members of the genus *Rickettsiella* was valid at the generic level but not at the species level. Three distinct groups were identified: (1) the *R. grylli* complex containing *R. grylli* and *R. popilliae*, with "*Rickettsiella armadillidii*" being recognized as a strain of *R. popilliae* rather than a synonym of *R. grylli*; (2) *R. chironomii*; and (3) "*R. buthi*", a strain previously considered to

TABLE BXII.γ.64. Differential characteristics of the recognized species of the genus *Rickettsiella*^a

Characteristics	<i>R. popilliae</i>	<i>R. chironomi</i>	<i>R. grylli</i>
<i>Morphology of infectious particles:</i>			
Rod	+	—	+
Disk	—	+	—
Crystalline body production	+	—	+
Serological reaction with anti- <i>R. popilliae</i> serum	+	—	—
<i>Natural hosts:</i>			
<i>Insecta</i>	+	+	+
<i>Coleoptera</i>	+	—	—
<i>Diptera</i>	+	+	—
<i>Orthoptera</i>	—	—	+
<i>Crustacea</i>	+	—	+
<i>Arachnida</i>	—	+	—
Genome size (Mb)	1.72	2.65	2.1
Mol% G + C of the DNA	38.9	36.7	36.3–38.4
<i>DNA thermal denaturation midpoint (°C)</i>			
In 0.1 × SSC	69.1	68.2	68.0–68.9
In 1 × SSC	85.3	84.4	84.2–85.1

^aSymbols: +, positive feature; —, negative feature.

belong to the species *R. chironomi*, from the scorpion *Buthus occitanus* (Arachnida, Scorpionida) (Frutos et al., 1994). According to Frutos et al., *Rickettsiella*, *Coxiella*, *Chlamydia*, and *Rickettsia* are taxonomically distinct genera, and the classification of the genus *Rickettsiella* into a separate order might be justifiable (Frutos et al., 1994). However, based on 16S rDNA sequence analysis, *R. grylli* is most closely related to *Coxiella burnetii* and *Legionella* species in the *Gammaproteobacteria*, whereas members of the order *Rickettsiales* belong to the *Alphaproteobacteria* (Roux et al., 1997) (Fig. BXII.γ.101). This finding is supported by the observation of electron-dense forms in the developmental cycles of both *Rickettsiella* and *Coxiella burnetii* (Avakyan et al., 1983; Popov et al., 1991). Based on 16S rDNA analysis, *Rickettsiella* strain GSU—which is present in the ovarian tissues and Malpighian tubules of female *Ixodes woodii* ticks (Acarina: Ixodidae)—has also been classified in the *Gammaproteobacteria* (Kurti et al., 2002) as has a *Rickettsiella*-like endosymbiont of the psyllid *Cecidotrioza sozanica* (Insecta) (Spaulding and von Dohlen, 2001). Both bacteria were closely related phylogenetically to *R. grylli*. However, because the 16S rDNA sequences of *R. popilliae* and *R. chironomi* have not yet been determined, it is not known whether all species of the genus *Rickettsiella* are misplaced in the order *Rickettsiales* and whether all current members of the genus are phylogenetically homologous (Roux et al., 1997).

The differential characteristics of the species of the genus *Rickettsiella* are presented in Table BXII.γ.64. Six specific names previously proposed (Table BXII.γ.63) are treated as synonyms of the first described species, *R. popilliae*, since no major differences have been encountered among them. The *Popillia*, *Melolontha*, *Tipula*, and *Cetonia* pathogens have common antigens (Krieg, 1958) (Croizier and Meynadier, 1971). The pathogens of *Tenebrio* and *Blatta* have not been extensively studied.

The separation of *R. popilliae* from *R. grylli* was initially based primarily on serological cross-reactions between the cricket and isopod pathogens and lack of reaction with *R. popilliae* antisera (Croizier and Meynadier, 1971). The developmental cycles of the cricket and isopod pathogens appear to be quite similar (Louis et al., 1977a, b), but the results of DNA hybridization studies indicate that *R. grylli* and *R. popilliae* are separate species, with “*R. armadillidii*”—the isopod isolate—showing greater DNA re-

latedness to *R. popilliae* than to *R. grylli* (Frutos et al., 1994). The hosts from which *R. grylli* have been isolated—*Orthoptera* and *Crustacea*—may not constitute a valid criterion for classification, but they might serve as a guide for the provisional placement of the less well-studied pathogens of the desert locust and amphipod (Table BXII.γ.63). *R. popilliae* and *R. grylli* have different genome sizes, but their DNAs have similar mol% G + C contents and DNA melting temperatures (Table BXII.γ.64).

The third species, *R. chironomi*, can be clearly separated from the other two by its unusual life cycle, in which flat disks are formed instead of rods. There is a remarkable similarity between the cycle of the midge pathogen and the cycles of the arachnid pathogens (Morel, 1977; Federici, 1980), although differences have been described (Louis et al., 1979). In 1976, the creation of the genus “*Porochlamydia*” was proposed to accommodate the phenotypically different isolates from the midge and the scorpion, with “*P. chironomi*” being present in the midge larvae and “*P. buthi*” infecting the scorpion *Buthus occitanus* (Arachnida, Scorpionida). This genus was not included in the Approved Lists or the 1984 issue of the *Manual*. The midge and scorpion isolates were later shown to have chromosomal DNAs markedly different in size (Frutos et al., 1989), to exhibit no DNA relatedness as demonstrated by the absence of cross-hybridization, and to have distinct DNA melting profiles. However, the differentiation of the scorpion isolate as a different species has not been officially proposed.

The stethorus beetle pathogen, for which the name *R. stethorae* has been assigned (Table BXII.γ.63), is clearly different in morphology from the above listed species of *Rickettsiella* (Hall and Badgley, 1957), but further studies are necessary to allow proper classification. The other isolates listed in Table BXII.γ.63 appear to be typical rickettsiellae, but additional information is required for appropriate placement in a species.

It should be noted that four of the species recognized in the present edition of the *Manual*, *R. popilliae*, *R. grylli*, *R. chironomi*, and *R. stethorae* were included on the Approved Lists of Bacterial Names. However, as indicated above, sufficient evidence is not available at present to warrant considering *R. stethorae* as a distinct species.

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DIFFERENTIATION OF THE SPECIES OF THE GENUS *RICKETTSIELLA*

Characteristics useful for the differentiation of *Rickettsiella* species are listed in Table BXII.γ.64.

List of species of the genus Rickettsiella

1. ***Rickettsiella popilliae*** (Dutky and Gooden 1952) Philip 1956, 267^{AL} (*Coxiella popilliae* Dutky and Gooden 1952, 749.) *pop. il' li. ae.* M.L. gen. n. *popilliae* of *Popillia*, the generic name of the Japanese beetle, one of its hosts.

This species includes the subjective synonyms "*Rickettsiella melolonthae*", "*Rickettsiella tenebrionis*", "*Rickettsiella cetoniidarum*", "*Rickettsiella tipulae*", "*Rickettsiella blattae*", "*Rickettsiella armadillidii*", and an unnamed isolate from various scarabaeid and carabid beetles (Table BXII.γ.63). The form usually obtained from infected host tissues by the separation procedures described above is a rod $0.2 \times 0.6 \mu\text{m}$ or slightly larger; the rods may be oval, curved or kidney-shaped, with rounded edges. Satisfactorily stained by the Giemsa or Macchiavello methods (Macchiavello, 1937) and presumably by the Gimenez (1964) method. The developmental cycle, the formation of crystalline bodies, and attempts at cultivation outside the host have been discussed above in the section on Further Descriptive Information.

R. popilliae is the etiological agent of blue disease of insect larvae, a name reflecting the discoloration of the infected larvae. Other names reflecting either the locality where the disease was discovered or the host have also been used. Infection is most commonly encountered among larvae but is not confined to this stage. The infection starts in the fat body but ultimately spreads to the blood and the other organs. Larvae have been infected experimentally by injection, by feeding, or by holding them in soil inoculated with suspensions of the organism. By injection, fewer than six organisms are sufficient to infect a Japanese beetle larva with its naturally occurring pathogen. The time required for appearance of symptoms is dependent on dosage and on the temperature of incubation of the larvae (optimum 27°C). With an injection of 10^4 organisms per larva, the symptoms appear after 19 d and death occurs in 19–26 d after the first appearance of symptoms. Larvae can be protected by injection of streptomycin (20 µg/larva) or sulfadiazine (200 µg/larva) but not penicillin or chlortetracycline (Dutky, 1959).

There is considerable variation in the time of appearance of symptoms and death of the larvae infected by the various strains of *R. popilliae*. It is not known to what extent this variation reflects strain virulence or host susceptibility. The organisms can be inactivated by incubation at 60°C for 10 min. The genome size is 1.72 Mb (Frutos et al., 1989, 1994).

The mol% G + C of the DNA is: 36.9 (T_m).

Type strain: no culture has been isolated.

2. ***Rickettsiella chironomi*** (ex Weiser 1963) Weiss, Dasch and Chang 1984b, 356^{VP} (Effective publication: Weiss, Dasch and Chang 1984a, 716.) *chi. ro. no' mi.* M.L. gen. n. *chironomi* of *Chironomus*, the generic name of the midge, one of its hosts.

This species includes unnamed isolates from the midge *Chironomus tentans* and the spider *Argyrodes gibbosus* (Table BXII.γ.63). The infectious stage has an unusual disk-shaped

morphology, $0.06 \times 0.6 \mu\text{m}$, as indicated above in the section on Further Descriptive Information. There is probably a greater diversity among the strains previously included in this species than among the strains of the other two species of *Rickettsiella*. "*R. buthi*", which infects the scorpion *Buthus occitanus* (Arachnida, Scorpionida) has a markedly different chromosomal DNA size (Frutos et al., 1989), exhibits no DNA relatedness to *R. chironomi*, and has a DNA melting profile distinct from that of *R. chironomi* (Frutos et al., 1994) (Table BXII.γ.63). However, the differentiation of the scorpion isolate as a different species has not been officially proposed.

In the midge, the disease affects primarily the last two instars and the pupal stages. The infection starts in the fat body and progresses to lobes around the intestine, which eventually rupture and fill the whole body with white fluid. Experimental infection results in death in 7–14 d (Weiser and Zizka, 1968). In arachnids, the hepatopancreas and intestinal diverticula are the organs primarily affected. The genome size of *R. chironomi* is 2.65 Mb, whereas the genome size of "*R. buthi*" is 1.55 Mb and its mol % G + C of the DNA is 41 (T_m) (Frutos et al., 1989, 1994).

The mol% G + C of the DNA is: 36.7 (T_m).

Type strain: no culture has been isolated.

3. ***Rickettsiella grylli*** (ex Vago and Martoja 1963) Weiss, Dasch and Chang 1984b, 356^{VP} (Effective publication: Weiss, Dasch and Chang 1984a, 716.) *gryl' li.* M.L. gen. n. *grylli* of *Gryllus*, the generic name of the cricket, one of its hosts.

This species includes the subjective synonym "*Rickettsiella schistocerae*" and an unnamed isolate from the amphipod *Crangonyx floridanus* (Table BXII.γ.63). The organism resembles *R. popilliae* in the morphology of its infectious particle and subsequent developmental stages and crystalline body production. It can be differentiated from *R. popilliae* by serological tests and, to a certain extent, by the natural hosts infected, although it does not have a high degree of host specificity (Table BXII.γ.63).

The disease in naturally infected hosts affects both larvae and adults and develops very slowly. In the cricket and locust, it is characterized by swelling of the abdomen and, at later stages, by turgidity of the intersegmentary membranes and by ventral inclination of the head. The time of death following first appearance of symptoms is quite variable. In the isopod, death is preceded by loss of weight and a whitish coloration of the intersegmentary membranes due to the accumulation of iridescent fluid in the body cavities. Diseased amphipods are also recognized by their opaque pale green iridescence. Most patently diseased amphipods die within six weeks.

The genome size is 2.1 Mb (Frutos et al., 1989, 1994). Based on the 16S rDNA sequence, *R. grylli* is most closely related to *Coxiella burnetii* within the *Gammaproteobacteria* (Roux et al., 1997).

The mol% G + C of the DNA is: 36.3–38.4 (T_m).

Type strain: no culture has been isolated.

Order VII. **Methylococcales** *ord. nov.*

JOHN P. BOWMAN

Me.thy.lo.coc.ca'les. M.L. fem. pl. n. *Methylococcus* type genus of the order; *-ales* ending to denote an order; M.L. fem. pl. n. *Methylococcus* order.

Morphology varies from **rods to cocci**. Cells are usually single or diploid, occasionally forming chains and tetrads. Do not form rosettes. If motile, possess single polar flagellum. **Gram negative**. The majority of species produce a polysaccharide **cyst-like resting stage**. Cysts may be desiccation resistant or sensitive. Dissimilatory methane oxidation is associated with **type I intracytoplasmic membranes**, which appear as stacks of vesicular disks derived from convolutions of the cytoplasmic membrane. Catalase and oxidase are usually produced. **Strictly aerobic respiratory metabolism** with oxygen as the electron acceptor. **Obligate utilizers of methane and other C₁ compounds for carbon and energy. Compounds with carbon-carbon bonds are not utilized**. High levels of organic solutes and ammonia ions inhibit growth. C₁ compounds are incorporated by the **ribulose monophosphate pathway**. The **tricarboxylic acid cycle is incomplete**, with 2-oxoglutarate dehydrogenase activity absent. Nitrogenase activity may be present. Contain mainly C₁₆ **fatty acids**, which are predominantly monounsaturated with several different C_{16:1} isomers present. **Ubiquinone-8 or 18-methyleneubiquinone-8** are the predominant lipoquinones. Habitat: isolated from aerobic and microaerobic habitats that have an influx of methane (from methanogenesis or hydrocarbon seeps), including benthic sediments of fresh water and marine bodies of water; oxyclines and thermoclines of lakes and oceans; wetland muds; coal mine drainage waters and coal surfaces; sewage sludge and wastewater; most soils; subsurface sediment; groundwater; and natural gas reserves. Some members form endosymbiotic relationships with invertebrates, including mytilid mollusks and pogonophora. Genome sizes range from 2.5–5.5 Mb. Member of the *Gammaproteobacteria*.

The mol% G + C of the DNA is: 43–65.

Type genus: **Methylococcus** Foster and Davis 1966, 1929, emend. Bowman, Sly, Nichols and Hayward 1993b, 748.

FURTHER DESCRIPTIVE INFORMATION

Bacteria that have the capability to oxidize and utilize methane as a sole carbon and energy source are referred to as methanotrophic bacteria or methanotrophs (Whittenbury et al., 1970b). The original description of the family *Methylococcaceae* (Whittenbury and Krieg 1984a) included two distinct groups of methanotrophs referred to in the literature as “type I” and “type II” methanotrophs. In addition, there is a third group called the “type X” methanotrophs (Hanson and Hanson, 1996) that essentially represents a subgroup of type I methanotrophs. Type I and II methanotrophs possess extensive biological differences that are morphological, biochemical, chemotaxonomic, and phylogenetic in nature. As type II methanotrophs belong to the *Alphaproteobacteria*, they are dealt with in a separate section of this volume. Currently, the family *Methylococcaceae* includes all type I methanotrophs (Bowman et al., 1995) in five genera: *Methylococcus*, *Methylosphaera*, *Methylobacter*, *Methylomicrobium*, and *Methylomonas*. Phenotypic properties and fatty acid components differentiating these genera are shown in Tables BXII.γ.65 and BXII.γ.66, respectively.

Methane oxidation The major characteristic that separates methanotrophs from other procaryotes is their ability to utilize methane as a sole carbon and energy source. Methanotrophs oxidize methane to CO₂ in a dissimilatory pathway that generates energy and allows access to metabolizable carbon units in the form of formaldehyde. The critical first step of this pathway is the oxidation of methane to methanol, which is performed by methane monooxygenase (MMO). Methanol is then oxidized to formaldehyde by a pyrrolquinol quinone-containing methanol dehydrogenase. The formaldehyde is then either used for cell carbon (see below) or oxidized to formate and then CO₂; the

TABLE BXII.γ.65. Phenotypic characteristics differentiating member genera of the Family *Methylococcaceae*^{a,b}

Characteristics	<i>Methylococcus</i>	<i>Methylobacter</i>	<i>Methylomicrobium</i>	<i>Methylomonas</i>	<i>Methylosphaera</i>
Cell morphology:					
Cocci					+
Cocci-rods	+				
Cocci-ellipsoid		+			
Rods			+	+	
Motility	D	D	+	+	–
Cyst formation	+	+	–	+	ND
Desiccation resistant	–	+	–	–	–
Pigmentation:					
Carotenoids	–	–	–	+	–
Melanin-like	D	D	–	–	–
Other	–	D	–	–	–
Growth at 25°C	–	+	+	+	–
Growth at 45°C	+	–	–	–	–
Requires seawater or Na ⁺	–	D	D	–	+
RuBisCo ^c	+	–	–	–	–
Nitrogen fixation	+	–	–	D	+
Major quinone ^d	MQ-8	Q-8	Q-8	MQ-8	ND
Mol% G + C of DNA (T _m)	59–65	48–55	48–60	50–59	43–46

^aFor symbols, see standard definitions.

^bData from Bowman et al. (1993b) and Bowman et al. (1997c).

^cRuBisCo, ribulose-1,5-bisphosphate carboxylase.

^dMQ-8, 18-methylene-quinone-8; Q-8, quinone-8. Data from Collins and Green (1985).

TABLE BXII.γ.66. Fatty acids of member genera of the Family *Methylococcaceae*^{a,b}

Fatty acids ^c	<i>Methylococcus</i>	<i>Methylobacter</i>	<i>Methylomicrobium</i>	<i>Methylomonas</i>	<i>Methylosphaera</i>
Saturated:					
C _{14:0}	1–6	7–10	1–2	19–25	2–3
C _{15:0}	1–13	0–4	–	0–1	1–2
C _{16:0}	33–56	8–9	11–24	4–9	14–15
C _{18:0}	tr–2	–	0–3	0–trc	tr
Monounsaturated:					
C _{16:1} ω8c	–	–	12–19	19–41	37–41
C _{16:1} ω7c	17–46	56–58	14–24	8–15	16–19
C _{16:1} ω7t	–	–	–	–	2–3
C _{16:1} ω6c	4–12	4–5	8–14	5–13	17–18
C _{16:1} ω5c	3–9	6–8	6–8	2–6	0–tr
C _{16:1} ω5t	0–6	10–11	8–28	8–17	–
C _{17:1} ω8c	0–tr	–	–	–	tr–1
C _{17:1} ω7c	0–2	–	–	0–1	tr–1
C _{17:1} ω7t	0–2	–	–	0–tr	–
C _{18:1} ω9c	0–3	–	–	0–tr	0–1
C _{18:1} ω7c	tr–7	–	0–3	tr–2	1–2
Other fatty acids:					
C _{15:0} iso	0–1	0–tr	–	0–2	–
C _{15:0} ante	0–tr	–	–	0–2	–
C _{16:0} cyc	0–1	–	–	–	–
C _{17:0} cyc	0–15	–	–	0–2	–
C _{19:0} cyc	0–3	–	–	0–tr	–

^aFor symbols, see standard definitions; tr, trace (fatty acid present at a level of less than 0.5%).

^bData from Bowman et al. (1991a), Bowman et al. (1993b), and Bowman et al. (1997c).

^cFatty acids are designated as follows: the number of carbon atoms:number of double bonds. The number (n) of carbon atoms of the closest double bond from the methyl end (ω) of the molecule is given by ωn. The suffixes iso, ante, and cyc indicate iso-branched, anteiso-branched, and cyclopropane fatty acids, respectively. The suffixes c and t indicate *cis* and *trans* geometry, respectively

final steps form reducing equivalents needed to drive methane oxidation.

In all type I methanotrophs, MMO is present in a membrane-bound form and is referred to as particulate MMO (pMMO). This enzyme consists of three polypeptides (sizes 45, 35, and 26 kDa) and contains copper (Semrau et al., 1995). Synthesis of a cytoplasmic version of MMO, referred to as soluble MMO (sMMO) (Dalton, 1992; Murrell, 1992; Lipscomb, 1994), appears to be a strain-specific trait (Koh et al., 1993; Miguez et al., 1997; Shen et al., 1997) that is relatively rare in type I methanotrophs. This enzyme is copper-repressible and contains three components. The first includes a 245 kDa hydroxylase that contains an unusual non-heme iron–oxygen linked active site. The second component is a 15 kDa regulatory protein, and the third is a ferredoxin-based reductase. Particulate MMO has a high affinity for methane, allowing for more efficient growth yields compared to sMMO, which appears to be an adaptation to copper-limiting growth conditions (Hanson and Hanson, 1996). However, the substrate specificity of sMMO is exceptionally broad, and it can oxidize an extraordinarily wide range of aliphatic, aromatic, and heterocyclic compounds (Haber et al., 1984). This property has been exploited biotechnologically, with methanotrophs proposed for organic transformations in industrial applications (Lidstrom and Stirling, 1990) and as a bioremediation agent (Hanson and Hanson, 1996).

The amino acid sequence of the *pmoA* gene, which encodes a pMMO protein component in type I methanotrophs, has been used in phylogenetic analyses. This gene is very similar to that encoding the ammonia monooxygenase (*amoA*) in *Nitrosococcus oceanus*, suggesting a common evolutionary link (Holmes et al., 1995a; Bodrossy et al., 1997). Several physiological comparisons, in fact, have been drawn between methanotrophs and ammonia-oxidizing bacteria, because both groups are capable of oxidizing methane and ammonia by similar means (Hanson and Hanson,

1996). The sequences of *pmoA* in type II methanotrophs are quite different from those of type I methanotrophs. The amino acid sequences of sMMO polypeptides (*mmoA*, *mmoB*, *mmoC*, and *mmoX*) show a high degree of conservation, however, and are similar between type I and II methanotrophs (Stainthorpe et al., 1991; Murrell, 1992). These genes have been used to develop oligonucleotide probes to specifically target methanotrophs (Holmes et al. 1995b; McDonald and Murrell, 1997; Miguez et al., 1997).

C₁ carbon assimilation Methanotrophs are specialized to utilize the C₁ compounds methane, methanol, and formaldehyde. Occasionally, other C₁ compounds can also be utilized, but this is the exception rather than the rule. Compounds with carbon–carbon bonds (acetate, D-glucose, etc.) are not utilized as energy sources by methanotrophs; however, they may be used as supplementary carbon sources when cells are grown in the presence of methane or other utilizable C₁ compounds (Hanson et al., 1992). So far, no evidence exists for the presence of facultative methanotrophic bacteria that are otherwise similar to the type I methanotrophs.

Formaldehyde is toxic to methanotrophic cells at relatively low concentrations and so is often not tested as a carbon source. In addition, freshly isolated methanotrophs often prove sensitive to methanol (due to accumulation of formaldehyde). Consequently, methanol utilization should be tested with only low methanol concentrations (10–30 mM) or with methanol provided as a vapor in a sealed container. Methanotrophs may be “trained” to tolerate higher concentrations of methanol (Lidstrom, 1988). Type I methanotrophs use the ribulose monophosphate (RuMP) pathway to incorporate formaldehyde into anabolic pathways. Type II methanotrophs, on the other hand, use the serine transhydroxymethylase (Serine) pathway to fix carbon (Strom et al., 1974). In the RuMP pathway, formaldehyde

is combined with ribulose-5-phosphate to form D-erythro-L-glycero-3-hexulose-6-phosphate through the action of the enzyme hexulose-6-phosphate synthase. Subsequently, 1-glyceraldehyde-3-phosphate is formed via 1-fructose-6-phosphate and 3-fructose-6-phosphate. Ribulose-5-phosphate is regenerated via rearrangement reactions from 3-fructose-5-phosphate. Hexulose-6-phosphate synthase is often used as an indicator of the RuMP pathway and thus can be used as a direct way to distinguish type I and II methanotrophs. Some type I methanotrophs, mostly *Methylococcus* strains, possess a low level of activity for α -hydroxypyruvate reductase, a key enzyme for the Serine pathway. The tricarboxylic acid cycle is incomplete and lacks the enzyme 2-oxoglutarate dehydrogenase. The cofactor for isocitrate dehydrogenase differs among type I methanotrophs: in some strains the enzyme activity is dependent on either NAD or NADP; in others it is dependent only on NAD and is unresponsive to NADP, and in still others it is NADP-dependent and unresponsive to NAD (Davey et al., 1972).

Intracytoplasmic membranes When cells are grown in the presence of methane (and sometimes methanol), characteristic intracytoplasmic membranes (ICMs) are formed from convolutions of the cytoplasmic membrane. ICMs are about 80–90 nm thick, appear as typical lipid bilayers, and in type I methanotrophs, are arranged as layered bundles of vesicular disks (Proctor et al., 1969; Davies and Whittenbury, 1970; De Boer and Hazeu, 1972; Hyder et al., 1979) (Fig. BXII.γ.102). Type II methanotrophs, on the other hand, have ICMs that extend peripherally throughout the cell and are arranged parallel to the cell wall. The relative ICM content in cells is enhanced in conditions of limiting oxygen and methane tensions or elevated copper ion concentrations and when a stable and rapid rate of methane transfer is available (Scott et al., 1981; Stanley et al., 1983).

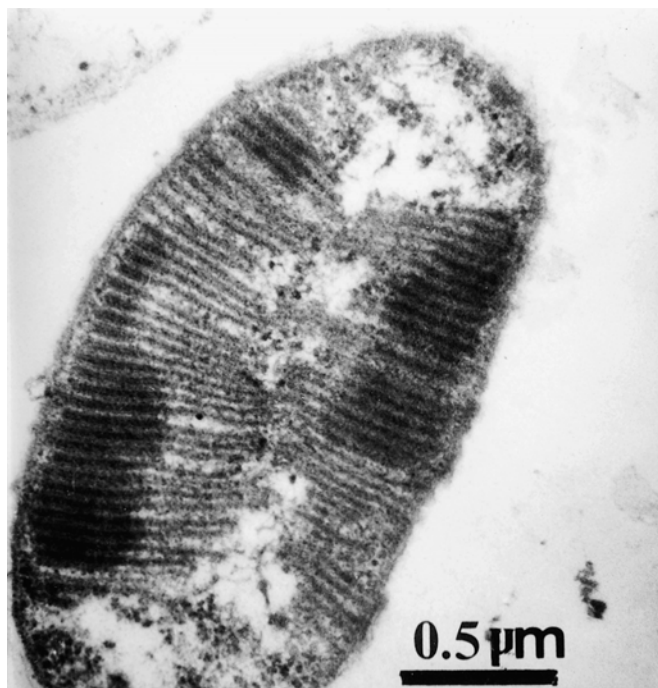


FIGURE BXII.γ.102. Electron micrograph of a ruthenium red-stained thin section of *Methylobacterium fodinarum*, showing typical type I methanotrophic intracytoplasmic membrane ultrastructure.

Resting stages Type I methanotrophs often produce single- or multiple-bodied cyst-like inclusions often regarded as being similar to the cysts observed in the genus *Azotobacter*. Encystment usually increases in the stationary growth phase and when cells are grown at low methane tensions (Whittenbury et al., 1970a). Cyst formation depends on cell density, calcium ion concentration, pH, and the rate of cell autolysis (Mulyukin et al., 1997). Cysts may be readily observed by the *Azotobacter* cyst-staining procedure developed by Vela and Wyss (1964). Cysts appear to be made up of a glucan-like polysaccharide, unlike those of the type II methanotroph *Methylocystis*, which are composed mostly of lipid (Sutherland and MacKenzie, 1977). Cyst production appears to allow some methanotrophs to resist methane starvation. In other species, it can also confer desiccation resistance. Heat resistance, however, is not present; it requires more specialized structures such as the exospore formed by the type II methanotroph *Methylosinus*. Excellent photographic descriptions of both multi- and single-bodied cysts have been published by Hazeu et al. (1980a).

Fatty acid composition Members of the *Methylococcaceae* contain mainly C_{16} fatty acids. Although this is not unusual among Gram-negative bacteria, type I methanotrophs species contain abundant levels of several $C_{16:1}$ fatty acid isomers that are rarely encountered in other prokaryotes, including $C_{16:1\ \omega 8c}$, $C_{16:1\ \omega 6c}$, $C_{16:1\ \omega 5c}$, and $C_{16:1\ \omega 5t}$ (Makula, 1978; Nichols et al., 1985; Bowman et al., 1991a). The combinations of these fatty acids are diagnostic for the identification of type I methanotrophic genera (Andreev and Gal'chenko, 1978; Bowman et al., 1991a; Bowman et al., 1993b, 1997c) (Table BXII.γ.66). By comparison, type II methanotrophs contain mostly $C_{18:1}$ fatty acids; however, as in the type I methanotrophs, the double bond in most is an unusual position ($C_{18:1\ \omega 8c}$) (Nichols et al., 1985). These unusual fatty acids are very useful as signatures for detection and quantitation of methanotroph populations in environmental samples (Nichols et al., 1987; Guckert et al., 1991; Jahnke et al., 1995; Sundh et al., 1997; Guezennec and Fiala-Medioni, 1996).

Lipopolysaccharide composition Studies of the carbohydrate moieties of the lipopolysaccharide of various type I methanotrophs have revealed a pattern typical of Gram-negative bacteria (i.e., D-glucose, L-fucose, and D-heptose) (Sutherland and Kennedy, 1986). The lipid moieties of lipopolysaccharide are composed mostly of $C_{16:0\ 3OH}$, with smaller quantities of β -hydroxylated and saturated components with carbon chain lengths of 10–15. The phospholipids found in type I methanotrophs are generally quite similar to one another and are composed mainly of phosphatidylethanolamine and phosphatidylglycerol, with lower levels of phosphatidylcholine and diphosphatidylglycerol (cardiolipin) (Makula, 1978; Andreev and Gal'chenko, 1983).

Quinones The isoprenoid lipoquinone (coenzyme Q) content of many type I methanotrophs consists mainly of ubiquinone (Q)-8 (94–99%), with small amounts of Q-7 also present. In several species, coenzyme Q is substituted with a methylene group identified as 18-methylene-Q-8 or MQ-8 (Collins and Green, 1985). The MQ-8 is accompanied by minor amounts of MQ-7.

Habitats Methane is an important greenhouse gas and the most abundant organic gas in the atmosphere (Crutzen, 1991). Methanotrophs are the largest global methane sink and, as a result, are ubiquitous in nature. They produce the highest and most active populations in environments where oxygen and methane are readily available (Reeburgh et al., 1993). Recent evidence suggests that methanotrophs make up a high propor-

tion of the total bacterial biomass (up to 40%) in many aquatic environments and surface sediments (Ross et al., 1997; P. I. Boon, personal communication). Methane is highly ^{13}C -depleted, and stable isotopic analyses indicate that a considerable proportion of carbon found in aquatic life at different trophic levels has its origin from methanotrophic bacteria (Boschker et al., 1998). Habitats from which methanotrophs have been successfully isolated include marshes and swamps, rivers and streams, rice paddies, oceans, ponds, lakes, soils of meadows and forests, ground-water, sewage sludge, and coal mines (Hanson and Hanson, 1996). Psychrophilic methanotrophs have been isolated from tundra soils and Antarctic lakes (Omel'chenko et al., 1996; Bowman et al., 1997c). Methanotrophs may be halophilic, although so far, only a few have been described with this property (Lidstrom, 1988; Lees et al., 1991; Sieburth et al., 1993; Bowman et al., 1997c). Methanotrophs have also been isolated from the effluent of an underground hot spring found within a natural gas

field (Bodrossy et al., 1995). Methanotrophs have been detected and/or isolated from acidic peat bog soils (Dedysh et al., 1998) and deep sea sediments (Guezennec and Fiala-Medioni, 1996). Methanotrophs can be endosymbiotic in invertebrates associated with marine hydrocarbon seeps (see below). There is also good evidence that methanotrophs are closely associated with aquatic macrophytes (King, 1994; Calhoun and King, 1998).

Endosymbionts Methanotrophic endosymbionts, in coexistence with sulfur-oxidizing chemoautotrophic endosymbionts (Childress et al., 1986; Cavanaugh et al., 1987; Fisher et al., 1993), are the metabolic basis of unusual invertebrate communities associated with hydrocarbon seeps in the Pacific and Atlantic Oceans. The invertebrate hosts include deep-sea mytilid mussels (family *Mytilidae*) (Childress et al., 1986), and the pogonophoran tubeworm *Siboglinum poseidoni* (Dando et al., 1994). Cold-water reefs of algae-free coral found in waters north of Norway, associated with hydrocarbon seeps, are also believed to use methane

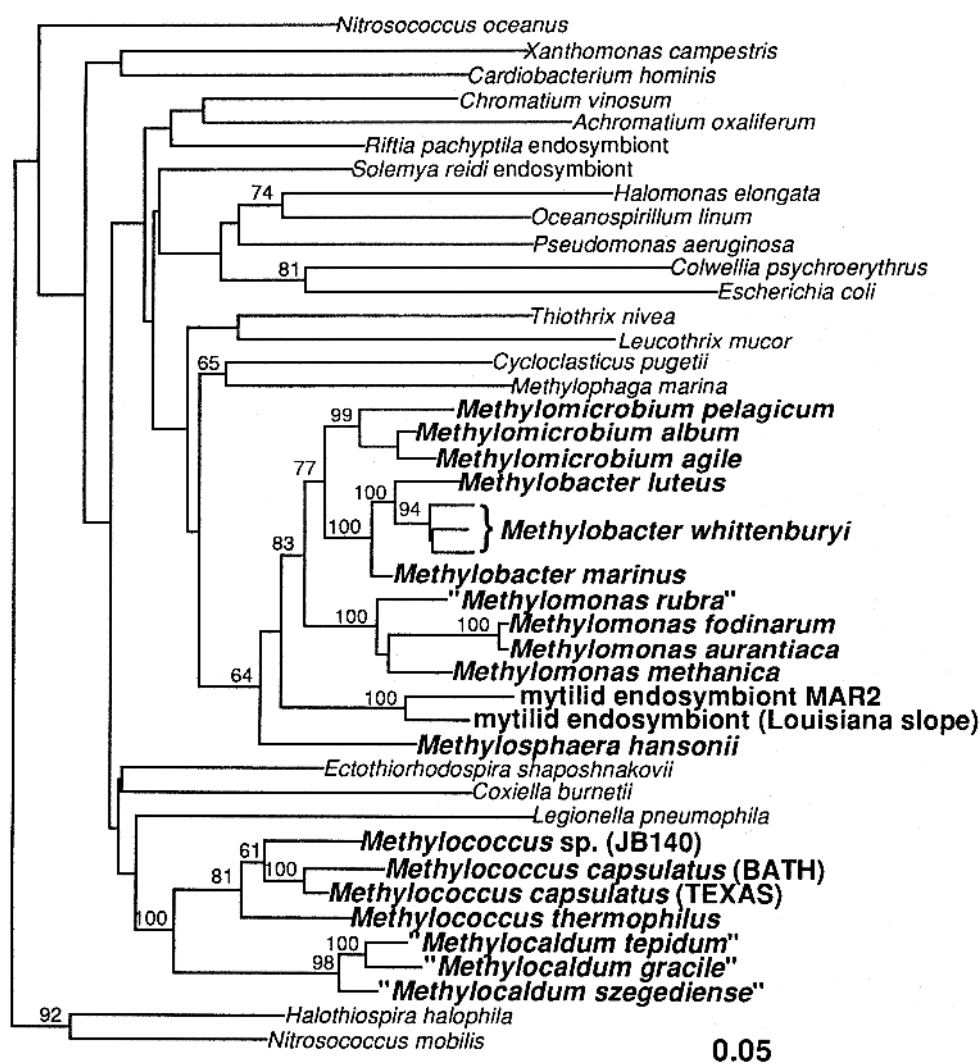


FIGURE BXII.γ.103. Unrooted phylogenetic tree based on 16S rRNA sequences, showing the position of the various genera belonging to the Order *Methylococcales*, Family *Methylococcaceae* (shown in bold type) in the *Gammaproteobacteria*. The tree is based on maximum likelihood values and joined by the neighborliness algorithm (outgrouped with the sequence for *Desulfovibrio desulfuricans*) using the PHYLIP v. 3.57c software package (Phylogenetic Inference Package, J. Felsenstein, University of Washington, Seattle). Values at the branch nodes are bootstrap percentage values generated from 250 replicated analyses. Bar = a taxonomic distance of 0.05.

as a major nutrient and thus presumably contain methanotroph symbionts (Hovland and Judd, 1988). The invertebrates can exist on methane as a primary carbon source, as shown by stable carbon isotope analyses (Southward et al., 1981; Cavanaugh, 1993). Methanotrophs growing in the gills of mytilids have been identified as type I methanotrophs by the presence of key enzymes of the RuMP pathway and characteristic fatty acids, as well as by oligonucleotide probes (Cavanaugh, 1993; Distel and Cavanaugh, 1994; Jahnke et al., 1995). Phylogenetic analyses indicate that mytilid symbionts form a distinct lineage within the *Methylococcaceae* (Distel and Cavanaugh, 1994). Cultivation of methanotroph symbionts has been unsuccessful to date.

TAXONOMIC COMMENTS

The use of 16S rRNA-based phylogenetic analysis has helped to resolve many of the previously identified nomenclatural problems affecting methanotrophs (Bratina et al., 1992; Brusseau et al., 1994; Bowman et al., 1995). On the basis of 16S rDNA sequencing, type I methanotrophs belong in the *Gammaproteobacteria* and in 1994–1995 were thought to comprise a single clade. However, in most phylogenetic trees based on currently available sequences, the *Methylococcales* is now made up of two separate clades (Fig. BXII.γ.103). The first cluster includes the “group X” methanotrophs—thermotolerant and thermophilic methanotrophs of *Methylococcus* and “*Methylodaldum*”. This clade represents a relatively deep lineage within the *Gammaproteobacteria*, with *Ectothiorhodospira* and *Legionella* being among the closest relatives. The second cluster contains mesophilic and psychrophilic type I methanotrophs, including the genera *Methylosphaera*, *Methylobacter*, *Methylomicrobium*, and *Methylomonas*. This cluster is most closely related to *Methylophaga*, *Cycloclasticus*, and various chemolithotrophic endosymbionts. From this, it could be argued that because the *Methylococcales* does not have phylogenetic coherency,

it should be divided into two families. However, sequence analysis still indicates that the type I and group X methanotrophic clades are similar (evolutionary distance 0.10–0.13). More importantly, both groups possess a number of traits, not the least being methane oxidation, that sets them apart from other members of the *Gammaproteobacteria*. It can be safely assumed that many novel methanotrophic species remain to be isolated and that their 16S rDNA sequences could hypothetically introduce topological changes to current phylogenetic trees. Thus, until a broader knowledge of methanotroph phylogeny and evolution is obtained it seems prudent to retain the family *Methylococcaceae* as it currently stands.

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Family I. *Methylococcaceae* Whittenbury and Krieg 1984b, 355^{VP} (Effective publication: Whittenbury and Krieg 1984a, 256) emend. Bowman, Sly, Nichols and Hayward 1993b, 745.

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Me.thy.lo.coc.ca'ce.ae. M.L. masc. n. *Methylococcus* type genus of the family; *-aceae* ending to denote a family; M.L. fem. pl. n. *Methylococcaceae* the *Methylococcus* family.

The description of the family is the same as that for the order.

Type genus: ***Methylococcus*** Foster and Davis 1966, 1929 emend. Bowman, Sly, Nichols and Hayward 1993b, 748

Differentiation of the Genera of the Family *Methylococcaceae*

A key to the genera of the family *Methylococcaceae* is provided below. In addition, Table BXII.γ.67 lists phenotypic features that differentiate the genera and Table BXII.γ.68 indicates differences in fatty acid composition between the genera.

Key to the genera of the family Methylococcaceae

- I. Can grow at at 45°C or higher. Form cysts and possess ribulose-1,5-diphosphate carboxylase activity. Cells are cocci or rods that may be motile. Colonies are light tan to dark brown. C_{16:0} and C_{16:1 ω7c} are the major fatty acid components in cells. The major lipoquinone is MQ-8. The mol% G + C content of the DNA ranges from 59–65 (*T_m*).
Genus *Methylococcus*
- II. Mesophilic. Elliptical rod-like morphology. Usually motile. Colonies on agar are light tan to brown with either a light brown pigment diffusing into agar media or a yellow pigment that may or may not diffuse into agar media. Cells are desiccation resistant. Nitrogenase activity is absent. C_{16:1 ω7c} is the major fatty acid component in cells. The major lipoquinone is Q-8. The mol% G + C content of the DNA ranges from 49–55 (*T_m*).
Genus *Methylobacter*
- III. Growth at temperatures > 40°C. Cells coccoid to long rods. Form cysts. Motile or non-motile. Not resistant to desiccation. Colonies buff to brown. Nitrogenase absent. Mol% G + C 57–59 (Bd).
Genus *Methylocaldum*
- IV. Mesophilic. Rod-like morphology. Motile. Grows at 25°C and may be slightly halophilic. Colonies on agar are non-pigmented. Nitrogenase activity is absent. Major fatty acids include C_{16:0}, C_{16:1 ω8c}, C_{16:1 ω7c}, and C_{16:1 ω5c}/C_{16:1 ω5t}. The major lipoquinone is Q-8. The mol% G + C content of the DNA ranges from 48–60 (*T_m*).
Genus *Methylomicrobium*
- V. Mesophilic. Rod-like morphology. Motile. Colonies on agar contain red, pink, or orange non-diffusible carotenoid pigments. Cells are desiccation sensitive. Nitrogenase activity may be present. C_{14:0} and C_{16:1 ω8c} are the major fatty acids in cells. The major lipoquinone is MQ-8. The mol% G + C content of the DNA ranges from 50–59 (*T_m*).
Genus *Methylomonas*
- VI. Growth at temperatures ≤37°C. Cells coccoid or fusiform; form packets. Motile. Form cysts. Colonies buff to brown. Nitrogenase absent. Major fatty acids C_{16:0}, C_{16:1 ω5t}, C_{16:1 ω6c}, C_{16:1 ω7c}, C_{16:1 ω8c}. Mol% G + C 54–55 (HPLC).
Genus *Methylosarcina*
- VII. Psychrophilic, growing only at –2–20°C. Cells are coccoidal, lack cysts, form gas vesicles, and are nonmotile. No pigmentation. No growth on solid agar media. Oxygen-tolerant nitrogenase activity present. Contains high levels of C_{16:1 ω8c} and lacks C_{16:1 ω5c}/C_{16:1 ω5t}. The mol% G + C content of the DNA ranges from 43–46 (*T_m*).
Genus *Methylosphaera*

TABLE BXII.γ.67. Phenotypic characteristics differentiating member genera of the Family *Methylococcaceae*^{a,b}

Characteristics	<i>Methylococcus</i>	<i>Methylosphaera</i>	<i>Methylobacter</i>	<i>Methylomicrobium</i>	<i>Methylomonas</i>
Cell morphology	Cocci-rods	Cocci	Cocci-ellipse	Rods	Rods
Motility	D	—	D	+	+
Cyst formation	+	—	+	—	+
Desiccation resistance	—	—	+	—	—
<i>Pigmentation:</i>					
Carotenoids	—	—	—	—	+
Melanin-like	D	—	D	—	—
Other	—	—	D	—	—
Growth at 25°C	—	—	+	+	+
Growth at 45°C	+	—	—	—	—
Requires seawater or Na ⁺	—	+	D	D	—
RuBisCo ^c	+	—	—	—	—
Nitrogen fixation	+	+	—	—	D
Major quinone ^d	MQ-8	ND	Q-8	Q-8	MQ-8
Mol% G + C of DNA (<i>T_m</i>)	59–65	43–46	48–55	48–60	50–59

^aSymbols, see standard definitions.^bData from Bowman et al. 1993b and Bowman et al., 1997c.^cRuBisCo, ribulose-1,5-bisphosphate carboxylase.^dMQ-8, 18-methylene-quinone-8; Q-8, quinone-8. Data from Collins and Green (1985).

TABLE BXII.γ.68. Fatty acids of member genera of the Family *Methylococcaceae*^{a, b}

Fatty acids	<i>Methylococcus</i>	<i>Methylosphaera</i>	<i>Methylobacter</i>	<i>Methylomicrobium</i>	<i>Methylomonas</i>
Saturated:					
C _{14:0}	1–6	2–3	7–10	1–2	19–25
C _{15:0}	1–13	1–2	0–4	–	0–1
C _{16:0}	33–56	14–15	8–9	11–24	4–9
C _{18:0}	tr–2	tr	–	0–3	0–tr
Monounsaturated:					
C _{16:1} ω _{8c}	–	37–41	–	12–19	19–41
C _{16:1} ω _{7c}	17–46	16–19	56–58	14–24	8–15
C _{16:1} ω _{7t}	–	2–3	–	–	–
C _{16:1} ω _{6c}	4–12	17–18	4–5	8–14	5–13
C _{16:1} ω _{5c}	3–9	0–tr	6–8	6–8	2–6
C _{6:1} ω _{5t}	0–6	–	10–11	8–28	8–17
C _{7:1} ω _{8c}	0–tr	tr–1	–	–	–
C _{7:1} ω _{7c}	0–2	tr–1	–	–	0–1
C _{7:1} ω _{7t}	0–2	–	–	–	0–tr
C _{8:1} ω _{9c}	0–3	0–1	–	–	0–tr
C _{8:1} ω _{7c}	tr–7	1–2	–	0–3	tr–2
Other fatty acids:					
C _{15:0}	0–1	–	0–tr	–	0–2
C _{15:0} cyc	0–tr	–	–	–	0–2
C _{16:0} cyc	0–1	–	–	–	–
C _{17:0} cyc	0–15	–	–	–	0–2
C _{19:0} cyc	0–3	–	–	–	0–tr

^a Data from Bowman et al. (1995).^b These values are % of total fatty acids; symbols: tr, trace component (<0.1%); –, not detected.

Genus I. *Methylococcus* Foster and Davis 1966, 1929^{AL} emend. Bowman, Sly, Nichols and Hayward 1993b, 748

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Me.thy.lo.coc'cus. Fr. n. *méthyle* the methyl radical; Gr. n. *coccus* a grain, berry; M.L. n. *Methylococcus* methyl coccus.

Cells appear as **cocci or rods**, 0.8–1.5 × 1.0–1.5 μm. Usually occur singly or in pairs. Form a polysaccharide-containing **cyst-like resting stage** that is desiccation sensitive. Poly-β-hydroxybutyrate granules present. **Gram negative**. If motile, cells are propelled by single polar flagellum. Catalase and oxidase positive. **Strictly aerobic respiratory metabolism** with oxygen as the electron acceptor. **Obligate utilizers of methane and methanol for carbon and energy**. Contain a **partially functional Benson–Calvin cycle** (phosphoribulokinase and ribulose-1,5-diphosphate carboxylase activity present); however, incorporation of CO₂ for cellular carbon occurs only in the presence of methane. **Fix atmospheric nitrogen** via an oxygen-sensitive nitrogenase. **Thermotolerant or moderately thermophilic** with optimal growth between 40–60°C. Major fatty acids include **C_{16:0}** and **C_{16:1} ω_{7c}**. **18-Methylenesubiquinone-8 (MQ-8)** is the predominant lipoquinone. Isolated from sediments of freshwater lakes and rivers, wetland muds, activated sludge, and wastewater. Belongs to the *Gammaproteobacteria* as part of the Order *Methylococcales*, Family *Methylococcaceae*.

The mol% G + C of the DNA is: 59–65 (*T_m*).

Type species: ***Methylococcus capsulatus* Foster and Davis 1966, 1929.**

FURTHER DESCRIPTIVE INFORMATION

Morphology and ultrastructure Cells of *Methylococcus capsulatus* are usually spherical in shape; however, in stationary growth phase cultures, a significant proportion of cells may appear as coccobacilli or short rods. *Methylococcus thermophilus* has a more pleomorphic morphology, with cells ranging from cocci to rods. Cells usually occur singly, in pairs, and sometimes in chains; however, rosette formations are absent. Although *Methylococcus capsulatus* strains lack flagella, *Methylococcus thermophilus* cells are usu-

ally vigorously motile upon isolation. Motility is spontaneously lost in many isolates of *Methylococcus thermophilus* (including the type strain) after several serial transfers. Motility is also more pronounced in younger cultures and in cultures grown on methanol.

The cell envelope of *Methylococcus* species is typical of Gram-negative bacteria. Hydroxy fatty acids from the outer membrane lipopolysaccharide have been analyzed in detail in *Methylococcus capsulatus*, *Methylococcus thermophilus*, and *Methylococcus* sp. JB140 (Bowman et al., 1991a). The major components include C_{10:0} 3OH, C_{12:0} 3OH, C_{14:0} 3OH, and C_{16:0} 3OH. Cells are covered by an exopolysaccharide capsule (Whittenbury et al., 1970a), which often causes a degree of cell aggregation.

The cells of *Methylococcus* species possess the intracytoplasmic membrane (ICM) and cyst formation that commonly occur in many members of the Family *Methylococcaceae*. The ICM formation and ultrastructure have been examined most extensively in *Methylococcus capsulatus* ATCC 33009 (strain "Bath") (Hyder et al., 1979). ICMs are formed when *Methylococcus capsulatus* is grown on methane or methanol, and ICM levels increase in proportion to methane oxidation rates. Growth in the absence of copper (levels less than 1.0 μmol/g dry weight) suppresses formation of ICMs, as well as drastically reducing pMMO activity and derepressing sMMO activity (Prior and Dalton, 1985).

Cysts Single-bodied spherical cysts similar to those observed in *Azotobacter chroococcum*, but comparatively simpler and less defined in structure, are formed by *Methylococcus capsulatus* and *Methylococcus thermophilus*, usually in the stationary growth phase (Whittenbury et al., 1970a; Malashenko et al., 1975b). Increased cyst formation is usually associated with increasing cell refractility,

but cyst formation is not as profuse as in species of *Methylobacter*. Cysts can be visualized by light microscopy after staining (Vela and Wyss, 1964). The cysts are mainly made up of glucan-type polysaccharides (Sutherland and MacKenzie, 1977), which are discarded as cells revert to a normal vegetative state. Production of cysts allow *Methylococcus capsulatus* to survive methane starvation for several weeks even though vegetative cells die rapidly in the absence of methane. The cysts, however, do not confer desiccation or heat resistance. In addition, *Methylococcus capsulatus* cells contain granules of poly- β -hydroxybutyrate, which also contribute to changes in cell refractility. Cells may also contain polyphosphate (volutin) inclusions.

Nutrition and growth conditions *Methylococcus* strains grow well in nitrate mineral salts (NMS) liquid media¹ *Methylococcus thermophilus* tends to produce dark brown colonies with an associated diffusible melanin-like brown pigment; the diffusible pigment is usually lost rapidly upon subculture. *Methylococcus capsulatus* colonies appear off-white to light tan in color and never produce diffusible pigments. Both *Methylococcus* species, depending on the strain, can readily form both rough and smooth colonial variants, with rough colonies possessing a somewhat cartilaginous consistency. This property appears to be particularly prevalent in isolates of *Methylococcus thermophilus* and phenotypically similar species (Bodrossy et al., 1997). With constant subculture, colonies tend to become consistently more butyrous and mucoid.

Methylococcus species are thermotolerant or moderately thermophilic, with optimal growth temperatures of 40–60°C. No strains have been found so far that can grow at temperatures higher than 65°C or below ~25°C.

No growth factors are required, and the strains are nonhalophilic. Some *Methylococcus capsulatus* strains can utilize methylamine, formate, and/or formamide as sole carbon and energy sources. However, *Methylococcus thermophilus* strains, the type strain of *Methylococcus capsulatus* ATCC 19069 (strain "Texas"), and the more heavily studied strain ATCC 33009 cannot utilize any of these compounds. No carbon-carbon bonded compounds can be used for growth.

Nitrogen sources are usually provided in the form of a nitrate or ammonium salt, although *Methylococcus* strains can also use yeast extract, Casamino acids (Difco), and amino acids. In addition, *Methylococcus capsulatus* and most strains of *Methylococcus thermophilus* can also fix atmospheric nitrogen by means of an oxygen-sensitive nitrogenase (Murrell and Dalton, 1983; Zhivotchenko et al., 1995b). A modified acetylene reduction procedure is used to detect nitrogenase activity in methanotrophic cultures (Takeda, 1988).

Some *Methylococcus capsulatus* strains can form sMMO, which allows these strains to grow fairly well in media lacking copper (Prior and Dalton, 1985). Evidence that a strain lacks sMMO is indicated by poor growth in copper-free mineral media; production of sMMO can also be rapidly detected by oxidation of naphthalene to α -naphthol (Brusseau et al., 1990). PCR detec-

tion using sMMO-specific primers or detection by filter or colony hybridization techniques can be used to definitively detect the presence of sMMO genes (Stainthorpe et al., 1991; Miguez et al., 1997).

Methylococcus species primarily uses the RuMP pathway to fix formaldehyde for cell carbon. However, the key enzyme for the serine pathway (α -hydroxypyruvate reductase) has also been detected (Strom et al., 1974). The tricarboxylic acid cycle is incomplete, lacking 2-oxoglutarate dehydrogenase activity, but an NAD-dependent isocitrate dehydrogenase is present. *Methylococcus* species also contain a partially functional Benson-Calvin pathway for CO₂ fixation, including activity of the enzymes ribulose-1,5-diphosphate carboxylase and phosphoribulokinase (Whittenbury and Krieg 1984a; Bowman et al., 1993b). The importance of this ability with respect to the cell carbon budget is still not understood, since autotrophic growth of *Methylococcus* species has not been obtained in the absence of methane.

Lipid composition The major fatty acids in *Methylococcus* species have been found to be C_{16:0} and C_{16:1} ω 7c (Table BXII.7.68 of the chapter describing the family *Methylococcaceae*). The high level of C_{16:0} is a characteristic that distinguishes this species from psychrophilic and mesophilic type I methanotrophs. The neutral lipid fraction of *Methylococcus capsulatus* is unusual among prokaryotes as it contains squalene, methylated sterols, and hopanoids (Bird et al., 1971; Neunlist and Rohmer, 1985). The unusual distribution of cyclic triterpenes and cyclopropane fatty acids (which become more prevalent with growth under reduced oxygen tension) is believed to improve the stability of outer and intracytoplasmic membranes (Jahnke and Nichols, 1986; Jahnke et al., 1992). The major respiratory coenzyme Q in *Methylococcus capsulatus* and *Methylococcus thermophilus* is 18-methylene-ubiquinone-8 (MQ-8) (Collins and Green, 1985).

Ecology Little information exists on the specific ecology of *Methylococcus* species; however, they have been isolated from a range of aquatic and terrestrial habitats, including soils and sediments of rivers and ponds, sewage sludge, and in the vicinity of various thermal environments (Malashenko et al., 1975a).

ENRICHMENT AND ISOLATION PROCEDURES

Methylococcus species can be enriched and isolated from the microaerobic and aerobic sediments of freshwater lakes and rivers, soil, activated sludge, and near certain thermal environments. A small amount of a sample is added to NMS liquid mineral medium in serum vials or in cotton-stoppered flasks placed within airtight containers. Methane is added directly to the vials and containers, usually by first removing a portion of the headspace. The best methane/air ratio is equivocal but should be in a range of 1:10–1:1. Methane should be of high purity, as natural gas may contain acetylene, which is a suicide substrate of pMMO and sMMO and will hinder or prevent growth, even at low concentrations. Static incubation should then proceed at 40–60°C. Successful enrichments of *Methylococcus* strains develop a white to brown turbidity that is often accompanied by a surface pellicle. Growth from the enrichments can be plated onto mineral salts agar plates, which are then incubated under 1:1 methane/air at 45–60°C.

One of the most problematic areas of methanotroph study is that of obtaining pure cultures. In practically all situations, methanotroph enrichments are severely contaminated by nonmethanotrophic (often methylotrophic) bacteria, which can easily overgrow and/or predate cultures. Due to the relatively slow growth of *Methylococcus* species and the humid conditions in

1. Nitrate mineral salts medium is slightly modified from Whittenbury et al. (1970b) and consists of (per liter of tap water): MgSO₄·7H₂O, 1.0 g; CaCl₂·6H₂O, 0.2 g; ferric-EDTA, 10 mg; NH₄Cl, 0.5 g or KNO₃, 1.0 g; trace element solution, 1.0 ml. The trace element solution contains (per liter of distilled water): CuSO₄·5H₂O, 0.2 g; ZnSO₄·7H₂O, 10 mg; MnCl₂·4H₂O, 3 mg; H₃BO₃, 30 mg; CoCl₂·6H₂O, 10 mg; and Na₂MoO₄·7H₂O, 10 mg; pH 4.0. The medium is adjusted to pH 6.8 before autoclaving. The autoclaved medium is allowed to cool, and 2 ml of 1.0 M sodium phosphate buffer (pH 6.8) is added. Addition of extra copper as CuSO₄·5H₂O promotes the growth of methanotrophs (Zhivotchenko et al., 1995a).

which plates are incubated, fungal contamination is also frequent unless containers are thoroughly cleaned with ethanol before each use. Fungicides, such as cycloheximide or nystatin, added to the medium are usually effective in reducing this problem.

A straightforward method for the isolation of *Methylococcus* and other methanotrophs uses mineral agar media that contain small amounts of yeast extract (0.05% w/v) and methanol (0.05% v/v) (Malashenko et al., 1975a), with incubation occurring under a methane/air atmosphere. The enrichment culture is then serially diluted onto the media to the point of extinction. Single colonies from these spread-plates are then transferred to liquid media. A number of passages from liquid media to spread-plates and back to liquid media are often required. The purity of the culture can be assessed by assuring that strains show no growth on nutrient media in the presence or absence of methane and that cells possess a reasonably consistent morphology. More details on enrichment, isolation, and potential problems have been detailed by Whittenbury et al. (1970b) and Hanson et al. (1992).

MAINTENANCE PROCEDURES

Cryopreservation (with cryoprotectants such as DMSO or glycerol) at -80°C and lyophilization are generally ineffective for storing *Methylococcus* strains, as dramatic losses in viability follow freezing (Nesterov et al., 1986). *Methylococcus* strains can be stored in liquid nitrogen (with 20–30% v/v glycerol) for up to 6 months before recoverability becomes problematic. The storage of cultures at low temperature (4°C) under a methane atmosphere has been found to be very effective for short-term storage of up to 6 months.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Nitrogen fixation in methanotrophs cannot be tested by the standard acetylene reduction technique because acetylene acts as a suicide substrate of methane monooxygenase. Instead, resting cultures are used. First, the methanotroph is cultured with methanol or methane in vials using nitrogen-free NMS media prepared with 0.2–0.3% (w/v) purified agar to make it semisolid, thereby reducing oxygen diffusion and creating a microaerobic environment that promotes nitrogen fixation. After adequate growth has occurred, acetylene is injected into the headspace (final concentration 2–5%, v/v) and the vials are incubated for several hours. During this time, samples are taken periodically and tested by gas chromatography-thermal conductivity to detect ethylene in the headspace (Murrell and Dalton, 1983; Takeda, 1988).

Putative soluble methane monooxygenase activity of methanotrophs can be rapidly assessed using the naphthalene oxidation technique (Brusseau et al., 1990). First, the methanotroph to be tested is cultured in copper-free NMS liquid medium. Glassware used for cultivation should be washed with 1% (v/v) nitric acid to strip off contaminating copper ions. An equal volume of a saturated naphthalene solution ($\sim 30\text{ mg/l}$ at 25°C) is added to a suspension of methanotrophic cells (degassed briefly of residual methane, using nitrogen). The mixture is incubated at room temperature for approximately one hour. To the suspension, 20 μl of a freshly prepared solution of 1% (w/v) tetratized *o*-dianisidine (Fast Blue salt B) is then added and mixed. The presumptive presence of soluble methane monooxygenase is indicated by the immediate production of a reddish semi-soluble diazo compound, formed from α -naphthol, which can be measured by spectrophotometry at 540 nm.

DIFFERENTIATION OF THE GENUS *METHYLOCOCCUS* FROM OTHER GENERA

Phenotypic characteristics and fatty acid profiles distinguishing *Methylococcus* from other type I methanotrophic genera are shown in Tables BXII.γ.67 and BXII.γ.68 of the chapter describing the family *Methylococcaceae*.

TAXONOMIC COMMENTS

Numerical taxonomic analysis suggested that several of the methanotrophic groups described by Whittenbury et al. (1970b) were related to *Methylococcus*. These groups were subsequently published as new species of *Methylococcus* (Romanovskaya et al., 1978), and their names subsequently appeared in validation list No. 7 as *Methylococcus bovis*, *Methylococcus chroococcus*, *Methylococcus luteus*, *Methylococcus vinelandii*, and *Methylococcus whittenburyi* (*Methylobacter capsulatus*). Subsequent studies using immunological, protein electrophoresis, and fatty acid analysis indicated that this expanded version of the genus *Methylococcus* was clearly heterogeneous (Andreev and Gal'chenko, 1978; Gal'chenko and Nesterov, 1981; Bezrukova et al., 1983; Meyer et al., 1986; Bowman et al., 1991a). Moreover, genomic characteristics demonstrated that the genus was made up of two major groups (Bowman et al., 1991b). This nomenclatural problem was resolved only by retention of *M. capsulatus* and *M. thermophilus* in the genus *Methylococcus*, with the description of *Methylococcus* appropriately emended (Bowman et al., 1993b) and transfer of the other species to the genus *Methylobacter*, which was created to accommodate them.

"*Methylococcus fulvus*" Malashenko, Romanovskaya and Kvashnikov 1972, 877 possesses phenotypic traits highly similar to those of *Methylobacter luteus* (Romanovskaya et al., 1978) and lacks a type strain.

The single strain making up the species *Methylococcus mobilis* Hazeu et al., 1980b, 676 has been lost, and this species should be regarded as invalid until a replacement strain has been isolated and described. *Methylococcus* species form a deeply branching clade within the *Gammaproteobacteria* (Fig. BXII.γ.103). Outside of this clade, the closest bacterial relative in terms of 16S rRNA sequence similarity is the photosynthetic species *Ectothiorhodospira shaposhnikovii* (evolutionary distance 0.09–0.10). The *Methylococcus* clade groups separately from the remaining type I methanotrophs, but evolutionary distances are roughly equivalent to the interposing genera (evolutionary distance 0.10–0.14).

DNA–DNA hybridization analysis indicates that *Methylococcus capsulatus* ATCC 19069 possesses high levels of hybridization with strain ATCC 33009 (Bowman et al., 1993b). A DNA hybridization range of 40–43% is found with the non-extant strain JB140; however, only negligible levels of DNA hybridization (<25%) is found between strains of *Methylococcus capsulatus* and *Methylococcus thermophilus*. Moderately thermophilic methanotrophs were recently described by Bodrossy et al. (1997) as the genus *Methylocaldum*, which includes the species *Methylocaldum szegediense* (the type species), *Methylocaldum tepidum*, and *Methylocaldum gracile*. (The latter was renamed from "*Methylomonas gracilis*" NCIMB 11128.) Bodrossy and colleagues have noted phenotypic similarity between *Methylococcus thermophilus* and *Methylocaldum* species; moreover, high levels of DNA hybridization have been reported between *Methylococcus thermophilus* IMV-2Yu and "*Methylomonas gracilis*" NCIMB 11912 (Bowman et al., 1993b). The phenotypic differences between *Methylococcus* and *Methylocaldum* appear to be trivial in nature and, based on the current description of *Methylo-*

coccus, it is nearly impossible to differentiate *Methylococcus* from *Methylocaldum*. Both groups possess similar fatty acid profiles, mol% G + C contents of the DNA, genomic characteristics, and *pmoA* sequences. On the other hand, 16S rDNA analysis suggests that the two groups are phylogenetically distinct (evolutionary distance ~0.08). This is in contrast to the above-mentioned similarities between the two groups, particularly the hybridization data for *Methylococcus thermophilus* IMV-2Yu and "*Methylomonas gra-*

cilis" NCIMB 11128 (Bowman et al., 1993b), and this anomaly suggests that the 16S rDNA sequence of *Methylococcus thermophilus* needs to be verified. It is certainly clear that, based on DNA hybridization data, *Methylocaldum gracile* is a subjective synonym of *Methylococcus thermophilus* and is thus misnamed. Further DNA hybridization analysis is needed to confirm or deny the genotypic distinctiveness of other *Methylocaldum* species from *Methylococcus thermophilus*.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *METHYLOCOCCUS*

Phenotypic and genotypic characteristics differentiating the species of *Methylococcus* are shown in Table BXII.γ.69, while additional characteristics are shown in Table BXII.γ.70. The phe-

notypic characteristics possessed by strains of *Methylococcus thermophilus* are essentially the same as those possessed by *Methylocaldum* strains.

List of species of the genus *Methylococcus*

1. *Methylococcus capsulatus* Foster and Davis 1966, 1929^{AL} *cap.su.la'tus*. M.L. adj. *capsulatus* encapsulated.

Cells are coccoidal and nonmotile. Colonies are white to light tan, round, convex, entire- or lobate-edged, with a cartilaginous or butyrous consistency. Other characteristics are as described for the genus and as listed in Tables BXII.γ.69 and BXII.γ.70. For further descriptive information, see Foster and Davis (1966) and Bowman et al. (1993b).

Optimum temperature for growth, ~45°C. pH range for growth, 5.5–8.5; optimum, 6.5–7.0.

Isolated from lakes and river mud, wetland muds, activated sludge, and wastewater.

The mol% G + C of the DNA is: 62.5 (Bd), type strain. Analysis of several other strains using the thermal denaturation method found a range of 62–65 mol% (Bowman et al., 1991b).

Type strain: Texas, ACM 1292, ATCC 19069 NCIMB 11853.

GenBank accession number (16S rRNA): X72770.

Additional Remarks: Another important strain is Bath (ATCC 33009, NCIMB 11132, VKM-TRMC, IMET 10544).

2. *Methylococcus thermophilus* Malashenko, Romanovskaya, Bogachenko and Shved 1975b, 779^{AL} *ther.mo'phil.us*. Gr. n. *therme* heat; Gr. adj. *philus* loving; M.L. adj. *thermophilus* heat loving.

Cells are coccoidal and usually motile on initial isolation. Motile strains usually lose motility after several subcultures. Colonies are light to dark brown, round, raised convex, with entire to irregular edges, and have a viscid or cartilaginous consistency. Produce a diffusible brown pigment, which is usually lost during the strain purification process. Other

characteristics are as described for the genus and as listed in Tables BXII.γ.69 and BXII.γ.70. For further descriptive information, see Bowman et al. (1993b).

Optimum growth temperature, 50–55°C. pH range for growth, 5.5–9.0; optimum, 7.0.

Isolated from therapeutic mud, moist subtropical-tropical soils, creek mud, and pond sediment.

TABLE BXII.γ.70. Other phenotypic characteristics of the species of the genus *Methylococcus*^{a,b}

Characteristics	1. <i>M. capsulatus</i>	2. <i>M. thermophilus</i>
Rod-like cells formed	d	+
Viscid-cartilaginous colonies	d	+
Poly-β-hydroxybutyrate inclusions	+	+
Polyphosphate inclusions	d	d
Capsular exopolysaccharide	+	+
Heat resistance (80°C, 20 min)	–	–
Desiccation resistance	–	–
Growth in the presence of:		
0.0075% KCN	+	+
0.02% sodium azide	–	–
0.001% crystal violet	–	–
0.001% malachite green	–	–
0.01% SDS	–	–
3.0% NaCl	–	d
Lysed by 0.2% SDS	–	–
Reduction of 0.01% triphenyltetrazolium chloride	–	–
Esculin hydrolysis	–	–
Catalase activity	+	+
Oxidase activity	+	d
Soluble methane monooxygenase activity	d	nd
Ribulose-1,5-diphosphate carboxylase activity	+	+
Phosphatase activity	d	–
Urease activity	–	d
Nitrate reduction (to nitrite)	d	+
Nitrogen fixation	+	d
Sole carbon and energy sources:		
Methane, methanol	+	+
Formate, methylamine, formamide, dimethylamine, trimethylamine, trimethylamine N-oxide	d	–

TABLE BXII.γ.69. Characteristics differentiating the species of the genus *Methylococcus*^{a,b}

Characteristic	1. <i>M. capsulatus</i>	2. <i>M. thermophilus</i>
Motile cells	–	+
Melanin-like pigment production	–	+
Growth at 55°C	–	+
Mol% G + C of DNA (<i>T_m</i>)	62.5–65.2	59.1–60.8

^aFor symbols, see standard definitions.

^bPhenotypic data from Bowman (1992) and Bowman et al. (1993b).

^aFor symbols see standard definitions; nd, not determined.

^bPhenotypic data from Bowman (1992) and Bowman et al. (1993b).

The species *Methylocaldum gracile* is a synonym of *Methylococcus thermophilus*, as indicated by DNA–DNA hybridization data (Bowman et al., 1993b). The phylogenetic discrepancy between strain IMV-2Yu and NCIM 11912 is attributed to a potentially erroneous *Methylococcus thermophilus* 16S rRNA sequence.

The mol% G + C of the DNA is: 59–61 (T_m).

Type strain: VKM-2Yu, V 2YU, ACM 3585.

GenBank accession number (16S rRNA): X73819.

Additional Remarks: GenBank accession number for *Methylocaldum gracile* is U89298.

Species Incertae Sedis

1. “**Methylococcus fulvus**” Malashenko, Romanovskaya and Kvashnikov 1972, 877.

fulvus. L. adj. *fulvus* tawny, yellowish brown.

See Taxonomic Comments.

2. **Methylococcus mobilis** Hazeu, Batenburg-van der Vegte and de Bruyn 1980b, 676^{VP} (Effective publication: Hazeu Batenburg-van der Vegte and de Bruyn 1980a, 211.)

mo' bi. lis. L. masc. adj. *mobilis* motile.

See Taxonomic Comments

The mol% G + C of the DNA is: unknown.

Genus II. Methylobacter Bowman, Sly, Nichols and Hayward 1993b, 749^{VP} emend. Bowman, Sly and Stackebrandt 1995, 184.

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Me.thy.lo.bac' ter. Fr. n. *méthyle* the methyl radical; Gr. n. *bacter* rod; M.L. n. *Methylobacter* methyl rod.

Cells are **spherical or elliptical**, 0.8–1.5 × 1.2–3.0 µm. Usually occur singly, in pairs, and occasionally in chains. **Gram negative**. When motile, form single polar flagellum. Form a **glucan-containing cyst-like resting stage**, which usually confers **desiccation resistance**. Poly-β-hydroxybutyrate granules present. **Strictly aerobic respiratory metabolism** with oxygen as the electron acceptor. **Obligate utilizers of methane for carbon and energy**. Carbon–carbon bonded compounds are not utilized. Oxidase and catalase positive. Uses the **ribulose monophosphate pathway** to fix formaldehyde for cell carbon. Key enzyme activity of the serine transhydroxymethylase pathway is absent. **Nitrogen fixation is absent**. Not susceptible to lysis by 0.2% sodium dodecyl sulfate. **Mesophilic**, with optimal growth between 30°C and 35°C. One species requires sodium ions and other factors for growth. The major fatty acid is C_{16:1 ω7c}. **Ubiquinone-8 (Q-8)** is the predominant lipoquinone. Habitat: Isolated from sediments of freshwater lakes and rivers, wetland muds, activated sludge, and wastewater. Belongs to the *Gammaproteobacteria* as part of the Order *Methylococcales*, Family *Methylococcaceae*.

Type species: **Methylobacter luteus** (Romanovskaya, Malashenko and Bogachenko 1978) Bowman, Sly, Nichols and Hayward 1993b, 749 (*Methylococcus luteus* Romanovskaya, Malashenko and Bogachenko 1978, 124; *Methylococcus bovis* Romanovskaya, Malashenko and Bogachenko 1978, 124.)

FURTHER DESCRIPTIVE INFORMATION

Morphology and ultrastructure Cells of *Methylobacter* species possess a characteristic elliptical, rod-like morphology, with a width of 0.8–1.5 × 1.2–3.0 µm. They occur mostly singly or in pairs, although chain formation is prevalent in some strains in the late exponential growth phase. When motile, the cells are propelled by a single polar flagellum. *Methylobacter luteus* is non-motile, while *Methylobacter whittenburyi* strains are usually motile when first isolated but can spontaneously lose this ability after extensive subculture. Motility appears most pronounced in young cultures of *M. whittenburyi* and *Methylobacter marinus*; older cultures are often devoid of motile cells (Romanovskaya et al., 1978; Bowman et al., 1993b). Cell walls have an ultrastructure that is typical of Gram-negative bacteria. Cells are surrounded by capsular material that is detectable by India ink staining.

Cells contain type I intracytoplasmic membranes, typical for other members of the family *Methylococcaceae*. *Methylobacter*

species form well-defined *Azotobacter*-type cysts, which give cells a refractile appearance (Whittenbury et al., 1970a). Cyst formation is most pronounced in stationary-growth-phase cultures and appears to be stimulated when oxygen becomes limiting for growth. The cysts are readily detectable by the *Azotobacter* cyst-staining procedure of Vela and Wyss (1964). The cysts are not heat resistant, but they are relatively desiccation resistant. Spontaneous loss of cysts has been noted in one study and usually leads to problems in maintaining culture viability (Lidstrom, 1988). Though the cysts may be a mechanism for surviving times of oxygen deprivation and desiccation, *Methylobacter* strains are still susceptible to methane starvation and eventually lose viability within 1–3 weeks without methane or methanol substrate. Poly-β-hydroxybutyrate granules tend to be formed by *Methylobacter* strains, particularly in younger cultures.

Cultural features *Methylobacter* strains form an even turbidity in NMS liquid medium, with little or no pellicle development. On agar, colonies are uniformly circular and convex with entire edges, possess a creamy consistency, and reach a diameter of 1–2 mm after 7–14 days under methane/air atmosphere (1:1, v/v). *M. whittenburyi* and *M. marinus* strains form colonies that are initially an off-white color but become increasingly tan to brown after prolonged incubation. At the same time, a tan diffusible melanin-like pigment is slowly released into the medium. *M. luteus* differs by forming lemon-yellow colonies, with some strains also releasing a water-soluble yellow pigment. Detailed chemical analysis of the yellow pigment has yet to be performed, but preliminary spectrophotometric scans indicated that it is not carotenoid in nature, nor does it undergo a bathochromic shift if exposed to strong alkaline solutions.

Methylobacter species are mesophilic, and most strains grow in the range of 15–40°C, with optimal growth occurring at about 30°C. The pH range for growth ranges from 5.5–9.0, with optimal growth occurring at about pH 7.0. No *Methylobacter* strains require growth factors, and all are nonhalophilic. The estuarine species *M. marinus* grows optimally with about 0.1 M NaCl in tap water or with half-strength seawater salts. One strain of *Methylobacter marinus* requires nicotinic acid for growth (Lidstrom, 1988).

Nutrition and metabolism *Methylobacter* strains are strictly aerobic obligate methanotrophs, with carbon and energy substrates limited to methane and methanol. A few strains, however, can

also utilize methylamine. No strains have been demonstrated to grow on carbon-carbon bonded compounds. Upon isolation, most *Methylobacter* strains may be sensitive to methanol and do not grow on media containing 0.1% (v/v) methanol. However, good growth occurs when methanol is provided only as a vapor. It is possible to acclimate strains to methanol by slowly adding increasingly higher concentrations of methanol to media while growing the strain on methane. Methane appears to be oxidized by only the particulate form of methane monooxygenase. The presence of soluble methane monooxygenase has not been demonstrated in *Methylobacter* strains so far (Stainthorpe et al., 1991; Smith et al., 1997c). Formaldehyde is fixed by the ribulose monophosphate pathway, and key enzymes for the serine pathway are absent. Ribulose-1,5-diphosphate carboxylase activity is also absent. *Methylobacter* strains can utilize nitrate and ammonia salts, yeast extract, Casamino acids, and various amino acids as nitrogen sources, but high levels (>0.5% w/v) of complex organic compounds are inhibitory to growth. Some strains, particularly those of *M. luteus*, can produce a urease and can use urea as a sole nitrogen source. No *Methylobacter* strain has been found to be capable of fixing atmospheric nitrogen.

Fatty acid composition The outer membrane lipopolysaccharide hydroxy fatty acids have been analyzed in detail in *M. luteus* and *M. whittenburyi*. The major components in *M. luteus* include C_{10:0} 3OH and C_{16:0} 3OH, with smaller quantities of C_{12:0} 2OH, C_{14:0} 3OH, and C_{15:0} 3OH. *M. whittenburyi* contains almost entirely C_{16:0} 3OH (Bowman et al., 1991a). The fatty acid profiles of *Methylobacter* species have been found to be very similar (Table BXII.γ.68 of the chapter describing the family *Methylococcaceae*, Andreev and Gal'chenko, 1978). The predominant fatty acid is C_{16:1} ω7c, accompanied by lower levels of C_{14:0}, C_{16:1} ω6c, C_{16:1} ω5c, and C_{16:0} (Bowman et al., 1991a). The lack of C_{16:1} ω8c and relatively low levels of C_{14:0} and C_{16:0} distinguish *Methylobacter* from other type I methanotrophs.

Ecology The specific ecology of *Methylobacter* species is generally unknown. Indirect immunofluorescence studies, however, indicate that they are common in surface sediments and in the water columns of various freshwater and brackish water environments making up as much as 10–40% of the total bacterial population (Malashenko et al., 1987; Gal'chenko et al., 1988; Gal'chenko, 1994; Saralov et al., 1984).

ENRICHMENT AND ISOLATION PROCEDURES

The enrichment and isolation approach described for the genus *Methylococcus* can also be applied for *Methylobacter* strains. However, the incubation temperature is in this case lowered to 25–30°C. Salt composition and pH of the media all should closely reflect the habitat being investigated.

MAINTENANCE PROCEDURES

Methods for maintenance are the same as given for the genus *Methylococcus*.

DIFFERENTIATION OF THE GENUS *METHYLOBACTER* FROM OTHER GENERA

Phenotypic, chemotaxonomic, and genotypic properties differentiating *Methylobacter* from other type I methanotrophs are

shown in Tables BXII.γ.67 and BXII.γ.68 of the chapter describing the family *Methylococcaceae*.

TAXONOMIC COMMENTS

The genus *Methylobacter* was formed from the genus *Methylococcus* (Bowman et al., 1993b). *Methylobacter* species are equivalent to a similarly named group first coined by Whittenbury et al. (1970b). DNA–DNA hybridization, fatty acid analyses, and phylogenetic analyses have further refined the genus, so that species lacking cyst formation and different fatty acid profiles have been moved into the genus *Methylomicrobium*. This has necessitated an emendation of the description of *Methylobacter* (Bowman et al., 1995).

Phylogenetic analysis using 16S rDNA sequences indicates that *Methylobacter* forms a relatively tight cluster within the *Gamma-proteobacteria*. *Methylobacter* is part of a larger phylogenetic cluster that includes *Methylosphaera*, *Methylomicrobium*, *Methylomonas*, and methanotrophic mytilid endosymbionts. The genus is most closely related to the genera *Methylomicrobium* and *Methylomonas* (evolutionary distance 0.04–0.06) (Bratina et al., 1992; Bowman et al., 1995).

The mol% G + C of the DNA of *Methylobacter* ranges from 48–55 (*T_m*). (Malashenko et al., 1975b; Meyer et al., 1986; Lysenko et al., 1988; Bowman et al., 1991b) (Table BXII.γ.71); this value is similar to those for *Methylomonas* and *Methylomicrobium*. DNA–DNA hybridization has been found to be useful in sorting out some of the taxonomic problems affecting the genus *Methylococcus*. Various *Methylococcus* species, including *Methylococcus vine-landii*, *Methylococcus chroococcus*, “*Methylococcus ucrainicus*”, *Methylococcus bovis* IMET 10593 (a non-pigmented strain), *Methylococcus whittenburyi*, and phenotypically similar isolates, all exhibit high levels of DNA–DNA hybridization (54–93% similarity) with each other indicating that they were a single genospecies (Lysenko et al., 1988; Bowman et al., 1993b). This argument is supported by inherent phenotypic similarity, similar protein electrophore patterns (Gal'chenko and Nesterov, 1981), immunological cross-reactivity (Bezrukova et al., 1983), and close phylogenetic affiliation. (See Fig. BXII.γ.103 in the description of the order *Methylococcales*). In addition, *Methylococcus luteus* shares very high levels of DNA–DNA hybridization (98% similarity) with the yellow-pigmented strain *Methylococcus bovis* IMV-B-3098 (Bowman et al., 1993b). Overall DNA–DNA hybridization levels among the three valid species of *Methylobacter* are 20–40%.

Recently, two species, *Methylobacter psychrophilus* (Omel'chenko et al., 1996) and “*Methylobacter alkaliphilus*” (Khmelenina et al., 1997), were proposed for strains isolated from tundra soil and soda lake habitats, respectively. *Methylobacter psychrophilus* possesses phenotypic properties and DNA base composition similar to those of *Methylosphaera hansonii*; both species consist of spherical nonmotile cells, lack cysts, possess gas vesicles, have psychrophilic growth requirements, and possess relatively low mol% G + C contents of their DNA (approximately 45–46). “*Methylobacter alkaliphilus*” possesses morphological traits typical of *Methylobacter* species but has a different fatty acid profile that includes high levels of C_{16:0} and, depending on the growth conditions, C_{16:0} iso (Khmelenina et al., 1997); C_{16:1} fatty acid is only a relatively minor component. A 16S rDNA sequence for this species would be useful in confirming that it belongs to the genus *Methylobacter*.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *METHYLOBACTER*

Phenotypic and genotypic characteristics differentiating the species of *Methylobacter* are shown in Table BXII.γ.71. Additional characteristics are shown in Table BXII.γ.72.

List of species of the genus *Methylobacter*

1. ***Methylobacter luteus*** (Romanovskaya, Malashenko and Bogachenko 1978) Bowman, Sly, Nichols and Hayward 1993b, 749^{VP} (*Methylococcus luteus* Romanovskaya, Malashenko and Bogachenko 1978, 124 *Methylococcus bovis* Romanovskaya, Malashenko and Bogachenko 1978, 124.)
lu' te.us. L. adj. *luteus* golden-yellow.

Cells are nonmotile cocci or elliptical rods, 1.2–1.5 × 1.5–2.0 µm. Other characteristics are as described for the genus and as listed in Tables BXII.γ.71 and BXII.γ.72. For further descriptive information, see Bowman et al. (1993b).

Optimum temperature for growth, ~30°C; range, 15–40°C. pH optimum, ~7.0; range, 5.5–9.0.

Isolated from sediments of freshwater lakes and ponds.

The mol% G + C of the DNA is: 50.3–51.1 (*T_m*).

Type strain: ACM 3304, IMET 10584, NCIMB 11914, VKM-53B.

GenBank accession number (16S rRNA): M95657.

Additional Remarks: “*Methylococcus fulvus*” Malashenko, Romanovskaya and Kvashnikov 1972, 877 is phenotypically

indistinguishable from *Methylobacter luteus* (Romanovskaya et al., 1978); moreover, it lacks an available strain, thereby making genotypic comparisons impossible.

2. ***Methylobacter marinus*** Bowman, Sly, Nichols and Hayward 1993b, 749^{VP}
mar' i.nus. L. adj. *marinus* from the sea.

Cells are motile elliptical rods, 0.8–1.2 × 1.5–2.0 µm. Requires 0.1 M NaCl and tap water (or half-strength natural or artificial seawater salts) for growth. May require nicotinic acid.

Other characteristics are as described for the genus and as listed in Tables BXII.γ.71 and BXII.γ.72. For further descriptive information, see Lidstrom (1988) and Bowman et al. (1993b).

Optimum temperature for growth, ~35°C; range, 15–40°C. pH optimum, ~7.0; range, 5.5–9.0.

Isolated from near a coastal sewage outfall in Pasadena, California.

TABLE BXII.γ.71. Characteristics differentiating the species of the genus *Methylobacter*^{a,b}

Characteristic	1. <i>M. luteus</i>	2. <i>M. marinus</i>	3. <i>M. whittenburyi</i>
Motility	–	+	d
Pigmentation	Y	Bb	Wb
NaCl and growth factor(s) required	–	+	–
Urease	+	–	d
Phosphatase	+	–	–
Mol% G + C of DNA (<i>T_m</i>)	50.3–51.1	54.3–55.0	48.7–53.7

^aFor symbols, see standard definitions; Y, yellow; Bb, buff-brown; Wb, white-brown.

^bPhenotypic data from Bowman et al. (1993b).

TABLE BXII.γ.72. Other phenotypic characteristics of the species of the genus *Methylobacter*^{a,b}

Characteristic	1. <i>M. luteus</i>	2. <i>M. marinus</i>	3. <i>M. whittenburyi</i>
Chains of cells	d	–	–
Poly-β-hydroxybutyrate present	+	+	+
Polyphosphate present	–	–	–
Capsule present	d	+	+
Cyst formation	+	d	+
Diffusible pigment formed	d	+	+
Growth at 37°C	d	+	d
<i>Growth in the presence of:</i>			
0.0075% KCN	d	d	d
0.02% sodium azide	–	d	d
0.001% crystal violet	–	d	–
0.001% malachite green	+	d	d
0.01% SDS	+	d	d
3% NaCl	d	+	–
Desiccation resistance	+	d	+
Heat resistance	–	–	–
Catalase	+	+	+
Oxidase	+	+	d
Serine pathway enzymes	–	–	–
Ribulose-1,5-diphosphate carboxylase	–	–	–
Nitrate (to nitrite) reduction	–	–	d
Nitrogen fixation	–	–	–
Reduction of 0.01% TTC ^c	d	–	d
<i>Carbon and energy sources:</i>			
Methanol as vapor	+	+	+
Methanol as solution (0.1% v/v)	d	–	d
Other C ₁ compounds ^d	–	–	–

^aFor symbols, see standard definitions.

^bPhenotypic data from Bowman et al. (1993b).

^cTTC, triphenyltetrazolium chloride.

^dCompounds tested (at 0.1%) included formate, formamide, methylamine, dimethylamine, and trimethylamine.

The mol% G + C of the DNA is: 54.3–55.0 (T_m).

Type strain: A4, ACM 4717.

GenBank accession number (16S rRNA): M95658.

3. **Methylobacter whittenburyi** (Romanovskaya, Malashenko and Bogachenko 1978) Bowman, Sly, Nichols and Hayward 1993b, 750^{VP} (*Methylococcus whittenburyi* Romanovskaya, Malashenko and Bogachenko 1978, 125) *whit.ten.bur' y.i.* L. n. *whittenburyi* after Whittenbury, in honor of British microbiologist Roger Whittenbury for his seminal research on methanotrophic bacteria.

Cells are motile elliptical rods, $1.2\text{--}1.5 \times 1.5\text{--}3.0 \mu\text{m}$. Other characteristics are as described for the genus and as listed in Tables BXII.γ.71 and BXII.γ.72. For further descriptive information, see Bowman et al. (1993b).

Optimum temperature for growth, $\sim 30\text{--}35^\circ\text{C}$; range, $15\text{--}40^\circ\text{C}$. pH optimum, ~ 7.0 ; range, $5.5\text{--}9.0$.

Isolated from sediments of freshwater lakes and ponds.

The mol% G + C of the DNA is: 48.7–53.7 (T_m).

Type strain: Y, 1521, ACM 3310, NCIMB 11128.

GenBank accession number (16S rRNA): L20843, X72773.

Species Incertae Sedis

1. “**Methylobacter alkaliphilus**” Khmelenina, Kalyuzhnaya, Starostina, Suzina and Trotsenko 1997, 260.

al.ka.li.phi' lus. M.L. n. *alcali* (from Arabic *al* the end; *qaliy* soda ash); Gr. adj. *philus* loving; M.L. adj. *alkaliphilus* loving alkaline conditions.

Coccoidal to elliptical cells, $1.2\text{--}1.3 \times 2.0\text{--}3.0 \mu\text{m}$. Motile by a single polar flagellum. Multiply by binary division. *Azotobacter*-type cysts and type I intracytoplasmic membranes are formed. Possess a protein surface layer (cup-shaped units).

Colonies on agar are white to cream, circular, with entire margins, and 1–2 mm in diameter after sufficient incubation in methane/air. No water-soluble pigments are released.

Halophilic and alkaliphilic. Grows at 0.15–10% NaCl, optimum growth at 2–4% NaCl. Optimum temperature, 30°C ; range, $8\text{--}37^\circ\text{C}$. Optimum pH, $9.0\text{--}9.5$; range, $7.0\text{--}10.5$.

Strictly aerobic. Obligate methanotroph: growth occurs only on methane and methanol. Carbon is assimilated by the ribulose monophosphate pathway. The serine-pathway enzyme α -hydroxypyruvate reductase is absent. The tricarboxylic-acid-cycle enzyme 2-oxoglutarate dehydrogenase is absent. Nitrate and ammonia are utilized as nitrogen sources via the glutamine and glutamate synthetase system.

Major fatty acids (when grown under optimal growth conditions, 0.7 M NaCl, pH 9.0) include $C_{16:0}$ (50.1%), $C_{16:0 \text{ cyc}}$ (11.6%), $C_{16:0 \text{ iso}}$ (8.7%), $C_{14:0}$ (8.0%), and $C_{16:1}$ (7.4%). At suboptimal growth conditions (low salt and neutral pH), cells produce more $C_{16:0 \text{ iso}}$ than $C_{16:0}$. Major phospholipids (under optimal growth conditions) include phosphatidylcholine, phosphatidylglycerol, and diphosphatidylglycerol (cardiolipin).

Isolated from soda lakes (pH 9.5, 1–3% salinity) in the Tuva region of Central Asia.

The mol% G + C of the DNA is: 47.6–47.9 (T_m).

Deposited strain: VKM B-2133.

2. **Methylobacter psychrophilus** Omel'chenko, Vasil'eva, Zavarzin, Savel'eva, Lysenko, Mityushina, Khmelenina and Trotsenko 2000, 423^{VP} (Effective publication: Omel'chenko, Vasil'eva, Zavarzin, Savel'eva, Lysenko, Mityushina, Khmelenina and Trotsenko 1996, 342.)

psy.chro' phi.lus. Gr. n. *psychros* cold; Gr. adj. *philus* loving; L. adj. *psychrophilus* cold loving.

Cells are coccoidal, $1.0\text{--}1.7 \mu\text{m}$ in diameter, occurring singly or in pairs. Possess a thin capsule. Multiply by binary division. Do not form cysts or other resting stages. Possess type I intracytoplasmic membranes and gas vesicles. Gas vesicles are more common in cells grown at higher cultivation temperatures and are cigar-shaped, 80–110 nm wide and up to 300 nm long.

Colonies are cream, circular, and convex, with an entire edge and a smooth shiny surface and possessing an aqueous consistency. Water-soluble pigments are not formed.

Psychrophilic. Optimum temperature, $3.5\text{--}10^\circ\text{C}$; range, $1.0\text{--}20^\circ\text{C}$. Optimum pH, $6.8\text{--}7.2$; range, $5.9\text{--}7.6$. Nonhalophilic. Growth factors are not required for growth.

Strictly aerobic obligate methanotroph. Growth occurs only on methane and methanol. Carbon is assimilated by the ribulose monophosphate pathway. Isocitrate dehydrogenase is NAD-specific, and 2-oxoglutarate dehydrogenase activity is absent. Ribulose-1,5-diphosphate carboxylase activity is absent.

The predominant fatty acid is $C_{16:1}$ with lower levels of $C_{14:0}$ and $C_{16:0}$.

Isolated from swampy soil of dwarf birch tundra in Russia.

The mol% G + C of the DNA is: 45.6 (T_m).

Deposited strain: Z-0021, VKM B-2103.

GenBank accession number (16S rRNA): AF152597.

Genus III. **Methylocaldum** Bodrossy, Holmes, Holmes, Kovács and Murrell 1998, 631^{VP} (Effective publication: Bodrossy, Holmes, Holmes, Kovács and Murrell 1997, 501)

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Me.thy.lo.cal.dum. N.L. n. *methylo* the methyl radical; L. adj. *caldus* hot; M.L. n. *Methylocaldum* heat-loving methylotherm.

Gram-negative, **coccoidal to long rods**. Moderately thermophilic; growth at temperatures $>40^\circ\text{C}$. **Utilize methane as sole carbon and energy source. No growth on compounds containing carbon-carbon bonds.** Enzymes for ribulose monophosphate and serine pathways of formaldehyde assimilation are present. No soluble

methane monooxygenase. Produce large cysts. Habitat: soil, hot springs.

The mol% G + C of the DNA is: 57–59.

Type species: **Methylocaldum szegediense** Bodrossy, Holmes, Holmes, Kovács and Murrell 1998, 631 (Effective publication: Bodrossy, Holmes, Holmes, Kovács and Murrell 1997, 501.)

List of species of the genus Methylocaldum

1. **Methylocaldum szegediense** Bodrossy, Holmes, Holmes, Kovács and Murrell 1998, 631^{VP} (Effective publication: Bodrossy, Holmes, Holmes, Kovács and Murrell 1997, 501.) *sze.ged.i.en.se*. Hung. n. Szeged the town of Szeged; M.L. adj. *szege diense* from the town of Szeged.

Gram-negative, coccoidal to long rods, $0.6\text{--}1.0 \times 1.0\text{--}1.5\text{ }\mu\text{m}$. Motile at some stages of growth. Chains may occur in late exponential phase. Pleomorphic in stationary phase. Nondiffusible pigments produced in later stages of growth. Colonies mucoid or cartilaginous, buff to light brown after 1 week. Older colonies may be shiny and smooth with entire margins or dry and rough with irregular margins. Utilizes methane as sole carbon and energy source. No growth on compounds containing carbon-carbon bonds. Sensitive to desiccation. Produces an extracellular polymeric substance in liquid culture. Optimum temperature 55°C (range $37\text{--}62^{\circ}\text{C}$). No growth on methanol, formate, formamide, or methylamine. Does not fix N_2 . Resistant to heating at 80°C . Enzymes for ribulose monophosphate and serine pathways of formaldehyde assimilation are present. No soluble methane monooxygenase. Produces large cysts. Habitat: soil, hot springs.

The mol% G + C of the DNA is: 57.

Type strain: OR2, NCIMB.

GenBank accession number (16S rRNA): U89300.

2. **Methylocaldum gracile** (ex Romanovskaya, Malashenko and Bogachenko 1978) Bodrossy, Holmes, Holmes, Kovács and Murrell 1998, 631^{VP} (Effective publication: Bodrossy, Holmes, Holmes, Kovács and Murrell 1997, 502.) *gra.cile*. L. adj. *gracile* slim, slender, thin.

Gram-negative, coccoidal to long rods, $0.4\text{--}0.5 \times 1.0\text{--}1.5\text{ }\mu\text{m}$. Motile. No chain formation. Pleomorphic in stationary phase. Nondiffusible pigments produced in later

stages of growth. Colonies are pasty or cartilaginous, buff to light brown after 1 week. Older colonies can turn dark brown. Utilizes methane as sole carbon and energy source. No growth on compounds containing carbon-carbon bonds. Optimum temperature 42°C (range $20\text{--}47^{\circ}\text{C}$). No growth above 50°C . Sensitive to desiccation and heating to 80°C . Does not fix N_2 . Enzymes for ribulose monophosphate and serine pathways of formaldehyde assimilation are present. No soluble methane monooxygenase. Produces large cysts. Habitat: soil, hot springs.

The mol% G + C of the DNA is: 59.

Type strain: VKM-14L, NCIMB 11912.

GenBank accession number (16S rRNA): U89298.

3. **Methylocaldum tepidum** Bodrossy, Holmes, Holmes, Kovács and Murrell 1998, 631^{VP} (Effective publication: Bodrossy, Holmes, Holmes, Kovács and Murrell 1997, 502.) *te.pi.dum*. M.L. adj. *tepidum* lukewarm.

Gram-negative, coccoidal to long rods, $1.0\text{--}1.2 \times 1.0\text{--}1.5\text{ }\mu\text{m}$. Chains may occur in late exponential phase. Pleomorphic in stationary phase. Nondiffusible pigments produced in later stages of growth. Colonies are buff and butyrous or cartilaginous after 1 week. Older colonies can turn light brown. Utilize methane as sole carbon and energy source. No growth on compounds containing carbon-carbon bonds. Sensitive to desiccation. Optimum temperature 42°C (range $30\text{--}47^{\circ}\text{C}$). No growth above 50°C . Does not fix N_2 . Enzymes for ribulose monophosphate and serine pathways of formaldehyde assimilation are present. No soluble methane monooxygenase. Produces large cysts. Habitat: soil, hot springs.

The mol% G + C of the DNA is: 59.

Type strain: LK6.

GenBank accession number (16S rRNA): U89297.

Genus IV. Methylobacterium Bowman, Sly and Stackebrandt 1995, 183^{VP}

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Me.thy.lo.mi.cro'bi.um. Fr. n. *méthyle* the methyl radical; Gr. adj. *micros* small; Gr. n. *bios* life; M.L. n. *Methylobacterium* a small organism able to utilize methyl units.

Rods $0.5\text{--}1.0 \times 1.5\text{--}2.5\text{ }\mu\text{m}$. Usually occur singly or in pairs. **Gram negative**. **Motile**, propelled by a single, unsheathed, polar flagellum. **Lack cysts** or other differentiated resting stages. Desiccation and heat-sensitive. Colonies are **nonpigmented**. **Have a strictly aerobic respiratory metabolism**, with oxygen as the electron acceptor. **Obligate utilizers of methane and methanol for carbon and energy**. Carbon-carbon bonded compounds are not utilized. Oxidase and catalase positive. Uses the **ribulose monophosphate pathway** to fix formaldehyde for cell carbon. Key enzyme activity of the serine transhydroxymethylase pathway is absent. Nitrogen fixation is absent. Not susceptible to lysis by 0.2% SDS. **Mesophilic** with optimal growth at $20\text{--}30^{\circ}\text{C}$. One species requires seawater salts for growth. Major fatty acids are **C_{16:1} ω7c**, **C_{16:1} ω5c**, **C_{16:1} ω5t**, and **C_{16:0}**. **Ubiquinone-8 (Q-8)** is the predominant lipoquinone. Isolated from sediments of freshwater lakes and rivers, wetland muds, agricultural and swampy soils, upper mixing layers of oceans, and estuarine waters. Belongs to the *Gammaproteobacteria* as part of the Order *Methylococcales*, Family *Methylococcaceae*.

The mol% G + C of the DNA is: 48–60.

Type species: **Methylobacterium agile** (Bowman, Sly, Nichols and Hayward 1993b) Bowman, Sly and Stackebrandt 1995, 183 (*Methylobacter agilis* Bowman, Sly, Nichols and Hayward 1993b, 749.)

FURTHER DESCRIPTIVE INFORMATION

Morphology and ultrastructure Cells of *Methylobacterium* species are regular short rods, $0.5\text{--}1.5 \times 1.5\text{--}2.5\text{ }\mu\text{m}$, occurring singly or in pairs. There is no propensity to form chains or rosettes. All species are actively motile by means of single polar flagellum. Cell walls are of the standard Gram-negative type and are surrounded by a thin slime capsule. Cells contain type I intracytoplasmic membranes typical for other members of the family *Methylococcaceae* but do not form glucan-containing cysts. Most strains contain poly-β-hydroxybutyrate and polyphosphate granules. Cells are not heat or desiccation-resistant and are somewhat sensitive to methane starvation, losing viability in only a few days when exposed to a methane-free atmosphere.

Lipopolysaccharide The carbohydrate fraction of the outer membrane lipopolysaccharide in *Methylobacterium album* includes D-glucose, L-fucose, and D-heptose (Sutherland and Kennedy, 1986). The hydroxy fatty acids from the lipopolysaccharide fraction of cell material of *Methylobacterium album* and *Methylobacterium agile* (Bowman et al., 1991a) are predominantly composed of C_{16:0 3OH}.

Cultural features *Methylobacterium* strains form an even turbidity in NMS liquid media without pellicle formation—a trait that is much more prevalent in *Methylomonas* and *Methylococcus* species. On agar, colonies are uniformly circular and convex, with entire edges, possessing a creamy consistency, and reaching a diameter of 1–2 mm after 7–14 days under a methane/air (1:1, v/v) atmosphere. Colonies are nonpigmented, with a translucent to opaque appearance, and no diffusible pigments are formed. *Methylobacterium* species are mesophilic, growing at 10–30°C. *Methylobacterium agile* and *Methylobacterium album* grow best at 25–30°C, while *Methylobacterium pelagicum* grows optimally at about 20–25°C. The pH range for growth ranges from 6.0–9.0, with optimal growth occurring at about pH 7.0. Neither *Methylobacterium agile* nor *Methylobacterium album* requires growth factors, and both are nonhalophilic. The marine species *Methylobacterium pelagicum* grows optimally in media containing either natural or artificial seawater (Sieburth et al., 1987).

Nutrition and metabolism *Methylobacterium* strains are strictly aerobic methanotrophs, with carbon and energy substrates limited to methane and some other C₁ compounds; however, one strain of *Methylobacterium agile* has the capacity utilize methylamine, dimethylamine, and trimethylamine (the type strain of *M. agile* can use only methylamine). No strains have been demonstrated to grow on carbon–carbon bonded compounds. Upon isolation, most *Methylobacterium* strains are able to tolerate methanol and grow quite well on it. Methane appears to be oxidized predominantly by a particulate methane monooxygenase. The presence of soluble methane monooxygenase has not been demonstrated in *Methylobacterium* strains so far (Stainthorpe et al., 1991). Formaldehyde is fixed by the ribulose monophosphate pathway, and key enzymes for the serine pathway, as well as ribulose-1,5-diphosphate carboxylase activity, are absent. *Methylobacterium* species can utilize nitrate and ammonia salts, yeast extract, Casamino acids, and various amino acids as nitrogen sources, although high levels (>0.5% w/v) of complex organic compounds are inhibitory to growth. Urease and nitrogen fixation activities are absent.

Fatty acid composition Fatty acid profiles of *Methylobacterium* species have been found to all be quite similar, with the predominant fatty acids including C_{16:1 ω5c}, C_{16:1 ω5t}, C_{16:1 ω8c}, C_{16:1 ω7c},

and C_{16:0} being the most abundant components (Table BXII.γ.68 of the chapter describing the family *Methylococcaceae*) (Bowman et al., 1991a). The low level of C_{14:0} and abundance of C_{16:1 ω5t} and C_{16:1 ω8c} distinguish *Methylobacterium* from *Methylobacter* and *Methylomonas* species. The high level of the *trans* fatty acid C_{16:1 ω5t} is unusual and does not appear to be due to stressful cultivation conditions. The primary respiratory lipoquinone has been found to be ubiquinone-8 (Q-8) (Collins and Green, 1985).

Ecology The ecology and distribution of *Methylobacterium* species is generally unknown. Indirect immunofluorescence studies indicate that *Methylobacterium pelagicum* appears to be the dominant methanotrophic inhabitant of the upper mixing layers of ocean waters but is significantly less common in estuarine waters (Sieburth et al., 1993).

ENRICHMENT AND ISOLATION PROCEDURES

The enrichment and isolation approach described for the genus *Methylobacterium* can be applied for *Methylobacter* strains, provided that the incubation temperature is lowered to 20–30°C. The salt composition and pH of the media should closely reflect the habitat being investigated. For isolation of *Methylobacterium pelagicum*, cycloheximide should be added to media to eliminate predation of cells by protozoa (Sieburth et al., 1987).

MAINTENANCE PROCEDURES

Methods for maintenance are the same as those described for the genus *Methylococcus*.

DIFFERENTIATION OF THE GENUS *METHYLOMICROBIUM* FROM OTHER GENERA

Phenotypic, chemotaxonomic, and genotypic properties differentiating *Methylobacterium* from other type I methanotrophs are shown in Tables BXII.γ.67 and BXII.γ.68 of the chapter describing the family *Methylococcaceae*.

TAXONOMIC COMMENTS

Phylogenetic analysis using 16S rDNA sequences indicates that *Methylobacterium* forms a tight cluster within the *Gammaproteobacteria* and is part of a larger phylogenetic cluster that includes *Methylosphaera*, *Methylobacter*, *Methylomonas*, and methanotrophic mytilid endosymbionts. The genus is most closely related to *Methylobacter* and *Methylomonas* (evolutionary distance 0.04–0.06) (Bowman et al., 1995).

DNA–DNA hybridization analysis indicates that the three known *Methylobacterium* species are genotypically distinct, with only relatively low levels of hybridization occurring between the species (Bowman et al., 1993b).

DIFFERENTIATION OF THE SPECIES OF THE GENUS *METHYLOMICROBIUM*

Phenotypic and genotypic characteristics differentiating the species of *Methylobacterium* are shown in Table BXII.γ.73 and additional characteristics are shown in Table BXII.γ.74.

List of species of the genus *Methylobacterium*

1. ***Methylobacterium agile*** Bowman, Sly, Nichols and Hayward 1993b) Bowman, Sly and Stackebrandt 1995, 183^{VP} (*Methylobacter agilis* Bowman, Sly, Nichols and Hayward 1993b, 749.)
a' gi. le. L. adj. *agilis* agile, nimble.

Characteristics are as described for the genus and as listed in Tables BXII.γ.73 and BXII.γ.74. For further descriptive information, see Bowman et al. (1993b).

Optimum temperature, 25–30°C range, 10–37°C. pH optimum, ~7.0; range, 6.0–9.0.

Isolated from freshwater sediments and swampy soil.

TABLE BXII.γ.73. Characteristics differentiating the species of the genus *Methylobacterium*^{a,b}

Characteristic	1. <i>M. agile</i>	2. <i>M. album</i>	3. <i>M. pelagicum</i>
Poly-β-hydroxybutyrate inclusions	d	+	—
Growth at 37°C	+	d	—
Growth occurs with 3.0% NaCl	—	—	+
NaCl (seawater) required for growth	—	—	+
Phosphatase activity	+	+	—
Nitrate reduction (to nitrite)	—	+	—
Methylamine used as a sole carbon and energy source	+	—	—
Mol% G + C of DNA (<i>T_m</i>)	58.1–59.6	54.4–56.1	48.5

^aFor symbols, see standard definitions.^bData from Bowman et al. (1993b).**TABLE BXII.γ.74.** Other phenotypic characteristics of the species of the genus *Methylobacterium*^{a,b}

Characteristics	1. <i>M. agile</i>	2. <i>M. album</i>	3. <i>M. pelagicum</i>
Polyphosphate inclusions	+ ^b	+	+
Capsular exopolysaccharide	+	d	+
Cyst formation	—	—	—
Motility	+	+	+
Pigmentation	—	—	—
Desiccation resistance	—	—	—
Heat resistance (80°C, 20 min)	—	—	—
Growth occurs in the presence of:			
0.0075% KCN	+	+	+
0.02% sodium azide	+	+	+
0.001% crystal violet	—	+	—
0.001% malachite green	+	d	—
0.01% SDS	d	—	—
Reduction of 0.01% triphenyltetrazolium chloride (TTC)	d	—	—
Lysed by 0.2% SDS	—	—	—
Catalase	+	+	+
Oxidase	+	d	+
Urease	—	—	—
Nitrogen fixation	—	—	—
Sole carbon and energy sources:			
Methane, methanol	+	+	+
Dimethylamine, trimethylamine	d	—	—
Formate, formamide, trimethylamine- <i>N</i> -oxide	—	—	—

^afor symbols, see standard definitions^bPhenotypic data from Bowman et al. (1993b).

The mol% G + C of the DNA is: 58.1–59.6 (*T_m*).

Type strain: A30, ACM 3308, ATCC 35068, NCIMB 11124.

GenBank accession number (16S rRNA): X72767.

2. ***Methylobacterium album*** (Bowman, Sly, Nichols and Hayward 1993b) Bowman, Sly and Stackebrandt 1995, 184^{VP} (*Methylobacter albus* Bowman, Sly, Nichols and Hayward 1993b, 749.)

al'bum. L. adj. *album* white.

Characteristics are as described for the genus and as listed in Tables BXII.γ.73 and BXII.γ.74. For further descriptive information, see Bowman et al. (1993b).

Optimum temperature, 25–30°C; range, 10–37°C. pH optimum, ~7.0; range, 6.0–9.0.

Isolated from freshwater sediments and swampy soil.

The mol% G + C of the DNA is: 54.4–56.1 (*T_m*).

Type strain: ACM 3314; IMET 10526, NCIMB 11123, VKM-BG8.

GenBank accession number (16S rRNA): X72777.

3. ***Methylobacterium pelagicum*** (Sieburth, Johnson, Eberhardt, Sieracki, Lidstrom and Laux 1987) Bowman, Sly and Stackebrandt 1995, 184^{VP} (*Methylomonas pelagica* Sieburth, Johnson, Eberhardt, Sieracki, Lidstrom and Laux 1987, 291; *Methylobacter pelagicus* Bowman, Sly, Nichols and Hayward 1993b, 749.)

pe.la'gi.cum. L. adj. *pelagicus* belonging to the sea.

Characteristics are as described for the genus and as listed in Tables BXII.γ.73 and BXII.γ.74. Requires seawater for optimal growth. For further descriptive information, see Sieburth et al. (1987) and Bowman et al. (1993b).

Optimum temperature, ~20–25°C; range, 10–30°C. pH optimum, ~7.0; range, 6.0–8.5. Isolated from the upper mixing zone of oceans and, to a lesser extent, from estuarine waters.

The mol% G + C of the DNA is: 48.5 (*T_m*).

Type strain: AA-23, ACM 3505, NCIMB 2265.

GenBank accession number (16S rRNA): X72775.

Genus V. *Methylomonas* Whittenbury and Krieg 1984b, 355^{VP} (Effective publication: Whittenbury and Krieg 1984a, 260) emend. Bowman, Sly, Nichols and Hayward 1993b, 746.

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Me.thy.lo.mo' nas. Fr. n. *méthyle* the methyl radical; Gr. n. *monas* unit; M.L. n. *Methylomonas* methyl monad or methyl unit.

Straight or slightly curved rods 0.5–1.0 × 1.0–4.0 µm, occurring singly or in pairs. Form a polysaccharide-containing **cyst** that is desiccation sensitive. Poly-β-hydroxybutyrate granules usually present. **Motile** by means of a single polar flagellum. **Gram negative**. Have a strictly aerobic respiratory metabolism, with oxygen as the electron acceptor. **Obligate utilizers of methane and methanol for carbon and energy**. No growth occurs on carbon–carbon bonded compounds. Fix formaldehyde by the **ribulose monophosphate pathway**. Ribulose-1,5-diphosphate carboxylase activity is absent. Some species fix atmospheric nitrogen via an oxygen-sensitive nitrogenase. Form **red, pink, and orange carotenoid non-water soluble pigments**. **Mesophilic**, growing between 10–40°C, optimal growth temperatures 25–30°C. Non-halophilic. Major fatty acids include C_{14:0} and C_{16:1 ω8c}. **18-Methyleneubiquinone-8 (MQ-8)** is the predominant lipoquinone. Habitat: Isolated from sediments of freshwater lakes and rivers, wetland muds, activated sludge and wastewater, and coal mine drainage waters. Belongs to the *Gammaproteobacteria* as part of the Order *Methylococcales*, Family *Methylococcaceae*.

The mol% G + C of the DNA is: 50–59 (T_m).

Type species: *Methylomonas methanica* Whittenbury and Krieg 1984b, 355 (Effective publication: Whittenbury and Krieg 1984a, 260.)

FURTHER DESCRIPTIVE INFORMATION

Morphology and ultrastructure The cells of *Methylomonas* species are regularly shaped rods that are either straight or slightly curved and are occasionally branched. Cells occur singly, in pairs, and sometimes as chains. All species are motile by single, unsheathed, polar flagellum. Cell walls are of the standard Gram-negative type, and well-defined slime capsules occur. *Methylomonas* species contain standard type I intracytoplasmic membranes (see Fig. BXII.γ.102 in the description of the order *Methylococcales*), and cells contain simple, single-bodied cysts similar to, but less poorly defined than, those typically observed in *Azotobacter chroococcum*. The cysts are similar to those observed in *Methylococcus* species and do not confer either desiccation or heat resistance. Most strains possess short shelf lives, dying off within a matter of days when lacking methane.

Lipopolysaccharides The hydroxy fatty acids from the outer membrane lipopolysaccharide have been analyzed in detail in *Methylomonas methanica*, *Methylomonas fodinarum*, and *Methylomonas aurantiaca* (Bowman et al., 1991a). The major component hydroxy fatty acid detected was C_{16:0 3OH}. Smaller proportions of C_{12:0 3OH}, C_{13:0 iso 3OH}, and C_{14:0 2OH} are also present, though their levels vary considerably between the species.

Cultural features In static liquid cultures, growth is usually restricted to a surface pellicle, which is pink-red or orange in color. On agar, *Methylomonas methanica* forms bright pink colonies that are circular, entire or lobate-edged, and butyrous in consistency, and that reach a diameter of 1–2 mm in diameter after 2 weeks under a methane/air atmosphere. The colonies of *Methylomonas fodinarum* are similar except that they have a bright orange color. The colonies of *Methylomonas aurantiaca* are larger, orange-pigmented, circular to irregular in shape, and possess a

raised convex elevation and a mucoid–viscid or cartilaginous consistency. Strains of *Methylomonas methanica* and *Methylomonas aurantiaca* often segregate into rough and smooth colonial variants, and the rough colonies usually have a somewhat cartilaginous consistency. Pigments formed by *Methylomonas* species are carotenoid in nature, lacking solubility in water but readily extractable in chloroform and acetone (Bowman et al., 1990a). Preliminary HPLC analysis of the pigments of *Methylomonas fodinarum* ACM 3268 shows the presence of as many as 24 different carotenoid components (S. Liaaen-Jensen, personal communication). *Methylomonas methanica* has been reported to form blue diffusible pigments when grown on iron deficient agar media (Whittenbury et al., 1970b).

All *Methylomonas* species have a mesophilic and nonhalophilic ecophysiology and do not require growth factors. Growth occurs between 10°C and 42°C. *Methylomonas fodinarum* strains have a temperature optimum of approximately 25–30°C. The temperature optima of *Methylomonas methanica* and *Methylomonas aurantiaca* range from 30–35°C. The pH range is 5.5–8.5, and best growth occurs at approximately pH 7.0.

Nutrition and metabolism *Methylomonas* species are obligate methanotrophs, with their sole carbon and energy sources restricted to methane and methanol. *Methylomonas aurantiaca* strains, however, can also utilize methylamine and, in some cases, dimethylamine. Other C₁ compounds are not utilized; these include formate, trimethylamine, and trimethylamine N-oxide. Methane is oxidized by a particulate methane monooxygenase; however, two *Methylomonas methanica* strains have been found to contain soluble methane monooxygenase (Koh et al., 1993; Shen et al., 1997). The ribulose monophosphate pathway is clearly the major route for fixation of formaldehyde, although *Methylomonas* strains may possess weak activity for α-hydroxypyruvate reductase, the key enzyme of the serine pathway. Ribulose-1,5-diphosphate carboxylase activity is absent, and isocitrate dehydrogenase activity is usually NAD-specific (Bowman et al., 1990a).

Nitrogen sources include nitrate, ammonia, yeast extract, Cas-amino acids, and various amino acids. The addition of complex nutrient sources, such as yeast extract, generally does not lead to growth stimulation and can be growth inhibitory at concentrations greater than 0.25% (w/v). Most *Methylomonas* strains are urease-positive and can use urea as a nitrogen source. *Methylomonas fodinarum* and *Methylomonas aurantiaca* are also able to fix atmospheric nitrogen by an oxygen-sensitive nitrogenase, whereas nitrogenase activity has been detected in only a few *Methylomonas methanica* strains. However, southern blot analysis has detected in several *Methylomonas methanica* strains (including the type strain ATCC 35067) a gene homologous to *nifH*, with a size of approximately 9 kb (Oakley and Murrell, 1993).

Fatty acid composition Fatty acid profiles of *Methylomonas* species are similar to one another, with the fatty acids C_{16:1 ω8c}, C_{14:0}, C_{16:1 ω7c}, and C_{16:1 ω5t} being the most abundant components. The relatively high levels of C_{14:0} and C_{16:1 ω8c} distinguish *Methylomonas* species from those of *Methylobacter* and *Methylomicrobium*. The primary respiratory lipoquinone has been found to be 18-methylene-ubiquinone-8 (MQ-8) (Collins and Green, 1985).

Ecology Knowledge of the specific ecology of *Methylomonas* species is currently limited. Indirect immunofluorescence studies suggest that *Methylomonas* species are common in surface sediments and the water columns of various fresh and brackish water environments and swampy soils (Reed and Dugan, 1978; Saralov et al., 1984; Malashenko et al., 1987; Gal'chenko et al., 1988). High populations of *Methylomonas methanica* are found in the littoral sediments of various subtropical rivers in Eastern Australia, with most probable number counts reaching 10^8 – 10^9 cells per gram of sediment. *Methylomonas aurantiaca* is extremely populous in activated sludge samples, while *Methylomonas fodinarum* is the most common methanotroph in Australian coal mine drainage water (Bowman, 1992).

ENRICHMENT AND ISOLATION PROCEDURES

The enrichment and isolation approach described for the genus *Methylococcus* can also be applied for *Methylomonas* strains, provided that the incubation temperature is lowered to 20–30°C. The salt composition and pH of the media should closely reflect the habitat being investigated.

MAINTENANCE PROCEDURES

Methods for maintenance are the same as those given for the genus *Methylococcus*.

DIFFERENTIATION OF THE GENUS *METHYLOMONAS* FROM OTHER GENERA

Phenotypic, chemotaxonomic, and genotypic properties differentiating *Methylomonas* from other type I methanotrophs are shown in Tables BXII.γ.67 and BXII.γ.68 of the chapter describing the family *Methylococcaceae*.

TAXONOMIC COMMENTS

The genus *Methylomonas* previously contained several species that were either obviously misclassified or invalid owing to lack of descriptions, validation, and/or type strains. A number of these nomenclatural and taxonomic problems have been resolved. Several invalid species are now recognized as being synonymous with *Methylobacillus glycogenes* (Urakami and Komagata, 1986b); these include “*Methylomonas methylovora*” (Kuono et al., 1973), “*Methylomonas methanolica*”, “*Methylomonas espxii*”, “*Methylomonas methanocatallesslica*”, and “*Methylomonas methanofructolica*” (Urakami and Komagata, 1986a). “*Methylomonas clara*” (Faust et al., 1977) is synonymous with *Methylophilus methylotrophus* (Jenkins et al.,

1987). All of these species are restricted methylotrophs utilizing methanol and methylamine as primary carbon and energy sources but without the capacity to utilize methane. “*Methylomonas methaninitrificans*” (Davis et al., 1964) Leadbetter 1974b, 269 and “*Methylomonas methanooxidans*” (Brown and Strawinski, 1958) Leadbetter 1974b, 269 in all probability are synonymous with the genus *Methylosinus* (Whittenbury et al., 1970a, 1970b), based on nitrogen fixing ability and morphology; however, this cannot be verified because no representative strains are extant. The species “*Methylomonas margaritae*” Takeda, Motomatsu, Hachiya, Fukuoka and Takahara 1974, 795 and “*Methylomonas flagellata*” Morinaga, Yamanaha, Otsuka and Hirose 1976, 1544 possess traits similar to *Methylomicrobium agile* or *Methylomicrobium album*, but there are no extant cultures of either species. The species “*Methylomonas gracilis*” Romanovskaya, Malashenko and Bogachenko 1978, 122 is synonymous with *Methylococcus thermophilus* (see comments for the genus *Methylococcus*).

Phylogenetic analysis using 16S rDNA sequences indicates that *Methylomonas* forms a relatively tight cluster within the *Gamma-proteobacteria* and is part of a larger phylogenetic cluster that includes *Methylosphaera*, *Methylobacter*, *Methylomicrobium*, and methanotrophic mytilid endosymbionts. The genus is most closely related to *Methylobacter* and *Methylomicrobium* (evolutionary distance 0.05–0.06) (Bratina et al., 1992; Bowman et al., 1995).

The mol% G + C of the DNA of *Methylomonas* cover a relatively broad range of 50–59 (T_m), much like those for *Methylomonas* and *Methylobacter* (Table BXII.γ.75). DNA–DNA hybridization analysis has found the three valid *Methylomonas* species to be genotypically distinct; however, significant DNA homology (30–50%) is present between *Methylomonas fodinarum* and *Methylomonas aurantiaca*. This corresponds with the high level of phylogenetic similarity shared by these species (See Fig. BXII.γ.103 in the description of the order *Methylococcales*). *Methylomonas methanica* is more distinct, sharing less than 25% DNA homology with its sister species. Very high levels of homology are present between *Methylomonas methanica* and “*Methylomonas rubra*”, indicating that “*Methylomonas rubra*” is a junior subjective synonym of *Methylomonas methanica*. This result agrees with the high levels of phenotypic similarity that these species share (Bowman et al., 1990a; 1993b). Phylogenetic analysis, however, indicates that *Methylomonas methanica* is somewhat distinct from other *Methylomonas* species (evolutionary distance 0.03–0.04) (Bratina et al., 1992). The genotypic relationship of “*Methylomonas rubra*” to *Methylomonas methanica* requires further verification.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *METHYLOMONAS*

Phenotypic and genotypic characteristics differentiating the species of *Methylomonas* are shown in Table BXII.γ.75 and additional characteristics are shown in Table BXII.γ.76.

TABLE BXII.γ.75. Characteristics differentiating the species of the genus *Methylomonas*^{a,b}

Characteristics	1. <i>M. methanica</i>	2. <i>M. aurantiaca</i>	3. <i>M. fodinarum</i>
Colony pigmentation	Pr	Or	Or
Growth at 37°C	d	+	–
Oxidase	+	–	–
Alkaline phosphatase	–	+	–
Nitrate reduction (to nitrite)	–	–	+
Methylamine as a sole carbon and energy source	+	+	+
Mol% G + C of DNA (T_m)	50.6–54.0	55.3–57.6	57.8–59.1

^aFor symbols, see standard definitions; Pr, pink-red; Or, orange.

^bData from Bowman et al. (1993b).

TABLE BXII.γ.76. Other phenotypic characteristics of the species of the genus *Methylomonas*^{a,b}

Characteristics	1. <i>M. methanica</i>	3. <i>M. aurantiaca</i>	2. <i>M. fodinarum</i>
Chains of cells	d	—	—
Poly-β-hydroxybutyrate inclusions	+	+	+
Polyphosphate inclusions	+	d	+
Capsular exopolysaccharide	d	+	+
Cyst formation	+	+	+
Motility	+	+	+
Desiccation resistance	—	—	—
Heat resistance (80°C, 20 min)	—	—	—
<i>Growth in the presence of:</i>			
0.0075% KCN	+	+	+
0.02% sodium azide	d	—	+
0.001% crystal violet	d	d	d
0.001% malachite green	d	d	+
0.01% SDS	—	d	—
3.0% NaCl	—	—	—
Lysed by 0.2% SDS	+	+	+
Reduction of 0.01% triphenyltetrazolium chloride (TTC)	—	+	+
Esculin hydrolysis	+	—	d
Catalase activity	+	+	+
Urease activity	+	d	+
Nitrogen fixation	d	+	+
<i>Sole carbon and energy sources:</i>			
Methane, methanol	+	+	+
Dimethylamine	—	d	—
Formate, formamide, trimethylamine, trimethylamine-N-oxide	—	—	—

^aFor symbols, see standard definitions.^bData from Bowman et al. (1993b).*List of species of the genus Methylomonas*

1. **Methylomonas methanica** Whittenbury and Krieg 1984b, 355^{VP} (Effective publication: Whittenbury and Krieg 1984a, 260.)

me.tha'ni.ca. M.L. adj. *methanica* related to or associated with methane.

Cells are motile rods. Gram negative. Characteristics are as described for the genus and as listed in Tables BXII.γ.75 and BXII.γ.76. For further descriptive information, see Bowman et al. (1993b).

Optimum temperature, ~25–30°C; range, 10–37°C. pH optimum ~7.0; range, 5.5–9.0.

Isolated from freshwater sediment, lake and pond water, and swampy soil.

The mol% G + C of the DNA is: 50.6–54.0 (*T_m*).

Type strain: ACM 3307, ATCC 35067, IMET 10543, NCIMB 11130.

GenBank accession number (16S rRNA): AF304196.

2. **Methylomonas aurantiaca** Bowman, Cox, Sly and Hayward 1990b, 470^{VP} (Effective publication: Bowman, Cox, Sly and Hayward 1990a, 285.)

au.ran.ti'a.ca. M.L. adj. *aurantiacus* orange-colored.

Cells are motile rods. Gram negative. Characteristics are as described for the genus and as listed in Tables BXII.γ.75 and BXII.γ.76. For further descriptive information, see Bowman et al. (1993b).

Optimum temperature, ~30–35°C; range, 10–40°C. pH optimum, ~7.0; range, 5.0–9.0.

Isolated from activated sludge, wastewater and swampy soil.

The mol% G + C of the DNA is: 57.6–59.1 (*T_m*).

Type strain: JB103, UQM 3406.

GenBank accession number (16S rRNA): X72776.

3. **Methylomonas fodinarum** Bowman, Cox, Sly and Hayward 1990b, 470^{VP} (Effective publication: Bowman, Cox, Sly and Hayward 1990a, 283.)

fo.di.na'rum. M.L. n. *fodinarum* of coal mine.

Cells are motile rods. Gram negative. Characteristics are as described for the genus and as listed in Tables BXII.γ.75 and BXII.γ.76. For further descriptive information, see Bowman et al. (1993b).

Optimum temperature, ~25–30°C; range, 10–35°C. pH optimum ~7.0; range, 5.5–9.0.

Isolated from coal mine drainage water and freshwater sediments.

The mol% G + C of the DNA is: 55.3–57.6 (*T_m*).

Type strain: LD2, UQM 3268.

GenBank accession number (16S rRNA): X72778.

4. **Methylomonas scandinavica** Khalyuzhnaya, Khmelenina, Kotelnikova, Holmquist, Pedersen and Trotsenko 2000, 949^{VP} (Effective publication: Khalyuzhnaya, Khmelenina, Kotelnikova, Holmquist, Pedersen and Trotsenko 1999, 571^{VP})

Cells are motile rods which are single or which form short chains. Gram negative. Single polar flagellum. Colonies are pigmented pink. Carbon sources include only methane and methanol metabolized using the RuMP pathway. Use ammonia and nitrate as nitrogen sources via the glutamate cycle and by reductive amination of pyruvate and 2-oxoglutarate. Optimum temperature, ~17°C; range 5–30°C. pH optimum 6.8–7.6; range 5.0–9.0. Generation time

for growth is about 15 h. Nonhalophilic. Cells contain phosphatidylglycerol and phosphatidylethanolamine but not cardiolipin. Major lipoquinone is MQ-8. Isolated from groundwater of the Aspö Hard Rock Laboratory tunnel, southeastern Sweden.

The mol% G + C of the DNA is: 53.8 (T_m).

Type strain: SR-5, VKM B-2140.

GenBank accession number (16S rRNA): AJ131369.

Species Incertae Sedis

1. **"Methylomonas margaritae"** Takeda, Motomatsu, Hachiya, Fukuoka and Takahara 1974, 795.

2. **"Methylomonas flagellata"** Morinaga, Yamanaha, Otsuka and Hirose 1976, 1544.

3. **"Methylomonas methaninitrificans"** (Davis, Coty and Stanley 1964) Leadbetter 1974b, 269 ("Pseudomonas methaninitrificans" Davis, Coty and Stanley 1964, 471.)

"Methylomonas methaninitrificans" is probably a strain of the genus *Methylosinus*.

4. **"Methylomonas methanooxidans"** (Brown and Strawinski 1958) Leadbetter 1974b, 269 ("Methanomonas methanooxidans" Brown and Strawinski 1958.)

"Methylomonas methanooxidans" is probably a strain of the genus *Methylosinus*.

Genus VI. *Methylosarcina* Wise, McArthur and Shimkets 2001, 620^{VP}

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Me.thyl.o.sar.ci'na. N.L. neut. n. *methylum* the methyl group; L. fem. n. *sarcina* pack, bundle; N.L. fem. n.

Methylosarcina methane-utilizing bundle of cells.

Gram-negative strictly aerobic short, coccoid, or fusiform rods.

Form aggregates in slime layer or capsule. Catalase positive; oxidase negative. **Form cysts.** **Only carbon sources methane and methanol.** Major fatty acids include C_{16:0}, C_{16:1 ω8c}, C_{16:1 ω7c}, C_{16:1 ω5t}.

The mol% G + C of the DNA is: 54.

Type species: ***Methylosarcina fibrata*** Wise, McArthur and Shimkets 2001, 620.

FURTHER DESCRIPTIVE INFORMATION

Cells of both species of *Methylosarcina* contain inclusion bodies and stacked membranes such as are found in other type I methanotrophs. *Methylosarcina fibrata* cells are often embedded in a fibrillar matrix. At the end of growth, cells of both species enlarge to produce cysts; these cysts do not survive pasteurization or drying for 7 d (Wise et al., 2001).

Both *Methylosarcina* species use only methane and methanol as carbon and energy sources; both use ammonia, nitrate, and a variety of organic nitrogen sources, and are unable to fix N₂. Both species reduce nitrate to nitrite but do not grow using nitrate as an electron acceptor and methane or methanol as the source of carbon and electrons under anaerobic conditions. Both possess the enzyme hexulose phosphate synthase of the ribulose monophosphate pathway for carbon assimilation and the *pmoA* gene for particulate methane monooxygenase (Wise et al., 2001).

List of species of the genus *Methylosarcina*

1. ***Methylosarcina fibrata*** Wise, McArthur and Shimkets 2001, 620^{VP}
fi.bra'ta. L. fem. adj. *fibra* a fiber or filament; M.L. fem. adj. *fibrata* covered with fibers or fibrils.

Description as for the genus with the following additional characteristics. Cells coccoid (diameter 0.8–1.5 μm). Form packets. Single cells, diplococci, and tetrads may be motile. Fibrils on cell surface.

The mol% G + C of the DNA is: 54.1 ± 0.2 (HPLC).

Type strain: AML-C10, ATCC 700909, DSM 13736.

GenBank accession number (16S rRNA): AF177296.

Analysis of 16S rDNA sequences showed that the type strains of *Methylosarcina fibrata* and *Methylosarcina quisquiliarum* are most closely related to each other and form a cluster distinct from other type I methyloprophs. The *Methylosarcina* 16S sequences were most similar to those of *Methylobacter*. Analysis of partial *pmoA* sequences showed that the *Methylosarcina* sequences were most closely related to those of *Methylobacterium album* (Wise et al., 1999, 2001).

ENRICHMENT AND ISOLATION PROCEDURES

These methanotrophs were obtained from liquid enrichments of landfill soil serially diluted in varying concentrations of a mineral salts medium adjusted to different pH values and incubated under atmospheres of differing air/CH₄/CO₂ mixtures. They were isolated by streaking onto agar of the same composition. *Methylosarcina fibrata* was more common in acidic (pH 4.8) enrichments; *Methylosarcina quisquiliarum*, in enrichments at pH 6.8 (Wise et al., 1999).

DIFFERENTIATION OF THE GENUS *METHYLOSARCINA* FROM OTHER GENERA

Wise et al. (2001) provide a comparison of the fatty acid profiles of the type strains of *Methylosarcina fibrata* and *Methylosarcina quisquiliarum* with those of members of the genera *Methylomonas*, *Methylobacter*, *Methylobacterium*, and *Methylococcus*.

2. ***Methylosarcina quisquiliarum*** Wise, McArthur and Shimkets 2001, 620^{VP}
quis.qui.li.d'rum. L. pl. fem. n. *quisquiliae* rubbish, trash, or refuse; L. gen. pl. fem. n. *quisquiliarum* of rubbish, trash, etc., denoting strains that were isolated from a landfill site.

Description as for the genus with the following additional characteristics. Cells pleomorphic: ovoid, fusiform, or rods (1 × 1–6 μm). Coccoid cells occur in packets. No fibrils on cell surface.

The mol% G + C of the DNA is: 54.3 ± 0.3 (HPLC).

Type strain: AML-D4, ATCC 700908, DSM 13737.

GenBank accession number (16S rRNA): AF177297.

Genus VII. Methylosphaera Bowman, McCammon and Skerratt 1998d, 327^{AL} (Effective publication: Bowman, McCammon and Skerratt 1997c, 1457)

JOHN P. BOWMAN

Me.thy.lo.sphae' ra. Fr. n. *méthyle* the methyl radical; Gr. n. *sphaira* sphere; M.L. n. *Methylosphaera* methyl sphere.

Cells are spherical, 1.5–2.0 µm in diameter. Occur singly or in pairs. Cysts or other differentiated **resting stages are not formed**. **Gas vesicles present**. Poly-β-hydroxybutyrate granules present. **Gram negative. Nonmotile. Nonpigmented. Strictly aerobic respiratory metabolism**, with oxygen as the electron acceptor. **Obligate utilizer of methane and methanol for carbon and energy**. Carbon–carbon bonded compounds are not utilized. Oxidase and catalase positive. The **ribulose monophosphate pathway** is used to fix formaldehyde for cell carbon. Key enzyme activity of the serine transhydroxymethylase pathway is absent. Nitrate, ammonium salts, and L-glutamine are used as nitrogen sources. Nitrate is reduced to nitrite. **Fix atmospheric nitrogen**. Ribulose-1,5-diphosphate carboxylase activity is not present. **Psychrophilic**, growing at –2–20°C with an optimal temperature range of 10–15°C. **No growth occurs on solid agar media. Seawater salts are required for optimal growth**. Major fatty acids include C_{16:0}, C_{16:1 ω8c}, C_{16:1 ω7c} and C_{16:1 ω6c}. Habitat: Isolated from hypolimnetic and benthic zones of an Antarctic meromictic lake of marine salinity. Belongs to the *Gammaproteobacteria* as part of the Order *Methylococcales*, Family *Methylococcaceae*.

The mol% G + C of the DNA is: 43–46 (T_m).

Type species: Methylosphaera hansonii Bowman, McCammon and Skerratt 1998d, 327 (Effective publication: Bowman, McCammon and Skerratt 1997c, 1457.)

FURTHER DESCRIPTIVE INFORMATION

Morphology and ultrastructure During all phases of growth *Methylosphaera hansonii* strains appear as featureless spherical cells that exhibit varying degrees of refractility under phase contrast microscopy. Although cells show signs of uneven binary division, evidence for budding division is still lacking. Cells possess cell walls of the standard Gram-negative type and possess type I intracytoplasmic membranes when grown under methane but not methanol. Strains are nonmotile and possess no flagella. They also do not form cysts or other types of resting stages typically found in other methanotrophic bacteria. Thus, *M. hansonii* cells have no desiccation or heat resistance and die rapidly in the absence of methane or methanol substrate. A strong accumulation of poly-β-hydroxybutyrate may occur in some cells. Preliminary electron microscopic examination indicates the occurrence of gas vesicles 0.05–0.1 × 0.2 µm (Bowman et al., 1997c; Bowman, unpublished data.).

Cultivation *M. hansonii* can be grown in nitrate mineral salts media supplemented with natural or artificial seawater salts. Strangely, strains exhibit no growth on solid agar media including those with high purity agars at relatively low concentrations. Agars tested included Bacto-agar (Difco), noble agar, agarose type V, and low melting point agarose at concentrations of 0.3–1.5% (w/v). This inability to grow on agar media has been previously observed with unidentified type I methylophils isolated from seawater off the British Isles (Lees et al., 1991). Those strains, however, were able to grow on semisolid noble agar whereas *M. hansonii* does not. When growing in statically incubated liquid media, *M. hansonii* forms a white turbidity with a distinct layer of sediment. It requires seawater for growth, with

growth occurring only at 0.25–1.5 × strength seawater salts. Poor growth occurs if only NaCl, rather than seawater, is provided, implying that strains require additional components found in seawater, although these components have yet been identified. All strains are psychrophilic, with optimum and maximum growth temperatures varying slightly between strains. Optimal growth occurs at 10–15°C; no growth occurs at 25°C. A doubling time of 20–24 h has been determined for strains growing at or close to their optimum temperature. In NMS–seawater liquid media, growth occurs between pH 6.0–8.0, and the optimum pH is approximately 7.5.

Nutritional and metabolic features *M. hansonii* is metabolically specialized, with only methane and methanol serving as suitable sources of carbon and energy. Methane monooxygenase activity appears to be restricted to the particulate (pMMO) form. No putative sMMO activity has been detected with the naphthalene oxidation assay. That this species lacks sMMO activity is supported by the fact that strains grow very poorly in media lacking copper. Growth on methanol (at 0.1%, v/v) is slower than that on methane but results in similar growth yields. However, no growth has been found to occur on a range of other C₁ compounds, including formate, formamide, methylamine, dimethylamine, trimethylamine, and trimethylamine N-oxide. Strains use the ribulose monophosphate pathway to fix formaldehyde for cell carbon, as demonstrated by the presence of hexulose-6-phosphate synthase activity. However, activity of α-hydroxypyruvate reductase, a key enzyme of the serine pathway, is absent. Ribulose-1,5-diphosphate carboxylase activity is also absent. Growth is strictly aerobic, and no growth occurs anaerobically in the presence of an alternative electron acceptor, such as nitrate or ferric chloride.

Strains utilize nitrate, ammonia and L-glutamine for nitrogen. L-Glutamine also stimulates growth yield and growth rate. More complex sources of nitrogen, such as yeast extract or Casamino acids (Difco), are less suitable, causing partial inhibition of growth when tested at 0.05–0.1% (w/v) and complete inhibition of growth at concentrations of 0.25–0.5% (w/v). Urease is not formed, and urea does not serve as a nitrogen source. Nitrogen fixation occurs and is not quite as oxygen sensitive as in other methanotrophs; at an oxygen level of about 10% (v/v), nitrogen fixation occurs in *Methylosphaera hansonii* but is substantially inhibited or abolished in *Methylosinus trichosporium* and *Methylococcus capsulatus* (Murrell and Dalton, 1983; Zhivotchenko et al., 1995b).

Genetic features The DNA base composition values of *Methylosphaera hansonii* are the lowest among the known methanotrophs, with a range of 43–46 mol% G + C. The type strain ACAM 549 (AM6) has a mol% G + C of 44.8.

Ecology Little is known about the ecology of *Methylosphaera*. Populations are known to become concentrated in the surface benthic sediments and the oxygen gradient zones of sulfate-depleted, marine-salinity lakes (Ace Lake and Burton Lake) found in the Vestfold Hills area of Eastern Antarctica. The waters of these lakes have an average annual temperature of about 2°C.

The distribution of *Methylosphaera hansonii* outside of these lakes is unknown; however, the Vestfold Hills lakes are of recent origin (8000–10000 years old), formed from a marine uplift (Franzmann, 1996) that followed the retreat of the ice sheet after the last ice age. This leads to the hypothesis that *Methylosphaera hansonii* may also be common in Antarctic coastal sediments.

ENRICHMENT AND ISOLATION PROCEDURES

Low temperature marine sediments and hypolimnetic zones of polar marine-salinity lakes (and possibly coastal marine zones) are the best sites for isolation of *Methylosphaera* strains. Enrichment is performed in mineral salts media prepared in serum vials with either natural or artificial seawater. A headspace of 1:1 methane/air is created and the vials incubated at 2–4°C. After 3–4 weeks, a white turbidity forms that is accompanied by white sedimented material. *Methylosphaera hansonii* enrichments contain a high population of nonmethanotrophic, usually methylophilic, co-contaminants. Known strains of *Methylosphaera* cannot be grown on solid agar and, consequently, the isolation procedure described for most methanotrophs (see enrichment and isolation details for the genus *Methylococcus*) is not possible. Isolation can be achieved by serially diluting enrichments to extinction in 96-well plastic microtiter trays (Bowman et al., 1997c) containing NMS seawater liquid media and incubated in containers containing 1:1 methane/air. Several strains may be purified in the same tray simultaneously. After sufficient incubation, the wells containing the highest dilutions that show growth are examined by microscopy. A number of separate transfers may be

required to eventually obtain morphologically homogenous cultures.

MAINTENANCE PROCEDURES

Long term preservation of *Methylosphaera hansonii* is currently problematic. The best procedure is to keep strains in serum vials with NMS seawater media under a methane/air atmosphere held at about 2°C, periodically subculturing them. Viability is rapidly lost following freezing, including snap freezing in liquid nitrogen.

DIFFERENTIATION OF THE GENUS *METHYLOSPHAERA* FROM OTHER GENERA

Phenotypic, genotypic, and chemotaxonomic traits that differentiate *Methylosphaera* from other type I methanotrophs are shown in Tables BXII.γ.67 and BXII.γ.68 of the chapter describing the family *Methylococcaceae*.

TAXONOMIC COMMENTS

16S rDNA based phylogenetic analysis indicates that *Methylosphaera hansonii* forms a separate branch at the periphery of the cluster of type I methanotrophs that includes the genera *Methylobacter*, *Methylomicrobium*, and *Methylomonas*. The closest relatives are methanotrophic mytilid endosymbionts (evolutionary distance 0.09–0.10).

DNA–DNA hybridization analysis indicates that strains from different lake isolation sites are closely related, showing 65–100% hybridization with ACAM 549.

List of species of the genus *Methylosphaera*

1. ***Methylosphaera hansonii*** Bowman, McCammon and Skerratt 1998d, 327^{VP} (Effective publication: Bowman, McCammon and Skerratt 1997c, 1457.)
han.son'i.i. M.L. gen. n. *hansonii* of Hanson, named after American microbiologist R.S. Hanson.

Cells are coccoidal and nonmotile. Colonies do not form on solid agar media. Liquid media cultures possess no pigment. Characteristics are as described for the genus and as

listed in Tables BXII.γ.67 and BXII.γ.68 of the chapter describing the family *Methylococcaceae*.

Optimum temperature for growth, ~10–15°C; range, –2–20°C. pH range for growth, 6.0–8.5; optimum, ~7.5.

Known habitat includes certain Antarctic marine-salinity meromictic lakes.

The mol% G + C of the DNA is: 43.5–45.9 (T_m).

Type strain: AM6, ACAM 549.

GenBank accession number (16S rRNA): U67929.

Order VIII. **Oceanospirillales** ord. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

O.ce.an.o.spi.ril.la'les. M.L. neut. n. *Oceanospirillum* type genus of the order; *-ales* ending to denote order; M.L. fem. n. *Oceanospirillales* the *Oceanospirillum* order.

The order *Oceanospirillales* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the order contains the families *Oceanospirillaceae*, *Alcanivoraceae*, *Ha-hellaceae*, *Halomonadaceae*, *Oleiphilaceae*, and “*Saccharospirillaceae*”.

Most genera halotolerant or halophilic. Motile except for *Al-*

cannivorax. Aerobic, microaerophilic, or facultatively anaerobic chemoorganotrophs.

Type genus: ***Oceanospirillum*** Hylemon, Wells, Krieg and Janasch 1973, 361^{AL}.

The distribution of *Methylosphaera hansonii* outside of these lakes is unknown; however, the Vestfold Hills lakes are of recent origin (8000–10000 years old), formed from a marine uplift (Franzmann, 1996) that followed the retreat of the ice sheet after the last ice age. This leads to the hypothesis that *Methylosphaera hansonii* may also be common in Antarctic coastal sediments.

ENRICHMENT AND ISOLATION PROCEDURES

Low temperature marine sediments and hypolimnetic zones of polar marine-salinity lakes (and possibly coastal marine zones) are the best sites for isolation of *Methylosphaera* strains. Enrichment is performed in mineral salts media prepared in serum vials with either natural or artificial seawater. A headspace of 1:1 methane/air is created and the vials incubated at 2–4°C. After 3–4 weeks, a white turbidity forms that is accompanied by white sedimented material. *Methylosphaera hansonii* enrichments contain a high population of nonmethanotrophic, usually methylophilic, co-contaminants. Known strains of *Methylosphaera* cannot be grown on solid agar and, consequently, the isolation procedure described for most methanotrophs (see enrichment and isolation details for the genus *Methylococcus*) is not possible. Isolation can be achieved by serially diluting enrichments to extinction in 96-well plastic microtiter trays (Bowman et al., 1997c) containing NMS seawater liquid media and incubated in containers containing 1:1 methane/air. Several strains may be purified in the same tray simultaneously. After sufficient incubation, the wells containing the highest dilutions that show growth are examined by microscopy. A number of separate transfers may be

required to eventually obtain morphologically homogenous cultures.

MAINTENANCE PROCEDURES

Long term preservation of *Methylosphaera hansonii* is currently problematic. The best procedure is to keep strains in serum vials with NMS seawater media under a methane/air atmosphere held at about 2°C, periodically subculturing them. Viability is rapidly lost following freezing, including snap freezing in liquid nitrogen.

DIFFERENTIATION OF THE GENUS *METHYLOSPHAERA* FROM OTHER GENERA

Phenotypic, genotypic, and chemotaxonomic traits that differentiate *Methylosphaera* from other type I methanotrophs are shown in Tables BXII.γ.67 and BXII.γ.68 of the chapter describing the family *Methylococcaceae*.

TAXONOMIC COMMENTS

16S rDNA based phylogenetic analysis indicates that *Methylosphaera hansonii* forms a separate branch at the periphery of the cluster of type I methanotrophs that includes the genera *Methylobacter*, *Methylomicrobium*, and *Methylomonas*. The closest relatives are methanotrophic mytilid endosymbionts (evolutionary distance 0.09–0.10).

DNA–DNA hybridization analysis indicates that strains from different lake isolation sites are closely related, showing 65–100% hybridization with ACAM 549.

List of species of the genus *Methylosphaera*

1. ***Methylosphaera hansonii*** Bowman, McCammon and Skerratt 1998d, 327^{VP} (Effective publication: Bowman, McCammon and Skerratt 1997c, 1457.)
han.son'i.i. M.L. gen. n. *hansonii* of Hanson, named after American microbiologist R.S. Hanson.

Cells are coccoidal and nonmotile. Colonies do not form on solid agar media. Liquid media cultures possess no pigment. Characteristics are as described for the genus and as

listed in Tables BXII.γ.67 and BXII.γ.68 of the chapter describing the family *Methylococcaceae*.

Optimum temperature for growth, ~10–15°C; range, –2–20°C. pH range for growth, 6.0–8.5; optimum, ~7.5.

Known habitat includes certain Antarctic marine-salinity meromictic lakes.

The mol% G + C of the DNA is: 43.5–45.9 (T_m).

Type strain: AM6, ACAM 549.

GenBank accession number (16S rRNA): U67929.

Order VIII. **Oceanospirillales** ord. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

O.ce.an.o.spi.ril.la'les. M.L. neut. n. *Oceanospirillum* type genus of the order; *-ales* ending to denote order; M.L. fem. n. *Oceanospirillales* the *Oceanospirillum* order.

The order *Oceanospirillales* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the order contains the families *Oceanospirillaceae*, *Alcanivoraceae*, *Ha-hellaceae*, *Halomonadaceae*, *Oleiphilaceae*, and “*Saccharospirillaceae*”.

Most genera halotolerant or halophilic. Motile except for *Al-*

cannivorax. Aerobic, microaerophilic, or facultatively anaerobic chemoorganotrophs.

Type genus: ***Oceanospirillum*** Hylemon, Wells, Krieg and Janasch 1973, 361^{AL}.

Family I. **Oceanospirillaceae** *fam. nov.*

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

O.ce.an.o.spi.ril.la'ce.ae. M.L. neut. n. *Oceanospirillum* type genus of the family; *-aceae* ending to denote family; M.L. fem. pl. n. *Oceanospirillaceae* the *Oceanospirillum* family.

The family *Oceanospirillaceae* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the family contains the genera *Oceanospirillum* (type genus), *Balneatrix*, *Marinomonas*, *Marinospirillum*, *Neptunomonas*, *Oceanobacter*, *Oleispira*, and *Pseudospirillum*. *Oceanobacter*, *Pseudospirillum* and *Oleispira* were proposed after the cut-off date for inclusion in this volume (June 30, 2001) and are not described here (see Satomi et al. (2002) and Yakimov et al. (2003a), respectively).

Motile by polar flagella. Aerobic; strictly respiratory except for *Neptunomonas*, which gives weak fermentation reactions. Aquatic; *Balneatrix* is found in fresh water, whereas other genera are marine.

Type genus: **Oceanospirillum** Hylemon, Wells, Krieg and Jannasch 1973, 361^{AL}.

Genus I. **Oceanospirillum** Hylemon, Wells, Krieg and Jannasch 1973, 361^{AL}*

BRUNO POT AND MONIQUE GILLIS

O.ce.an.o.spi.ril'illum. M.L. n. *oceanus* ocean; Gr. n. *spira* a spiral; M.L. dim. neut. n. *spirillum* spirillum a small spiral; *Oceanospirillum* a small spiral (organism) from the ocean (seawater).

Rigid, helical cells 0.4–1.2 µm in diameter. Motile by bipolar tufts of flagella. A polar membrane underlies the cytoplasmic membrane at the cell poles in all species so far examined by electron microscopy. **Intracellular poly-β-hydroxybutyrate is formed. All species form thin-walled coccoid bodies that predominate in old cultures.** Gram negative. **Aerobic**, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor. Nitrate respiration does not occur. Nitrate can be reduced to nitrite in all oceanospirilla. Optimum temperature for growth, 25–32°C. **Oxidase positive.** Indole and aryl sulfatase negative. Casein, starch, hippurate, and esculin are not hydrolyzed. **Sea-water is required for growth. Carbohydrates are neither oxidized nor fermented.** Amino acids or the salts of organic acids serve as carbon sources. Growth factors are not usually required. Isolated from coastal seawater, decaying seaweed, and putrid infusions of marine mussels.

The mol% G + C of the DNA is: 45–50.

Type species: **Oceanospirillum linum** (Williams and Rittenberg 1957) Hylemon, Wells, Krieg and Jannasch 1973, 374 (*Spirillum linum* Williams and Rittenberg 1957, 82.)

FURTHER DESCRIPTIVE INFORMATION

All species of *Oceanospirillum* consist of helical cells; however, variants having less curvature may arise after prolonged transfer. For example, the type strain of *O. japonicum* consisted initially of long, helical cells with several turns (Watanabe, 1959), but now consists of slightly curved or S-shaped cells. The cells have a constant and characteristic type of clockwise (right-handed) helix. Only one phylogenetically unrelated species (*O. pusillum*) has

a counterclockwise (left-handed) helix (Terasaki, 1972). Photographs showing the size and shape of various species of oceanospirilla and *O. minutulum* (now *Marinospirillum minutulum*) are presented in Fig. BXII.γ.104.

An unusual elaboration of the plasma membrane, the “polar membrane”, occurs in all of the species so far examined (Beveridge and Murray, unpublished results). It is attached to the inside of the plasma membrane by bar-like links and is located, most commonly, in the region surrounding the polar flagella (Murray and Birch-Andersen, 1963). Such a membrane has been found mainly in genera of helical bacteria, such as *Spirillum*, *Campylobacter*, *Aquaspirillum*, *Ectothiorhodospira*, and *Rhodospirillum*.

All species have intracellular poly-β-hydroxybutyrate, but granules may not be evident in cells having a small diameter and chemical analysis may be required to demonstrate the polymer.

All species have bipolar tufts of flagella and all species show extensive formation of coccoid bodies (sometimes termed “microcysts”) in old cultures. These bodies have thin walls and resemble spheroplasts; however, they are resistant to lysis in distilled water (Kelly, 1959). Whether coccoid bodies are resistant to desiccation is not known. Three main modes of formation of coccoid bodies were described by Williams and Rittenberg (1957), as follows: (a) two cells may entwine and apparently fuse. The cells become shorter and thicker and a protuberance develops at the point of fusion. This gradually enlarges and absorbs the organisms to form the coccoid body. More than one coccoid body may develop from a pair of entwined spirilla; (b) a spirillum may become shorter and thicker and a protuberance arises from the center of the cell or from each end of the cell. The protuberances enlarge and eventually merge into a single coccoid body as the helical cell is absorbed; (c) a spirillum may undergo a gradual shortening and rounding to form a coccoid body. The majority of coccoid bodies present in old cultures appears to be viable and can “germinate” when placed into a fresh medium (Williams and Rittenberg, 1956). Germination is by unipolar or bipolar growth of a helical cell from the coccoid body, with the latter being absorbed into the developing helical cell.

Seawater is required for the growth of all species. Media prepared with natural seawater or with 2.75% NaCl have been used for enrichment and isolation (Williams and Rittenberg, 1957; Terasaki, 1963, 1970, 1980). Commonly used culture media for

*Editorial Note: The genus *Oceanospirillum* has recently undergone taxonomic re-evaluation. *O. minutulum* has been transferred to the genus *Marinospirillum* as *Marinospirillum minutulum* (Satomi et al., 1998); see the Taxonomic Comments section in this chapter. *O. commune* and *O. vagum* are homotypic synonyms of *Marinomonas communis* and *Marinomonas vaga*, respectively. After the cut-off date for inclusion of taxonomic changes in this volume of the *Systematics*, Satomi et al. (2002) emended the description of *Oceanospirillum* and proposed the transfer of *O. jannaschii* to the genus *Marinobacterium* as *Marinobacterium jannaschii*, the transfer of *O. japonicum* to the new genus *Pseudospirillum* as *Pseudospirillum japonicum*, the transfer of *O. kriegii* to the new genus *Oceanobacter* as *Oceanobacter kriegii*, and the transfer of *O. pusillum* to the new genus *Terasakiella* as *Terasakiella pusilla*.

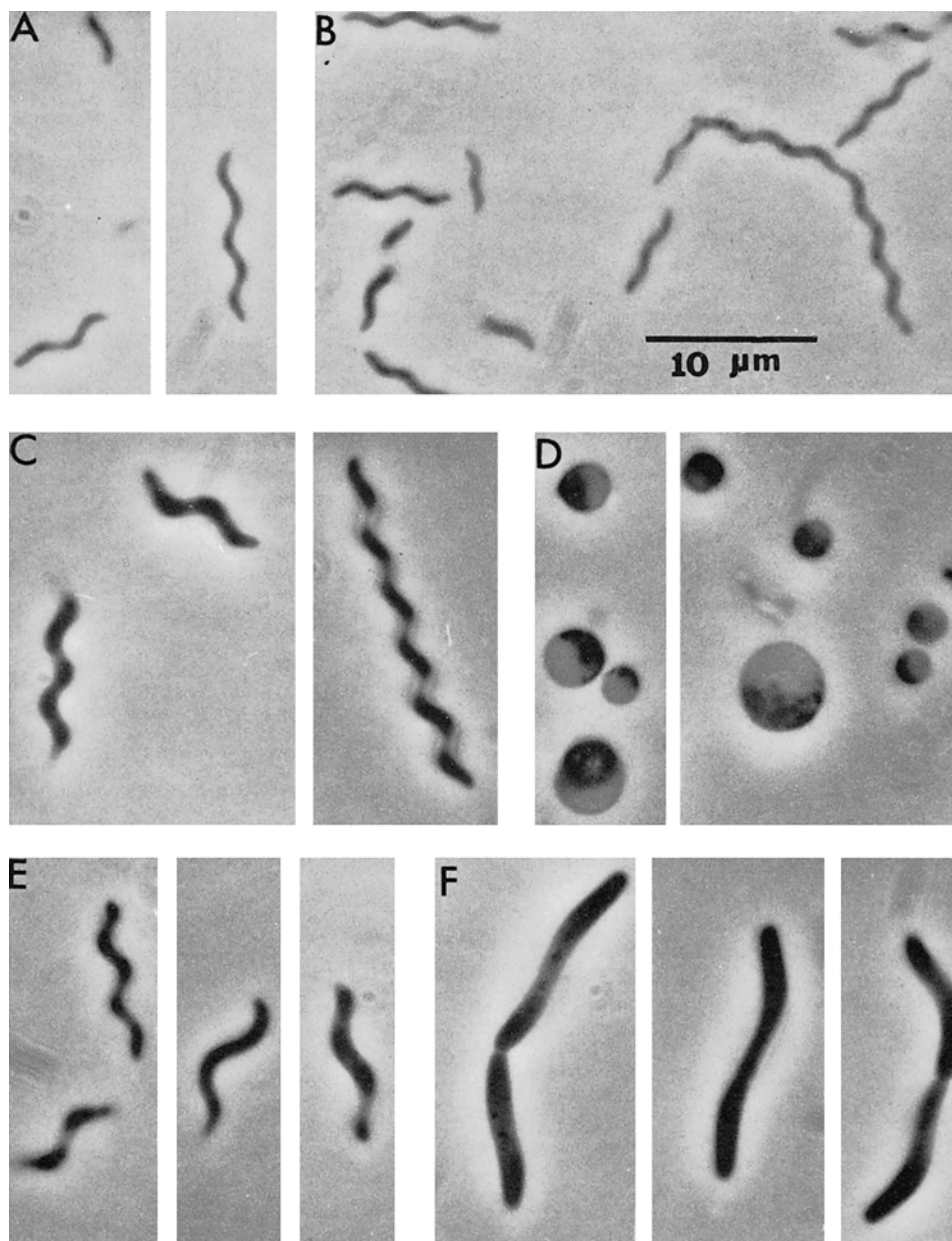


FIGURE BXII.γ.104. Phase contrast photomicrographs of several species of the genus *Oceanospirillum*. All photomicrographs were taken at the same magnification. A, *Marinospirillum minutulum* ATCC 19193. B, *O. linum* ATCC 11336. C, *O. maris* ATCC 27509. D, coccoid bodies of *O. maris* formed after 7 d of incubation. E, *O. beijerinckii* subsp. *beijerinckii* ATCC 12754. F, *O. japonicum* ATCC 19191. Reproduced with permission from N.R. Krieg, *Bacteriological Reviews*, 40: 55–115, 1976, ©American Society for Microbiology.)

oceanospirilla are nutrient broth prepared with natural seawater and PSS, or MPSS broth¹ prepared with artificial seawater².

Oceanospirilla generally produce moderate to abundant, turbid growth in 2–3 d in PSS seawater broth (Hylemon et al., 1973). In seawater-nutrient broth, membranous masses are often formed at the surface and can be dispersed with shaking to yield turbid cultures (Terasaki, 1972).

Colonies of oceanospirilla generally develop within 2–3 d on PSS seawater agar and are usually white, circular, and convex, ranging from pinpoint to 1.5 mm in diameter (Hylemon et al., 1973). Colonies on seawater-nutrient agar are generally pinpoint in size at 48 h but become larger (up to 2.0 mm in diameter) at 7 d; they are usually convex or umbonate, glistening, opaque, pale yellow, and butyrous (Terasaki, 1972). Rough (R) colonies may arise on prolonged transfer; for example, the colonies of the type strain of *O. japonicum* are presently of the R type. Some species produce a water-soluble, yellow-green fluorescent pigment on PSS seawater agar.

Most species grow best at a temperature of 30–32°C; however, *O. maris* subsp. *hirosimense* grows best at 25°C (Terasaki, 1972).

1. See the genus *Aquaspirillum* for recipes for these media.

2. Artificial sea water for use in PSS broth, g/l of distilled water: NaCl, 27.5; MgCl₂, 50; MgSO₄, 2.0; CaCl₂, 0.5; KCl, 1.0; and FeSO₄, 0.01.

The nutrition of oceanospirilla is generally simple. Most species grow in simple defined media with amino acids or the salts of organic acids as carbon sources and ammonium ions as the nitrogen source. However, *O. linum* is specifically stimulated by methionine in a medium containing succinate and malate as carbon sources, and *O. maris* subsp. *williamsae* has a growth factor requirement that has not yet been identified. A listing of the carbon sources for oceanospirilla is given below in Table BXII.γ.77. Some apparent contradictions occur between the results obtained from different laboratories, although the results within each laboratory are reproducible. These differences are likely attributable to differences in definitions of what constitutes a positive growth response, and in some cases to the use of different strains.

The use of antisera in agglutination tests with a limited number of strains has indicated that the species of *Oceanospirillum* can be distinguished serologically (McElroy and Krieg, 1972). The antisera were prepared against whole cells and adsorbed with heated cells, leaving only antibodies against thermolabile antigens.

Oceanospirilla have been isolated from coastal seawater (Williams and Rittenberg, 1957), decaying seaweed (Jannasch, 1963), and putrid infusions of marine mussels (Terasaki, 1963, 1970, 1980). By direct microscopic counts of the bacteria present in clear and turbid seawaters near Port Aransas, Texas, Oppenheimer and Jannasch (1962) found that spirilla comprised only 0.1–2.5% of the total bacteria present. Whether oceanospirilla occur in the open sea is not known. Based on chemostat experiments, Jannasch (1963) suggested that the growth of oceanospirilla might be restricted to environments of higher nutrient concentration than is found in ordinary seawater, such as in zones surrounding decaying particulate matter. With regard to occurrence of oceanospirilla in putrid infusions of marine mussels, the source is most likely marine mud adherent to the mussels (Terasaki, 1970).

ENRICHMENT AND ISOLATION PROCEDURES

The enrichment and isolation method used by Williams and Rittenberg (1957) is as follows. A seawater sample is mixed with an equal volume of Giesberger's base medium (NH_4Cl , 0.1%; K_2HPO_4 , 0.05%; MgSO_4 , 0.05%) plus 1.0% calcium lactate. After incubation and appearance of spirilla, a portion of the initial culture is sterilized and mixed with an equal volume of sterile Giesberger's medium lacking NH_4Cl . This mixture is then inoculated from the unsterilized portion of the initial culture. One to three subcultures done in this manner are sufficient to establish the spirilla as the predominant type. For isolation, the enrichment is diluted 1:100 to 1:100,000 with sterile seawater. The dilution bottles are shaken vigorously and allowed to stand at room temperature for 20 min to allow migration of spirilla to the surface of the dilution. Isolation is then accomplished by streaking the surface water onto a suitable agar medium such as nutrient agar prepared with seawater and containing 0.3% yeast autolysate. Plates are incubated at 30°C and after 24 h examined for distinctive, granular, umbonate or pulvinate colonies with a ground-glass appearance.

The method of Terasaki (1970) has yielded excellent results for the isolation of oceanospirilla from putrid infusions. Marine mussels are smashed with a hammer and placed in a Petri dish with a teaspoon of marine mud. Sterilized seawater is poured into the dish until the mussels sink completely in the solution. The infusion is incubated at 27–28°C and examined for the de-

velopment of spirilla after 1, 2, 4, and 7 d. Isolation is accomplished by streaking dilutions onto suitable agar media.

For enrichment by use of continuous cultures, see Jannasch (1967).

MAINTENANCE PROCEDURES

Oceanospirilla may be maintained in semisolid PSS seawater medium (containing 0.15% agar to give a jelly-like consistency) at 30°C with weekly transfer (Hylemon et al., 1973). Cultures may also be maintained as stabs in seawater-nutrient agar at room temperature with monthly transfer (Terasaki, 1972).

Preservation is most easily accomplished by suspending a dense concentration of cells in seawater-nutrient broth containing 10% (v/v) dimethylsulfoxide, with subsequent freezing in liquid nitrogen. A method for freeze-drying oceanospirilla has been reported by Terasaki (1975).

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Characterization methods for oceanospirilla have been described in detail by Terasaki (1972, 1979) and Hylemon et al. (1973). The comments given in this *Manual* for the genus *Aquaspirillum* also apply to the genus *Oceanospirillum*, except that media containing natural or artificial seawater must be used for all characterization tests.

DIFFERENTIATION OF THE GENUS *OCEANOSPIRILLUM* FROM OTHER GENERA

See the genus *Aquaspirillum*, in Volume 2 Part C in this *Manual*, for characteristics of *Oceanospirillum* that distinguish the genus from other morphologically or physiologically similar genera.

TAXONOMIC COMMENTS

In the eighth edition of *Bergey's Manual of Determinative Bacteriology* (Krieg, 1974), a single genus, *Spirillum*, contained all of the various aerobic and microaerophilic spirilla, including freshwater and marine species. However, the DNA base composition for the genus ranged from 38 to 65 mol% G + C and appeared to be unusually broad for a bacterial genus. Moreover, three groups were evident within the genus: (a) the aerobic, freshwater spirilla that could not tolerate 3% NaCl (mol% G + C 50–65); (b) the aerobic, marine spirilla that required seawater for growth (mol% G + C = 42–48); and (c) the large, microaerophilic spirilla that belong to the species *S. volutans* (mol% G + C = 38). Accordingly, Hylemon et al. (1973) divided the genus into the three genera *Spirillum*, *Aquaspirillum*, and *Oceanospirillum*, with the marine organisms comprising the latter genus. This subdivision was used in the first edition of *Bergey's Manual of Systematic Bacteriology* (Krieg, 1984a). Although this scheme proved useful for practical purposes, it was only gradually that the phylogenetic aspects of the three subdivisions were revealed.

In an analysis of the 16S rRNA oligonucleotide catalogs of the species *O. minutulum* (now *Marinospirillum minutulum*) and *Oceanospirillum maris*, Woese et al. (1982) found that both organisms belonged to group III of the phototrophic bacteria as defined by Gibson et al. (1979), but they were not closely related to each other. Later, Woese et al. (1985) studied three additional species of *Oceanospirillum*—*O. japonicum*, *O. linum*, and *O. beijerinckii*. These species, together with the families *Enterobacteriaceae* and *Vibrionaceae*, constituted the core of 'subgroup 3' of the *Gammaproteobacteria* (Stackebrandt et al., 1988).

An organism known as "*Spirillum lunatum*" (Williams and Rit-

TABLE BXII.γ.77. Carbon sources used by *Oceanospirillum* species and *Marinospirillum minutulum*

Characteristic	<i>O. linum</i>		<i>O. beijerinckii</i> subsp. <i>beijerinckii</i>		<i>O. beijerinckii</i> subsp. <i>pelagicum</i>	<i>O. maris</i> subsp. <i>maris</i>	<i>O. maris</i> subsp. <i>hirosimense</i>	<i>O. multiglobuliferum</i>	<i>O. jannaschii</i>	<i>O. japonicum</i>	<i>O. kriegii</i>	<i>O. pusillum</i>	<i>M. minutulum</i>		
Method ^{a,b}	A ^c	B ^d	A	B	B ^d	A ^c	B	B	A ^f	A	B	A ^f	B	A	B
<i>Carbon source:</i>															
Citrate	—	—	—	—	d	—	—	+	+	—	—	+	+	—	—
Aconitate	—	nd	—	nd	nd	—	nd	nd	—	—	nd	—	nd	—	nd
Isocitrate	—	nd	—	nd	nd	—	nd	nd	nd	—	nd	nd	nd	—	nd
α-Ketoglutarate	—	nd	—	nd	nd	—	nd	nd	d	—	nd	+	nd	d	nd
Succinate	—	—	—	+	+	—	+	+	+	—	+	+	+	+	+
Fumarate	—	—	—	+	+	—	+	+	+	—	+	+	+	+	+
Malate	—	—	+	—	+	d	+	+	+	+	+	+	+	+	+
Oxaloacetate	—	nd	—	nd	nd	+	nd	nd	nd	+	nd	nd	nd	+	nd
Pyruvate	—	—	+	—	+	—	+	+	+	+	+	+	+	+	+
Lactate	—	—	—	—	+	—	+	+	+	+	+	+	+	+	+
Malonate	—	—	—	—	—	—	—	—	—	—	—	d	+	—	—
Tartrate	nd	—	nd	—	+	nd	+	—	—	nd	nd	—	—	d	—
Acetate	d	—	—	—	+	—	+	+	+	+	+	+	+	—	+
Propionate	—	—	—	—	+	—	+	+	+	—	+	+	+	+	+
Butyrate	nd	—	—	—	nd	nd	nd	+	+	nd	+	d	+	nd	+
Caproate	—	nd	—	nd	nd	—	nd	nd	d	—	nd	d	nd	—	nd
β-Hydroxybutyrate	—	nd	—	nd	nd	—	nd	nd	+	—	nd	+	nd	—	nd
p-Hydroxybenzoate	—	nd	—	nd	d	—	d	nd	—	—	nd	+	nd	—	nd
Ethanol	—	—	—	—	d	—	d	—	+	—	—	+	—	—	—
n-Propanol	—	—	—	—	d	—	d	—	+	—	—	+	—	—	—
n-Butanol	—	—	—	—	—	—	—	—	d	—	—	+	—	—	—
Glycerol	—	—	—	—	nd	—	nd	—	—	—	—	—	+	—	—
L-Histidine	—	nd	—	nd	nd	—	nd	nd	—	—	nd	—	nd	—	nd
L-Tyrosine	—	nd	—	nd	nd	—	nd	nd	—	—	nd	—	nd	—	nd
L-Phenylalanine	—	nd	—	nd	nd	—	nd	nd	d	—	nd	—	nd	—	nd
L-Alanine	—	nd	—	nd	nd	—	nd	nd	+	+	nd	+	nd	—	nd
L-Glutamate	—	nd	—	nd	nd	+	nd	nd	d	+	nd	d	nd	+	nd
L-Aspartate	—	nd	—	nd	nd	—	nd	nd	d	—	nd	—	nd	—	nd
L-Glutamine	—	nd	—	nd	nd	—	nd	nd	nd	+	nd	nd	nd	d	nd
Asparagine	—	nd	—	nd	nd	—	nd	nd	nd	—	nd	nd	nd	—	nd
L-Proline	—	nd	—	nd	nd	d	nd	nd	+	—	nd	+	nd	+	nd
L-Hydroxyproline	—	nd	—	nd	nd	—	nd	nd	nd	—	nd	nd	nd	—	nd
L-Ornithine	—	nd	—	nd	nd	—	nd	nd	+	—	nd	—	nd	—	nd
L-Citrulline	—	nd	—	nd	nd	—	nd	nd	+	—	nd	—	nd	—	nd
L-Arginine	—	nd	—	nd	nd	—	nd	nd	+	—	nd	d	nd	—	nd
L-Lysine	—	nd	—	nd	nd	—	nd	nd	d	—	nd	—	nd	—	nd
Putrescine	—	nd	—	nd	nd	—	nd	nd	+	—	nd	+	nd	—	nd
L-Methionine	—	nd	—	nd	nd	—	nd	nd	nd	—	nd	nd	nd	—	nd
L-Serine	—	nd	—	nd	nd	—	nd	nd	d	—	nd	—	nd	—	nd
L-Cysteine	—	nd	—	nd	nd	—	nd	nd	nd	—	nd	nd	nd	—	nd
Glycine	—	nd	—	nd	nd	—	nd	nd	d	—	nd	—	nd	—	nd
L-Leucine	—	nd	—	nd	nd	—	nd	nd	d	—	nd	—	nd	—	nd
L-Isoleucine	—	nd	—	nd	nd	—	nd	nd	d	—	nd	—	nd	—	nd
L-Valine	—	nd	—	nd	nd	—	nd	nd	d	—	nd	—	nd	—	nd
L-Tryptophan	—	nd	—	nd	nd	—	nd	nd	—	—	nd	—	nd	—	nd

^aMethod A (Hylemon et al., 1973): A turbidimetrically standardized cell suspension in synthetic seawater was inoculated into a defined, vitamin-free medium containing the carbon sources (0.1%) and ammonium sulfate as the nitrogen source. Growth responses were measured turbidimetrically after one 72-h serial transfer from the initial cultures, using a Klett colorimeter with the blue (420 nm) filter and 16-mm cuvettes. Symbols: +, 10 or more Klett units of turbidity for all strains tested; —, less than 10 Klett units of turbidity; d, differs among strains; nd, not determined.

^bMethod B (Terasaki, 1972, 1979): A cell suspension washed in basal, defined, vitamin-free medium (Williams and Rittenberg, 1957) containing natural seawater and lacking carbon sources. The cells were inoculated into similar media containing the test compounds (0.05%) and ammonium chloride as the nitrogen source. After 7 d, growth was estimated turbidimetrically. Symbols: +, a turbidity of 0.025 absorbance units or greater for all strains tested; —, a turbidity of less than 0.025; d, differs among strains; nd, not determined.

^cStrain ATCC 12753 failed to grow with any sole carbon source, while strain ATCC 11336 grew only with acetate. Both strains grew abundantly when succinate plus malate were supplied as carbon sources and L-methionine as the nitrogen source.

^dStrain OF3 (Terasaki, 1972, 1973) differs from the results given in the table in that it grows with a large variety of sole carbon sources: citrate, succinate, fumarate, malate, pyruvate, lactate, acetate, propionate, and butyrate. Whether this strain should be included in the species *O. linum* is uncertain.

^eThe results are given for *O. maris* subsp. *maris*. *O. maris* subsp. *williamsae* fails to grow with any sole carbon (or sole nitrogen) source and, therefore, appears to have an auxotrophic growth requirement. This requirement has not yet been defined.

^fAs reported by Bowditch et al. (1984a).

tenberg, 1957) was included in the genus *Oceanospirillum* by Hy-
lemon et al. (1973), but this posed taxonomic problems. The
characteristics of the type strain (ATCC 11337 or NCMB 54) did
not fit the original description of the species, and Linn and Krieg
(1978) found that NCMB strain 54 consisted of a mixture of two
dissimilar organisms. The first type was a short, vibrioid rod that
possessed a single polar flagellum, grew in either the presence
or absence of seawater, catabolized sugars, did not form coccoid
bodies, and had a mol% G + C of 63–64. The second type was
a larger, helical organism that possessed bipolar flagellar tufts,
required seawater, failed to attack sugars, formed coccoid bodies,
and had a mol% G + C of 45. The smaller organism did not
appear to belong to either *Oceanospirillum* or *Aquaspirillum* and
it remains unclassified. The larger organism had characteristics
more in accord with the original description of “*S. lunatum*” but
differed in certain respects; it has been classified as a new sub-
species of *O. maris*: *O. maris* subsp. *williamsae*.

Bowditch et al. (1984a, b) added four species to the genus
Oceanospirillum, mainly based on immunological relationships.
These species were *Oceanospirillum commune*, for the organism
previously named *Marinomonas communis* (Van Landschoot and
De Ley, 1983, 1984), *Oceanospirillum vagum* for *Marinomonas vaga*
(Van Landschoot and De Ley, 1983, 1984), and two species *Oce-
anospirillum jannaschii* and *Oceanospirillum kriegii* for two groups
of unnamed marine bacteria I-I and H-I, respectively. As a result,
the genus definition of *Oceanospirillum* needed to be changed
drastically, with the unfortunate loss of most of the readily de-
terminable phenotypic features from the genus definition (Krieg,
1984a) and the extension of the upper mol% G + C limit for
the genus from 51 to 57. By this extension, a considerable overlap
of mol% G + C range was introduced between the genera *Aqua-
spirillum* (49–65 mol% G + C) and *Oceanospirillum* (42–51 mol%
G + C). In this way, one of the most reliable genotypic features
discriminating both genera was lost. Phylogenetic data (Pot et
al. 1989, Pot, 1996; Satomi et al., 1998), however, have since
shown that all four species cannot be regarded as members of
the genus *Oceanospirillum*.

On the basis of a polyphasic approach including DNA–DNA
and DNA–rRNA hybridizations, Pot et al. (1989) showed that
only five species, including the type species, constituted a sepa-
rate rRNA branch in the *Gammaproteobacteria* and redefined the
genus *Oceanospirillum* to contain *O. linum*, *O. maris*, *O. beijerinckii*,
O. multiglobuliferum, and, more distantly, *O. japonicum*. Based on

DNA–DNA hybridizations (as suggested by Krieg, 1984a) and
numerical comparison of whole-cell proteins, *O. maris* subsp. *hi-
roshimense* and *O. beijerinckii* subsp. *pelagicum* were created for the
former species *O. hiroshimense* and *O. pelagicum*. *O. pusillum* was
shown to belong to the *Alphaproteobacteria*, and *O. commune* and
O. vagum were relegated to their original generic positions as
Marinomonas communis and *Marinomonas vaga*, respectively. The
two species *O. jannaschii* and *O. kriegii* were shown to be phylo-
genetically too remote to be considered members of the genus
Oceanospirillum, and, together with *O. minutulum*, they constituted
separate rRNA branches in the *Gammaproteobacteria*.

Subsequently, this phylogenetic heterogeneity was confirmed
by studies of fatty acid, quinone, and polyamine compositions
(Hamana et al., 1994; Sakane and Yokota, 1994; Bertone et al.,
1996). All species, except *O. pusillum*, contained ubiquinone-8
(Q-8) as a major respiratory quinone (Table BXII.γ.78). Like
other spirilla from the *Alphaproteobacteria* (see the genus *Aqua-
spirillum* in this book), *O. pusillum* contained over 90% Q-10. The
thirteen strains of *Oceanospirillum* that have been investigated for
their fatty acid composition by Sakane and Yokota were divided
into three groups (Table BXII.γ.79 and BXII.γ.80). Group I in-
cluded the 10 strains belonging to *O. linum*, *O. maris* subsp. *hi-
roshimense*, *O. maris* subsp. *williamsae*, *O. beijerinckii* subsp. *beijer-
inckii*, *O. beijerinckii* subsp. *pelagicum*, *O. multiglobuliferum*, and *O.
japonicum*, all of which have a low mol% G + C (42.5–48.4).
Group II included the two type strains of *O. jannaschii* and *O.
kriegii* and had a high mol% G + C content (54.8–54.9). Group
III included only *O. pusillum* and could be clearly distinguished
from other marine spirilla in having C_{14:0 3OH} as the major 3-
hydroxy fatty acid, besides Q-10 (Table BXII.γ.80). Bertone et al.
(1996) confirmed the separate position of *O. japonicum*, *O. jan-
naschii*, and *O. kriegii*.

All *Oceanospirillum* species including *O. jannaschii* and *O. kriegii*
contain both putrescine and spermidine. The relative content
(Table BXII.γ.81) of putrescine is very small when compared
with the level found in members of the *Alphaproteobacteria*. The
relative concentration of putrescine for *O. pusillum* corresponds
with that of other members of the *Alphaproteobacteria*. The absence
of 2-hydroxy putrescine and homospermidine is a unifying char-
acter for the *Gammaproteobacteria*. The polyamine profile of *Oce-
anospirillum* I and II is not different, nor are their fatty acid
profiles.

Later, 16S rDNA sequence analysis of all *Oceanospirillum* spe-

TABLE BXII.γ.78. Cellular quinone systems in *Oceanospirillum* species and *Marinospirillum minutulum*^a

Species ^b	Strain	Group	Quinone system				
			Q-6	Q-7	Q-8	Q-9	Q-10
<i>O. linum</i>	IFO 15448 ^T	Ic	3	4	91	2	
	IFO 15449	Ic	1	2	96	1	
<i>O. beijerinckii</i> subsp. <i>beijerinckii</i>	IFO 15445 ^T	Id	1	12	83	4	
<i>O. beijerinckii</i> subsp. <i>pelagicum</i>	IFO 13612 ^T	Id	2	7	91	1	
<i>O. maris</i> subsp. <i>hiroshimense</i>	IFO 13616 ^T	Ic	1	4	94	1	
<i>O. maris</i> subsp. <i>williamsae</i>	IFO 15468 ^T	Ic	12	7	80	1	
<i>O. multiglobuliferum</i>	IFO 13614 ^T	Ic	1	4	94	1	
<i>O. jannaschii</i>	IFO 15466 ^T	IIb	3	7	89	1	
<i>O. japonicum</i>	IFO 15446 ^T	Ib	1	14	84	1	
	IFO 15447	Ib		9	88	3	
<i>O. kriegii</i>	IFO 15467 ^T	IIa	2	5	89	4	
<i>O. pusillum</i>	IFO 13613 ^T	III			1	6	93
<i>Marinospirillum minutulum</i>	IFO 15450 ^T	Ia	1	1	97	1	

^aAfter Sakane and Yokota (1994).

^b*Oceanospirillum maris* subsp. *maris* has not been investigated.

TABLE BXII.γ.79. Cellular concentrations of non-polar fatty acids in *Oceanospirillum* species and *Marinospirillum minutulum*^a

Species ^b	Strain	Non-polar fatty acid ^c											
		C _{12:0}	C _{12:1}	C _{14:0}	C _{14:1}	C _{15:0}	C _{16:0}	C _{16:1}	C _{17:0}	C _{17:1}	C _{18:0}	C _{18:1}	C _{20:0}
<i>O. linum</i>	IFO 15448 ^T	3	2	1			16	48				30	
	IFO 15449	4	4	1			16	45				29	
<i>O. beijerinckii</i> subsp. <i>beijerinckii</i>	IFO 15445 ^T	4		4		1	32	50				9	
<i>O. beijerinckii</i> subsp. <i>pelagicum</i>	IFO 13612 ^T	4		2	1		22	46				23	3
<i>O. maris</i> subsp. <i>hirosimense</i>	IFO 13616 ^T	4	2	1			27	49			1	15	
<i>O. maris</i> subsp. <i>williamsae</i>	IFO 15468 ^T	4	4	2			31	47			1	11	
<i>O. multiglobuliferum</i>	IFO 13614 ^T	3	2	2			28	44				20	
<i>O. jannaschii</i>	IFO 15447	2		1			19	46			1	31	
<i>O. japonicum</i>	IFO 15466 ^T	3		1			25	57				14	
	IFO 15446 ^T	3		1			22	53				21	
<i>O. kriegii</i>	IFO 15467 ^T	7	4	1	1	2	16	36	2	3	1	27	
<i>O. pusillum</i>	IFO 13613 ^T	3		1	3		15	18			1	58	
<i>Marinospirillum minutulum</i>	IFO 15450 ^T	2			4		35	26				32	

^aAfter Sakane and Yokota (1994).^b*Oceanospirillum maris* subsp. *maris* has not been investigated.^cThe percentage of the acid relative to the total non-polar acids.**TABLE BXII.γ.80.** Cellular concentrations of 2- and 3-hydroxy fatty acids in *Oceanospirillum* species and *Marinospirillum minutulum*^a

Species ^b	Strain	3-hydroxy fatty acids ^c						2-hydroxy fatty acid ^d
		C _{10:0}	C _{12:0}	C _{14:0}	C _{14:1}	C _{16:0}	C _{18:0}	
<i>O. linum</i>	IFO 15448 ^T	100						—
	IFO 15449	100						—
<i>O. beijerinckii</i> subsp. <i>beijerinckii</i>	IFO 15445 ^T	63			30	6		—
<i>O. beijerinckii</i> subsp. <i>pelagicum</i>	IFO 13612 ^T	60			30	9		—
<i>O. maris</i> subsp. <i>hirosimense</i>	IFO 13616 ^T	100						—
<i>O. maris</i> subsp. <i>williamsae</i>	IFO 15468 ^T	100						—
<i>O. multiglobuliferum</i>	IFO 13614 ^T	100						—
<i>O. jannaschii</i>	IFO 15447	100						—
<i>O. japonicum</i>	IFO 15466 ^T	4	96					—
	IFO 15446 ^T	3	97					—
<i>O. kriegii</i>	IFO 15467 ^T	19	54			27		—
<i>O. pusillum</i>	IFO 13613 ^T			87		2	10	+ (C _{18:1})
<i>Marinospirillum minutulum</i>	IFO 15450 ^T		61	3	36			—

^aAfter Sakane and Yokota (1994).^b*Oceanospirillum maris* subsp. *maris* has not been investigated.^cThe percentage of the acid relative to the total 3-hydroxy acids.^d—, absent; +, present

cies confirmed the above findings. Kawasaki et al. (1997), in a phylogenetic study of helically shaped bacteria in the *Alphaproteobacteria*, showed that *O. pusillum* was not related to other taxa of spirilla and constituted a separate branch in the *Alphaproteobacteria*, also confirming previous findings of Woese et al. (1985) and Pot (1996). Therefore, *O. pusillum* cannot belong to the genus *Oceanospirillum* (Kawasaki et al., 1997). Phenotypic characteristics support the removal of *O. pusillum* from the genus: a single flagellum at each pole, a counterclockwise type of helix, and a mol% G + C of 51, which is slightly higher than the range of 42–48 for the rest of the genus. As a formal transfer has not been proposed, *O. pusillum* is therefore listed below as “Species assigned but phylogenetically not belonging in *Oceanospirillum*”.

Satomi et al. (1998) determined and compared 16S rDNA sequences of all the *Oceanospirillum* species. They found that *O. linum*, *O. maris*, *O. beijerinckii*, and *O. multiglobuliferum* constituted a single rRNA cluster, separate from the branches formed by *Marinobacter*, *Marinobacterium*, and *Marinomonas*. Based on their findings, they also excluded *O. japonicum* from the genus *Oceanospirillum*. Phenotypically, *O. japonicum* is different from other *Oceanospirillum* species since it does not form coccoid bodies and its flagella appear to be crescent shaped with less than one helical

turn, whereas those of other species have one or more helical turns. Moreover, *O. japonicum* grows best at 35–37°C. As a formal new description has not been proposed, *O. japonicum* is therefore listed below as a “Species assigned but phylogenetically not belonging in *Oceanospirillum*”.

In the same study, it was shown that *O. minutulum* clustered on a separate branch together with new isolates from kusaya gravity (Satomi et al., 1998). For this branch, a new genus *Marinospirillum* has been proposed, containing the two species *M. minutulum* and *M. megaterium* (Satomi et al., 1998).

Compared to the other *Oceanospirillum* species, *O. jannaschii* and *O. kriegii* both have a higher mol% G + C (54.8–54.9) and occupy a separate phylogenetic position (Satomi et al., 1998). Many phenotypic characteristics discriminate these species from the genus *Oceanospirillum* (Table BXII.γ.82). Although not discussed separately by the authors, *O. jannaschii* occurred on the same branch as *Marinobacterium* (González et al., 1997). Further taxonomic research, including DNA–DNA hybridizations, should be performed to substantiate the exact level of genotypic relationship between *O. jannaschii* and *Marinobacterium georgiense*. *O. kriegii* constituted a separate branch in the 16S rRNA dendrogram (Satomi et al., 1998).

TABLE BXII.γ.81. Cellular concentrations of polyamines in *Oceanospirillum* and *Marinospirillum minutulum*^{a,b,c}

Species ^d	Strain	Medium ^e	Dap	H-Put	Put	Cad	Spd	HSpd
<i>O. linum</i>	IFO 15448 ^T	199SW	—	—	0.01	—	0.65	—
	IFO 15449	199SW	—	—	0.02	—	0.80	—
<i>O. beijerinckii</i> subsp. <i>beijerinckii</i>	IFO 15445 ^T	199SW	—	—	0.01	—	0.64	—
<i>O. beijerinckii</i> subsp. <i>pelagicum</i>	IFO 13612 ^T	199S	—	—	0.06	—	0.48	—
		199SW	—	—	0.01	—	0.65	—
<i>O. maris</i> subsp. <i>hiroshimense</i>	IFO 13616 ^T	199S	—	—	0.02	—	0.86	—
		199SW	—	—	0.03	—	0.90	—
<i>O. maris</i> subsp. <i>williamsae</i>	IFO 15468 ^T	199SW	—	—	0.03	—	0.90	—
<i>O. multiglobuliferum</i>	IFO 13614 ^T	199S	—	—	0.08	—	0.40	—
		199SW	—	—	0.01	—	0.45	—
<i>O. jannaschii</i>	IFO 15466 ^T	199SW	—	—	0.02	—	0.80	—
<i>O. japonicum</i>	IFO 15446 ^T	199SW	—	—	0.01	—	1.11	—
	IFO 15447	199SW	—	—	0.01	—	1.23	—
<i>O. kriegii</i>	IFO 15467 ^T	199SW	—	—	0.03	—	0.84	—
<i>O. pusillum</i>	IFO 13613 ^T	199S	—	—	0.15	—	1.40	—
		199SW	—	—	0.24	—	1.19	—
<i>Marinospirillum minutulum</i>	IFO 15450 ^T	199SW	—	—	0.10	—	0.72	—

^aCells were harvested at stationary growth phase.^bAfter Hamana et al. (1994).^cAbbreviations: Dap, diaminopropane; H-Put, 2-hydroxyputrescine; Put, putrescine; Cad, cadaverine; Spd, spermidine; HSpd, homospermidine; —, not detectable (<0.005).^d*Oceanospirillum maris* subsp. *maris* has not been investigated.^eMedia:199, polyamine-free growth medium from Flow Lab., Irvine, U.K., pH 7.0.; 199S, medium 199 dissolved in 70% synthetic seawater, pH 7.0; 199SW, medium 199 dissolved in seawater, pH 7.0.

Based on 16S rRNA gene sequence analysis, the genus *Oceanospirillum* should therefore be limited to *O. linum*, *O. maris* subsp. *maris*, *O. maris* subsp. *hiroshimense*, *O. maris* subsp. *williamsae*, *O. beijerinckii* subsp. *beijerinckii*, *O. beijerinckii* subsp. *pelagicum*, and *O. multiglobuliferum*. Consequently, *O. japonicum*, *O. jannaschii*, *O. kriegii*, and *O. pusillum* should be removed from the genus. The genus definition described above has been adapted accordingly. The precise taxonomic affiliation of the last four species needs to be further determined.

Note added in proof: Satomi et al. (2002) have formally revised the taxonomic status of the genus *Oceanospirillum* and proposed the formal transfer of the species assigned to but not belonging to the genus *Oceanospirillum* to new and existing genera. (See Kimura et al., 2002 in Further Reading section below.)

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FURTHER READING

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DIFFERENTIATION OF THE SPECIES OF THE GENUS *OCEANOSPIRILLUM*

Morphological and physiological characteristics of the species of *Oceanospirillum* are indicated in Tables BXII.γ.77 and BXII.γ.82.

Chemotaxonomic characteristics of the species are indicated in Tables BXII.γ.78, BXII.γ.79, BXII.γ.80, and BXII.γ.81.

List of species of the genus *Oceanospirillum*

- 1. *Oceanospirillum linum*** (Williams and Rittenberg 1957) Hylemon, Wells, Krieg and Jannasch 1973, 374^{AL} (*Spirillum linum* Williams and Rittenberg 1957, 82.)
li' num. L. n. *linum* flax, thread.

The morphological characters are depicted in Fig. BXII.γ.104 and listed in Table BXII.γ.82. The physiological and chemotaxonomic characters are indicated in Tables BXII.γ.78, BXII.γ.79, BXII.γ.80, BXII.γ.81, and BXII.γ.82.

Sole carbon sources are listed in Table BXII.γ.77. Growth in defined media is usually poor; however, abundant growth occurs in defined media containing malate plus succinate as carbon sources and methionine as the nitrogen source. Nitrate is not used.

Strain OF3, isolated by Terasaki (1972, 1973), differs from other strains of *O. linum* in that it can grow well in defined media with a variety of sole carbon sources and

TABLE BXII.7.82. Morphological and physiological characteristics of the species of *Oceanospirillum* and of *Marinospirillum minutulum*^a

Characteristics	<i>O. linum</i>	<i>O. beyerlinckii</i>	<i>O. maris</i>	<i>O. multiglobulosum</i>	<i>O. janaschii</i>	<i>O. japonicum</i>	<i>O. kriegii</i>	<i>O. pusillum</i>	<i>Marinospirillum minutulum</i>
Type of helix ^b	C	C	C	C	SR	C	SR	CC	C
Wavelength of helix, µm	1.8–4.0	6.3–7.2	3.5–7.0	3.5–5.0	nd	7.0–20.0	nd	1.7–2.0	2.0–2.8
Helix diameter, µm	0.8–1.4	1.5–3.0	1.4–2.8	1.0–2.0	nd	2.0–5.0	nd	1.0–1.2	0.6–1.5
Length of helix, µm	4.0–30.0	2.0–15.5	2.5–40.0	2.0–10.0	nd	5.0–75.0	nd	1.2–4.0	3.0–8.0
Cell diameter, µm	0.4–0.6	0.6–1.2	0.6–1.1	0.5–0.9	nd	0.8–1.4	nd	0.3–0.5	0.3–0.4
Flagellar arrangement ^c	BT	BT	BT	BT	M1-2	BT	MS	BS	BT
Polar membrane present	nd	+ d	+ d	nd	nd	+	nd	nd	+
Intracellular poly-β-hydroxybutyrate formed	+	+	+	+	+	+	+	+	+ ^e
Cocoid bodies predominant:									
After 3–4 weeks	+	+	+	+	nd	+	nd	+	+
After 24–48 h	–	–	–	+	nd	–	nd	–	–
Range of NaCl (%) for growth in peptone water after 7 d ^f	0.5–8.0	0.5–8.0	nd	0.5–4.0	nd	0.5–8.0	nd	0.5–8.0	0.5–8.0
Growth in PSS-seawater broth containing ^g									
9.75% total NaCl	+	+	+	nd	nd	nd	nd	nd	+
12.75% total NaCl	–	–	–	nd	nd	nd	nd	nd	d
Temperature range for growth (°C) ^h	11–39	8–41	2–35	6–37	+	10–43	+	6–40	11–37
Optimum growth temperature 25°C rather than 30–32°C	–	–	d ⁱ	–	nd	–	nd	–	–
Phosphatase	+	+	d ⁱ	+	+	w	–	w	–
Nitrate reduced to nitrite	–	–	–	–	+	+	–	+	+
Oxidase	+	+	+	+	+	+	+	+	+
Catalase	+ or w	+ or w	d ⁱ	+	nd	w or –	nd	w or –	–
Anaerobic growth with nitrate	–	–	–	–	nd	–	nd	–	–
Denitrification	–	–	–	–	–	–	–	–	–
Acid produced from sugars	–	–	–	–	nd	–	–	–	–
Hydrolysis of esculin, hippurate, starch, or casein	–	–	–	–	nd	–	nd	–	–
Indole test	–	–	–	–	nd	–	nd	–	–
Sulfatase (0.1% phenolphthalein disulfate) ^g	–	–	–	nd	nd	nd	nd	nd	–
Gelatin liquefaction at 20°C after: ^{i,j}									
7 d	–	d	–	–	–	d	–	–	–
28 d	+	d	–	–	–	d	–	–	–
42 d	+	d	–	–	–	d	–	–	–

(continued)

TABLE BXII-γ.82. (cont.)

Characteristics	<i>O. linum</i>	<i>O. beijerinckii</i>	<i>O. maris</i>	<i>O. multiglobuliferum</i>	<i>O. jannaschii</i>	<i>O. japonicum</i>	<i>O. kriegii</i>	<i>O. pusillum</i>	<i>Marinospirillum minutulum</i>
<i>Gelatin hydrolysis at 30°C. after:</i> 4 d ^{4a,g}	—	—	—	nd	—	—	—	nd	—
Reduction of 0.3% H ₂ SeO ₃ ^g	+	—	—	nd	nd	nd	nd	nd	+
Water-soluble brown pigment formed in the presence of ^g									
0.1% tyrosine	+	+	d	nd	nd	nd	nd	nd	—
0.1% phenylalanine	—	—	—	nd	nd	nd	nd	nd	—
0.1% tryptophan	d	—	d	nd	nd	nd	nd	nd	d
Growth in the presence of ^g									
0.1% oxgall	+	+	+	nd	nd	nd	nd	nd	+
0.1% glycine	+	—	+	nd	nd	nd	nd	nd	+
Growth on: ^g									
Eosin methylene blue agar	—	—	d	nd	nd	nd	nd	nd	—
MacConkey agar	—	—	—	nd	nd	nd	nd	nd	—
Triple-Sugar iron agar	+	—	+	nd	nd	nd	nd	nd	+
Sellers agar	—	—	—	nd	nd	nd	nd	nd	—
Methyl red-Voges-Proskauer broth	+	—	+	nd	—	nd	—	nd	—
Water-soluble yellow-green fluorescent pigment formed ^h	+	—	d	nd	—	nd	—	nd	—
Deoxyribonuclease ^g	—	+	—	nd	nd	nd	nd	nd	—
Ribonuclease ^g	d	+	—	nd	nd	nd	nd	nd	—
Urease ^g	—	—	—	nd	nd	nd	nd	nd	—
Auxotrophic growth requirement	+ ^k	—	—	—	nd	—	nd	—	—
Mol% (G + C) of DNA	48–50	47–49	45–47	46	56–57	45	54–56	51	42–44

^aPhenotypic data are from Krieg (1984a), Baumann et al. (1972), and Bowditch et al. (1984a); +, present in all strains; —, lacking in all strains; d, depending on the strain; w, weak reaction; nd, not determined.

^bC, clockwise helix; CC, counterclockwise helix (determined by focusing on the bottom of the cells).

^cSR, straight rod; BT, bipolar tufts, BS, bipolar single; M1-2, 1–2 flagella at one pole; MS, monopolar single.

^dCharacteristic has not been tested in *O. maris* subsp. *hirosimense* and *O. beijerinckii* subsp. *pelagicum*.

^eNo granules are visible microscopically in *Marinospirillum minutulum* but chemical analysis indicates presence of the polymer.

^fData from Terasaki (1972, 1979).

^gData from Hylemon et al. (1973).

^h*Oceanospirillum kriegii* grows at 35°C but not at 40°C; *Oceanospirillum jannaschii* does not grow at either 35°C or 40°C.

ⁱ*Oceanospirillum maris* subsp. *hirosimense* has an optimal growth temperature of 25°C; catalase reaction can be negative, weak, or positive; phosphatase reaction can be positive or negative. For detailed information: see Hylemon et al. (1973), Terasaki (1972, 1979), and Table BXII-γ.83.

^jGelatin liquefaction was not tested in *O. maris* subsp. *maris*. It was negative in *O. maris* subsp. *hirosimense*, positive in *O. beijerinckii* subsp. *beijerinckii*, and differed depending on the strain in *O. beijerinckii* subsp. *pelagicum*.

^k*Oceanospirillum linum* grows poorly or not at all in a defined medium with a single carbon source and ammonium ions as the nitrogen source; however, abundant growth occurs in a defined medium containing succinate plus malate as carbon source and methionine as the nitrogen source. (See also footnote ^a in Table BXII-γ.82.) *Oceanospirillum maris* subsp. *williamsae* ATCC 2954¹ does not grow in vitamin-free medium and requires an unidentified growth factor.

ammonium ions as the nitrogen source (see footnote d, Table BXII.γ.77). Other characteristics of this strain are similar to those of the type strain (Terasaki, 1973), but whether it should be included in this species is uncertain.

The species includes organisms previously assigned to the two species *Spirillum linum* and *Spirillum atlanticum* by Williams and Rittenberg (1957). The two species were combined into the single species *O. linum* by Hylemon et al. (1973), based on a high degree of similarity in phenotypic characters and in DNA base composition.

Isolated from coastal seawater.

The mol% G + C of the DNA is: 48–50 (T_m).

Type strain: ATCC 11336, DSM 6292, NCMB 56.

2. ***Oceanospirillum bejerinckii*** (Williams and Rittenberg 1957) Hylemon, Wells, Krieg and Jannasch 1973, 374^{AL} (*Spirillum bejerinckii* Williams and Rittenberg 1957, 90.) *bei.jer.inck'i.i.* M.L. gen. n. *bejerinckii* of Beijerinck; named after Prof M.W. Beijerinck of Delft, Holland.

The morphological characters are depicted in Fig. BXII.γ.104 and listed in Table BXII.γ.82. The physiological and chemotaxonomic characters are indicated in Tables BXII.γ.78, BXII.γ.79, BXII.γ.80, BXII.γ.81, and BXII.γ.82. Sole carbon sources are listed in Table BXII.γ.77. Ammonium ions can serve as a sole nitrogen source; nitrate is not used.

Isolated from coastal seawater.

The mol% G + C of the DNA is: 47 (T_m).

Type strain: ATCC 12754, DSM 7166, NCMB 52.

- a. ***Oceanospirillum bejerinckii* subsp. *bejerinckii*** (Williams and Rittenberg 1957) Hylemon, Wells, Krieg and Jannasch 1973, 375^{AL} (*Spirillum bejerinckii* Williams and Rittenberg 1957, 90.)

Differs from *O. bejerinckii* subsp. *pelagicum* as indicated in Table BXII.γ.83.

The mol% G + C of the DNA is: 47 (T_m).

Type strain: ATCC 12754, DSM 7166, NCMB 52.

- b. ***Oceanospirillum bejerinckii* subsp. *pelagicum*** (Terasaki 1973) Pot, Gillis, Hoste, Van de Velde, Bekaert, Kersters and De Ley 1989, 32^{VP} (*Spirillum pelagicum* Terasaki 1973, 65.)

pe.la'gi.cum. L. neut. adj. *pelagicum* belonging to the sea.

Differs from the *O. bejerinckii* subsp. *bejerinckii* as indicated in Table BXII.γ.83.

Isolated from putrid infusions of marine mussels.

The mol% G + C of the DNA is: 49 (T_m).

Type strain: ATCC 33337, DSM 6288, IFO 13612, NCMB 2228.

GenBank accession number (16S rRNA): AB006761.

3. ***Oceanospirillum maris*** Hylemon, Wells, Krieg and Jannasch 1973, 376^{AL} *ma'ris.* L. n. *mare* the sea; L. gen. n. *maris* of the sea.

The morphological characters are depicted in Fig. BXII.γ.104 and listed in Table BXII.γ.82. The physiological and chemotaxonomic characters are indicated in Tables BXII.γ.78, BXII.γ.79, BXII.γ.80, BXII.γ.81, and BXII.γ.82. Sole carbon sources are listed in Table BXII.γ.77.

Isolated from coastal seawater (Jannasch, 1967).

The mol% G + C of the DNA is: 45–46 (T_m).

Type strain: ATCC 27509, DSM 6286, LMG 5213.

GenBank accession number (16S rRNA): AB006771.

TABLE BXII.γ.83. Differentiating characteristics for the subspecies of *Oceanospirillum bejerinckii*^a

Characteristic	<i>O. bejerinckii</i> subsp. <i>bejerinckii</i>	<i>O. bejerinckii</i> subsp. <i>pelagicum</i>
Temperature range for growth (°C)	14–37	8–41
Growth with:		
Lactate, citrate, malate, pyruvate, acetate, propionate, tartrate	–	+
<i>p</i> -Hydroxybenzoate, ethanol, <i>n</i> -propanol	–	d
Gelatin liquified at 20°C	+	d
Growth after 7 d in peptone water with:		
0.5–6.0% NaCl	+	–
0.5–8.0% NaCl	–	+
Mol% G + C of DNA	47	49

^aPhenotypic data from Terasaki (1972, 1979); +, present in all strains; –, lacking in all strains; d, depending on the strain

- c. ***Oceanospirillum maris* subsp. *maris*** Hylemon, Wells, Krieg and Jannasch 1973, 376^{AL}

Differs from the *O. maris* subsp. *williamsae* and the *O. maris* subsp. *hiroshimense* as indicated in Table BXII.γ.84.

The mol% G + C of the DNA is: 46 (T_m).

Type strain: ATCC 27509, DSM 6286, LMG 5213.

GenBank accession number (16S rRNA): AB006771.

- d. ***Oceanospirillum maris* subsp. *hiroshimense*** (Terasaki 1973) Pot, Gillis, Hoste, Van de Velde, Bekaert, Kersters and De Ley 1989, 33^{VP} (*Spirillum hiroshimense* Terasaki 1973, 62.)

hi.ro.shi.men'se. M.L. neut. adj. *hiroshimense* pertaining to Hiroshima, Japan.

Characters are as described for the species. Differs from the *O. maris* subsp. *maris* and the *O. maris* subsp. *williamsae* as indicated in Table BXII.γ.84.

Isolated from putrid infusions of marine mussels.

The mol% G + C of the DNA is: 47 (T_m).

Type strain: IFO 13616, DSM 9524.

- e. ***Oceanospirillum maris* subsp. *williamsae*** Linn and Krieg 1984, 355^{VP}

will'iam.sae. M.L. gen. n. *williamsae* of Williams; named after Marion A. Williams, who was the first to describe species of marine spirilla.

Characters are as described for the species. Differs from the *O. maris* subsp. *maris* and the *O. maris* subsp. *hiroshimense* as indicated in Table BXII.γ.84.

Isolated from a mixture of organisms comprising NCMB strain 54 by Linn and Krieg (1978).

The mol% G + C of the DNA is: 45 (T_m).

Type strain: ATCC 29547, IFO 15468.

GenBank accession number (16S rRNA): AB006763.

4. ***Oceanospirillum multiglobuliferum*** (Terasaki 1973) Terasaki 1979, 143^{AL} (*Spirillum multiglobuliferum* Terasaki 1973, 69.)

mul.ti.glo.bu.li'fe.rum. L. adj. *multus* much, many; L. dim. n. *globulus* a small sphere, globule; L. v. *fero* to bear, carry; M.L. neut. adj. *multiglobuliferum* bearing many globules.

The morphological characters are listed in Table BXII.γ.82. The physiological and chemotaxonomic characters are indicated in Tables BXII.γ.78, BXII.γ.79, BXII.γ.80, BXII.γ.81, and BXII.γ.82. Sole carbon sources

TABLE BXII.γ.84. Differentiating characteristics for the subspecies of *Oceanospirillum maris*^a

Characteristic	<i>O. maris</i> subsp. <i>maris</i>	<i>O. maris</i> subsp. <i>hirosimense</i>	<i>O. maris</i> subsp. <i>williamsae</i>
Optimal growth temperature (°C)	30–32	25	30–32
Catalase reaction	strongly +	d	weakly +
Phosphatase activity	—	+	—
DNase activity	—	—	+
RNase activity	—	—	+
Growth with 1% glycine	+	—	—
Growth in vitamin-free, defined growth medium	+	—	—
Growth with:			
L-Glutamate, oxaloacetate	+	—	—
Succinate, pyruvate, lactate, tartrate, acetate, propionate	—	+	—
Mol% G + C of DNA	46	47	45

^aPhenotypic data from Hylemon et al. (1973); +, present in all strains; —, lacking in all strains; d, depending on the strain

are listed in Table BXII.γ.77. Differs from other species by forming unusually large numbers of coccoid bodies even in 24- to 48-h-old broth cultures. Ammonium ions can serve as a sole nitrogen source; nitrate is not used.

Isolated from putrid infusions of marine mussels.

The mol% G + C of the DNA is: 46 (T_m).

Type strain: IFO 13614.

GenBank accession number (16S rRNA): AB006764.

Species assigned but phylogenetically not belonging in *Oceanospirillum*

1. ***Oceanospirillum jannaschii*** Bowditch, Baumann and Bauman 1984b, 503^{VP} (Effective publication: Bowditch, Baumann and Bauman 1984a, 227.)

jan.nasch'i.i. M.L. gen. n. *jannaschii* of Jannasch; named after H.W. Jannasch.

Straight rods. Some morphological, physiological, and chemotaxonomic characters are listed in Tables BXII.γ.78, BXII.γ.79, BXII.γ.80, BXII.γ.81, and BXII.γ.82 (see also Baumann et al., 1972). Sole carbon sources are listed in Table BXII.γ.77. Utilize 39–46 organic compounds including fatty acids, tricarboxylic acid cycle intermediates, alcohols, amino acids, and amines. Utilizes γ-aminovaleate and histamine (Baumann et al., 1972).

Isolated from seawater after enrichment (Baumann et al., 1972).

The mol% G + C of the DNA is: 56–57 (T_m).

Type strain: ATCC 27135, DSM 6295, IFO 15466.

GenBank accession number (16S rRNA): AB006765.

2. ***Oceanospirillum japonicum*** (Watanabe 1959) Hylemon, Wells, Krieg and Jannasch 1973, 375^{AL} (*Spirillum japonicum* Watanabe 1959, 78.)

ja.pon'i.cum. M.L. neut. adj. *japonicum* pertaining to Japan.

The morphological characters are depicted in Fig. BXII.γ.104 and listed in Table BXII.γ.82; however, the type strain presently has morphological features that differ from the original description, in that the cells are no longer helical with several waves but instead are curved, straight, or S-shaped; moreover, colonies of this strain are presently of the R (rough) type. Therefore, it is likely that the type strain has undergone alteration since its isolation in 1959. Three reference strains isolated by Terasaki (1972, 1973) have morphological features that more nearly correspond

to those given in the original description (strains IF4, IF8, and UF3), and also they form colonies of the S (smooth) type.

The physiological and chemotaxonomic characters are indicated in Tables BXII.γ.78, BXII.γ.79, BXII.γ.80, BXII.γ.81, and BXII.γ.82. Sole carbon sources are listed in Table BXII.γ.77.

Isolated from putrid infusions of marine mussels.

The mol% G + C of the DNA is: 45 (T_m).

Type strain: ATCC 19191, DSM 7165.

3. ***Oceanospirillum kriegii*** Bowditch, Baumann and Bauman 1984b, 503^{VP} (Effective publication: Bowditch, Baumann and Bauman 1984a, 227.)

krie'gi.i. M.L. gen. n. *kriegii* of Krieg; named after N.R. Krieg.

Straight rods. Some morphological and physiological characters are listed in Tables BXII.γ.78, BXII.γ.79, BXII.γ.80, BXII.γ.81, and BXII.γ.82 (see also Baumann et al., 1972). Sole carbon sources are listed in Table BXII.γ.77. Utilize 29–33 organic compounds including D-glucose and D-fructose but no other pentose, hexose, or disaccharide; also utilize tricarboxylic acid cycle intermediates, alcohols, amino acids, and amines. Produces an extracellular lipase (Baumann et al., 1972).

Isolated from sea water after enrichment (Baumann et al., 1972).

The mol% G + C of the DNA is: 54–56 (T_m).

Type strain: ATCC 27133, DSM 6294, IFO 15467, NCMB 2042.

GenBank accession number (16S rRNA): AB006767.

4. ***Oceanospirillum pusillum*** (Terasaki 1973) Terasaki 1979, 142^{AL} (*Spirillum pusillum* Terasaki 1973, 67.)

pu.sil' lum. L. dim. neut. adj. *pusillum* very small.

The morphological characters are listed in Table BXII.γ.82. The physiological and chemotaxonomic characters are indicated in Tables BXII.γ.78, BXII.γ.79, BXII.γ.80, BXII.γ.81, and BXII.γ.82. Sole carbon sources are listed in Table BXII.γ.77. Ammonium ions can serve as a sole nitrogen source; nitrate is not used.

Isolated from putrid infusions of marine mussels.

Belongs to the *Alphaproteobacteria*.

The mol% G + C of the DNA is: 51 (T_m).

Type strain: ATCC 33338, DSM 6293, IAM 14442, IFO 13613, NCMB 2229.

GenBank accession number (16S rRNA): AB006768.

Genus II. **Balneatrix** Dauga, Gillis, Vandamme, Ageron, Grimont, Kersters, De Mahenge, Peloux and Grimont 1993b, 624^{VP} (Effective publication: Dauga, Gillis, Vandamme, Ageron, Grimont, Kersters, De Mahenge, Peloux and Grimont 1993a, 42)

CATHERINE DAUGA

Bal'ne.a.trix. L. fem. n. *balneatrix* bathet.

Straight or curved rods, 0.5–0.7 × 2.8–5 µm, sometimes elongated and flexuous. Gram negative. **Motile by single polar flagellum. Aerobic**, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor. Produce convex, round, smooth colonies on solid media. Optimum temperature, 30°C; range, 20–46°C. Optimum pH, 6.5–7.5. Growth factors are not required. **No growth occurs in medium containing more than 1% NaCl**. Chemoorganotrophic. Acid is produced from glucose and certain other carbohydrates. **Oxidase positive**. Weakly catalase positive. **Indole positive**. Urease negative. Reduce nitrate to nitrite. Freshwater organisms; occasionally pathogenic for humans.

The mol% G + C of the DNA is: 54.

Type species: *Balneatrix alpica* Dauga, Gillis, Vandamme, Ageron, Grimont, Kersters, De Mahenge, Peloux and Grimont 1993b, 624 (Effective publication: Dauga, Gillis, Vandamme, Ageron, Grimont, Kersters, De Mahenge, Peloux and Grimont 1993a, 43).

FURTHER DESCRIPTIVE INFORMATION

Balneatrix strains grow after a 2-d incubation at 30°C. The center of the colony is pale yellow after 2–3 d, and pale brown after 4–5 d.

Balneatrix strains are nonfermentative, as verified by oxidation–fermentation medium (Casalta et al., 1989). All strains tested are Voges–Proskauer negative by the Barrit reference method. Nutritional tests can be done using Biotype-100 strips (BioMerieux, Marcy l'Etoile, France) with M70 minimal medium (Véron, 1975) supplemented with 0.15% agar. Strains of *Balneatrix* are nutritionally diverse, and some strains can grow with citrate, D-alanine, ethanolamine, D-glucosamine, L-histidine, DL-lactate, D(+)-malate (see Table BXII.γ.85). Arginine dihydrolase, lysine and ornithine decarboxylases, acetamide, and starch hydrolysis are invariably negative. Tetrathionate is not reduced. Gelatin is weakly hydrolyzed. Citrate can serve sometimes as the sole carbon source for growth. Tween-80 hydrolysis also differs among strains. Tributyrin is hydrolyzed, and the egg yolk reaction (lecithinase) is positive. The gamma-glutamyl transferase test is positive. There is no ONPG hydrolysis and no extracellular DNase (Casalta et al., 1989).

Balneatrix strains are susceptible *in vitro* to a variety of antimicrobial agents, including β-lactam, macrolides, and aminoglycoside antibiotics, sulfamethoxazole–trimethoprim, chloramphenicol, deoxycycline, minocycline, ofloxacin, and nalidixic acid. They are resistant to clindamycin and vancomycin.

At this writing, *Balneatrix* strains have been isolated only from thermal water and clinical specimens at a spa therapy center in southern France. Thirty-five cases of pneumonia and two cases of meningitis have been caused by *Balneatrix* strains (Hubert et al., 1991). According to epidemiological data, the bacteria were present in the hot water spring spa, and favorable growing conditions were found only in vapor baths. After disinfection of water pipes by chlorination, no further cases of infection were observed.

ENRICHMENT AND ISOLATION PROCEDURES

Balneatrix strains can be isolated by traditional culture techniques and incubation under aerobic conditions at 20–41°C. Environ-

TABLE BXII.γ.85. Nutritional characteristics of *Balneatrix alpica*, as determined with Biotype-100 strips^a

Substrates	Growth response ^b
N-acetyl-glucosamine, <i>cis</i> -aconitate, L-alanine, DL-γ-amino- <i>n</i> -butyrate, DL-δ-amino- <i>n</i> -valerate, L-aspartate, betaine, D-fructose, fumarate, D-gluconate, D-glucose, L-glutamate, glutarate, glycerol, DL-β-hydroxybutyrate, <i>myo</i> -inositol, α-ketoglutarate, L-malate, maltose, maltotriose, D-mannitol, D-mannose, L-proline, propionate, putrescine, D-ribose, L-serine, D-sorbitol, succinate, L-tyrosine	+
<i>trans</i> -Aconitate, adonitol, D-arabinose, D-arabitol, L-arabitol, benzoate, caprate, caprylate, D-cellobiose, <i>m</i> -coumarate, dulcitol, <i>i</i> -erythritol, esculin, L-fucose, D-galactose, D-galacturonate, gentisate, β-gentiobiose, D-glucuronate, DL-glycerate, <i>m</i> -hydroxybenzoate, <i>p</i> -hydroxybenzoate, histamine, itaconate, 2-keto-D-gluconate, 5-keto-D-gluconate, lactose, lactulose, D-lyxose, malonate, maltitol, D-melezitose, D-melibiose, 1- <i>O</i> -methyl-α-galactopyranoside, 1- <i>O</i> -methyl-β-galactopyranoside, 3- <i>O</i> -methyl-D-glucopyranose, 1- <i>O</i> -methyl-α-D-glucopyranoside, 1- <i>O</i> -methyl-β-D-glucopyranoside, mucate, palatinose, phenylacetate, 3-phenylpropionate, protocatechuate, quinate, D-raffinose, L-rhamnose, D-saccharate, saccharose, L-sorbose, D(–)-tartrate, L(+)-tartrate, <i>meso</i> -tartrate, D-tagatose, D-trehalose, tricarballoylate, trigonelline, tryptamine, L-tryptophan, D-turanose, xylitol, D-xylose	–
D-Alanine	d (7)
Citrate	d (1)
Ethanolamine	d (2)
D-Glucosamine	d (2)
L-Histidine	d (6)
DL-Lactate	d (7)
D(+)-Malate	d (6)

^aSymbols : +, all strains positive ; –, all strains negative ; d, differs among strains. Numbers in parentheses indicate the number of strains that are positive.

^bBased on results for eight strains.

mental and clinical strains can grow in 24 h in trypto-casein soy broth (Sanofi Diagnostics Pasteur, Marnes-la-Coquette, France). Brain–heart infusion and IsoVitaleX chocolate agar (Sanofi Diagnostics Pasteur) can be used for blood and CSF (cerebrospinal fluid) samples, respectively. Strains can be subcultured on trypto-casein soy agar or nutrient agar. Antimicrobial disks can be tested on Mueller–Hinton agar (Sanofi Diagnostics Pasteur).

MAINTENANCE PROCEDURES

The organisms survive at 30°C in trypto-casein soy broth for several weeks. They can survive for several years when frozen at –80°C in brain–heart infusion containing 50% glycerol.

DIFFERENTIATION OF THE GENUS *BALNEATRIX* FROM OTHER GENERA

The characteristics which differentiate *Balneatrix* from other genera are listed in Table BXII.γ.86.

Because *Balneatrix* shares some properties with *Flavobacterium*, such as isolation requirements, pigmented colonies, and biochemical reactions, it was initially compared with this genus; how-

TABLE BXII.γ.86. Phenotypic characteristics differentiating the genera *Balneatrix*, *Oceanospirillum*, *Marinomonas*, *Halomonas*, and *Flavobacterium*^a

Characteristic	<i>Balneatrix</i>	<i>Flavobacterium</i> ^b	<i>Halomonas</i> ^c	<i>Marinomonas</i> ^d	<i>Oceanospirillum</i> ^e
Cell diameter, μm	0.5–0.7	0.5–3	0.6–1.9	0.5–4	0.3–1.4
Shape:					
Curved rods	+			+	
Straight rods	+	+	+	+	
Helical					+
Flagellar arrangement:					
Number	1–2	none	1–2	single	single or tufts
Arrangement	polar		polar	bipolar	bipolar
Salt tolerance (%NaCl, w/v)	0–1	nd	0.1–32	0.7–3.5	2.75–9.75
Maximum temperature (°C)	46	42	45	40	39–43
Nitrate reduced to nitrite	+	D	+	–	D
Anaerobic growth with nitrate	–	–	+	–	–
Indole	+	D	d	nd	–
Urease	–	D	d	–	–
Esculin hydrolyzed	–	+	d	nd	–
Acid from carbohydrates	+	+	+	+	–
Mol% G + C of DNA	54	31–42	60–61	45–50	45–51

^aSymbols: +, positive for all species; –, negative for all species; nd, not determined; D, differs among species; d, differs among strains.

^bData from Holmes et al., 1984.

^cCharacteristics of *Halomonas elongata*. Data from Vreeland, 1984.

^dCharacteristics of *Marinomonas vaga* and *Marinomonas communis*, which were misnamed *Oceanospirillum vagum* and *O. commune* (Pot et al., 1989). Data from Baumann et al., 1972.

^eData from Krieg, 1984b.

ever, the mol% G + C value of 54 for the DNA of *Balneatrix* does not correspond to that of *Flavobacterium*.

Despite its environmental habitat and oxidative ability, *Balneatrix* is easily differentiated from such genera as *Oceanospirillum*, *Marinomonas*, or *Halomonas* based on salt tolerance, shape, flagellar arrangement, and ability to catabolize several carbohydrates. *Balneatrix* can be quickly differentiated from these genera by the inability to grow with more than 1% NaCl.

TAXONOMIC COMMENTS

Strains of *Balneatrix* are closely related by DNA–DNA hybridization and belong to the same species. Strain 4-87, isolated from cerebrospinal fluid, has been selected as the type strain.

rRNA–DNA hybridization and 16S rDNA sequencing have shown that the genus *Balneatrix* belongs to the class *Gammaproteobacteria*. Phylogenetically, *Balneatrix* is located within a subline including many *Oceanospirillum* species (Fig. BXII.γ.105). In this group, bacteria are heterogeneous and difficult to classify. Like *Balneatrix*, *Halomonas halmophila* was first misidentified as *Flavobacterium halmophilum* (Franzmann et al., 1988). *Halomonas aquamarina*, *Halomonas halophila*, and *Halomonas marina* were first assigned to the genus *Deleya* (Dobson et al., 1993).

Phylogenetically, *Balneatrix* branches in the vicinity of *Oceanospirillum* species, but *O. vagum*, *O. commune*,* *O. jannaschii*, and *O. kriegii* are misnamed according to DNA–rRNA hybridization (Pot et al., 1989). This, in addition to the fact that *Balneatrix alpica* is the only freshwater pathogenic bacterium in the subline, argues that *Balneatrix* is a new genus.

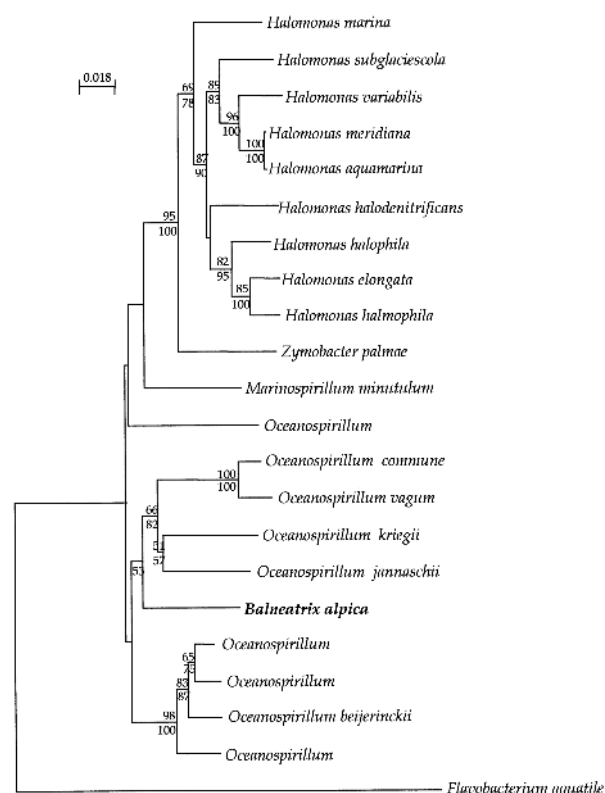


FIGURE BXII.γ.105. Relationships between the marine bacteria of the class *Gammaproteobacteria* and *Balneatrix*. *Flavobacterium aquatile*, which shares some phenotypic characteristics with *Balneatrix*, served as an outgroup. The tree shown is the result of distance analysis (NEIGHBOR routine in PHYLIP 3.5c, Kimura 2-parameters). The same tree topology was obtained with parsimony analysis (heuristic search option in PAUP 3.1.1 with 1000 replicates of the random-addition sequence). Percentages of bootstrap replicates supporting a branching pattern are given above and below the corresponding branches; the value above the branch is the parsimony analysis bootstrap percentage, and that below is from the distance analysis (100 bootstrap replicates).

*Editorial Note: *Oceanospirillum vagum* and *O. commune* are junior objective synonyms of *Marinomonas vaga* and *M. communis*.

List of species of the genus *Balneatrix*

1. ***Balneatrix alpica*** Dauga, Gillis, Vandamme, Ageron, Grimont, Kersters, De Mahenge, Peloux and Grimont 1993b, 624^{VP} (Effective publication: Dauga, Gillis, Vandamme, Ageron, Grimont, Kersters, De Mahenge, Peloux and Grimont 1993a, 43.)
al'pi.ca. L. fem. adj. *alpica* pertaining to the Alps.

The description is the same as that given for the genus. Other characteristics are listed in Table BXII.γ.85. The type strain has all the properties given for the species and hydrolyses Tween-80.

The mol% G + C of the DNA is: 54 (HPLC).

Type strain: 4-87, CIP 103589.

GenBank accession number (16S rRNA): Y17112.

Genus III. *Marinomonas* Van Landschoot and De Ley 1984, 91^{VP} (Effective publication: Van Landschoot and De Ley 1983, 3071)

ANTONIO SANCHEZ-AMAT AND FRANCISCO SOLANO

Ma.ri.no.mo'nas. L. adj. *marinus* pertaining to the sea; Gr. n. *monas* a unit, monad; M.L. *Marinomonas* sea monad.

Gram-negative, straight or curved rods. Motile by means of polar flagella at one or both poles. **Aerobic**, having a strictly respiratory type of metabolism. **Oxidase positive or negative. Na⁺ is required for growth. Do not accumulate poly-β-hydroxybutyrate.** Do not require organic growth factors. **Do not produce extracellular amylase. Utilize acetate but not butyrate or valerate.** Utilize glutamate, sorbitol, and malate. Commonly isolated from seawater.

The mol% G + C of the DNA is: 45–50.

Type species: ***Marinomonas communis*** (Baumann, Baumann, Mandel and Allen 1972) Van Landschoot and De Ley 1984, 91 (Effective publication: Van Landschoot and De Ley 1983, 3071) (*Alteromonas communis* Baumann, Baumann, Mandel and Allen 1972, 420.)

FURTHER DESCRIPTIVE INFORMATION

Species of *Marinomonas* are rods, either curved such as *M. communis*, or straight such as *M. vaga* and *M. mediterranea* (Fig. BXII.γ.106). They are motile; *M. communis* and *M. vaga* show a single, unsheathed, polar flagellum inserted at one or both poles of the cell. *M. mediterranea* shows only a single flagellum at one pole of the cell (Fig. BXII.γ.106).

Marinomonas strains grow in typical marine media such as Marine Agar (Zobell, 1941). In this complex medium, only *M. mediterranea* shows pigmentation; it synthesizes a brown to black melanin pigment derived from L-tyrosine as main precursor. This pigment is observed both in the colonies and diffusing into the surrounding medium (Fig. BXII.γ.107). *Marinomonas* strains also grow in basal marine agar (BMA) containing 0.2% D-glucose as carbon and energy source (Baumann et al., 1972). This basal medium is used in studies of carbon and energy source utilization. No organic growth factor is required; however, growth of *M. mediterranea* is favored in media containing organic compounds as the nitrogen source. The most suitable chemically defined medium for growing this species is MMM¹. All culture media for *Marinomonas* must contain a marine salts base composition. *Marinomonas* species show an absolute requirement for Na⁺ ion.

The physiological characteristics of the *Marinomonas* species are listed in Tables BXII.γ.87 and BXII.γ.88. Thirty-three strains of *M. communis* and 17 *M. vaga* strains isolated from the Pacific Ocean have been phenotypically characterized (Baumann et al., 1972). In contrast, only a single *M. mediterranea* strain—isolated from the Mediterranean Sea as its species name indicates—has been described (Solano et al., 1997). Unless otherwise indicated, the characterization of this strain has been also performed according to the protocols of Baumann et al. (1972), described as well by Gauthier and Breittmayer (1992).

Members of the genus *Marinomonas* are aerobic and use oxygen as a universal electron acceptor. No *Marinomonas* species is able to denitrify, although *M. mediterranea* is able to reduce nitrate to nitrite. *M. communis* is oxidase positive whereas the other two species are oxidase negative. In some cases permeabilization of *M. vaga* strains with toluene will allow a positive oxidase reaction to occur. It has been suggested that this result is due to a low level of cytochrome *c* in *M. vaga* cells (Baumann et al., 1984d); however, *M. mediterranea* always gives a negative result in cytochrome *c* oxidase tests, even when the cells are toluene-treated (Solano et al., 1997).

In assays for extracellular degradative activities, *Marinomonas* species give negative results, with the exception of *M. mediterranea*, which shows lipase and gelatinase activities. Gelatin hydrolysis was detected by comparison of the solidification of Marine Broth, supplemented with 12% gelatin and inoculated with *M. mediterranea*, with an uninoculated control. Under these conditions, a weak gelatinase activity was detected (Solano et al., 1997).

Marinomonas species are able to utilize acetate, but not butyrate or valerate, as a sole carbon and energy source. All of the *M. communis* and *M. vaga* strains metabolize *m*-hydroxybenzoate, *p*-hydroxybenzoate, and quinate by means of *meta* cleavage of the intermediate protocatechuate (Baumann et al., 1972). In contrast, *M. mediterranea* is unable to metabolize *m*-hydroxybenzoate. All *Marinomonas* species can use D-glucose. *Marinomonas communis* and *M. vaga* metabolize D-glucose and D-fructose via the Entner–Doudoroff pathway (Sawyer et al., 1977a). Aspartate kinase activity has been detected in these two species (Baumann and Baumann, 1974). All species can use glutamate as a sole carbon and energy source, but none can use glycine or L-tyrosine. Data for other carbon sources are listed in Tables BXII.γ.87 and BXII.γ.88.

1. MMM is a chemically defined medium containing (per liter): NaCl, 20 g; MgSO₄·7H₂O, 7.0 g; MgCl₂·6H₂O, 5.3 g; KCl, 0.7 g; CaCl₂, 1.25 g; FeSO₄·7H₂O, 25 mg; CuSO₄·5H₂O, 5 mg; K₂HPO₄, 75 mg; sodium glutamate, 2 g; and Tris base, 6.1g. The pH of this medium is adjusted to 7.4.

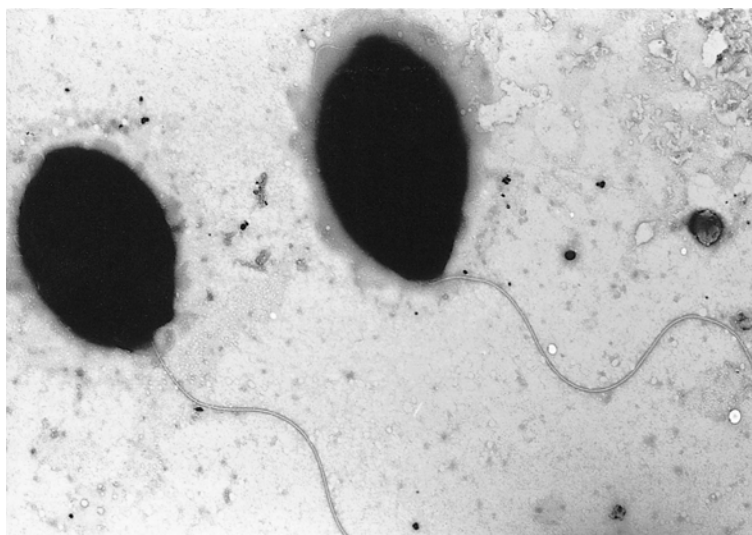


FIGURE BXII.γ.106. Electron micrograph of *Marinomonas mediterranea* MMB-1. Negatively stained with phosphotungstic acid ($\times 20,000$).

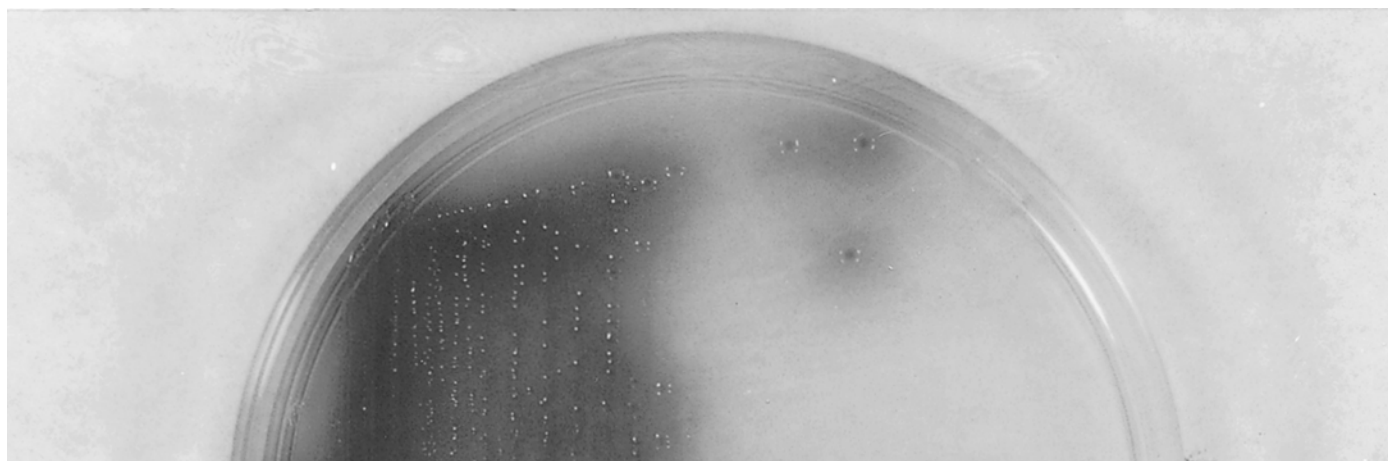


FIGURE BXII.γ.107. Colonies of *Marinomonas mediterranea* MMB-1 on Marine Agar after four days of incubation at 25°C ($\times 2.1$).

Marinomonas communis and *M. vaga* can serve as prey for the growth of marine bdellovibrios (Taylor et al., 1974).

The mol% G + C content of the DNA of *M. communis* and *M. vaga* has been calculated using the buoyant density method. The mol% G + C of the DNA of *M. mediterranea* strain MMB-1 was determined by HPLC analysis after DNA hydrolysis and dephosphorylation (Solano et al., 1997). In this strain, the chromatogram revealed two additional peaks besides those of the four expected standard deoxynucleosides. The base modifications in the compounds giving these peaks remain unidentified, although their UV adsorption spectra suggest that they are methylated

derivatives of the purine deoxynucleosides. The method used to estimate the mol% G + C content in the other two species of *Marinomonas* does not allow one to know whether this feature is shared by all the *Marinomonas* species.

Resistance of *M. mediterranea* to some antimicrobial agents was tested in order to evaluate them as genetic markers. This species was sensitive to ampicillin (50 µg/ml), chloramphenicol (10 µg/ml), gentamicin (10 µg/ml), kanamycin (40 µg/ml), streptomycin (10 µg/ml), rifampicin (50 µg/ml), and tetracycline (10 µg/ml). Resistance to the latter compound was assayed in media lacking Mg^{2+} .

TABLE BXII.γ.87. Differential characteristics of the species of the genus *Marinomonas*^a

Characteristic	1. <i>M. communis</i>	2. <i>M. mediterranea</i>	3. <i>M. vaga</i>
<i>Cell morphology:</i>			
Straight rod	—	+	+
Curved rod	+	—	—
<i>Growth at:</i>			
5°C	—	—	—
25°C	+	+	+
35°C	+	—	+
40°C	+	—	—
NO ₃ [−] reduction to NO ₂ [−]	—	+	—
Cytochrome <i>c</i> oxidase	+	—	—
Pigmentation	—	+	—
Lipase	—	+	—
Gelatinase	—	+	—
<i>Utilization of:</i>			
D-Fructose	+	—	+
<i>m</i> -Hydroxybenzoate	+	—	+
α-Ketoglutarate	+	—	+

^aSymbols: see standard definitions.

Several molecular techniques have been applied to *M. mediterranea* (Solano et al., 2000). Vectors containing the RP4 *mob* site are mobilizable by conjugation. Conjugation can be performed at 25°C on the surface of plates containing the medium LB2216². Plasmids, such as pKT230, with the p15 origin of replication are stable in this species and hence can be used as cloning vectors. On the contrary, plasmids containing the *oriR6K* behave as suicidal vectors in *M. mediterranea*, which makes them useful as transposon delivery vectors. Tn5 and Tn10 derivatives have been assayed for this purpose. Higher transposition frequencies were obtained with Tn10 derivatives encoding kanamycin and gentamicin resistance than with Tn10 transposons with a different marker, or with Tn5 transposons. Other genetic markers such as ampicillin and chloramphenicol resistance were not properly expressed in *M. mediterranea* even if they were present in plasmids. *M. mediterranea* is normally lac(−), but after transposon mutagenesis with one mini-Tn10 derivative containing the *lacZ* reporter gene, approximately 55% of the colonies showed blue coloration when grown in media containing X-Gal, thus indicating the usefulness of *lacZ* as a reporter gene in *M. mediterranea*. Mutants of *M. mediterranea* can also be obtained by nitrosoguanidine mutagenesis (Solano et al., 1997). UV radiation is not an efficient mutagen for this strain, however.

A relevant feature in *M. mediterranea* is its capacity to synthesize melanins. These pigments are made from L-tyrosine as precursor and by the involvement of the enzyme tyrosinase (EC 1.14.18.1) (Solano et al., 1997). Tyrosinase is a copper protein that belongs to the group of polyphenol oxidases (PPOs). The other important copper enzyme in this group is laccase (EC 1.10.3.2). *M. mediterranea* also shows this activity, due to a multipotent enzyme showing both tyrosinase and laccase activities (Sanchez-Amat and Solano, 1997).

Some other aerobic, Gram-negative, marine proteobacteria are able to synthesize brown to black pigments in complex media. However, aside from *M. mediterranea*, actual PPO activity has been described only in the strain 2-40 (Kelley et al., 1990; Solano and Sanchez-Amat, 1999). In some cases, melanins result from spon-

taneous oxidation of intermediates of L-tyrosine catabolism, such as homogentisate in *Shewanella colwelliana* (Ruzafa et al., 1994; Kotob et al., 1995). In this case, melanin-like pigments are synthesized only in aerated media with a high amount of tyrosine. These pigments, designated pyomelanins, are very similar in their chemical properties to melanins obtained by activity of the enzyme tyrosinase. To differentiate between these two possibilities, actual PPO activity should be determined spectrophotometrically or revealed by staining after PAGE (Solano et al., 1997).

A high number of *M. communis* and *M. vaga* strains have been isolated using enrichment methods with different compounds as carbon sources (Baumann et al., 1972). These isolations led to the view that *Marinomonas* is a usual component of the bacterial flora in marine waters; however, there are no data about the ecological distribution of *M. mediterranea*. The single strain of this species was detected after screening thousands of colonies of marine microorganisms for the ability to synthesize melanins. No other strains with pigmentation identical to *M. mediterranea* were detected, which suggests that it may not be very abundant. It is important to bear in mind that it is not easy to differentiate between *Marinomonas* and other aerobic marine bacteria using only phenotypic characteristics (see below). Strains for which 16S rDNA sequence analyses clearly indicate they belong to the genus *Marinomonas* have been isolated from North Sea bacterioplankton (Eilers et al., 2000). In addition, a clone showing a close relationship to the genus *Marinomonas* was detected in a 16S rDNA sequence clone library obtained from bacterial communities associated with the seagrass *Halophila stipulacea* (Weidner et al., 2000).

ENRICHMENT AND ISOLATION PROCEDURES

Marinomonas species can be isolated by direct plating of seawater samples on a complex medium such as Marine Agar (Zobell, 1941). *M. communis* and *M. vaga* are able to use *m*-hydroxybenzoate, and this property can be used for an enrichment method. Seawater (500 ml) is transferred to a 2-liter Erlenmeyer flask and supplemented with 25 ml 1M Tris-HCl (pH 7.5), 0.5 g NH₄Cl, 38 mg K₂HPO₄, 14 mg FeSO₄·7H₂O, and 0.5 g *m*-hydroxybenzoate. The culture is incubated at 20–25°C for up to 10 d. When growth is observed, the culture is streaked onto BMA containing 0.1% (w/v) *m*-hydroxybenzoate in order to obtain pure cultures (Baumann et al., 1984d).

MAINTENANCE PROCEDURES

Long-term preservation of *Marinomonas* species can be achieved by lyophilization. A suitable protocol for preparation of the cells has been described by Gauthier and Breittmayer (1992). The lyophilized cultures are reconstituted by adding 0.5 ml of Marine Broth. A few drops are streaked onto Marine Agar and the remaining is transferred to a tube containing 4 ml of Marine Broth. It is advisable to avoid high aeration of the culture during the first hours of incubation. Growth is observed after 2–3 d.

An alternative method of preservation is freezing. Glycerol is added to an overnight culture in Marine Broth to a final concentration of 20% and the culture is immediately stored at −75°C. Using this protocol, frozen *M. mediterranea* cells have remained viable for more than five years.

Strains can be maintained by serial transfer on Marine Agar for routine work in the lab. After 2–3 days of growth at 25°C, the plates can be preserved for 3–4 weeks at 15°C. It is not recommended to keep the cultures at 4°C, because viability is lost much faster than when the cultures are stored at 15°C.

2. LB2216 is obtained by mixing—after autoclaving—equal amounts of LB medium (containing 1.5% NaCl) and Marine Agar 2216.

TABLE BXII.γ.88. Other characteristics of the species of the genus *Marinomonas*^a

Characteristics	1. <i>M. communis</i> ^b	2. <i>M. mediterranea</i>	3. <i>M. vaga</i> ^b
Motility	+	+	+
Flagellar arrangement:			
Polar	+	+	+
Peritrichous	—	—	—
Anaerobic growth	—	—	—
Na ⁺ requirement	+	+	+
Poly-β-hydroxybutyrate accumulation	—	—	—
Denitrification	—	—	—
Luminescence	—	—	—
Amylase	—	—	—
Utilization of:			
D-Glucose, D-mannose, D-sorbitol, citrate, acetate, succinate, malate, glycerol	+	+	+
DL-β-Hydroxybutyrate	+	+	d
β-Alanine, L-serine, L-lysine, L-arginine, spermine	+	nd	d
Ethanol, propanol	+	nd	—
D-Galactose, L-rhamnose, L-arabinose	d	nd	+
Maltose, propionate	d	—	d
DL-Glycerate, ethanolamine	d	nd	—
Cellobiose, N-acetylglucosamine, erythritol, heptanoate	—	nd	+
Lactose, butyrate, valerate, methanol, L-glycine, L-tyrosine	—	—	—
D-Ribose, D-arabinose, caproate, caprylate, pelargonate, ribitol, trigonelline	—	nd	d
Butanol	—	nd	—
Mol% G + C of DNA	45.9–48	46.3 ± 0.9	46.4–49.3

^aSymbols: see standard definitions; nd, not determined.

^b*M. communis* and *M. vaga* are able to use the following compounds as principal sources of carbon and energy: D-gluconate, fumarate, DL-lactate, aconitate, D-mannitol, γ-aminobutyrate, sarcosine, putrescine, saccharate, pyruvate, *m*-inositol, L-alanine, D-alanine, L-ornithine, L-histidine, L-proline, betaine.

The following compounds are not used as sole or principal sources of carbon and energy by *Marinomonas communis* or *M. vaga*: sucrose, melibiose, salicin, D-fucose, inulin, cellulose, mucate, L-threonine, L-leucine, L-isoleucine, L-valine, L-tryptophan, D-tryptophan, L-norleucine, DL-α-aminobutyrate, DL-α-aminovalerate, δ-aminovalerate, formate, isobutyrate, isovalerate, oxalate, malonate, maleate, glutarate, adipate, pimelate, suberate, azelate, sebacate, D-tartrate, *meso*-tartrate, glycolate, levulinate, D-mandelate, L-mandelate, citraconate, itaconate, mesaconate, ethyleneglycol, 2,3-butanediol, methanol, isopropanol, isobutanol, *n*-butanol, propyleneglycol, benzoylformate, DL-kynurenine, kynurenate, anthranilate, *m*-aminobenzoate, *p*-aminobenzoate, methylamine, benzylamine, histamine, tryptamine, butylamine, α-amylamine, 2-amylamine, pentylamine, creatine, hippurate, pantothenate, acetamide, nicotinate, nicotinamide, allantoin, adenine, guanine, cytosine, thymine, uracil, *n*-dodecane, phenylethanedil, phenol, naphthalene.

11–89% of the strains included in *M. communis* and *M. vaga* are able to use the following compounds as principal sources of carbon and energy: D-xylose, D-trehalose, D-glucuronate, D-galacturonate, caprate, L-tartrate, L-aspartate, L-phenylalanine, L-citrulline.

DIFFERENTIATION OF THE GENUS *MARINOMONAS* FROM OTHER GENERA

The characteristics that differentiate *Marinomonas* species from other aerobic, marine, Gram-negative bacteria are listed in Table BXII.γ.89. Further characterization of more strains might help to establish other phenotypic criteria for differentiating these genera; however, all of these genera share many physiological and biochemical features and their differentiation can be problematic. In this regard, some chemotaxonomic markers such as isoprenoid quinone and fatty acid composition have been tested (Akagawa-Matsushita et al., 1992a; Ivanova et al., 2000c). Molecular analysis, such as rRNA–DNA hybridization and particularly rDNA sequencing, have shown that *Marinomonas* species form a

well-defined cluster (Van Landschoot and De Ley 1983; Solano and Sanchez-Amat, 1999). 16S rDNA sequencing of additional *Marinomonas* strains could be useful for designing nucleic acid probes for the identification and quantification in natural habitats of microorganisms belonging to this genus.

TAXONOMIC COMMENTS

There are many Gram-negative, nonfermentative, heterotrophic, marine bacteria with polar flagella. Those with mol% G + C contents below 50 were temporarily assigned to the genus *Alteromonas*, and those with a mol% G + C of 57–64 were included in the genus *Pseudomonas* (Baumann et al., 1972). Further characterization of the genus *Alteromonas* revealed a great diversity

TABLE BXII.γ.89. Differential characteristics of the genus *Marinomonas* and other aerobic, Gram-negative, marine proteobacteria with polar flagellation^a

Characteristic	<i>Marinomonas</i>	<i>Oceanospirillum</i> ^b	<i>Alteromonas</i> ^c	<i>Marinobacterium</i> ^d	<i>Marinobacter</i> ^e	<i>Neptunomonas</i> ^f	<i>Pseudoalteromonas</i> ^g	<i>Pseudomonas</i> ^g
Poly-β-hydroxybutyrate granules	—	+	—	—	—	+	—	D
Amylase	—	—	+	—	—	—	+ ^h	D
Utilization of:								
Carbohydrates	+	—	+	+	—	+	+	+
Malate	+	D	—	+	—	—	—	+
D-Sorbitol	+	—	—	—	—	—	—	—
Mol% G + C of DNA	45–50	42–51	44–47	54.9	52.7	46	37–50	55–64

^aFor symbols see standard definitions.

^bData from Krieg (1984b).

^cData from Gauthier et al. (1995a).

^dData from González et al. (1997).

^eData from Gauthier and Brittmayer (1992).

^fData from Hedlund et al. (1999a).

^gData from Baumann et al. (1972, 1983b).

^h*Pseudoalteromonas nigrifaciens* is negative.

(Van Landschoot and De Ley, 1983; Gauthier et al., 1995a). Two of the species originally included in the genus *Alteromonas* were *A. communis* and *A. vaga*. Immunological analysis of Fe-containing superoxide dismutase and glutamine synthetase established a relationship between these species and the genus *Oceanospirillum*, and hence their assignment to this genus under the species names *Oceanospirillum commune* and *O. vagum* was proposed (Bowditch et al., 1984a). DNA–rRNA hybridization studies applied to some Gram-negative bacteria, including those initially assigned to the genus *Alteromonas*, revealed that *A. communis* and *A. vaga* constituted a separate rRNA branch (Van Landschoot and De Ley, 1983). Taking into consideration this fact and also phenotypic traits, the genus *Marinomonas* was created to contain the two species as *M. communis* and *M. vaga*. Phylogenetic analysis based on 16S rDNA sequences of these two species and a new strain, MMB-1, revealed that they were closely related, with percentage identities greater than 95% (Fig. BXII.γ.108). These data supported creation of the genus *Marinomonas*, in which the strain MMB-1 constituted the third species, namely *M. mediterranea* (So-

lano and Sanchez-Amat, 1999). The data also showed that these species were more closely related to members of the genus *Oceanospirillum* than to the species remaining in the modified genus *Alteromonas* or *Pseudoalteromonas*.

FURTHER READING

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Van Landschoot, A. and J. De Ley. 1983. Intra- and intergeneric similarities of the rRNA cistrons of *Alteromonas*, *Marinomonas* (gen. nov.) and some other Gram-negative bacteria. *J. Gen. Microbiol.* 129: 3057–3074.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *MARINOMONAS*

Table BXII.γ.87 shows the main differential characteristics of the three *Marinomonas* species described. Other characteristics are listed in Table BXII.γ.88.

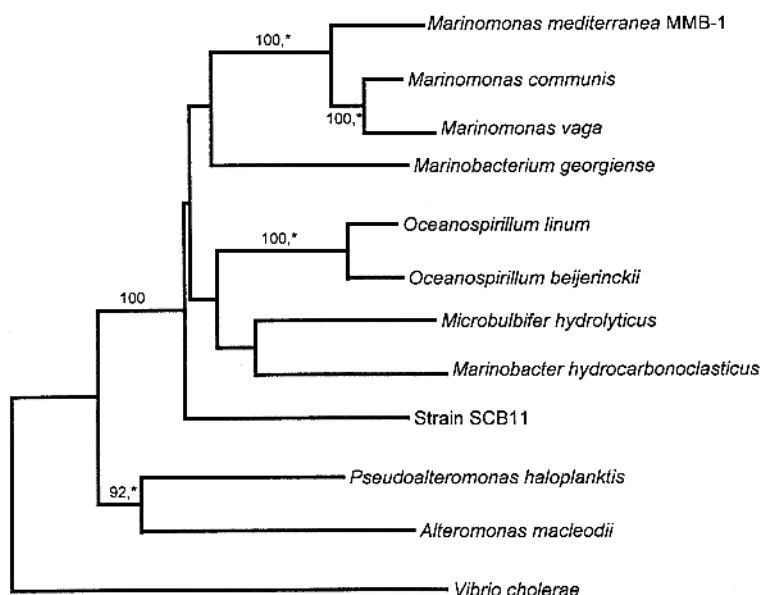


FIGURE BXII.γ.108. Phylogenetic position of *Marinomonas mediterranea* strain MMB-1 in the context of other marine bacteria. The tree was generated by the neighbor-joining method from complete 16S rDNA sequences. Numbers indicate the support for these branches by bootstrap analysis. Asterisks indicate branches also identified by maximum parsimony method. Only values higher than 90% are included (bar = 1% dissimilarity). (Reprinted with permission from F. Solano and A. Sanchez-Amat, *International Journal of Systematic Bacteriology* 49: 1241–1246, 1999 ©International Union of Microbiological Societies.)

List of species of the genus *Marinomonas*

1. ***Marinomonas communis*** (Baumann, Baumann, Mandel and Allen 1972) Van Landschoot and De Ley 1984, 91^{VP} (Effective publication: Van Landschoot and De Ley 1983, 3071) (*Alteromonas communis* Baumann, Baumann, Mandel and Allen 1972, 420.)

com.mu'nis. L. adj. *communis* common.

The characteristics are as described for the genus and as listed in Tables BXII.γ.87 and BXII.γ.88. Curved rods, 0.7–1.5 × 1.8–3.0 μm. Motile by means of polar flagella at one or both poles. Oxidase positive. Metabolize *m*-hydroxybenzoate, *p*-hydroxybenzoate, and quinate, but not benzoate or *o*-hydroxybenzoate, by means of *meta* cleavage of the intermediate protocatechuate.

The mol% G + C of the DNA is: 45–48 (Bd).

Type strain: ATCC 27118, DSM 5604, LMG 2864.

2. ***Marinomonas mediterranea*** Solano and Sanchez-Amat 1999, 1245^{VP}

med.i.terr.an'e.a. L. gen. n. *mediterranea* of the Mediterranean, referring to the isolation of the type strain from the Mediterranean Sea in the region of Murcia, on the southeastern Spanish coast.

The characteristics are as described for the genus and as listed in Tables BXII.γ.87 and BXII.γ.88. Gram-negative rods, 0.5–0.9 × 1.1–1.8 μm. Motile by a single polar flagellum. Cytochrome oxidase negative. No denitrification

occurs. Lipase positive; gelatinase weakly positive. Utilize D-glucose, D-mannose, D-sorbitol, citrate, β-hydroxybutyrate, succinate, glycerol, and malate. Produce melanin. Contain a pluripotent polyphenol oxidase and a tyrosinase activated by SDS.

The mol% G + C of the DNA is: 46.3 ± 0.9 (HPLC).

Type strain: MMB-1, ATCC 700492, CECT 4803.

GenBank accession number (16S rRNA): AF063027.

3. ***Marinomonas vaga*** (Baumann, Baumann, Mandel and Allen 1972) Van Landschoot and De Ley 1984, 91^{VP} (Effective publication: Van Landschoot and De Ley 1983, 3071) (*Alteromonas vaga* Baumann, Baumann, Mandel and Allen 1972, 420.)

va'ga. L. adj. *vaga* wandering.

The characteristics are as described for the genus and as listed in Tables BXII.γ.87 and BXII.γ.88. Straight rods 0.7–1.5 × 1.8–3.0 μm. Motile by means of polar flagella at one or both poles. Oxidase negative. Metabolize *m*-hydroxybenzoate, *p*-hydroxybenzoate and quinate, but not benzoate or *o*-hydroxybenzoate, by means of *meta* cleavage of the intermediate protocatechuate.

The mol% G + C of the DNA is: 46–50 (Bd).

Type strain: ATCC 27119.

GenBank accession number (16S rRNA): X67025.

Genus IV. Marinospirillum Satomi, Kimura, Hayashi, Shouzen, Okuzumi and Fujii 1998,
1346^{VP}

MASATAKA SATOMI, BON KIMURA AND TATEO FUJII

Ma.ri.no.spi.ril' lum. L. adj. *marinus* of the sea; Gr. n. *spira* a spiral; M.L. dim. neut. n. *spirillum* a small spiral; *Marinospirillum* a small spiral from the sea.

Helical cells. Motile by bipolar tufts of flagella. Thin-walled coccoid bodies are formed in aging cultures. Endospores are not formed. Aerobic, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor. **Na⁺ is required for growth.** Chemoheterotrophic. **Accumulate poly-β-hydroxybutyrate. Oxidase positive. Carbohydrates are neither oxidized nor fermented.** The major isoprenoid quinone type is Q-8. Belongs to the class *Gammaproteobacteria*. One species has been isolated from putrid infusions of marine mussels; another has been isolated from kusaya gravy.

The mol% G + C of the DNA is: 42–45.

Type species: Marinospirillum minutulum (Watanabe 1959) Satomi, Kimura, Hayashi, Shouzen, Okuzumi and Fujii 1998, 1346 (*Oceanospirillum minutulum* (Watanabe 1959) Hylemon, Wells, Krieg and Jannasch 1973, 373; "*Spirillum minutulum*" Watanabe 1959, 83.)

FURTHER DESCRIPTIVE INFORMATION

Because *Marinospirillum minutulum*—like most members of the genus *Oceanospirillum*—was isolated from putrid infusions of marine mussels, its source is most likely marine mud adherent to the mussels (Terasaki, 1970).

Marinospirillum megaterium was isolated from kusaya gravy (Fujii et al., 1990), which is used for producing Japanese traditional dried fish, and its true ecological niche is unknown. Under microscopic observation, this species is usually observed as the dominant microbial population in the gravy, characterized by large helical cells, but is unculturable on agar surfaces. The main chemical characteristics of kusaya gravy are as follows: salinity of approximately 3% NaCl concentration, low dissolved oxygen concentration, and large amounts of volatile basic nitrogen compounds (200–500 mg/100 g) (Satomi et al., 1997).

ENRICHMENT AND ISOLATION PROCEDURES

Terasaki (1970) enriched and isolated *M. minutulum* using the following methods. Marine mussels were smashed with a hammer and placed in a Petri dish with a teaspoon of marine mud. Sterilized seawater was poured into the dish until the mussels sank completely in the solution. The infusion was incubated at 27–28°C and examined for the development of spirilla after 1, 2, and 7 d. Isolation was accomplished by streaking dilutions onto suitable agar media.

Fujii et al. (1990) enriched and isolated *M. megaterium* as follows. A loopful of kusaya gravy was used to inoculate the inlet tube of a horizontal glass tube, in which the inlet and outlet ends

were bent upwards. After an appropriate time, the isolates accumulated predominantly in the outlet because of their high motility. By repeating this selective accumulation and purification process, the isolates could almost be purified (by microscopic observation). Isolation was accomplished by streaking a large amount of cells onto the suitable agar medium, because the ability of *M. megaterium* to form colonies on an agar surface was low. The ability to form colonies on the agar surface was lost after subsequent transfers of the culture.

MAINTENANCE PROCEDURES

Marinospirillum species may be maintained on stab cultures in TSSY semi-solid agar, which has the following composition (g/l): Trypticase peptone (BBL), 17.0; Phytone peptone (BBL), 3.0; Yeast nitrogen base (Difco), 1.0; NaCl, 30.0; and agar, 2.0; pH 8.0. Cultures are incubated at 25°C with biweekly transfer. Long-term preservation of *M. minutulum* by freeze-drying has been reported by Terasaki (1975).

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Characterization methods for marinospirilla have been described in detail by Terasaki (1972, 1973) and Satomi et al. (1998). Media containing natural or artificial seawater or 2.5–3.0% sodium chloride must be used for all characterization tests. Media for *M. megaterium* must be supplemented with 0.1–0.3% agar because the organism requires semisolid media for growth.

DIFFERENTIATION OF THE GENUS *MARINOSPIRILLUM* FROM OTHER GENERA

The genus *Marinospirillum* cannot be differentiated phenotypically from *Oceanospirillum*, but each *Marinospirillum* species can be differentiated from *Oceanospirillum*, by positive nitrate reduction (*M. minutulum*) and a prominent microaerophilic nature with no growth on agar surfaces (*M. megaterium*).

TAXONOMIC COMMENTS

Phylogenetic analysis based on 16S rDNA gene sequences of the genus *Marinospirillum*, consisting of two species *M. minutulum* and *M. megaterium*, showed that this genus was closely related to the genera *Oceanospirillum* and *Halomonas* in the *Gammaproteobacteria* group (Fig. BXII.γ.109). Although members of the genera *Marinospirillum* and *Oceanospirillum* share helical cell morphology, there is a large phylogenetic distance between *Marinospirillum* and *Oceanospirillum*.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *MARINOSPIRILLUM*

Table BXII.γ.90 lists the characteristics that differentiate *M. minutulum* from *M. megaterium*.

List of species of the genus Marinospirillum

1. ***Marinospirillum minutulum*** (Watanabe 1959) Satomi, Kimura, Hayashi, Shouzen, Okuzumi and Fujii 1998, 1346^{VP} (*Oceanospirillum minutulum* (Watanabe 1959) Hylemon,

Wells, Krieg and Jannasch 1973, 373; "*Spirillum minutulum*" Watanabe 1959, 83.)
mi.nu'tu.lum. L. dim. neut. adj. *minutulum* very little.

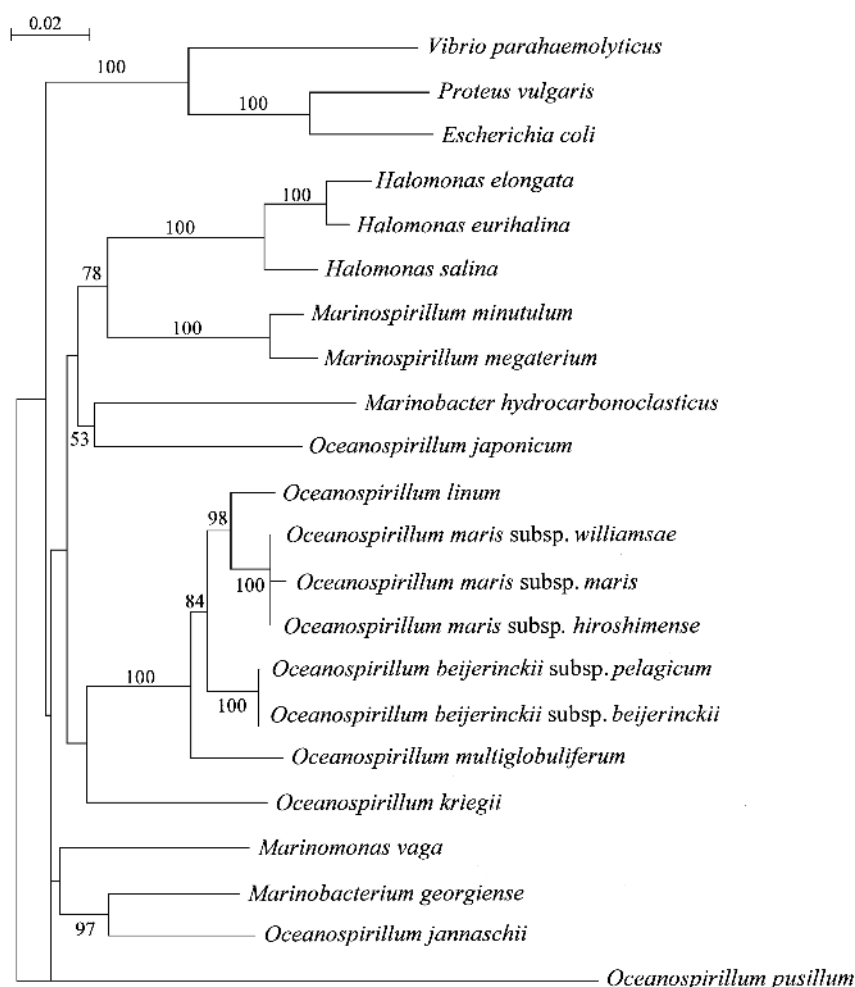


FIGURE BXII.γ.109. Phylogenetic trees of the genus *Marinospirillum* and related bacteria based on the nucleotide sequences of the 16S rRNA gene. The tree was constructed using the neighbor-joining method, and the nucleotide substitution rates (K_{nu} values) were computed by Kimura's 2-parameter model. The scale bar indicates the genetic distance of 0.02 K_{nu}. The numbers at the nodes indicate the percentages of occurrence in 1000 bootstrapped trees. Only values greater than 40% are shown. *Oceanospirillum pusillum* was used as an outgroup. Accession numbers for sequences used: *Marinospirillum minutulum* (AB006769), *M. megaterium* (AB006770), *Oceanospirillum linum* (M22365), *O. maris* subsp. *maris* (AB006771), *O. maris* subsp. *hirosimense* (AB006762), *O. maris* subsp. *williamsae* (AB006763), *O. beijerinckii* subsp. *beijerinckii* (AB006760), *O. beijerinckii* subsp. *pelagicum* (AB006761), *O. multiglobuliferum* (AB006764), *O. japonicum* (AB006766), *O. kriegii* (AB006767), *O. jannaschii* (AB006765), *O. pusillum* (AB006768), *Halomonas elongata* (X67023), *Halomonas eurihalina* (X87218), *Halomonas salina* (X87217), *Marinomonas vaga* (X67025), *Marinobacter hydrocarbonoclasticus* (X67022), *Marinobacterium georgiense* (U58339), *Vibrio parahaemolyticus* (X56580), *Escherichia coli* (J01859), *Proteus vulgaris* (X07652).

TABLE BXII.γ.90. Characteristics differentiating the species of the genus *Marinospirillum*^{a,b}

Characteristic	1. <i>M. minutulum</i> ^c	2. <i>M. megaterium</i> ^d
Cell diameter, μm	0.3–0.4	0.8–1.2
Microaerophilic	–	+
Catalase	+	– or W
Nitrate reduced to nitrite	+	–
Growth in the presence of:		
1% oxgall	+	–
1% glycine	+	–

^aSymbols: see standard definitions.

^bData from Satomi et al. (1998).

^cATCC 19193.

^dStrains H7 and Sp5.

Rigid, helical cells with clockwise turns (Terasaki, 1973). Cell diameter, 0.3–0.4 μm; wavelength, 2.0–2.8 μm; helix diameter, 0.6–1.5 μm; length of helix, 3.0–8.0 μm. Coccoid bodies are predominant at 3–4 weeks but not after 24–48 h growth. NaCl is required for growth: optimum concentration, 2.5–3.0%; range 0.2–10%. Optimum temperature, 25–30°C; range, 4–30°C. Optimum pH, 8.0; range, 7.0–10.5. Aerobic. Grow in liquid media under static or aerobic conditions with shaking, or on plates. Catalase positive. Do not hydrolyze gelatin, hippurate, or starch. Reduce nitrate to nitrite. Do not produce DNase, RNase, urease, or phosphatase. The type strain was isolated from putrid infusions of marine mussels.

The mol% G + C of the DNA is: 42–44 (HPLC).

Type strain: ATCC 19193, DSM 6287.

GenBank accession number (16S rRNA): AB006769.

2. **Marinospirillum megaterium** Satomi, Kimura, Hayashi, Shouzen, Okuzumi and Fujii 1998, 1346^{VP}
me.ga.te'ri.um. Gr. adj. *mega* large; Gr. n. *teras, teratis* monster, beast; M.L. n. *megaterium* big beast.

Rigid, helical cells, $0.8\text{--}1.2 \times 5\text{--}15 \mu\text{m}$, having clockwise turns; thin-walled coccoid bodies $2.0\text{--}2.5 \mu\text{m}$ in diameter. NaCl is required for growth: optimum concentration, 3%;

range, 0.5–9.0%. Optimum temperature, $20\text{--}25^\circ\text{C}$; range, $4\text{--}25^\circ\text{C}$. Optimum pH, 8; range, 7.5–9.0. Microaerophilic: no growth occurs in liquid media under static or aerobic conditions with shaking, or on plates. Catalase weakly positive or negative. Do not hydrolyze gelatin, hippurate, or starch. Do not reduce nitrate. Do not produce DNase, RNase, urease, or phosphatase. The type strain was isolated from kusaya gravy.

The mol% G + C of the DNA is: 44–45 (HPLC).

Type strain: H7, JCM 10129.

GenBank accession number (16S rRNA): AB006770.

Genus V. *Neptunomonas* Hedlund, Geiselbrecht, Bair and Staley 1999b, 1325^{VP} (Effective publication: Hedlund, Geiselbrecht, Bair and Staley 1999a, 258)

BRIAN P. HDLUND

Nep.tu.no.mo' nas. Rom. myth n. *Neptune* the Roman god of the sea; Gr. n. *monas* a unit, monad; M.L. n. *Neptunomonas* Neptune's monad.

Rod-shaped or slightly curved bacteria. Approximately $0.7\text{--}0.9 \times 2.0\text{--}3.0 \mu\text{m}$. (See Fig. BXII.γ.110). Cells may produce a capsule that is visible by India ink staining. Coccoid bodies may predominate in old cultures and are associated with a loss of viability. Poly-β-hydroxybutyrate is accumulated and may form small inclusions that are visible by phase-contrast microscopy. **Motile by a single polar flagellum.** Aerobic; however, fermentation tests with some sugars and sugar alcohols are weakly positive. Acid is produced slowly from some carbohydrates. Na⁺ is required for growth. Oxidase and catalase positive. Phosphatase positive. Utilize some carbohydrates, polycyclic aromatic hydrocarbons, amino acids, organic acids, and sugar alcohols as sole carbon sources for chemoorganotrophic growth. Temperature range for known strains, $\leq 4\text{--}30^\circ\text{C}$. Indigenous to coastal marine sediments. Belong to the *Oceanospirillum* group of the class *Gammaproteobacteria* based on phylogenetic analyses of 16S rDNA gene sequences.

The mol% G + C of the DNA is: 46.

Type species: ***Neptunomonas naphthovorans*** Hedlund, Geiselbrecht, Bair and Staley 1999b, 1325 (Effective publication: Hedlund, Geiselbrecht, Bair and Staley 1999a, 258.)

FURTHER DESCRIPTIVE INFORMATION

Only one species of *Neptunomonas* is currently known, *N. naphthovorans*.

Polycyclic aromatic hydrocarbon (PAH) catabolism by *N. naphthovorans* strains NAG-2N-113 and NAG-2N-126 has been studied in some detail (Hedlund et al., 1999a). Both strains grow on naphthalene or 2-methylnaphthalene as a sole carbon and energy source. Only strain NAG-2N-113 grows weakly on phenanthrene. Both strains degrade 1-methylnaphthalene and acenaphthene under certain conditions; however, the degradation of these molecules is not coupled to growth. Colonies produce indigo when

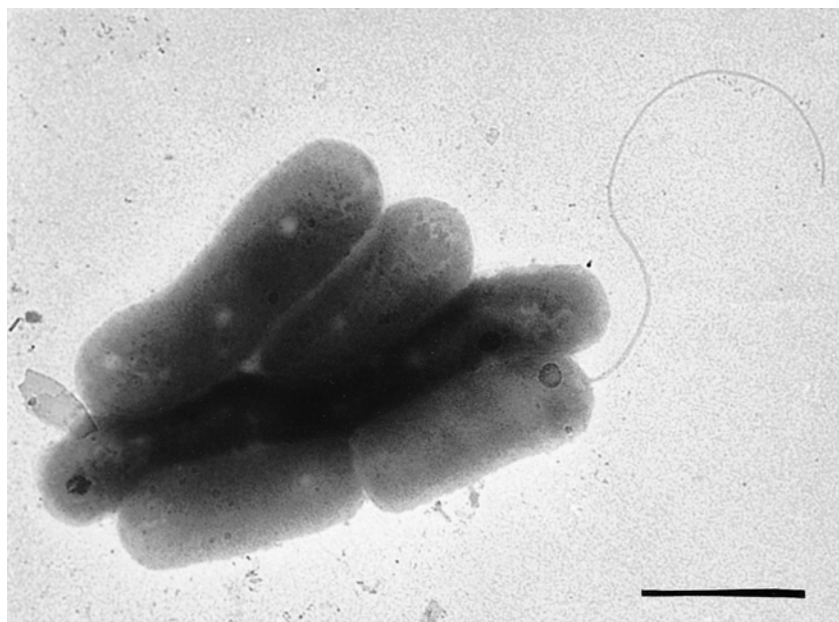


FIGURE BXII.γ.110. Electron micrograph of negatively stained *Neptunomonas naphthovorans* strain NAG-2N-126. Bar = $2 \mu\text{m}$. (Reproduced with permission from B. Hedlund et al., Applied and Environmental Microbiology 65: 251–259, 1999 ©American Society for Microbiology, Washington, D.C.)

naphthalene-induced cells are exposed to indole vapors, a characteristic *Neptunomonas* shares with naphthalene-degrading strains of *Pseudomonas*, *Burkholderia*, and *Marinobacter*, among others. This reaction is catalyzed by the naphthalene dioxygenase. A portion of the gene encoding the naphthalene dioxygenase large subunit has been sequenced from strains NAG-2N-113 and NAG-2N-126; they bear roughly 50–65% sequence identity to *Pseudomonas* and *Burkholderia* naphthalene dioxygenases. However, the low mol% G + C for the *Neptunomonas* dioxygenase gene indicates it is not newly acquired from organisms with a high mol% G + C by horizontal gene transfer. *Neptunomonas* does not produce colored intermediates characteristic of the *meta*-cleavage pathway when grown on naphthalene or other aromatic compounds.

A variety of carbon sources are used by *Neptunomonas* as sole carbon and energy sources, including naphthalene, 2-methylnaphthalene, *p*-hydroxybenzoate, D-fructose, D-glucose, citrate, DL- β -hydroxybutyrate, glutarate, succinate, DL-lactate, pyruvate, L-arginine, L-serine, L-glutamate, acetate, L-proline, DL-alanine, glycerol, mannitol, and D-arabitol.

All strains of this species have been isolated from a single area in Puget Sound near Seattle, WA USA.

ENRICHMENT AND ISOLATION PROCEDURES

Neptunomonas strains have been isolated from coastal marine sediment contaminated with coal tar creosote, which consists mainly of polycyclic aromatic hydrocarbons. Two alternative methods have been used. The first method involves using a most probable number (MPN) approach to enumerate marine bacteria capable of growth on naphthalene and subsequently obtaining isolates from the positive MPN tubes. All manipulations have been done at 4°C with media that have been equilibrated to that temperature for several hours, but whether this is a necessary precaution is not yet known. Serial dilutions (10-fold) of the sediment are made into 5- or 10-ml portions of the artificial seawater solution ONR7a,¹ and naphthalene crystals are added as a sole carbon and energy source. The tubes are typically incubated near *in situ* temperature without shaking and monitored daily for turbidity. It should be noted that *Neptunomonas* strains do not remain viable for long periods in ONR7a with naphthalene; therefore, sampling the dilution tubes within days of the culture becoming turbid is recommended. Samples from positive tubes are diluted with ONR7a, and the dilutions are spread onto plates of ONR7a solidified with 0.8% agarose or onto a complex marine medium, such as Marine Broth 2216 solidified with 1.5% agar. Naphthalene crystals are added to the Petri dish lids, the plates are inverted, and the plates are sealed with Parafilm. If *Neptunomonas* is outnumbered by other naphthalene-degrading bacteria such

as *Cycloclasticus*, *Neptunomonas* strains can be selected based on their ability to grow well in complex media such as Difco Marine Broth 2216.

A second isolation method employs a direct plating approach. Sediment is diluted into ONR7a and spread onto plates of ONR7a solidified with 0.8% agarose. Naphthalene crystals are added to the Petri dish lids, the plates are inverted, and then sealed.

MAINTENANCE PROCEDURES

Strains of *Neptunomonas* can be maintained for up to 1 month at 4°C on Parafilm-sealed plates of solidified Difco Marine Broth 2216. Additionally, freezer stocks containing glycerol or DMSO can be stored at –80°C for at least 2 years. Lyophilized cultures have also been revived; however, their long-term viability has not yet been evaluated.

DIFFERENTIATION OF THE GENUS *NEPTUNOMONAS* FROM OTHER GENERA

The characteristics that differentiate *Neptunomonas* from phylogenetically similar genera are listed in Table BXII.γ.91.

Several other genera of naphthalene-degrading marine bacteria exist, including *Cycloclasticus* (Dyksterhouse et al., 1995), *Marinobacter* (Gauthier et al., 1992), *Vibrio* (Geiselbrecht et al., 1996), *Pseudoalteromonas* (Hedlund et al. 1996a), and *Sphingomonas* (Zylstra, personal communication). Differential characteristics of these bacteria are shown in Table BXII.γ.92. However, it is generally recommended that 16S rRNA genes be sequenced in order to confidently identify new isolates of marine PAH-degraders.

TAXONOMIC COMMENTS

Based on phylogenetic analyses using nearly complete 16S rRNA gene sequences, *Neptunomonas* is a member of the *Oceanospirillum* group of the class *Gammaproteobacteria*. However, the exact phylogenetic relationship among *Neptunomonas*, *Oceanospirillum*, *Marinomonas*, and *Marinobacterium* is uncertain, as evidenced by the low bootstrap value for the node connecting *Neptunomonas* to the tree (Fig. BXII.γ.111).

The phylogeny and taxonomy of the *Oceanospirillum* group is further complicated by nomenclatural problems associated with some members of the group. Four species, *O. linum*, *O. maris*, *O. beijerinckii*, and *O. multiglobuliferum* form a monophyletic cluster based on rRNA hybridization (Pot et al., 1989), multilocus enzyme electrophoresis (Pot et al., 1989), quinone analysis (Sakane and Yokota, 1994), fatty acid analysis (Sakane and Yokota, 1994), and 16S rRNA gene phylogenetic analysis (Hedlund et al., 1999a). Other taxa, such as *O. kriegii*, and *O. jannaschii*, are not as closely related by any measurement, and it is likely each will be assigned to a new genus. *O. japonicum* is phylogenetically distinct from the true members of the genus *Oceanospirillum*; however, it does share many phenotypic similarities with them. *O. pusillum* obviously should not be included in the genus *Oceanospirillum* since it belongs to the class *Alphaproteobacteria* (Satomi et al., 1998). Therefore, the taxonomy of the *Oceanospirillum* group will likely change dramatically in the near future.

FURTHER READING

Hedlund, B.P., A.D. Geiselbrecht, T.J. Bair and J.T. Staley. 1999. Polycyclic aromatic hydrocarbon degradation by a new marine bacterium, *Neptunomonas naphthovorans* gen. nov., sp. nov. Appl. Environ. Microbiol. 65: 251–259.

1. ONR7a consists of several solutions that are prepared separately and mixed when cool. First, a 10× seawater salts solution is prepared, consisting of (g/l distilled water): NaCl, 228; Na₂SO₄, 40; KCl, 7.2; NaBr, 0.83; NaHCO₃, 0.31; H₃BO₃, 0.27; and NaF, 0.026. This 10× solution is diluted to 1×, and the following are added (g/l 1× solution): Na₂HPO₄·7H₂O, 0.089; NH₄Cl, 0.27; and TAPSO buffer (3-[N-tris(hydroxymethyl)methylamino]-2-hydroxypropanesulfonic acid), 1.3. After the solution has been autoclaved and cooled to at least 50°C, 20.0 ml of a 50× divalent cation solution and 5.0 ml of a 200× Fe(II) solution are added per liter. The 50× divalent cations solution, which is prepared and autoclaved separately, contains (g/l): MgCl₂·6H₂O, 559.1; CaCl₂·2H₂O, 72.8; and SrCl₂·6H₂O, 1.21. The 200× solution of Fe(II) is prepared by dissolving 0.04 g of FeCl₂·4H₂O with a few drops of concentrated HCl; water is then added to 100 ml, and the solution is sterilized by filtration. Long-term storage of the complete ONR7a solution is not recommended.

List of species of the genus *Neptunomonas*

1. ***Neptunomonas naphthovorans*** Hedlund, Geiselbrecht, Bair and Staley 1999b, 1325^{VP} (Effective publication: Hedlund, Geiselbrecht, Bair and Staley 1999a, 258.)
naph.tho.vo'rans. Chem. n. *naphtho*-combining form of *naphthalene*, a white, crystalline hydrocarbon; L. v. *vor*o to devour; M.L. part. adj. *naphthovorans* naphthalene-devouring.

The characteristics are as described for the genus and as listed in Table BXII.γ.91. The type strain was isolated from Eagle Harbor, a creosote-contaminated EPA Superfund site in Puget Sound, Washington, USA.

The mol% G + C of the DNA is: 46.3 (T_m).

Type strain: NAG-2N-126, ATCC 700637.

GenBank accession number (16S rRNA): AF053734.

TABLE BXII.γ.91. Characteristics differentiating the genus *Neptunomonas* from related genera^a

Characteristic	<i>Neptunomonas</i>	<i>Marinobacterium</i>	<i>Oceanospirillum</i> ^b	<i>O. japonicum</i>	<i>O. kriegii</i>
<i>Cell shape:</i>					
Curved rods	+				+
Straight rods	+	+			+
Spirilla			+	+	
Flagellar arrangement	single polar	single polar	bipolar tufts	bipolar or single tufts	single polar
Cell diameter, μm	0.7–0.9	0.5–0.7	0.3–1.2	0.8–1.4	0.8–1.2
Poly-β-hydroxybutyrate accumulation	+	–	+	+	+
NO ₃ [–] reduced to NO ₂ [–]	–	–	D	–	–
Catalase	+	+	D	w or –	NR
Amylase	–	–	–	–	NR
Gelatinase	–	–	D	d	NR
Lipase	–	+	NR	NR	+
Phosphatase	+	NR	D	w	NR
Acid from carbohydrates	+ ^c	–	–	–	–
Carbohydrates fermented	+ ^c	–	–	–	–
Temperature range, °C	≤4–30	4–41	2–43	10–43	20–35
Water-soluble, brown pigment	+	NR	D	–	NR
mol% G + C of DNA	46	54.9	45–50	45	54.8

^aFor symbols see standard definitions; NR, not reported; w, weakly positive.

^bThis includes the members of the genus *Oceanospirillum* as described in this book.

^cCarbohydrate oxidation/fermentation tests were done in PSS media (Hylemon et al., 1973). Acidification took up to two weeks to occur.

TABLE BXII.γ.92. Differentiation of *Neptunomonas* from some other marine naphthalene-degrading bacteria^{a,b}

Characteristic	<i>Neptunomonas</i>	<i>Cycloclasticus</i>	<i>Marinobacter</i>	<i>Pseudoalteromonas</i>	<i>Vibrio</i>
Indigo from indole ^c	+	–	+	d	–
Utilization of phenanthrene ^c	– or w	+	– ^d	d	+
Na ⁺ requirement	+	+	D	+	+ ^c
Growth on marine broth 2216	+	– or w	+	+	+
Utilization of carbohydrates	+	–	–	+	+
D-glucose	+	–	–	+	+
D-fructose	+	–	–	d	D
Utilization of amino acids	+	+ ^f	+ ^g	+	+
Acid from D-glucose	+ or w	NG	–	–	+
mol% G + C	46	39	57.3–57.7 ^d	37–50	38–51

^aFor symbols see standard definitions; NG, no growth in test medium; w, weakly positive.

^bMarine bacteria are defined here as requiring Na⁺ ions for growth. Taxa that have not been fully described are not considered here.

^cThe results for these phenotypes are only given for naphthalene-degrading strains. Otherwise, characteristics are generalized to encompass the whole genus.

^dFor these features we used the results of Spröer et al. (1998), which differ from those obtained previously (Gauthier et al., 1992).

^eAll *Vibrio* strains require Na⁺ except *V. cholerae*.

^f*Cycloclasticus* utilizes only L-glutamate.

^g*Marinobacter* utilizes only L-proline and L-glutamate.

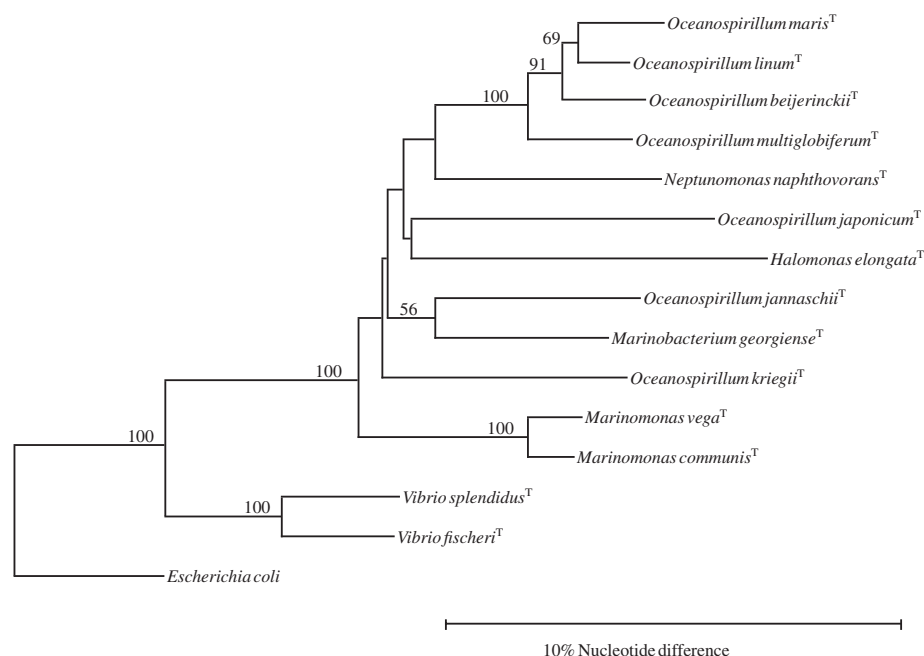


FIGURE BXII.γ.111. Dendrogram showing the relationship between *Neptunomonas* and its close relatives. The dendrogram was created in TreeCon using Jukes-Cantor distance with the Kimura 2-parameter correction. Numbers at branch nodes represent bootstrap values for 100 replications. Numbers below 50 are not shown. (Reproduced with permission from B. Hedlund et al., *Applied and Environmental Microbiology* 65: 251–259, 1999 ©American Society for Microbiology, Washington, D.C.)

Family II. *Alcanivoraceae* fam. nov.

PETER N. GOLYSHIN, SHIGEAKI HARAYAMA, KENNETH N. TIMMIS AND MICHAIL M. YAKIMOV

Al.ca.ni.vo.ra'.ce.ae. M.L. masc. n. *Alcanivorax* type genus of the family; *-aceae* ending to denote family; M.L. masc. pl. n. *Alcanivoraceae* the *Alcanivorax* family.

Rods 0.6–0.8 × 1.6–2.5 μm in size, depending on the substrate used for growth. Some isolates are capable of anaerobic growth. Reduce nitrate to nitrite. **Principal carbon and energy sources are linear-chain alkanes and their derivatives**, with carbon chain length between 9 and 20. Only a few simple organic compounds are used as carbon and energy source (formate, acetate, propionate, methyl-pyruvate, and α-ketoglutarate). **Moderately halo-**

philic, NaCl content optimal for growth is between 3 and 10%. Strains of all recognized species and recent isolates are mesophilic, with the temperature optima about 25–30°C. Some isolates produce glucose lipid surfactants.

Type genus: ***Alcanivorax*** Yakimov, Golyshin, Lang, Moore, Abraham, Lünsdorf and Timmis 1998, 346.

Genus I. Alcanivorax Yakimov, Golyshin, Lang, Moore, Abraham, Lünsdorf and Timmis 1998, 346^{VP}

PETER N. GOLYSHIN, SHIGEAKI HARAYAMA, KENNETH N. TIMMIS AND MICHAIL M. YAKIMOV

Al.ca.ni.vo'.rax. M.L. masc. n. *alcanum* alkane, aliphatic hydrocarbon; L. adj. *vorax* voracious, gluttonous; M.L. masc. n. *Alcanivorax* alkane-devouring.

Rods 0.6–0.8 × 1.6–2.5 μm when grown in pyruvate-supplemented medium and 0.6–0.8 × 1.0–1.5 μm when grown on *n*-alkanes. Nonmotile. Gram negative. Colonies 2–3 days old are circular, colorless, and transparent; in older cultures, they turn opaque and light yellow if grown on alkanes. **Aerobic and microaerophilic**, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor. Nitrate is reduced to nitrite, but anaerobic growth does not occur on nitrate. **Oxidase and catalase positive.** **Growth occurs on aliphatic hydrocarbons as sole or principal carbon sources.** Some strains produce glucose lipid surfactants that exist in two forms. **Moderately**

halophilic; optimum NaCl concentration 3% (range 0.5–15%). Isolated from seawater and seawater sediments enriched with crude oil, petroleum hydrocarbons, or single aliphatic hydrocarbons.

The mol% G + C of the DNA is: 53–66.

Type species: ***Alcanivorax borkumensis*** Yakimov, Golyshin, Lang, Moore, Abraham, Lünsdorf and Timmis 1998, 347.

FURTHER DESCRIPTIVE INFORMATION

Some strains produce a glycine-containing biosurfactant precursor that is linked to the cell surface and increases the hydro-

phobicity of the cells and their affinity to oil droplets suspended in the water phase. Some strains produce another form—the mature extracellular glucose lipid biosurfactant, which lacks the terminal glycine residue, and which forms micelles with water-insoluble oil fractions, thereby increasing their bioavailability.

These organisms are distributed in natural marine environments around the world, and their predominance in microbial communities accidentally polluted, or experimentally spiked with oil, suggests an important role in natural petroleum biodegradation (Harayama et al., 1999).

Strains of “*Alcanivoraceae*” were isolated or detected in 16S rDNA clone libraries from different geographic locations (Table BXII.γ.93).

Genome format The genome size estimated for *Alcanivorax borkumensis* SK2T and AP1 is about 3 Mbp. Up to now, no plasmids have been detected in these bacteria.

Isolates of *Alcanivorax* are amenable to conjugative DNA transfer, and transposon mutagenesis (miniTn5-mutagenesis has been characterized by a random positioning of the inserts in the chromosome of *A. borkumensis* strain SK2T with an appropriate antibiotic marker (chloramphenicol, streptomycin/spectinomycin). Although isolates are sensitive to kanamycin, the relatively high salinity of common *Alcanivorax* growth media interferes with the use of this antibiotic marker for selection purposes.

The fluorescent *in situ* hybridization probe 5'-CCA AGA ATA CTA AGA TTC CC—complementary to rRNA between nucleotides 821-842 (*E. coli* numbering)—labeled with Cy3 to detect “*Alcanivoraceae*” in environmental samples has given satisfactory results under stringent conditions (35% (w/vol) formamide; hybridization at 46°C for 2 h). Probes that distinguish *Fundibacter* and *Alcanivorax* species hybridize to 16S rRNA between nucleo-

tides 203 and 223 (*E. coli* numbering); these probes are 5'-GCG AGC TCA TCC ATC TGC A and 5'-GCG AGC TCA TCC ATC AGC A, respectively (mismatches are shown in boldface type).

ENRICHMENT AND ISOLATION PROCEDURES

The principal sources of these organisms are seawater and seawater sediments, enriched with crude oil, petroleum hydrocarbons, or single aliphatic hydrocarbons. *n*-Hexadecane can serve as the sole carbon source for enrichment of the seawater/sediment samples; addition of nitrogen and phosphorus is required. The flat, colorless, transparent colonies of *Alcanivorax*-related strains develop on the agar surface within 3–5 d after plating. After incubation for longer than 1 week, colonies turn opaque and cream-colored.

DIFFERENTIATION OF THE GENUS *ALCANIVORAX* FROM OTHER GENERA

Characteristics that differentiate the genus from other related or phenotypically similar genera are given in Table BXII.γ.94.

TAXONOMIC COMMENTS

The analysis of 16S rRNA gene sequencing data of both recognized and recently isolated strains belonging to the genera *Alcanivorax* and *Fundibacter* (Yakimov et al., 1998; Bruns and Berthe-Corti, 1999) suggest these bacteria belong to a distinct taxonomic group, presumably a new family, “*Alcanivoraceae*”, phylogenetically equidistant from the characterized lineages of the class *Gammaproteobacteria*, “*Alteromonadaceae*”, *Halomonas* spp., belonging to *Halomonadaceae* and members of family *Oceanospirillaceae* (Fig. BXII.γ.112).

A controversy exists regarding the taxonomic position of the

TABLE BXII.γ.93. List of strains relevant to the genus *Alcanivorax* and clones derived from 16S rRNA gene libraries from different geographic locations

Strain/clone designation	Origin	Notes
<i>Alcanivorax borkumensis</i> SK2, DSM 11573 ^T	Isle of Borkum, North Sea	Type and only species of <i>Alcanivorax</i>
<i>Alcanivorax jadensis</i> T9, DSM 12178 ^T	Jadebusen, North Sea	Formerly known as <i>Fundibacter jadensis</i> T9
<i>Alcanivorax borkumensis</i> AP1, SK, SK1, SK4A, SK7, MM1 ^a	Isle of Borkum, North Sea	Isolates obtained from the enrichments along with the type strain SK2
<i>Alcanivorax</i> sp. ME104 ^a	Mediterranean Sea, Harbor of Messina, Sicily	98% 16S rRNA gene similarity with the type strain
<i>Alcanivorax borkumensis</i> ME103, ME106, ME111 ^a	Mediterranean Sea, Harbor of Messina, Sicily	100% 16S rRNA gene similarity with the type strain
<i>Alcanivorax borkumensis</i> LE4	Singapore shore	100% 16S rRNA gene similarity with the type strain
<i>Alcanivorax</i> sp. SCB54	California (E. Pacific)	Partially sequenced environmental clone
<i>Alcanivorax</i> sp. ME105, ME112 ^a	Mediterranean Sea, Harbor of Messina, Sicily	95–98% 16S rRNA gene similarity with the type strain
MBIC4324–4330, 4335–4439, 4346–4351 ^b	Sea of Japan	
MBIC4322, 4323, 4332, 4334 ^b	Palau (S. Pacific)	
MBIC4197–4100 ^b	Canada (NE Pacific)	
MBIC4064, 4066, 4067 ^b	Adriatic Sea, Croatia	
MBIC4331, 4168 ^b	Okinawa (W. Pacific)	
MBIC4089, 4090 ^b	Miyagi (Central Pacific)	
MBIC4426, 4427 ^b	Shizuoka (W. Pacific)	
MBIC4103, 4104 ^b	Hawaii (Central Pacific)	
“ <i>Alcanivorax profundus</i> ” MRS3300 ^c	Deep area (3300m) of Mediterranean Sea located between Straits of Messina and South Ionian Sea: 36°30'N–15°50'E	98% 16S rRNA gene similarity with <i>Alcanivorax jadensis</i> strain T9
<i>Alcanivorax borkumensis</i> 9B1, 7B ^c	Coral Sea (S. Pacific), harbor of Noumea, New Caledonia	100% 16S rRNA gene similarity with the type strain

^aAvailable in the strain collection of the Department of Microbiology of the GBF (German Research Center for Biotechnology), Mascheroder Weg 1, 38124 Braunschweig, Germany (Email pgo@GBF.de).

^bAvailable in the strain collection of Marine Biotechnology Institute, 3-75-1, Kamaishi Laboratory, Heita, Kamaishi, Iwate 026-0001, Japan (Email shigeaki.harayama@kamaishi.mbio.co.jp).

^cAvailable in the strain collection of the Istituto Sperimentale Talassografico CNR, Spianato S. Raineri 86, 98122 Messina, Italy (Email iakimov@its.me.cnr.it).

TABLE BXII.γ.94. Comparison of key phenotypic characteristics of *Alcanivorax* spp. and other heterotrophic marine bacteria belonging to the *Gammaproteobacteria*^{a,b}

Characteristic	<i>Alcanivorax</i> spp.	<i>Halomonas</i> spp.	<i>Marinobacter</i> spp.	<i>Marinomonas</i> spp.	<i>Oceanospirillum</i> spp.
Typical Gram-negative cell wall	+	+	+	+	+
Morphology	Rods	Rods	Rods	Rods	Helical
Number and arrangement of flagella	None	Peritrichous	None or single polar	Single polar	Bipolar tufts
Optimal temperature for growth (°C)	20–30	30	32	20–25	25–32
Growth at 4°C	±	+	–	–	±
Growth at 45°C	–	NR	+	+	–
Optimal NaCl concentration (%) for growth	3–5	3–8	3–6	0.7–3.5	0.5–8
Maximal NaCl concentration (%) for growth	15	32.5	20	NR	9.75
Nitrate reduced to nitrite	+	+	+	–	–
Oxidase test (Kovacs)	+	+	+	–	+
Arginine dihydrolase activity	–	–	–	–	NR
Ornithine decarboxylase	–	+	–	–	NR
Gelatin liquefaction	–	±	–	–	–
Starch hydrolysis	–	–	–	–	–
Lysine decarboxylase	–	+	–	–	NR
Urease activity	–	±	–	–	–
<i>Utilization as sole carbon source:</i>					
DL-Alanine	–	+	–	+	±
L-Arginine	–	+	–	±	–
Aspartate	–	+	–	NR	–
Cellobiose	–	+	–	±	–
D-Fructose	–	+	–	+	–
D-Gluconate	–	+	–	+	NR
D-Glucose	–	+	–	+	–
L-Glutamate	–	+	+	+	±
Glycerol	–	+	–	+	–
Hexadecane	+	+	+	+	NR
p-Hydroxybenzoate	–	–	+	+	±
D-Mannitol	–	+	–	+	–
D-Mannose	–	+	–	+	–
L-Serine	–	+	–	±	–
Succinate	–	+	+	+	+
Sucrose	–	+	–	–	–
Accumulation of poly-β-hydroxybutyrate	±	+	–	–	+
Mol% G + C of the DNA	53–66	60–61	53	44–48	42–51

^a +, Positive reaction or growth; ±, variable reaction; negative reaction or growth; NR, not reported.

^b Modified from Yakimov et al. (1998).

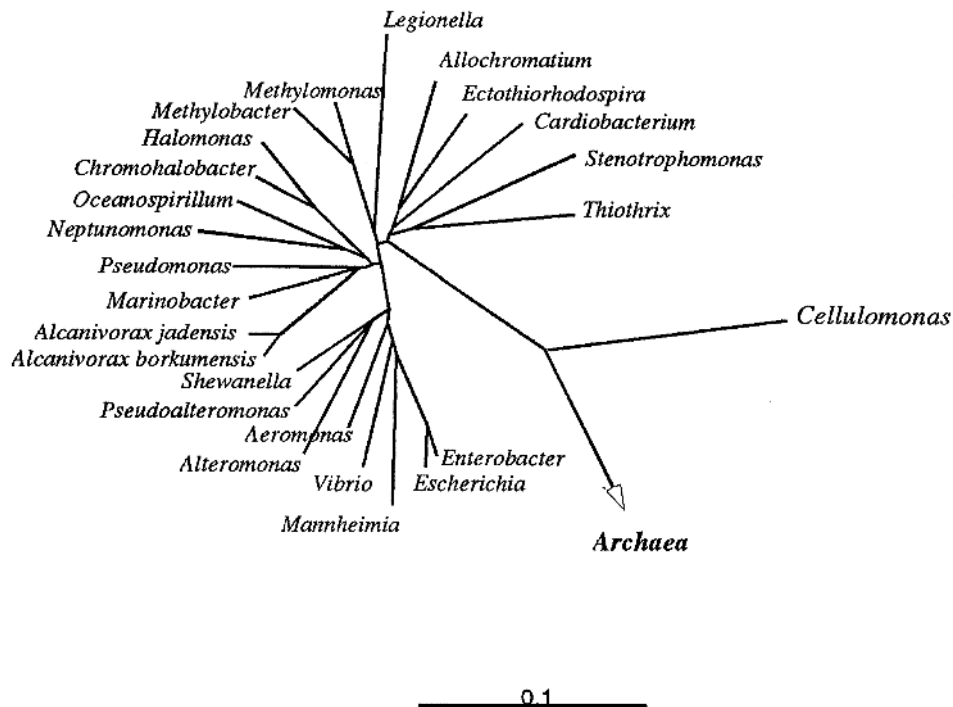


FIGURE BXII.γ.112. Phylogenetic position of species of genus *Alcanivorax* among principal genera within *Gammaproteobacteria*. Phylogenetic tree was rooted from an archaeon, *Pyrodicticum occultum*, and *Cellulomonas* was used as an outgroup. The scale bar represents 0.1 fixed point mutation per sequence position. *Fundibacter jadensis* was transferred to the genus *Alcanivorax* as *Alcanivorax jadensis* (González and Whitman, 2002; Fernández-Martínez et al., 2003).

DNA/DNA hybridization studies between *A. borkumensis* and *F. jadensis* clearly indicate, however, that these two organisms do not belong to the same species (Steiner and Tindall, personal communication).

The sequencing of *gyrB* genes from above type cultures and a large number of recently isolated—but not validly published—*Alcanivorax* strains suggest existence of at least five genotypically distinct groups, presumably at the level of species (Fig. BXII.γ.113).

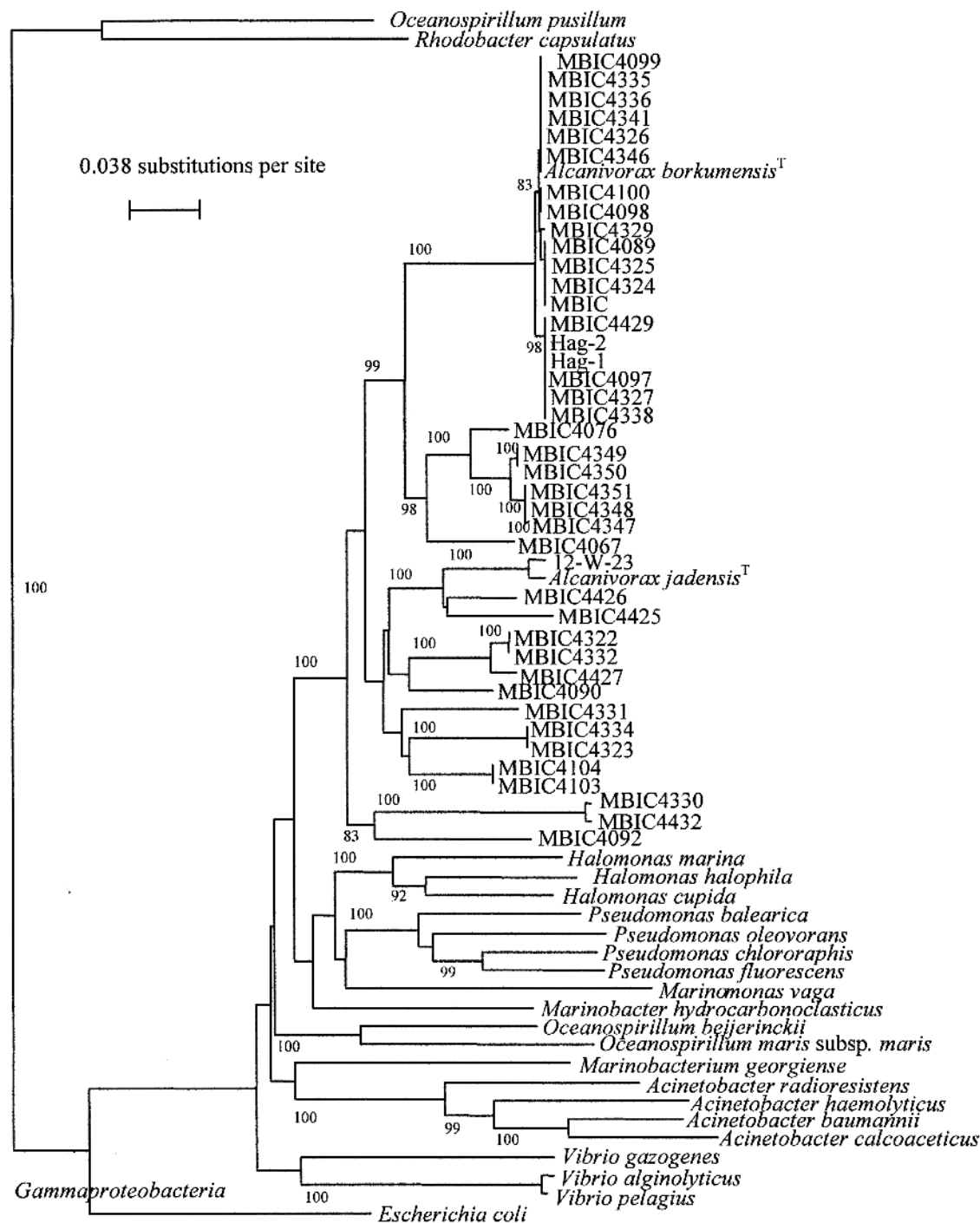


FIGURE BXII.γ.113. Phylogenetic tree derived from the similarities between the *gyrB* sequences. Bootstrap values over 80% are shown at the branch points. Strains are listed in Table BXII.γ.93. *Fundibacter jadensis* was transferred to the genus *Alcanivorax* as *Alcanivorax jadensis* (González and Whitman, 2002; Fernández-Martínez et al., 2003).

List of species of the genus *Alcanivorax*

1. *Alcanivorax borkumensis* Yakimov, Golyshin, Lang, Moore, Abraham, Lünsdorf and Timmis 1998, 347^{VP}
borku.me' n. sis. M.L. adj. *borkumensis* from the island of Borkum, a small island in Western-Elms harbor in the North Sea, located close to the German-Dutch border.

The characteristics are as described for the genus, with the following additional information. Optimum temperature 20–30°C; range, 10–35°C. Optimum NaCl concentration 3–10%; range, 1.0–12.5%. Reduce nitrate to nitrite, though no anaerobic growth occurs. Do not produce agarase, amylase, arginine dihydrolase, ornithine decarboxylase, lysine decarboxylase, gelatinase, or esculinase. Exhibit Tween 80 hydrolysis. Utilize aliphatic hydrocarbons (formate, acetate, propionate, methylpyruvate, α -ketoglutarate) and some short-chain fatty acids as sole or principal carbon source. Produce glucose lipids (Fig. BXII.γ.114) that exist in two forms: a glycine-containing biosurfactant precursor and the mature extracellular glucose lipid. Principal fatty acids are C_{16:0}, C_{16:1}, and C_{18:1}. Resistant to ampicillin, tetracycline, nalidixic acid; sensitive to streptomycin/spectinomycin, chloramphenicol, kanamycin.

The mol% G + C of the DNA is: 53.4 (HPLC).

Type strain: SK2, ATCC 700651, CIP 105606, DSM 11573.

GenBank accession number (16S rRNA): Y12579.

2. *Alcanivorax jadensis* Fernández-Martínez, Pujalte, García-Martínez, Mata, Garay, and Rodríguez-Valera 2003, 337^{VP}
(*Fundibacter jadensis* Bruns and Berthe-Corti 1999, 447.)
ja.den' sis. M. L. adj. *jadensis* referring to the region Jade, which forms part of the bay “Jadebusen”, which belongs to the German North Sea coast.

Rods 0.3–0.7 μ m wide and 0.8–1.8 μ m long. Nonmotile. Pili are present. Gram negative. Colonies are circular, 0.25–1.25 mm in diameter and ivory-pigmented. The temperature spectrum ranges from 10–40°C, with optimal growth occurring at 30°C. Shows weak halotolerance and grows at NaCl concentrations ranging from 0.5–15% (w/v); opti-

um, 3%. It is able to grow both aerobically and anaerobically, and reduces nitrate. It grows with the carbon sources pyruvate and acetate if a synthetic medium is used and also with tetradecane, hexadecane, and pristane if the medium is supplemented with vitamins and mineral salts. The cells are resistant to ampicillin, cefazolin, cephalotin, erythromycin, linomycin, nalidixic acid, novobiocin, ofloxacin, oxacillin, penicillin, and tetracycline. Oxidase and catalase positive. Growth occurs on aliphatic hydrocarbons as the sole or principal carbon source (tetradecane, hexadecane, and pristane). Pyruvate and acetate can be used. The organisms emulsify alkanes, but do not produce glucose lipid.

The mol% G + C of the DNA is: 63.6 (T_m).

Type strain: T9, ATCC 700854, DSM 12178.

GenBank accession number (16S rRNA): AJ001150.

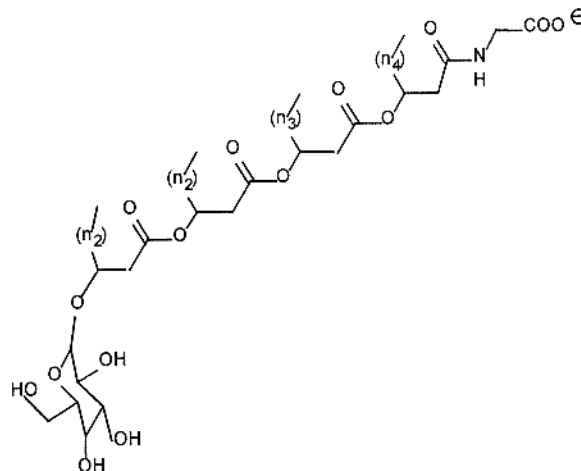


FIGURE BXII.γ.114. Cell-associated glucose lipid from *A. borkumensis* SK2^T.

Family III. *Hahellaceae* fam. nov.

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Ha. hel.la' ce. ae. M.L. fem. n. *Hahella* type genus of the family; *-aceae* ending to denote family;
M.L. fem. pl. n. *Hahellaceae* the *Hahella* family.

The family *Hahellaceae* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the family contains the genera *Hahella* (type genus) and *Zooshikella*. *Zooshikella* was proposed after the cut-off date for inclusion in this volume (June 30, 2001) and is not described here (see Yi et al. (2003)).

Aerobic or facultatively anaerobic chemoorganotrophs. Require NaCl for growth.

Type genus: **Hahella** Lee, Chun, Moon, Ko, Lee, Lee and Bae 2001a, 664^{VP}.

Genus I. **Hahella** Lee, Chun, Moon, Ko, Lee, Lee and Bae 2001a, 664^{VP}

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Ha. hel'la. M.L. fem. n. named after Yung Chil Hah, a Korean bacteriologist who pioneered microbiological research in Korea.

Gram-negative motile rod. Facultatively anaerobic; produces acid from sugars. Reduces nitrate to nitrite. Hydrolyzes esculin and gelatin. **Produces extracellular polysaccharides and red pigment. Requires NaCl.**

The mol% G + C of the DNA is: 55.

Type species: **Hahella chejuensis** Lee, Chun, Moon, Ko, Lee, Lee and Bae 2001a, 665.

FURTHER DESCRIPTIVE INFORMATION

Analysis of 16S rDNA sequences showed that *Hahella chejuensis* is a member of the *Gammaproteobacteria* but is not closely related to any known genera (Lee et al., 2001a).

ENRICHMENT AND ISOLATION PROCEDURES

The strain was isolated directly from diluted marine sediment spread on ZoBell's medium (Lee et al., 2001a).

MAINTENANCE PROCEDURES

The strain was stored at -80° in 20% (w/v) glycerol (Lee et al., 2001a).

DIFFERENTIATION OF THE GENUS *HAHELLA* FROM OTHER GENERA

Lee et al. (2001a) provide a table of characteristics that differentiate *Hahella chejuensis* from other marine/halophilic bacteria in the *Gammaproteobacteria*.

List of species of the genus *Hahella*1. **Hahella chejuensis** Lee, Chun, Moon, Ko, Lee, Lee and Bae 2001a, 665^{VP}

che.ju.en'sis. M.L. adj. *chejuensis* pertaining to Cheju Island, Republic of Korea, geographical origin of the type strain of the species.

Young cells $0.5\text{--}0.7 \times 1.6\text{--}9.0 \mu\text{m}$; older cells $0.7\text{--}0.8 \times 1.4\text{--}1.7 \mu\text{m}$. Halophilic; produces red pigment soluble in methanol. Growth at 1–8% NaCl (optimum 2% NaCl), 10–45°C, and pH 6–10 (optimum pH 7). Grows on and produces acid from adonitol, D(+)-fructose, D(+)-glucose, in-

ositol, D(+)-maltose, D(–)-mannitol, D(+)-mannose, D(–)-sorbitol, sucrose, and D(+)-trehalose. Does not grow on L(+)-arabinose, citrate, D(+)-galactose, D(+)-lactose, malate, malonate, D(+)-melibiose, D(+)-raffinose, L(+)-rhamnose, and D(+)-xylose. Major fatty acids octadecenoic, *cis*-9-hexadecenoic/*iso*-2-hydroxypentadecanoic, hexadecanoic, and 3-hydroxydodecanoic.

The mol% G + C of the DNA is: 55 (T_m).

Type strain: 96CJ10356, KCTC 2396, IMSNU 11157.

GenBank accession number (16S rRNA): AF195410.

Family IV. **Halomonadaceae** Franzmann, Wehmeyer and Stackebrandt 1989, 205^{VP} emend. Dobson and Franzmann 1996, 558

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Ha.lo.mo.na.da'ce.ae. M.L. fem. n. *Halomonas* type genus of the family; *-aceae* ending to denote family; M.L. fem. pl. n. *Halomonadaceae* the *Halomonas* family.

The family *Halomonadaceae* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rDNA sequences; the family contains the genera *Halomonas* (type genus), *Carnimonas*, *Chromohalobacter*, *Cobetia*, *Deleya*, and *Zymobacter*. *Cobetia* was proposed after the cut-off date for inclusion in this volume (June 30, 2001) and is not described here (see Arahal et al. (2002a)).

Halotolerant or halophilic, except for *Zymomonas*. Motile by means of flagella. Aerobic or facultatively anaerobic chemotrophs.

Type genus: **Halomonas** Vreeland, Litchfield, Martin, and Elliot 1980, 494^{VP} emend. Dobson and Franzmann 1996, 557.

Genus I. **Halomonas** Vreeland, Litchfield, Martin, and Elliot 1980, 494^{VP} emend. Dobson and Franzmann 1996, 557

RUSSELL H. VREELAND

Ha.lo.mo'nas. Gr. n. *hals*, halos salt of the sea; Gr. n. *monas* a unit, monad; M.L. fem. n. *Halomonas* salt(-tolerant) monad.

Straight or curved rod shaped cells, generally $0.6\text{--}0.8 \times 1.6\text{--}1.9 \mu\text{m}$. One species (*H. halodenitrificans*) presents coccoid cells. Species may form elongated, flexuous filaments under certain conditions. Endospores are not formed. **Gram negative. Rod shaped species are motile** by lateral, polar, or peritrichous flagella. Colonies are white or yellow, turning light brown with age. **Possess**

a mainly respiratory type of metabolism with oxygen as the terminal electron acceptor. Some species are also capable of anaerobic growth in the presence of nitrate. Some species have been reported to grow under anaerobic conditions in the absence of nitrate if supplied with glucose (but not other carbohydrates or amino acids). Nitrate is reduced to nitrite; nitrogen

gas is not formed. All species tested are catalase positive; 17 of the 21 known species are also oxidase positive. Chemoorganotrophic. Carbohydrates, amino acids, polyols, and hydrocarbons can serve as sole carbon sources in mineral media. Ammonium sulfate can serve as a sole nitrogen source. **Intracellular granules are not produced.** **Halotolerant** (also described as slight to moderate halophiles), **able to grow in NaCl concentrations ranging from 0.1–32.5% (w/v).** The major respiratory quinone is ubiquinone 9. The major fatty acids are C_{16:1}, C_{17:0 cyc}, C_{16:0}, C_{18:1}, and C_{19:0 cyc}. In addition to having the signature sequences found in the *Halomonadaceae*, *Halomonas* species contain the following four signature bases (by *E. coli* numbering): C at position 1424, U at position 1439, A at position 1462, and C at position 1464. **Isolated from saline environments around the world**, including solar salt facilities, intertidal estuaries, the open ocean, and hypersaline lakes (Dead Sea, Israel; Organic Lake, Antarctica).

The mol% G + C of the DNA is: 52–68.

Type species: Halomonas elongata Vreeland, Litchfield, Martin and Elliot 1980, 495.

FURTHER DESCRIPTIVE INFORMATION

During exponential growth, 20 of the 21 *Halomonas* species consist of mixtures of straight and curved rods. Upon entry into stationary phase, several *Halomonas* species also form elongated, flexuous filaments of various lengths. The percentage and length of these elongated cells depends upon the type of growth medium. In complex Casamino acids medium (CAS)¹ with 8% NaCl, all of the cells may be elongated, with many filaments containing irregular loops and bends. In a chemically defined mineral salts medium (MS)² containing 8% NaCl, as few as 25% of the cells may be elongated, and the filaments seldom produce irregular looping or extensive bending.

Electron microscopy of thin sections from optimum growth conditions reveals a typical Gram-negative type of cell wall (Vreeland and Martin, 1980).

The number and arrangement of flagella on short cells depends upon the species examined. Cells generally possess 4–7 flagella. In some strains, the flagella are arranged laterally, while others possess only polar flagella. One biovar of this genus has both lateral and polar flagellation. Motile *Halomonas* cells describe a helix when viewed by light microscopy. Motility is generally lost rapidly under low oxygen concentrations. Flagella have not been observed on elongated cells or on cells grown in low NaCl concentrations (<0.2%).

On solid CAS or MS medium containing 8% NaCl, most *Halomonas* colonies are white to cream-colored, smooth, glistening, opaque, and ~2 mm in diameter after 24 h at 30°C. The colonies

generally become yellow to light brown and spread following prolonged incubation.

Halomonas species require Na⁺ for growth—at least 0.1% w/v NaCl added to CAS medium or 0.3% w/v NaCl added to MS medium—and will grow in a wide range of NaCl concentrations (0.1–32% NaCl w/v). When subjected to osmotic shock from either rapid dilution or increased NaCl concentration, *Halomonas* cells respond as osmometers and exhibit swelling or plasmolysis. Osmotically shocked *Halomonas* cells do not lose viability and soon reestablish osmotic balance and resume growth. This reestablishment of osmotic balance is mediated by an increased concentration of ectoine, which is the primary compatible solute found in *H. elongata*. The optimum NaCl concentration is 2.2–8.0% at 30°C. The Na⁺ requirement of the type species can be satisfied by NaCl, NaNO₃, or NaBr (Vreeland and Martin, 1980). The NaCl requirement of the other species has not been tested in detail. NaCl tolerance is affected by growth temperature, but apparently not by the carbon source being used (Vreeland and Martin, 1980). In terms of the ability to promote salt tolerance, temperatures can be arranged from greatest degree of promotion to least as follows: 30°C, 23°C, 37°C, 15°C, 45°C, 4°C (Vreeland et al., 1980).

In CAS medium containing 8% NaCl, the *Halomonas* species that have been tested grow at pH values from 5.0–9.0. Alkaliphilic strains generally grow from pH 7.0–11.0 with an optimum at 9.5. The pH tolerance has not been tested on MS medium and has not been tested at different temperatures.

The physiological characteristics of all properly recognized *Halomonas* species are listed in Tables BXII.γ.95 and BXII.γ.96. Although considered mainly aerobic, some *Halomonas* species have been reported to grow on glucose under anaerobic conditions in the absence of nitrate or other terminal electron acceptors. If the glucose is replaced by other carbohydrates or an amino acid, nitrate is required for anaerobic growth.

Members of the genus are nutritionally versatile. The following compounds often serve as sole carbon sources for growth: glucose, gluconate, glycerol, fructose, mannose, sucrose, cellobiose, succinate, mannitol, alanine, glutamine, glutamate, asparagine, aspartate, lysine, histidine, phenylalanine, tyrosine, tryptophan, proline, arginine, leucine, isoleucine, valine, methionine, cysteine, serine, and threonine.

Halomonas species have been isolated from a solar salt facility on the island of Bonaire, Netherlands Antilles (Vreeland et al., 1980) from the Dead Sea and Canada (Huval et al., 1995), the Antarctic (Franzmann et al., 1987; James et al., 1990a), Great Salt Lake, Utah (Fendrich, 1988), estuaries (Hebert and Vreeland, 1987), and the Pacific Ocean (Baumann et al., 1983a). Representatives of the genus may be even more widespread in nature.

ENRICHMENT AND ISOLATION PROCEDURES

Halomonas species may be isolated using a wide variety of media. Brine samples can be spread onto the surface of the medium and incubated at close to the environmental temperature and under high humidity for 2–7 d. *Halomonas* colonies are white to cream colored and easily distinguished from red-pigmented halophiles. Alternatively, the brine samples can be added directly to filter pads saturated with medium. This method has the advantage of allowing the use of small (9 mm) Petri plates, facilitating shipment to distant field areas. A disadvantage of the technique is that the white *Halomonas* colonies can be difficult to see

1. CAS medium is a modification of the medium of Abram and Gibbons, as described by Gibbons (1969), and contains (g/l of distilled water): yeast extract, 1.0; casamino acids (Difco) (not "vitamin-free"), 7.5; Proteose peptone No. 3 (Difco), 5.0; sodium citrate, 3.0; MgSO₄·7H₂O, 20.0; K₂HPO₄, 7.5; and NaCl (or solar salt), 80.0. The pH is adjusted to 8.0 ± 0.1 with NaOH prior to sterilization and is 7.5 ± 0.1 after autoclaving at 121°C for 20 minutes (Vreeland et al., 1980, 1984). CAS medium is stable for several weeks when stored in the dark. It should be discarded if any crystal formation is seen.

2. MS medium is similar to that described by Vreeland and Martin (1980). It contains (g/l): MgCl₂·6H₂O, 5.3; KCl 0.75; and (NH₄)₂SO₄, 4.1. It is supplemented with a carbon source (10–50 mM) and NaCl (2.92–198.7 g/l). Phosphate (K₂HPO₄·3H₂O) is added as a sterile 10× concentrate after the medium has been autoclaved to give a final concentration of 0.5 g/l. The medium may also be supplemented with CaCl₂ (0.11 g/l) to enhance growth. The pH of the medium is adjusted to 7.2 with 1 N KOH prior to autoclaving and is 7.0 after sterilization.

on white filter pads; also, the pad fiber makes colonial isolation tedious.

A more direct isolation technique is to take advantage of the halotolerance of all of the species by setting up a salt-mediated selection medium. High salt brines can be mixed with low salt or distilled water, then spread onto a low salt medium. Alternatively, low salt samples can be added to high salt media to select for salt-tolerant organisms.

MAINTENANCE PROCEDURES

Halomonas strains can be maintained on agar slants containing 8% (w/v) NaCl. Stock cultures are transferred every 6 months, allowed to grow at 30°C for 1–7 d, and then stored in the dark at 4°C (Vreeland et al., 1980). Some media that have proven useful for growing *Halomonas* are HSC medium³, MH medium⁴, and AOL medium⁵.

Most species of *Halomonas* have been lyophilized, but survival rates may vary with the strain and species. *Halomonas elongata* strains ATCC 33173 and 33174 have survived for up to 2 years. The growth from a fresh CAS slant or broth is suspended in 0.5 ml of a solution containing Proteose peptone No. 3 (Difco) (0.5%), yeast extract (1.0%), and NaCl (0.2%). A small amount of the suspension (0.1–0.2 ml) is transferred to a lyophilization vial, dipped into liquid N₂ for 5–10 min, and lyophilized. Preserved cultures are reconstituted by suspending the cells in a small amount of CAS lacking salt. After 15–30 min, the cells are transferred to CAS broth containing NaCl and incubated at 30°C.

Halomonas strains may also be stored frozen on ceramic Protec® or Microbank® beads at –80°C. When using ceramic beads, it is best to grow the cultures under optimum growth conditions and then add the culture to fresh medium and place it onto the pre-sterilized beads. The cultures may then be re-incubated at 37°C for several hours to overnight to allow time for the cells to attach to the beads. After this incubation, the

3. HSC medium is prepared as two separate solutions (A and B). Solution A contains NaCl or solar salt (25%, w/v); the pH is adjusted to 11 with 1 N NaOH, and "vitamin-free" casein (Difco) is added to a concentration of 15 g/l. This mixture is incubated at 30°C overnight to allow protein acidic groups to be exposed. The pH is then adjusted to 7.9 ± 0.1 with sterile NaOH or HCl. Solution B also contains 25% (w/v) solar salt or NaCl and is supplemented with yeast extract (0.2%), sodium citrate (0.6%), MgSO₄·7H₂O (4.0%), and ferric ammonium sulfate (0.01%). This solution is adjusted to pH 7.9 with 1 N NaOH and sterilized by autoclaving. Just before use, solutions A and B are mixed 1:1 (v/v). If a solid medium is to be used, agar (4.0%) is added to solution B prior to autoclaving.

4. MH medium contains 1% yeast extract, 0.5% Proteose Peptone no. 3 (Difco), and 0.1% glucose. It may be supplemented with synthetic sea salts. The concentrations of sea salts that have been used include: 0.5, 5.0, 10, 20, and 25% (w/v). The pH of this medium is usually adjusted to 7.2 prior to sterilization. Note that the salt concentrations listed for this medium do not represent the concentration of individual salts, such as NaCl, KCl, and MgCl₂; consequently, media made with sea salts obtained from different suppliers may differ in terms of the actual ion contents. Given the tolerance of the various *Halomonas* species, however, this should not be a problem.

5. AOL medium (Franzmann et al., 1987) is a synthetic medium that mimics the Antarctic hypersaline lake from which several species of *Halomonas* have been isolated. It consists of NaCl, 80.0 g; MgSO₄·7H₂O, 9.5 g; KCl, 5.0 g; CaCl₂·2H₂O, 0.2 g; (NH₄)₂SO₄, 0.1 g; KNO₃, 0.1 g; and yeast extract, 1.0 g; suspended in 960 ml of distilled water. The pH is adjusted to 7.0 and the medium is sterilized by autoclaving. When the solution cools to 60°C, 20 ml of Huntner mineral base solution (Atlas, 1997) is added, followed by 1.0 ml of a filter-sterilized vitamin solution containing (mg/100 ml): cyanocobalamin, 10.0; biotin, 2.0; thiamine hydrochloride, 10.0; calcium pantothenate, 5.0; folic acid, 2.0; nicotinamide, 5.0; and pyridoxine hydrochloride, 10.0. The medium is then completed by adding 20.0 ml of a sterile phosphate solution containing 0.25 g K₂HPO₄ and 0.25 g KH₂PO₄ per 100 ml of distilled water. The medium may be solidified with 15 g/l agar. An AOL-peptone medium may also be made by adding 5.0 g/l peptone to AOL medium.

fluid may be drawn off and the culture frozen. *Halomonas* cultures have been preserved for up to 10 years using this technique; however, the quality of the culture has been found to deteriorate slowly under these conditions (R. Vreeland, unpublished observations).

Halomonas may also be stored frozen on ceramic beads under liquid N₂. Due to the variation in survival rates for other storage methods, liquid N₂ storage is the method of choice for long-term preservation.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

A common characteristic of all *Halomonas* species is their ability to grow at a wide variety of salt concentrations. Huval and Vreeland (unpublished studies) have determined that the phenotypic characteristics of these bacteria are often affected by the salt concentration of the growth and testing medium. These effects are particularly noticeable at the low and high salt extremes. This aspect has also been encountered by Hebert and Vreeland (1987) during their characterization of *Halomonas halodurans*. Consequently, prior to conducting any taxonomic analysis on isolates suspected of belonging to *Halomonas*, the optimum growth conditions with respect to NaCl concentration, temperature, and pH must be determined. These characteristics are best tested by use of complex media, each containing 0, 3.5, 8.0, 15.0, 20.0, or 32.0% (w/v) NaCl or solar salt. Temperature tolerance and qualitative effects of temperature on salt tolerance can be tested by incubating tubes of each salt concentration at temperatures from 4–45°C. After 2–3 weeks, all tubes without visible growth should be incubated for an additional 2–3 weeks at the optimum temperature to determine whether any apparent inhibitory conditions have actually been lethal (Vreeland, 1993).

DNA may be extracted from *Halomonas* by the method of Marmur (1961) with some modification. *Halomonas* suspensions and the sodium dodecyl sulfate (SDS) used for lysis must be heated to 50°C. Following SDS addition (2% final concentration), the lysate is held at 50°C for 15 min. Pronase is not added to the suspension. Water-saturated phenol (also at 50°C) is then added to fully denature the protein. The preparation may then be cooled to room temperature and the rest of the Marmur procedure followed. This modification is necessary since some *Halomonas* strains possess a nuclease, which, unless inactivated by heating, causes rapid breakdown of DNA following cell lysis.

All other characterization tests may be performed using conventional techniques (Holding and Collee, 1971; Vreeland, 1993). All media are supplemented with NaCl at the optimum concentration, as determined above. Uninoculated controls are necessary to ensure that any reactions detected are not artifacts caused by the NaCl.

DIFFERENTIATION OF THE GENUS *HALOMONAS* FROM OTHER GENERA

Halomonas is easily differentiated from such genera as *Zymomonas*, *Cellvibrio*, *Oceanospirillum*, or *Serpens* on the basis of salt tolerance, fermentative ability, flexibility, high mol% G + C value, and the ability to catabolize several carbohydrates. *Halomonas* can be quickly differentiated from other genera by its ability to survive exposure to, and to grow in, very high (20%) NaCl concentrations.

TAXONOMIC COMMENTS

The genus *Halomonas* currently contains 23 recognized, validly named species, which have been included because of the simi-

TABLE BXII.γ.95. Carbon sources utilized by *Halomonas* species^a

Characteristic	1. <i>H. elongata</i>	2. <i>H. aquamarina</i>	3. <i>H. campisalis</i>	5. <i>H. cupida</i>	6. <i>H. desiderata</i>	7. <i>H. eurihalina</i>	8. <i>H. halmophila</i>	9. <i>H. halodenitrificans</i>	10. <i>H. halodurans</i>	11. <i>H. halophila</i>	13. <i>H. magadii</i>	14. <i>H. marina</i>	15. <i>H. marisflavae</i>	16. <i>H. maura</i>	17. <i>H. meridiana</i>	18. <i>H. pacifica</i>	19. <i>H. pantelleriensis</i>	20. <i>H. salina</i>	21. <i>H. subglaciescola</i>	22. <i>H. variabilis</i>	23. <i>H. venusta</i>
L-Arabinose	+	—	—	+	+	+	—	—	—	+	+	—	+	+	d	—	—	d	—	—	—
Cellobiose	+	—	—	d	D	D	—	—	—	+	+	—	+	+	d	—	—	d	—	—	—
D-Glucose	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	d	—	—	+
Lactose	+	—	—	d	—	D	—	—	+	d	+	—	+	+	d	—	—	d	—	—	—
Maltose	+	d	+	—	+	+	+	—	—	+	—	—	+	+	+	+	+	d	—	—	d
Mannose	+	—	—	+	D	+	+	—	+	+	+	—	+	+	+	—	+	d	—	—	—
Sucrose	+	d	+	d	+	+	+	—	—	+	+	—	+	+	+	—	+	d	—	—	d
Acetate	+	+	+	+	+	+	—	—	+	+	+	+	+	+	+	+	+	+	—	+	+
Citrate	+	d	—	+	+	—	—	—	—	+	+	+	+	+	—	+	+	d	—	—	+
Gluconate	+	+	—	+	+	+	—	—	—	+	+	+	+	+	+	+	+	+	—	—	+
Lactate	+	+	+	+	+	—	—	—	—	+	+	+	+	+	+	+	+	d	+	—	+
Propionate	+	+	—	+	—	+	+	—	—	+	—	—	—	+	+	+	+	d	—	—	+
Succinate	+	+	—	+	+	+	—	—	—	+	+	+	—	—	+	+	+	d	+	—	+
L-Alanine	+	+	—	+	+	—	—	—	—	+	+	+	—	+	+	+	+	d	+	—	+
L-Arginine	+	+	—	+	—	+	+	—	—	+	+	d	—	+	+	+	+	d	+	—	+
L-Histidine	+	—	—	+	—	—	+	—	—	d	+	—	—	+	—	+	—	—	—	—	+
L-Isoleucine	+	+	—	d	—	+	+	+	—	d	+	d	—	—	+	—	—	d	—	—	d
L-Lysine	+	+	—	+	—	D	+	+	—	d	+	d	—	—	d	+	—	d	+	—	+
L-Serine	+	—	—	+	—	D	+	+	—	+	+	+	—	+	d	+	—	d	—	—	—
L-Tryptophan	—	—	—	—	—	+	—	—	+	d	+	—	—	—	+	—	—	d	—	—	d
Glycine	+	—	—	d	—	D	—	—	+	—	—	—	—	—	D	+	—	+	—	—	—
L-Leucine	+	d	—	+	—	+	+	—	+	d	+	+	—	+	+	+	—	d	—	—	+

^aFor symbols see standard definitions.

larity of their 16S rDNA sequences and the presence of respiratory ubiquinone-9. Although the genus is currently claimed to constitute a monophyletic group of organisms, it contains at least three main evolutionary branches. Subgroup 1 is composed of *H. halmophila*, *H. elongata* (the type species), *H. eurihalina*, *H. salina*, *H. halophila*, *H. pacifica*, *H. cupida*, and *H. halodenitrificans*. The 16S rDNA sequence similarities between pairs of these species range from 93.1 to 99.2%. The second subgroup contains both biovars of *H. subglaciescola* (ACAM 21 and ACAM 12), *H. halodurans*, *H. venusta*, *H. aquamarina*, *H. meridiana*, and *H. variabilis*, and the 16S rDNA sequence similarities among these species range from 94.4 to 100%. The third subgroup is composed of a single species, *H. marina*. When 16S rDNA sequences are used to distinguish the various phylogenetic subgroups within the genus, the situation improves only marginally. At the most restrictive similarity levels (98% and above), the genus seems to contain at least 10 individual phylogenetic groups (Fig. BXII.γ.115). Unfortunately, resolution of these subgroups deteriorates rapidly below this level. In fact, sequence homology levels existing between some members of the current genus are lower than those between members of the *Halomonas* and species currently listed in a different genus (Dobson and Franzmann, 1996).

Confusion about the current taxonomy of this group has arisen because the data that have been used to describe some of these individual species have been inconsistent. For instance, in the initial descriptions of *Halomonas subglaciescola* (Franzmann et al., 1987) and *Halomonas meridiana* (James et al., 1990a), data were used to compare these species to the type species *H. elongata*. Unfortunately, these two reports had only 55 kinds of characteristics in common, and within these common tests, 15 (27%) gave different results for *H. elongata*. Moreover, 17 of the 55 (31%) test results reported by James et al. (1990a) differed from

those reported by Franzmann et al. (1987) for the type strain of *H. subglaciescola*. There is presently no information on the degree to which such discrepancies would affect the taxonomic placement of the species involved.

A similar situation exists with the organisms that were originally part of genera such as *Deleya* and *Halovibrio*, and with the species "*Paracoccus halodenitrificans*", all of which were merged with the *Halomonas* based entirely upon 16S rDNA sequence similarities and a single chemotaxonomic feature. Very few of the phenetic characters originally used to describe these taxa are equivalent to those used to describe the rest of the *Halomonas* species. In fact, over 80 of the nearly 100 phenotypic descriptors for *Deleya* involve utilization of sole carbon sources, with virtually no information about the presence or absence of specific enzyme systems or abilities. Thus, it is difficult to provide a coherent and useable phenotypic description of the genus *Halomonas* and to provide reliable identifying features for the individual species.

At the chemotaxonomic level, the genus *Halomonas* represents a reasonably coherent group of bacteria; however, they do not possess specific or unique cell components that can be considered to be taxonomically distinct markers. All of the species possess ubiquinone-9 as their major respiratory quinone. When cells are grown on AOL with peptone, the major fatty acids of the *Halomonas* species that have been examined are C_{16:1} ω_{7c}, C_{16:0}, C_{17:0} cyclo, C_{18:1}, and C_{19:0} cyclo 11-12. The major polar lipids are phosphatidylethanolamine, phosphatidylglycerol, and an unidentified periodate-Schiff positive phospholipid (Franzmann and Tindall, 1990). Vreeland et al. (1984) have shown that the type strain *H. elongata* also contains cardiolipin. This particular lipid may be detected in increased concentration in cells grown in higher salt concentration.

Common molecular features among the species include the

TABLE BXII-γ.96. Additional characteristics of *Halomonas* species^a

Characteristic	1. <i>H. elongata</i>	2. <i>H. aquamarina</i>	3. <i>H. campisalis</i>	5. <i>H. cupida</i>	6. <i>H. desiderata</i>	7. <i>H. eurihalina</i>	8. <i>H. halnophila</i>	9. <i>H. halodentitificans</i>	10. <i>H. halodurans</i>	11. <i>H. halophila</i>	13. <i>H. magadii</i>	14. <i>H. marina</i>	15. <i>H. marisflavae</i>	16. <i>H. maura</i>	17. <i>H. meridiana</i>	18. <i>H. pacifica</i>	19. <i>H. pantelleriensis</i>	20. <i>H. salina</i>	21. <i>H. subglaciicola</i>	22. <i>H. variabilis</i>	23. <i>H. venusta</i>
Rods/curved rods	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Cocci																					
Gram reaction	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Motile																					
Pigmentation ^b	C	C		C		C	Y	W	W	C	C	C	YO	C	W	C	CP	C	W	B	C
Growth conditions:																					
0% NaCl	+	+	+	+	+	D	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
0.5% NaCl	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
3.5% NaCl	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
8.0% NaCl	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
15% NaCl	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
25% NaCl	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
<10°C	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
15°C	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
25°C	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
30°C	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
37°C	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
45°C	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Oxidase present	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Nitrate reduced to nitrite	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Urease produced	d	d	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Starch hydrolysis	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
mol% G +	60–61	57–58	66	60–63	66	59–65	62.9	64–66	63.2	66.7	62.2	62–64	59	64.1	58.8–59.1	67–68	65	60.7–64.2	60.9–62.9	57	52–54
C content	–	+	–	–	–	d	–	–	–	d	+	–	–	–	+	–	–	–	–	–	–

^aFor symbols see standard definitions.^bB, light brown; C, cream-beige; CP, cream to pink; W, white; Y, yellow; YO, yellow-orange.

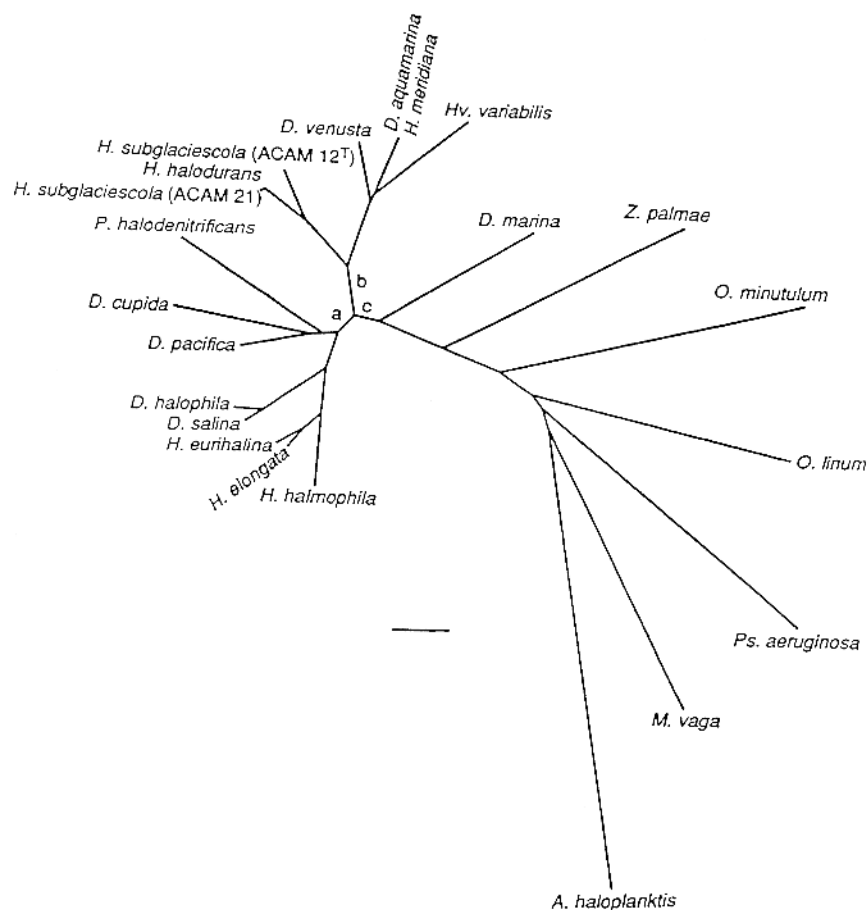


FIGURE BXII.γ.115. Distance tree containing the members of the genus *Halomonas* and other members of the class *Gammaproteobacteria*. Bar = 0.01 distance calculated by the Jukes-Cantor equation. Lower case letters a, b, and c refer to short internodes connecting the main branches of the monophyletic *Halomonas*. Abbreviations: *H.*, *Halomonas*; *O.*, *Oceanospirillum*; *Z.*, *Zymobacter*; *M.*, *Marinomonas*; *Ps.*, *Pseudomonas*; *A.*, *Alteromonas*. Redrawn and reprinted with permission from S. Dobson and P.D. Franzmann, *International Journal of Systematic Bacteriology* 46: 550–558, 1996 ©International Union of Microbiological Societies.

presence of specific bases at 16 specific sites along the 16S rRNA molecule (Table BXII.γ.97). A taxonomic study of six species of *Halomonas* has recently been conducted by Huval et al. (1995). The DNA-DNA reassociation data presented in that article shows *Chromohalobacter israelensis* and *Chromohalobacter canadensis* possess 49% and 55% similarity, respectively, with the type species *Halomonas elongata*. *Halomonas halodurans* demonstrates DNA similarity of 28% with *H. elongata*. In contrast, *H. halophila* shows no detectable similarity with the type species of this genus (Table BXII.γ.98). Dobson et al. (1993) have shown that *H. aquamarina* shares a 41% DNA similarity with *H. venusta*, 16% similarity with *H. cupida*, and only 12% similarity with *H. pacifica* and *H. marina*. DNA-DNA hybridizations have not yet been conducted using any other species in the genus.

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TABLE BXII.γ.97. 16S rRNA signatures of the genus *Halomonas*^a

Position number	Sequence of nucleotide
76–93	6 bp stem
484	A
486	C
640	G
660 ^b	A-
745	U
668–	A-
738–	U-
669–	A-
737	U
776	U
1124	U
1297	U
1298	C
1423	A
1424	C
1439	U
1462	A
1464	C

^aData from Dobson and Franzmann (1996).

^bDashes indicate that the specified nucleotide base pairs with the next one listed.

TABLE BXII.γ.98. DNA–DNA similarities between the type strains of *Halomonas elongata*, *Chromohalobacter canadensis* and strains of related species^a

Source target DNA	% Hybridization with radiolabeled DNA	
	<i>H. elongata</i> ATCC 33173 ^T	<i>C. canadensis</i> NRCC 41227 ^T
<i>H. elongata</i> ATCC 33173 ^T	100	57
<i>H. elongata</i> ATCC 33174	72	49
<i>H. halodurans</i> ATCC 29868 ^T	28	34
<i>H. halophila</i> CCM 662 ^T	0	7
<i>C. canadensis</i> NRCC 41227 ^T	55	100
<i>C. israelensis</i> ATCC 43985 ^T	49	20

^aData from Huval et al. (1995).

FURTHER READING

Dobson, S.J. and P.D. Franzmann. 1996. Unification of the genera *Deleya* (Baumann et al. 1983), *Halomonas* (Vreeland et al. 1980), and *Halovibrio* (Fendrich 1988) and the species *Paracoccus halodenitrificans* (Robinson and Gibbons 1952) into a single genus, *Halomonas*, and placement of the genus *Zymobacter* in the family *Halomonadaceae*. *Int. J. Syst. Bacteriol.* 46: 550–558.

Vreeland, R.H., C.D. Litchfield, E.L. Martin and E. Elliot. 1980. *Halomonas elongata*, a new genus and species of extremely salt-tolerant bacteria. *Int. J. Syst. Bacteriol.* 30: 485–495.

Vreeland, R.H. 1991. The Family *Halomonadaceae*. In Balows, Trüper, Dworkin, Harder and Schleifer (Editors), *The Prokaryotes*, 2nd edition, Springer-Verlag, New York. pp. 3181–3188.

DIFFERENTIATION OF THE SPECIES OF THE GENUS *HALOMONAS*

Characteristics differentiating the species of the genus *Halomonas* are listed in Table BXII.γ.96.

List of species of the genus Halomonas

1. ***Halomonas elongata*** Vreeland, Litchfield, Martin and Elliot 1980, 495^{VP}
e.lon'ga.ta. L. fem. part. adj. *elongata* elongated, stretched out.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. Morphological features are shown in Figs. BXII.γ.116 and BXII.γ.117. Additional characteristics of the species include the ability to grow at 0.5% and 30% NaCl. Grows at pH 5.0–9.0 in complex medium with 8% NaCl at 30–37°C. Grows at 15°C. Catalase, phosphatase, lysine decarboxylase, β-galactoside, and indole are produced. Some strains may be positive for esculin hydrolysis. Hydrogen sulfide negative. Does not pos-

sess DNase or phenylalanine deaminase. Hydrolyzes Tween 20, but not Tween 80. This species has been reported to produce acid from fructose, D-glucose, lactose, D-maltose, and sucrose, but not from D-galactose, inulin, or D-mannitol. In addition to those listed in Table BXII.γ.95, the following compounds serve as sole carbon sources: esculin, D-galactose, L-rhamnose, D-salicin, D-xylose, fumarate, pyruvate, β-hydroxybutyrate, L-aspartic acid, L-glutamic acid, L-ornithine, L-valine, proline, L-asparagine, and tyrosine. The species does not utilize inulin, raffinose, trehalose, malonate, L-tryptophan, L-threonine, or methionine as sole carbon sources. Sensitive to gentamicin (10 μg), mercuric chloride (1:5000), neomycin (30 μg), streptomycin (10 μg), and



FIGURE BXII.γ.116. Phase contrast micrograph of *Halomonas elongata* strain 1H9 (ATCC 33173), showing both long and short cell forms (× 2000).

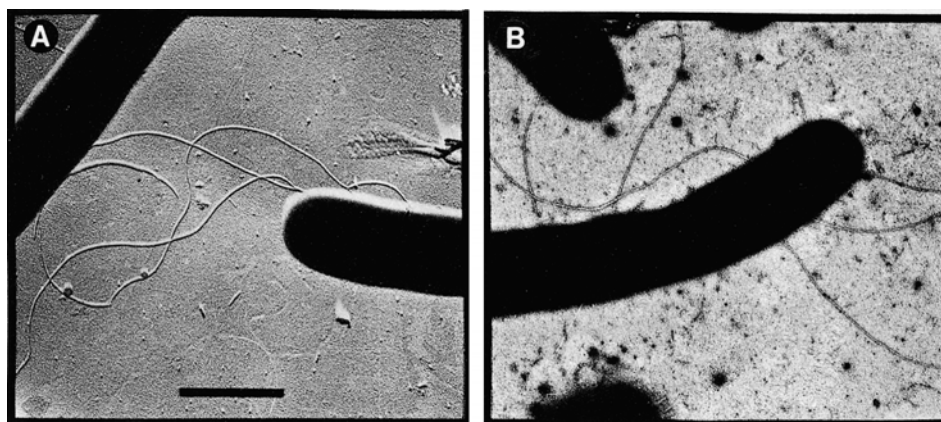


FIGURE BXII.γ.117. Electron micrographs of *Halomonas elongata*, showing flagellar arrangement. A, strain 1H9 (ATCC 33173); carbon shadowed. B, strain 1H15 (ATCC 33174); negatively stained (Bar = 1.0 μ m).

chloramphenicol (30 μ g). Resistant to ampicillin (10 μ g), bacitracin (10 i.u.), cephalothin (30 μ g), erythromycin (15 μ g), nalidixic acid (30 μ g), novobiocin (20 μ g), vibriostat O/129 (10 μ g), penicillin G (10 i.u.), and tetracycline (30 μ g).

Sodium salts, such as sodium glutamate, NaNO_3^- , and NaBr, can substitute for NaCl. Na_2SO_3 and NaI cannot. The chloride salts of magnesium, lithium, potassium, and ammonia cannot substitute for NaCl in growth media. Possesses peptidoglycan. Based upon amino acid analyses, the peptidoglycan contains leucine in a crossbridge (Vreeland et al., 1984). Present in solar salterns, estuaries, salt marshes, and other environments containing NaCl. This species originally contained two biovars (ATCC 33174 = 1H15 [Vreeland et al., 1980] and DSM 3043 = 1H11 [Vreeland et al., 1980]). Both biovars have now been reclassified as *Chromohalobacter salexigens* (Arahal et al., 2001a).

The mol% G + C of the DNA is: 60.5 ± 0.5 (Bd, T_m).

Type strain: 1H9 of Vreeland et al., 1980, ATCC 33173, DSM 2581.

GenBank accession number (16S rRNA): M93355, X67023.

2. ***Halomonas aquamarina*** (ZoBell and Upham 1944) Dobson and Franzmann 1996, 556^{VP} (*Alcaligenes aquamarinus* (ZoBell and Upham 1944) Hendrie, Holding and Shewan 1974, 537; *Deleya aquamarina* (ZoBell and Upham 1944) Akagawa and Yamasato 1989, 466; "*Achromobacter aquamarinus*" ZoBell and Upham 1944, 264.)
a.qua.ma.ri'na. L. n. *aqua* water; L. adj. *marina* of the sea; L. adj. *aquamarina* pertaining to seawater.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. This species may or may not produce H_2S and phenylalanine deaminase. The species does not hydrolyze esculin or produce DNase or β -galactosidase. This species utilizes malate, pyruvate, β -hydroxybutyrate, caprylate, L-aspartic acid, L-glutamic acid, and proline as sole sources of carbon. The species also utilizes butyrate, isobutyrate, valerate, isovalerate, caproate, heptanoate, adipate, pimelate, azelate, sebacate, levulinate, erythritol, *meso*-inositol, *n*-propanol, *n*-butanol, isobutanol, cytosine, and uracil. This is the only *Halomonas* species known to hydrolyze starch (Table BXII.γ.96) and utilize suberate. Accumulates polyhydroxyalkanoates. Some strains

of the species may utilize D-galactose, trehalose, L-ornithine, and L-valine. The species does not grow on inulin, L-rhamnose, D-salicin, D-xylose, formate, malonate, oxalacetate, D-tartrate, benzoate, L-threonine, or tyrosine as the sole carbon source. There are no data available about the sensitivity of this species to antimicrobials. Isolated from the Pacific Ocean. The similarity of the 16S rRNA sequence of this species to the other *Halomonas* species ranges from 93.6–100%.

The mol% G + C of the DNA is: 60.5 ± 0.5 (Bd, T_m).

Type strain: ATCC 14400, DSM 30161, IAM 12550, NCMB 557.

GenBank accession number (16S rRNA): M93352.

3. ***Halomonas campisalis*** Mormile, Romine, Garcia, Ventosa, Bailey and Peyton 2000, 949^{VP} (Effective publication: Mormile, Romine, Garcia, Ventosa, Bailey and Peyton 1999, 556.)
cam.pi.sa'lis. L. masc. n. *campus* plain, field; L. masc. n. *sal* salt; gen. *salis* of salt; L. masc. n. *campisalis* of the plain of salt, of the salt plain.

General characteristics are as described for the genus and are presented in Tables BXII.γ.95 and BXII.γ.96. Additional characteristics include the ability to grow at temperatures from 4°C to 50°C. Optimal growth occurs at 30°C. This species reproduces at pH between 6.0 and 11.0 with optimal growth at pH 9.5. The optimal salt concentration is 1.5 M with a growth range from 0.2 to 4.5 M. In addition to those listed in Table BXII.γ.95, the species utilizes *n*-acetyl-glucosamine, ethanol, D(–)-fructose, D(+)–glucosamine, glycerol, pyruvate, and yeast extract. The species does not utilize benzoic acid, D(+)–galactose, D(+)–mannose, methanol, pectin, D(–)-ribose, L-rhamnose, L(–)-sorbitol, whey, and D(+)–xylose. The organism does not grow fermentatively. The species gives negative tests for arginine dihydrolase, phenylalanine deaminase, indole production, and Voges–Proskauer.

This species is susceptible to ampicillin, chloramphenicol, and rifampin. It is resistant to kanamycin, neomycin, and streptomycin. The concentrations of antibiotics used were not provided in the original description.

The species was isolated from a soil sample just below the crystalline salt surface of a salt flat south of Alkali Lake in eastern Washington State, USA.

The mol% G + C of the DNA is: 66 (T_m).

Type strain: 4A, ATCC 700597.

GenBank accession number (16S rRNA): AF054286.

4. **Halomonas cupida** (Baumann, Baumann, Mandel and Allen 1972) Dobson and Franzmann 1996, 556^{VP} (*Deleya cupida* (Baumann, Baumann, Mandel and Allen 1972) Baumann, Bowditch and Baumann 1983a, 801; *Alcaligenes cupidus* Baumann, Baumann, Mandel and Allen 1972, 426.) *cu'pi.da*. L. adj. *cupidus* desiring.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. This species possesses peritrichous flagella; it has been described as utilizing 69–85 organic compounds, possibly more than any other *Halomonas* species. In addition to those listed, the species also utilizes D-galactose, D-xylose, fumarate, malate, pyruvate, β-hydroxybutyrate, benzoate, caprylate, L-aspartic acid, L-glutamic acid, proline, tyrosine, melibiose, saccharate, galacturonate, glucuronate, N-acetylglucosamine, butyrate, isobutyrate, valerate, isovalerate, caproate, heptanoate, pelargonate, caprate, glycolate, DL-glycerate, aconitate, erythritol, meso-inositol, n-propanol, isobutanol, benzoylformate, p-hydroxybenzoate (but not the *ortho* or *meta* isomers), putrescine, spermine, betaine, sarcosine, hippurate, and allantoin. The species may or may not grow on L-rhamnose, D-salicin, D-trehalose, malonate, and L-ornithine. This organism does not use formate, oxalate, D-tartrate, L-valine, or L-threonine as the sole source of carbon. The similarity of the 16S rDNA of this species to other *Halomonas* species ranges from 92.6–96.5%. Isolated from marine habitats.

The mol% G + C of the DNA is: 60–63 (Bd).

Type strain: ATCC 27124, DSM 4740.

GenBank accession number (16S rRNA): L42615.

5. **Halomonas desiderata** Berendes, Gottschalk, Heine-Dobbernack, Moore and Tindall 1997, 242^{VP} (Effective publication: Berendes, Gottschalk, Heine-Dobbernack, Moore and Tindall 1996, 165.) *de.si.de.ra'ta*. L. part. adj. *desiderata* wished for, the strain wished for.

Cells are motile with peritrichous flagella and are 0.4–0.6 × 1.0–2.6 μm. The colony type has not been described, but the species does not produce pigments. Requires sodium ions. Optimum pH is 9.7 and the optimum temperature for growth is 37–42°C. The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96.

Facultatively anaerobic in the presence of nitrate. Obligately alkaliphilic; optimum pH 9.5. Grows at 45°C. Denitrifying. Does not grow below pH 7 or above pH 11. Does not grow at 50°C. Supplemental vitamins or amino acids are not needed. Catalase positive and phenylalanine deaminase positive. Arginine dihydrolase negative. Does not hydrolyze starch, gelatin, casein, Tween 80, Tween 60, hippurate, or pullulan. Produces poly-β-hydroxybutyrate as a reserve material but does not produce levan from sucrose. Grows on MacConkey agar, *Salmonella-Shigella* agar, and Cetrimide agar. At pH 9.7, the species is resistant to up to 1% bile salts.

Utilizes D-xylose, ribose, trehalose, glycerol, DL-β-hydroxybutyrate, and benzoate as sole sources of carbon. Does not utilize inositol. Not fermentative; indole and Voges–

Proskauer negative. The major respiratory quinone is ubiquinone-9. The major fatty acids are C_{18:1} (52–65% depending on the strain), C_{16:0} (13–23% depending on the strain), and C_{16:1 ω7c}. Other fatty acids detected in various strains include C_{10:0}, C_{10:0 3OH}, C_{12:0}, C_{12:0 3OH}, C_{14:0}, C_{17:0 cyclo}, C_{18:0}, and C_{19:0 cyclo}. The major phospholipids are phosphatidylglycerol, phosphatidylethanolamine, diphosphatidylglycerol, and an uncharacterized phosphoglycolipid. The similarity of the 16S rDNA of this species to other *Halomonas* species ranges from 89.7–94.8%. DNA–DNA hybridization data show similarities below 60–70%. Isolated from municipal sewage.

The mol% G + C of the DNA is: 66 (HPLC).

Type strain: DSM 9502.

GenBank accession number (16S rRNA): X92417.

6. **Halomonas eurihalina** (Quesada, Valderrama, Bejar, Ventosa, Gutierrez, Ruiz-Berraquero and Ramos-Cormenzana 1990) Mellado, Moore, Nieto and Ventosa 1995b, 715^{VP} (*Volcaniella eurihalina* Quesada, Valderrama, Bejar, Ventosa, Gutierrez, Ruiz-Berraquero and Ramos-Cormenzana 1990, 265.) *eu.ri.ha.li'na*. Gr. adj. *euris* wide, broad; Gr. adj. *halinos* salted; M.L. adj. *eurihalina* growing at a wide range of salt concentrations.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. Colonies grown on MH medium with 7.5% NaCl at 32°C for 72 h are 4–5 mm in diameter, circular, convex, smooth, opaque, cream-colored with entire margins, and very mucoid. This species requires Na⁺ ions. Sodium can be supplied as NaCl, Na₂SO₄, or NaBr. Under optimum conditions, the species grows at pH 5.0–9.0. Catalase positive. Phosphatase, lysine decarboxylase, and β-galactosidase negative. Does not hydrolyze esculin. Produces acid from fructose and maltose. Some strains may also produce acid from D-galactose, D-glucose, lactose, mannitol, and sucrose. Utilizes fumarate, β-hydroxybutyrate, L-valine, L-threonine, proline, and tyrosine as sole carbon sources. Some strains also utilize D-galactose, raffinose, D-salicin, D-trehalose, L-ornithine, and methionine as sole carbon sources. This species hydrolyzes gelatin and grows on KCN, MacConkey, and Cetrimide agar. Some strains of this species produce H₂S and hydrolyze Tween 20. Sensitive to chloramphenicol (30 μg). Some strains may be sensitive to gentamicin (10 μg) and neomycin (30 μg). Resistant to most other antimicrobials. Isolated from hypersaline habitats (soils and salterns) and from ocean water.

The mol% G + C of the DNA is: 59.1–65.7 (T_m).

Type strain: ATCC 49336, DSM 5720.

GenBank accession number (16S rRNA): L42620, X87218.

7. **Halomonas halmophila** (Elazari-Volcani 1940) Franzmann, Wehmeyer and Stackebrandt 1989, 205^{VP} (Effective publication: Franzmann, Wehmeyer and Stackebrandt 1988, 19) emend. Dobson, James, Franzmann and McMeekin 1990, 462 (*Flavobacterium halmophilum* Elazari-Volcani 1940, 85.) *hal.mo.phi'lum*. Gr. n. *halme* brine, seawater; Gr. adj. *philos* loving; M.L. n. adj. *halmophilum* seawater loving.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. Does not produce acid from sugars, D-mannitol, or inulin and does not produce indole. The species is sensitive to nalidixic acid

(30 µg) and resistant to ampicillin (10 µg), gentamicin (10 µg), neomycin (30 µg), streptomycin (10 µg), and tetracycline (30 µg). This species was transferred into the genus *Halomonas* when 16S rRNA cataloging showed an S_{AB} value of 0.66 to the type species of the genus *Halomonas*. The similarity of the 16S rDNA of this species to other *Halomonas* species ranges from 92.2–97.9%. The mol% G + C content of the DNA has been reported to be 49.7. The exact site from which this bacterium was isolated is not known, but was probably somewhere in the Dead Sea.

The mol% G + C of the DNA is: 49.7 (method unknown).

Type strain: ATCC 19717, DSM 5349.

GenBank accession number (16S rRNA): M59153.

The name of this species has been incorrectly spelled as *Flavobacterium halmephilum* (Weeks, 1974) and as *Flavobacterium halmephilum* (Weeks, 1974).

8. ***Halomonas halodenitrificans*** (Robinson and Gibbons 1952) Dobson and Franzmann 1996, 556^{VP} (*Paracoccus halodenitrificans* (Robinson and Gibbons 1952); *Micrococcus halodenitrificans* Robinson and Gibbons 1952, 154.)

ha.lo.de.ni.tri'fi.cans. Gr. n. *hals*, *halis* salt; M.L. v. *denitrifico* to denitrify; M.L. part. adj. *halodenitrificans* salt (requiring) denitrifying.

This is the only member of this genus that has a coccoid morphology. The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. Most of the characteristics have been taken from the description of Kocur (1984). When grown on nutrient agar containing 6% NaCl, the colonies of this species are described as circular, entire, convex, butyrous, glistening, cream-colored, and opaque. The optimum growth temperature is 25–30°C; the growth range is 5–32°C. The species cannot grow in media containing less than 3% NaCl. Optimum growth occurs with 4.4–8.8% NaCl; slow growth occurs with up to 20% NaCl in media. Catalase positive; phosphatase is present. Hydrogen sulfide is not produced. Esculin is not hydrolyzed. Lysine decarboxylase, phenylalanine deaminase, and β-galactosidase negative. Indole and Voges–Proskauer negative. Does not hydrolyze Tween 80. Acid is not produced from either D-glucose or sorbitol. Acid production from other sugars or polyols has not been tested. L-valine, L-threonine, proline, and L-asparagine serve as sole carbon sources. Does not utilize L-aspartic acid, L-glutamic acid, or methionine as sole carbon sources. Sensitive to mercuric chloride and polymyxin B, although no concentrations have been specified. Resistant to gentamicin (10 µg), vibriostat O/129 (10 µg), penicillin G (10 i.u.), streptomycin (10 µg), tetracycline (30 µg), and chloramphenicol (30 µg). The similarity of the 16S rDNA of this species to each member of the genus *Halomonas* ranges from 91.5–95.6%. Isolated from meat-curing brines.

The mol% G + C of the DNA is: 64–66 (T_m , Bd).

Type strain: ATCC 13511, CCM 286, DSM 735, NCMB 700.

GenBank accession number (16S rRNA): L04942.

9. ***Halomonas halodurans*** Hebert and Vreeland 1987, 350^{VP} *ha.lo.du'rans.* Gr. n. *hals*, *halos*; the sea, salt; L. v. *durare* to last endure; M.L. masc. adj. *Halodurans* salt tolerating.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. The original description of this species was based on tests using media

supplemented with 2.6% NaCl. Later descriptions of this species by Hebert and Vreeland (1987) were performed at the optimum NaCl concentration for growth of the species (8.0%). Hebert and Vreeland (1987) noted several differences in the results of the phenotypic testing and attributed them to the differences in the salt content of the media. The biochemical properties listed in Tables BXII.γ.95 and BXII.γ.96 for this species are those corresponding to tests conducted with media containing 8% NaCl.

The cells are motile by single polar flagellum that is 8–10 µm in length and 12–14 nm wide. Colonies are circular, convex, smooth, entire, 2–3 mm in diameter, and nonpigmented following 48 h of growth at 20–35°C. This species does not grow in 30% NaCl. Grows at pH 6.0–8.0 in 8% NaCl.

Catalase positive; phosphatase, and lysine decarboxylase are present. Phenylalanine deaminase and β-galactosidase are absent. Hydrolyzes esculin. Tolerant to KCN. Hydrolyzes Tween 80. Hydrogen sulfide, indole, and acetyl-methyl carbinol (Voges–Proskauer) are not produced. Produces acid from D-fructose, D-galactose, D-glucose, lactose, salicin, sucrose, and D-xylose. Does not produce acid from D-mannitol, D-mannose, or L-rhamnose. Utilizes D-galactose, inulin, ribose, malonate, pyruvate, benzoate, L-ornithine, L-threonine, proline, and L-asparagine as sole sources of carbon. The species cannot use L-rhamnose, malate, β-hydroxybutyrate, or methionine as sole carbon sources. Sensitive to erythromycin (15 µg), nalidixic acid, (30 µg), streptomycin (10 µg), tetracycline (30 µg), and polymyxin B (concentration unknown). The species is resistant to vibriostat O/129 (10 µg), penicillin G (10 i.u.), and chloramphenicol (30 µg). The similarity of the 16S rDNA of this species to other species of *Halomonas* ranges from 93.7–99.9%. The level of DNA–DNA homology between this species *H. elongata* and *H. canadensis* is shown in Table BXII.γ.98. Isolated from Great Bay Estuary (New Hampshire, USA).

The mol% G + C of the DNA is: 63.2 ± 1.1 (T_m , Bd).

Type strain: ATCC 29686, DSM 5160.

GenBank accession number (16S rRNA): L42619.

10. ***Halomonas halophila*** (Quesada, Ventosa, Ruiz-Berraquero and Ramos-Cormenzana 1984) Dobson and Franzmann 1996, 556^{VP} (*Deleya halophila* Quesada, Ventosa, Ruiz-Berraquero and Ramos-Cormenzana 1984, 290.)

ha.lo.phi'la. Gr. n. *halos* salt; Gr. adj. *philus* loving; M.L. adj. *halophila* salt loving.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. When grown with 7.5% sea salt for 4 days at 32°C, the colonies are circular, low convex, smooth, opaque, and cream-colored. Grows in 30% NaCl and at pH 5.0–9.0. Tests positive for catalase and phosphatase. DNase and phenylalanine deaminase negative. Produces hydrogen sulfide and hydrolyzes esculin, but does not hydrolyze Tween 80. Indole and Voges–Proskauer negative. Produces acid from D-galactose, D-glucose, and D-maltose. Some strains may also produce acid from sucrose, D-trehalose, and D-xylose. Acid is not produced from lactose and mannitol. In addition to the carbon sources listed in Table BXII.γ.95, this species also utilizes D-galactose, D-salicin, D-trehalose, D-xylose, fumarate, malate, pyruvate, L-aspartic acid, and L-glutamic acid as sole carbon sources. Some strains may also use L-orni-

thine. The species does not utilize L-rhamnose, hippurate, D-tartrate, benzoate, caprylate, or L-valine as sole carbon sources. The species is sensitive to ampicillin (10 µg), cephalothin (30 µg), erythromycin (15 µg), chloramphenicol (30 µg), polymyxin B, and rifampin A (no concentrations have been specified for the latter two anti-microbials). Resistant to bacitracin (10 i.u.) and lincomycin (concentration unknown). The similarity of the 16S rDNA of this species to each member of the *Halomonas* ranges from 93.8–99.2%. Isolated from saline soils.

The mol% G + C of the DNA is: 66.7 (T_m).

Type strain: CCM3662, DSM 4770.

GenBank accession number (16S rRNA): L42619 and M93353.

11. ***Halomonas magadii*** Duckworth, Grant, Jones, Meijer, Márquez and Ventosa 2000b, 1415^{VP} (Effective publication: Duckworth, Grant, Jones, Meijer, Márquez and Ventosa 2000a, 59.)
ma.ga'di.i. M.L. gen. n. *magadi* of Magadi, named for lake Magadi, a saline soda lake in Kenya.

The characteristics are as described for the genus and are listed in Tables BXII.γ.95 and BXII.γ.96. Other descriptive characters are listed below. Rods 4.0–6.0 × 0.6–0.8 µm. Catalase positive, produces circular, low convex opaque cream-colored colonies. Alkaliphilic, growth occurs between pH 7.0 and 11.0 with an optimum around 9.5. Grows at salt concentrations from 0% to 20% between 25°C and 40°C with an optimum at 37°C. Phosphatase and phenylalanine deaminase negative. Does not hydrolyze gelatin or casein. Indole negative. No anaerobic growth in the presence or absence of nitrate. Produces H₂S from cysteine. When grown in alkaline medium, the species utilizes amygdalin, esculin, D-fructose, D-fucose, D-galactose, D-melibiose, L-raffinose, D-ribose, salicin, starch, trehalose, D-xylose, adonitol, D-glucuronolactone, D-glucosamine, dulcitol, erythritol, ethanol, glycerol, D-mannitol, *m*-inositol, propanol, D-sorbitol, N-acetylglucosamine, propionide, quinate, D-saccharate, salicylate, suberate, tartrate, L-asparagine, DL-aspartic acid, DL-phenylalanine, L-glutamine, α-aminovalerate, aconitate, α-ketoglutarate, butyrate, caprylate, fumarate, DL-glycerate, D-glucuronate, glutamate, *p*-hydroxybenzoate, hippurate, DL-malate, malonate, oxalate, L-ornithine, L-leucine, L-methionine, L-proline, L-threonine, L-valine, and pyruvate. Sarcosine, ethionine, creatine, DL-aminobutyrate, Tween 80, inulin, and L-rhamnose are not used. Sensitive to erythromycin (5 mg), streptomycin (10 mg), polymyxin (300 IU), sulphafurazole (100 mg), chloramphenicol (25 mg), and oleandomycin (5 mg). Not sensitive to gentamicin (10 mg), kanamycin (30 mg), fusidic acid (10 mg), tetracycline (25 mg), rifampicin (2 mg), bacitracin (10 IU), neomycin (30 mg), novobiocin (5 mg), and vancomycin (30 mg). Listed as insensitive to sulphamethoxazole, nitrofurantoin, trimethoprim, ampicillin, penicillin G, and methicillin, but no concentrations were specified.

The major polar lipids are phosphatidylglycerol, phosphatidyl glycerol phosphate, diphosphatidylglycerol, phosphatidylethanolamine, and a small amount of an unidentified glycolipid. Cells also contain ubiquinones 6 and 9.

The mol% G + C of the DNA is: 62 (T_m).

Type strain: 21M1, NCIMB 13595.

GenBank accession number (16S rRNA): X92150.

12. ***Halomonas marina*** (Cobet, Wirsén and Jones 1970) Dobson and Franzmann 1996, 556^{VP} (*Deleya marina* (Cobet, Wirsén and Jones 1970) Baumann, Bowditch and Baumann 1983a, 801; *Pseudomonas marina* (Cobet, Wirsén and Jones 1970) Baumann, Baumann, Mandel and Allen 1972, 423; "*Arthrobacter marinus*" Cobet, Wirsén and Jones 1970, 159.) *ma.ri'na.* L. adj. *marina* of the sea.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. The cells possess polar flagella. Baumann et al. (1983a) note that this species has flagella that tend to bend back toward the cell, giving arrangements that appear to be lateral or peritrichous. The species has been described as utilizing 35–47 organic compounds as sole sources of carbon. In addition to those listed in Table BXII.γ.95, this species utilizes D-galactose, fumarate, malate, pyruvate, β-hydroxybutyrate, L-aspartic acid, L-glutamic acid, L-proline, butyrate, valerate, isovalerate, caproate, heptanoate, pelargonate, caprate, DL-glycerate, α-ketoglutarate, aconitate, *n*-butanol, sarcosine, and creatine. Some strains may also utilize malonate and L-ornithine as sole carbon sources. The species does not use L-rhamnose, ribose, D-salicin, D-trehalose, D-xylose, hippurate, D-tartrate, benzoate, L-valine, or L-threonine. The similarity of the 16S rDNA of this species to other *Halomonas* species is only 92.3–94.8%. Isolated from marine habitats.

The mol% G + C of the DNA is: 63 (Bd).

Type strain: C 25374, DSM 4741.

GenBank accession number (16S rRNA): M93354.

13. ***Halomonas marisflavae*** Yoon, Choi, Lee, Kho, Kang, and Park 2001a, 1176^{VP}
ma.ri.s fla'vae. L. gen. neut. n. *maris* of the sea; L. neut. adj. *flavum* yellow; L. gen. neut. n. *marisflavae* of the Yellow Sea, Korea.

The characteristics are as described for the genus and are listed in Tables BXII.γ.95 and BXII.γ.96. The species produces rod or oval shaped cells that are 0.9–1.3 × 1.7–2.3 µm following 3 days growth at 28°C. Cells are motile with a single polar flagellum. Colonies are yellow-orange, smooth, glistening, circular, and slightly irregular on Tryptic Soy Agar (TSA) and on Marine Agar (MA). The colonies are described as being convex on TSA and concave on MA. Optimal growth occurs between 0.5% and 12% NaCl (w/v) and at 25–30°C. The optimal pH for growth is 7.0–8.0. Grows anaerobically on MA medium although the exact conditions for this characteristic were not specified. Catalase positive. Hydrolyzes esculin and gelatin, but not casein, hypoxanthine, Tween 80, tyrosine, or xanthine. Indole and arginine deaminase negative. Assimilates and produces acid from D-glucose, arabinose, mannose, mannitol, gluconate, malate, and citrate. The species is also described as producing acid from glycerol, erythritol, ribose, D-xylose, galactose, D-fructose, arbutin, salicin, maltose, melibiose, trehalose, gentiobiose, D-turanose, D-lyxose, and D-furanose. Also produces a weak acid reaction from adonitol, mannitol, and xylitol.

The predominant respiratory quinone is ubiquinone-9. The major fatty acids are C_{18:1}, C_{16:0}, C_{16:1 ω7c} and/or C_{15:0 iso 2OH}.

Isolated from the Yellow Sea, Korea.

The mol% G + C of the DNA is: 59 (HPLC).

Type strain: SW32, JCM 10873, KCCM 80003.

GenBank accession number (16S rRNA): AF251143.

14. **Halomonas maura** Bouchotroch, Quesada, Del Moral, Llamas and Bejar 2001, 1630^{VP}
mau'ra. L. adj. *maurus* northwest African.

The characteristics are as described for the genus and are listed in Tables BXII.γ.95 and BXII.γ.96. This species produces one of the longest rods ($6.0\text{--}9.0 \times 0.5\text{--}0.7 \mu\text{m}$) in the genus. It and *Halomonas eurihalina* are the only species known to produce large amounts of exopolysaccharide. Colonies are circular, convex, mucoid, cream-colored, and 2–3 mm in diameter after 24 hours growth at 32°C. Colonies expand to over 5 mm in diameter after 72 hours. Grows in salt concentrations between 1% and 15% with optimal growth occurring between 7.5 and 10% (w/v). The salt concentrations given actually reflect the percentage of total sea salts and not specifically NaCl. However, the original description of this species states that the organism is able to grow optimally in medium containing 1.2M NaCl (7.0% w/v) and 0.2M $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ (4.5%). The description also states that $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ may be substituted for magnesium sulfate. The species is capable of anaerobic growth in the presence of nitrate but not fumarate. Catalase positive. Does not produce acids from sugars. Reduces selenite and oxidizes gluconate. The species produces H_2S from peptones and hydrolyzes Tween 20. Indole, methyl-red, and Voges-Proskauer tests are negative. The species does not utilize esculin, tyrosine, DNA, gelatin, casein, lecithin, and Tween 80, nor does it hydrolyze blood, produce phenylalanine deaminase, grow on cetrimide agar or under anaerobic conditions on normal medium. In addition, the type strain of the species produces phosphatase and grows on MacConkey and KCN agars and is ONPG negative.

In addition to the compounds listed in Table BXII.γ.95, the species utilizes D-galactose, *myo*-inositol, and D-sorbitol as sole sources of carbon and energy. The species does not utilize formate, D-salicin, or L-sorbose. The type strain differs somewhat in these characters in that it also utilizes citrate, ethanol, fumarate, D-fructose, glycerol, D-rhamnose, and D-ribose. The type strain also does not utilize lactose or D-trehalose. The species also utilizes the amino acids listed in Table BXII.γ.95, as well as L-ornithine and L-valine, as sole sources of carbon and nitrogen. The organism does not utilize cysteine or L-tryptophan in this manner. As a whole, the species is susceptible to amoxycillin, ampicillin, cephalothin, cefoxitin, chloramphenicol, nalidixic acid, nitrofurantoin, polymyxin, rifampicin, and trimethoprim-sulfamethoxazole. The species is resistant to tetracycline and Vibriostat O/129. The type strain is also sensitive to gentamicin, penicillin, sulfonamide, and tobramycin while being resistant to erythromycin. Unfortunately, no antibiotic concentrations were provided in the original species description. These data may be provided by Bouchotroch et al. (1999) which was cited by Bouchotroch et al. (2001).

The major fatty acids (up to 81% total) of the type strain are $\text{C}_{18:1 \omega 7\text{c}}$, $\text{C}_{16:1 \omega 7\text{c}}/\text{C}_{15:0 \text{ iso } 2\text{OH}}$, and $\text{C}_{16:0}$. The bacterium produces exo-polysaccharide designated EPS S-31 which is composed of neutral sugars glucose, mannose, and galactose in a ratio of 1:4:2.5. EPS S-31 emulsifies crude oil.

The species was isolated from a solar saltern in Asilah Morocco.

The mol% G + C of the DNA is: 62.2–64.1 (T_m).

Type strain: S-31, CECT 5298, DSM 13445.

GenBank accession number (16S rRNA): AJ271864.

15. **Halomonas meridiana** James, Dobson, Franzmann and McMeekin 1990b, 470^{VP} (Effective publication: James, Dobson, Franzmann and McMeekin 1990a, 277.)
me.ri.di.a'na. L. adj. *meridiana* of the south.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. The colonies produced by this species are smooth, circular, convex, and white to off-white, with entire margins. Yellow colonies may be produced on media containing alanine or pyruvate. The cells may possess either lateral or polar flagella. The species grows in media of pH 6.0–9.0; some strains may grow at pH 5.0. Catalase positive. Test results for phosphatase, DNase, lysine decarboxylase, phenylalanine deaminase, and β-galactosidase are all negative. Most strains have been reported to hydrolyze Tween 80 and Tween 20. Produces acid from D-fructose, maltose, and sucrose. Some strains may produce acid from D-galactose, D-glucose, lactose, and mannitol. Utilizes D-xylose, fumarate, β-hydroxybutyrate, L-valine, proline, and L-asparagine as sole sources of carbon. Most strains also utilize D-galactose and D-rhamnose. Some strains may utilize raffinose, D-salicin, D-trehalose, L-ornithine, L-threonine, and methionine as sole carbon sources. Does not use malonate. Resistant to most antibiotics, including ampicillin (10 μg), bacitracin (10 i.u.), cephalothin (30 μg), erythromycin (10 μg), mercuric chloride 1:5000 (200 mg/l), nalidixic acid (30 μg), novobiocin (20 μg), vibriostat O/129 (10 μg), penicillin G (10 i.u.), streptomycin (10 μg), and tetracycline (30 μg). Some strains show sensitivity to gentamicin (10 μg), neomycin (30 μg), and chloramphenicol (30 μg). The similarity of the 16S rDNA of this species to each member of the *Halomonas* ranges from 93.6–100%. Isolated from Antarctic lakes.

The mol% G + C of the DNA is: 58.8–59.1 ± 0.8 (T_m).

Type strain: ACAM 246, ATCC 49692, DSM 5425, UQM 3352.

GenBank accession number (16S rRNA): M93356.

16. **Halomonas pacifica** (Baumann, Baumann, Mandel and Allen 1972) Dobson and Franzmann 1996, 556^{VP} (*Deleya pacifica* (Baumann, Baumann, Mandel and Allen 1972) Baumann, Bowditch and Baumann 1983a, 801; *Alcaligenes pacificus* Baumann, Baumann, Mandel and Allen 1972, 426.)
pa.ci'fic.a. M.L. adj. *pacifica* pertaining to the Pacific ocean.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. In addition, the species has been shown to utilize fumarate, malate, pyruvate, β-hydroxybutyrate, benzoate, L-aspartic acid, L-glutamic acid, L-ornithine, proline, N-acetylglucosamine, butyrate, isobutyrate, valerate, isovalerate, caproate, heptanoate, glutarate, DL-glycerate, 2,3-butyleneglycol, benzoate, *p*-hydroxybenzoate (but not the *ortho* and *meta* isomers), phenylacetate, δ-aminovalerate, DL-kynurenine, kynurenate, anthranilate, benzylamine, putrescine, spermine, histamine, butylamine, betaine, sarcosine, and nicotinate as carbon sources. Some strains may use caprylate as well. The species does not use D-galactose, L-rhamnose, ribose, D-salicin, D-trehalose, D-xylose, malonate, D-tartrate, L-valine, or L-threonine as sole carbon sources. The similarity of the 16S rDNA of this species to each member of the *Halomonas* ranges from 93.2–96.6%. Isolated from marine habitats.

The mol% G + C of the DNA is: 67–68 (Bd).

Type strain: ATCC 27122, DSM 4742.

GenBank accession number (16S rRNA): L42616.

17. **Halomonas pantelleriensis** Romano, Nicolaus, Lama, Manca and Gambacorta 1997, 601^{VP} (Effective publication: Romano, Nicolaus, Lama, Manca and Gambacorta 1996, 332.)

pan.tel.le.ri.en'sis. from the place of isolation, Pantelleria Island in the south of Sicily, Italy.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. Colonies are cream to pink, making this the only species of the genus with pink colonies. Colonies are smooth, granulate, entire, circular, raised, and 0.11 mm in diameter. The species produces PHB. In defined medium, the species requires biotin or thiamine (50 µg/l) in addition to the carbon sources listed in Table BXII.γ.95. Optimum growth occurs in media containing 10% NaCl, 33–35°C, pH 9.0. The species is alkalophilic and grows at a pH range of 7.5–11.0. Catalase and α-glucosidase positive; β-galactosidase negative. Decomposes tyrosine. Hydrolyzes hippurate, but does not attack casein, gelatin or L- or D-N'-benzoylarginine-*p*-nitroanilide. The species gives a weak reaction for phenylalanine deaminase and is not sensitive to lysozyme. In addition to the substrates listed in Table BXII.γ.95, the species also utilizes fructose, D-xylose, galactose, trehalose, salicin, sorbitol, glycerol, pyruvate, heptanoate, and valine. The organism cannot utilize raffinose, ethanol, or valerate. The species is sensitive to chloramphenicol (30 µg) and erythromycin (30 µg). It is insensitive to lincomycin (15 µg), gentamicin (10 µg), ampicillin (25 µg), tetracycline (50 µg), penicillin (10 µg), neomycin (30 µg), bacitracin (10 µg), and novobiocin (30 µg). The similarity of the 16S rDNA of this species to other members of the *Halomonas* ranges from 93.1–95.8%. This species represents a distinct phylogenetic line within the larger genus, and is most closely related to *H. marina* and *H. halodenitrificans*. Isolated from hard sand on the island of Pantelleria, south of Sicily, Italy.

The mol% G + C of the DNA is: 65.02 (HPLC).

Type strain: ATCC 700273, DSM 9661.

GenBank accession number (16S rRNA): X93493.

18. **Halomonas salina** (Valderrama, Quesada, Bejar, Ventosa, Gutierrez, Ruiz-Berraquero and Ramos-Cormenzana 1991) Dobson and Franzmann 1996, 556^{VP} (*Delaya salina* Valderrama, Quesada, Bejar, Ventosa, Gutierrez, Ruiz-Berraquero and Ramos-Cormenzana 1991, 382.)
sa.li'na. L. adj. *salina* salted, saline.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. Of all of the *Halomonas* species, *H. salina* is currently the easiest to distinguish at the phenotypic level. The species has several characteristics that are not shared by the other halomonads. It forms coccoid cells when grown in low salts, but produces typical elongated filaments when grown in optimum (7.5% total sea salts) and higher salts, but not in 30% salts. The cells are nonmotile, produce a capsule, and accumulate poly-β-hydroxybutyrate. The colonies produced after 48 h at 32°C on MH medium are small (1–2 mm in diameter), convex, smooth, glossy, and translucent yellow to cream and have entire margins. This is also the only *Halomonas* species known to reduce selenite and to require Mg²⁺, in addition to Na⁺. Unlike other *Halomonas* species, *H. salina* can only use cations supplied as either Cl[−] or SO₄^{2−} salts. Members of the species grow at pH 6.0–9.0. Some strains also grow

at pH 5.0. Catalase positive. Some strains also possess DNase, phenylalanine deaminase, and β-galactosidase. Some strains produce hydrogen sulfide, are KCN tolerant and hydrolyze Tween 80 and Tween 20. Some strains grow on MacConkey and cetrimide agars. Does not possess phosphatase, lysine decarboxylase, or lecithinase. Some strains may produce acid from D-mannitol and sucrose. Acid is not produced from other sugars or poly-alcohols. The members of this species utilize D-salicin, fumarate, malate, and L-aspartic acid as sole sources of carbon. Some strains may also utilize D-galactose, inulin, raffinose, L-rhamnose, sorbose, D-trehalose, D-xylose, formate, pyruvate, D-tartrate, β-hydroxybutyrate, and L-glutamic acid. The species does not utilize oxalate, benzoate, or caprylate as sole carbon sources. Sensitive to ampicillin (10 µg), cephalothin (30 µg), erythromycin (15 µg), nalidixic acid (15 µg), penicillin G (10 i.u.), and chloramphenicol (30 µg). Some strains are also sensitive to gentamicin (10 µg), streptomycin (10 µg), and tetracycline (30 µg). Isolated from hypersaline habitats such as soils, salt ponds, salt lakes, and from the sea.

The mol% G + C of the DNA is: 60.7–64 (T_m).

Type strain: ATCC 49509, DSM 5928.

GenBank accession number (16S rRNA): L42617, X87217.

19. **Halomonas subglaciescola** Franzmann, Burton and McMeekin 1987, 32^{VP}

sub.gla'ci.es.co.la. L. pref. *sub* below; L. n. *glacies* ice; L. suff. *cola* to dwell; M.L. adj. *subglaciescola* dwelling below ice.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. When motile, the cells are peritrichously flagellated. On solid media, the colonies produced are white to cream. Yellow colonies may form on media enriched with alanine or hydroxy-L-proline. This species cannot be grown on simple media, such as CAS or MH. Grows at temperatures as low as −5°C. The species has two biovars. Biovar I is motile and produces long filamentous cells >10 µm long. Biovar II is nonmotile and produces cells <10 µm long. The type strain belongs to Biovar I. Some characteristics of the species are listed in Tables BXII.γ.95 and BXII.γ.96. The species grows at pH 5.0–9.0 and does not grow above 25°C. The species does not grow in the presence of 30% NaCl. Catalase and phenylalanine deaminase positive. Phosphatase, lysine decarboxylase, and β-galactosidase negative. Does not produce indole or hydrogen sulfide; does not hydrolyze esculin. The species does not produce acid from either adonitol or sorbitol. There is no data available for acid production from other compounds. This species has been reported to grow on L-threonine and proline as primary carbon sources. It does not utilize sucrose, L-aspartic acid, L-glutamic acid, methionine, or L-asparagine as sole carbon sources. Sensitive to mercuric chloride (200 mg/l). Some strains are sensitive to Vibriostat O/129 (10 µg), penicillin G (10 i.u.), and chloramphenicol (30 µg). Resistant to gentamicin (10 µg), neomycin (30 µg), streptomycin (10 µg), and tetracycline (30 µg). The similarity of the 16S rDNA of both biovars of this species to other *Halomonas* species ranges from 93.7–96.1% for Biovar I and from 93.6–99.9% for Biovar II. The 16S rDNA sequence of Biovar II shows only 98.2% similarity to that of Biovar I. However, Biovar II shows 99.9% sequence similarity to *Halomonas halodurans*, indicating that Biovar II is likely a variant of the later species. Isolated from Antarctic hypersaline lakes.

The mol% G + C of the DNA is: 60.9–62.9 ± 1.0 (method unknown).

Type strain: ATCC 43668, DSM 4683, UQM 2926.

GenBank accession number (16S rRNA): M93358.

20. **Halomonas variabilis** (Fendrich 1989) Dobson and Franzmann 1996, 556^{VP} (*Halovibrio variabilis* Fendrich 1989, 205.) *va.ri.a'bi.lis*. L. adj. *variabilis* changeable, variable; referring to variation of the cell diameter with changing salt concentrations.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. The cells are curved rods (described as vibron shaped). Coccoid bodies may form in older cultures. Cells have monotrichous flagellation. Colonies are circular, entire, smooth, slimy, raised, and produce a light brown pigment. The cells grow in NaCl concentrations from 1.2–4.9 M, with an optimum salt concentration of 1.0 M. The optimum pH for growth is 7.5; minimum, 6.5; maximum, 8.4. Temperature range for growth: 15–37°C; optimum, 33°C. Minimum doubling time, 7.5 h. The similarity of the 16S rDNA of this species to other *Halomonas* species ranges from 91.5–96.8%. Isolated from the North Arm of the Great Salt Lake Utah (USA).

The mol% G + C of the DNA is: 57.0 (method unknown).

Type strain: DSM 3051.

GenBank accession number (16S rRNA): M93357, X90483.

21. **Halomonas venusta** (Baumann, Baumann, Mandel and Allen 1972) Dobson and Franzmann 1996, 556^{VP} (*Delaya venusta* (Baumann, Baumann, Mandel and Allen 1972) Baumann, Bowditch and Baumann 1983a, 801; *Alcaligenes venustus* Baumann, Baumann, Mandel and Allen 1972, 426.) *ve.nus'tus*. L. adj. *venustus* lovely, beautiful.

The characteristics are as described for the genus and as listed in Tables BXII.γ.95 and BXII.γ.96. The cells possess peritrichous flagella. The species utilizes 52–76 organic compounds. Also utilized, in addition to compounds listed in Table BXII.γ.95, are: galacturonate, glucuronate, N-acetylglucosamine, butyrate, isobutyrate, valerate, isovalerate, caproate, heptanoate, caprate, glutarate, glycolate, α-ketoglutarate, aconitate, *meso*-inositol, propyleneglycol, 2,3-butyleneglycol, *n*-propanol, *n*-butanol, isobutanol, L-mandelate, benzoylformate, benzoate, *o*-, *m*-, and *p*-hydroxybenzoate, phenylate, quinate, δ-aminovalerate, anthranilate, ethanolamine, putrescine, betaine, sarcosine, acetamide, and allantoin. The similarity of the 16S rDNA of this species to other *Halomonas* species ranges from 93.2–98%.

The mol% G + C of the DNA is: 58.7 (Bd).

Type strain: ATCC 27125, DSM 4743.

GenBank accession number (16S rRNA): L42618.

Genus II. *Carnimonas* Garriga, Ehrmann, Arnau, Hugas and Vogel 1998, 684^{VP}

MARGARITA GARRIGA, MATTHIAS A. EHLMANN, JACINT ARNAU, MARTA HUGAS AND RUDI F. VOGEL

Car.ni'mo.nas. L. gen. n. *carnis* of meat; Gr. n. *monas* a unit, monad; *Carnimonas* a monad of meat.

Straight or slightly curved rods, 0.5–0.6 × 1.0–1.7 μm, occurring singly or in pairs. Gram negative. Does not form spores. **Non-motile. Oxidase and catalase positive. Aerobic, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor. Moderately halotolerant.** Optimum temperature for growth is 28–30°C. **No growth occurs at 5°C or 37°C.** Chemorganotrophic. **Acid, but no gas, is produced from D-glucose, D-xylose, and various other carbohydrates. β-Galactosidase (ONPG) activity occurs.** The main respiratory quinone is ubiquinone-9. Main components in the polar lipid composition are diphosphatidylglycerol, phosphatidylglycerol, and phosphatidylethanolamine. Major fatty acids are C_{16:0}, C_{16:1}, C_{18:1}, and C_{19:0 cyc}. Belongs to the class *Gamma*proteobacteria. Forms dark spots on the surface of raw, cured meat products.

The mol% G + C of the DNA is: 56.

Type species: *Carnimonas nigrificans* Garriga, Ehrmann, Arnau, Hugas and Vogel 1998, 685.

FURTHER DESCRIPTIVE INFORMATION

On tryptone soy agar, colonies appear white, convex, shiny, and circular. No pigmentation is visible in any culture media. Optimum temperature for growth is 28–30°C. No growth occurs at 5°C or 37°C. Growth occurs in the presence of 8% NaCl, but not at higher levels.

Indole production, reduction of nitrate, and the Voges–Proskauer reaction are negative. All strains produce acid from glucose, xylose, melibiose, maltose, and saccharose. No gas is pro-

duced from glucose. Additional physiological data are provided in Table BXII.γ.99.

Garriga et al. (1998) have reported that the main polar lipids are diphosphatidylglycerol, phosphatidylglycerol, and phosphatidylethanolamine. In addition, there are three unidentified components. The fatty acid profile contains major amounts of several saturated and unsaturated straight chain fatty acids and only minor fractions of 3-hydroxylated fatty acids. The main fatty acids are palmitic acid (C_{16:0}), which comprises 40% of total fatty acids, and the cyclopropanic acid C_{19:0 cyc}, which comprises 21.07%. Oleic acid (C_{18:1 ω9t}) comprises 7.45% and a C_{18:1 ω7c/ω9t/ω12t} group comprises 12.9%. Moreover, a C_{16:1} fatty acid comprises 6.7%, whereas only minor amounts of C_{12:1 3OH} (1.85%) and traces of C_{16:0 3OH} (0.46%) are found. No match was found after comparison with the microbial ID, TSBA library (Newark, DE, USA).

No pathogenic effect on mice is detectable 8 d after intraperitoneal injection of up to 1.9 × 10¹⁰ CFU of *Carnimonas nigrificans*. Under similar conditions, the bacteria used as controls (*Escherichia coli* HM-42, *Pseudomonas aeruginosa* HS-116, *Staphylococcus aureus* HS-93) show a concentration-dependent lethality.

The presence of dark spots on the surface of raw, cured meat products was first described by Hugas and Arnau (1987). In 1993, Arnau and Garriga identified a Gram-negative bacterium (now *C. nigrificans*) as being responsible for this defect. The defect, a rust-like color turning to black within hours, can be reproduced on comminuted pork meat containing salt (40 g/kg) and dex-

trose (20 g/kg) after inoculation of an overnight culture and storage in an aerobic environment at 30°C. The browning effect is increased by some amino acids, i.e., glycine, L-arginine, L-glutamine, and L-monosodium glutamate, when added (20 g/kg) to a meat mixture containing salt and dextrose. N-Acetyl-L-cysteine, L-cysteine, potassium metabisulfite, and propyl-3,4,5-trihydroxybenzoate (5 g/kg) are useful in the prevention of this defect (Arnau and Garriga, 2000). The browning produced by *C. nigrificans* with dextrose and amino acids and the inhibitory properties of some of the substances studied show similarities to the Maillard reaction. However, important differences exist in the temperature pattern, the oxygen effect, and carbohydrates involved.

ENRICHMENT AND ISOLATION PROCEDURES

The isolation of *C. nigrificans* from raw, cured meat products must be done at the beginning of browning; no recovery is possible later. The organism is capable of growing on Cetrimide agar and MacConkey agar; consequently, these selective media could be used for isolation procedures. Incubation should be performed under aerobic conditions at 30°C for 2–3 days. After isolation, *C. nigrificans* grows well on the usual nutrient media (e.g., tryptone soy broth, brain–heart infusion broth).

MAINTENANCE PROCEDURES

Cultures in routine use can be maintained in tryptone soy broth or agar at 4°C. Cultures may be preserved for long-term storage in tryptone soy broth with 20% glycerol at –80°C.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Comparative sequence analysis of 16S rRNA gene sequences has revealed a diagnostic sequence that can be used as a target site for specific amplification. The sequence of the oligonucleotide used as specific primer for *C. nigrificans* strain CTCBS1 is 5'-TAA CGT CCT TCA TGC CGG-3' (binding position 469–486 in the *E. coli* numbering system). This primer (bs1) has been checked for its specificity against more than 10,000 16S rRNA sequences by using the probe-checking software provided by the Ribosomal Database Project (Maidak et al., 1996). PCR and cycle conditions for the species-specific reaction with primer bs1 and universal

primer 616V are as follows: one initial cycle 94°C (120 s), followed by 32 cycles of 94°C (45 s), 50°C (90 s), and 72°C (120 s). Master mixes are prepared with reaction buffer containing 10 mM Tris-HCl, 1.5 mM MgCl₂, 50 mM KCl, pH 8.3, 200 nM dNTP, 0.5 pmol each primer, and Taq DNA polymerase (Boehringer, Mannheim, Germany).

Specific amplification of a 480 bp fragment using primer bs1 and 616V occurs only in *C. nigrificans* strains.

DIFFERENTIATION OF THE GENUS *CARNIMONAS* FROM OTHER GENERA

The closest relatives of *Carnimonas nigrificans* are *Zymobacter palmae* and the genus *Halomonas* (at least *Chromohalobacter*). Some physiological and morphological properties, e.g., relationship to oxygen, growth temperature, hydrolysis of starch, oxidase reaction, as well as flagellation and motility, are useful in differentiating *Carnimonas* from these organisms (Table BXII.γ.99 and Table BXII.γ.100).

The mol% G + C content of 56, the presence of ubiquinone 9 as the major respiratory lipoquinone, and the lack of C_{19:0 cyc} in the non-polar lipid profile are sufficient to exclude these organisms from *Oceanospirillum*. The mol% G + C content of *Carnimonas nigrificans* CTCBS1 differs by at least 4.2% from those of *Halomonas* and *Chromohalobacter*, and there are distinct differences in the fatty acid profiles of these organisms; the levels of C_{16:0} in the profiles of members of the genus *Halomonas* range from 15.5 to 32%, whereas the level in *Carnimonas nigrificans* CTCBS1 is 40%. Additionally, in contrast to *Halomonas*, no C_{17:0 cyc} could be detected in *Carnimonas nigrificans* CTCBS1.

TAXONOMIC COMMENTS

The phylogenetic tree, based on 16S rDNA sequences, is shown in Fig. BXII.γ.118. The topology of all organisms illustrated in the tree is based on maximum likelihood analysis of 16S rDNA sequences (Fig. BXII.γ.118) and is consistent with a tree previously published by Dobson and Franzmann (1996). The closest relatives of strain *Carnimonas nigrificans* are *Zymobacter palmae* with 93.3% sequence similarity and, to a lesser extent, species of the genus *Halomonas* (at least 91.9%) and *Chromohalobacter* (91.5%).

The phylogenetic tree shows that the closest relationship is

TABLE BXII.γ.99. Features differentiating the genus *Carnimonas* from other genera^a

Characteristic	<i>Carnimonas</i>	<i>Chromohalobacter</i>	<i>Halomonas</i>	<i>Oceanospirillum</i>	<i>Zymobacter</i>
Cells are rod-shaped	+	+	+	D	+
Cells are helical	–	–	–	D	–
<i>Relationship to oxygen:</i>					
Aerobic	+	+	+	+	–
Facultatively anaerobic	–	–	–	–	+
Acid production from hexoses	+	+	+	–	+
Motile	–	+	D	+	+
Oxidase	+	–	D	+	–
Growth at 37°C	–	+	+	+	+
Growth with 0% NaCl	+	–	D	–	–
Tolerates 15% NaCl	–	+	+	–	–
<i>Hydrolysis of:</i>					
Esculin	+	–	D	–	–
Starch	+	–	–	–	–
Violet to brown colonies on culture media	–	+	–	–	–
<i>Habitat:</i>					
Raw, cured meat	+	–	–	–	–
Palm sap	–	–	–	–	+
Coastal sea water	–	–	–	+	–
Solar salt facilities, intertidal estuaries, and hypersaline lakes	–	+	+	–	–

^a Symbols: see standard definitions.

TABLE BXII.γ.100. Other characteristics of the genus *Carnimonas*^a

Characteristic	Reaction or result
α-Glucosidase (PNPG), phenylalanine deaminase	+
Arginine dihydrolase, urease, DNase, lecithinase	—
Nitrate reduction	—
Hydrolysis of gelatin, casein, Tween 80	—
<i>Acid production from:</i>	
Amygdalin, esculin, galactose, melibiose, sucrose, salicin	+
L-Arabinose	—
Fructose, mannose, mannitol, melezitose, raffinose, ribose, trehalose	d
<i>Utilization of</i> ^b	
Adipate, L-arabinose, caprate malate, mannitol, N- acetylglucosamine	—
Citrate, mannose, maltose, gluconate	+

^aSymbols: see standard definitions.^bOn media containing vitamins and yeast extract.

to *Zymobacter palmae* ATCC 51623^T. *Zymobacter palmae* is the single species of the genus *Zymobacter* and the closest phylogenetic relative to strain CTCBS1 (Okamoto et al., 1993). The G + C content of its DNA is 55.8 mol%, in the range of that determined for CTCBS1. The profiles of the major fatty acids of the two strains are similar; however, a significant amount (5%) of a C_{12:0} fatty acid that might be unique to *Zymobacter palmae* is not detected in strain CTCBS1 or the other closely related genera.

FURTHER READING

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- Hugas, M. and J. Arnau. 1987. Aparición de manchas de color en la corteza y grasa del jamón durante el post-salado. In Arnau, Hugas and Monfort (Editors), *Jamón Curado, Aspectos Técnicos*, Institut de Recerca i Tecnologia Agroalimentàries, Monells, Spain. pp. 179–182.

List of species of the genus *Carnimonas*

- Carnimonas nigrificans*** Garriga, Ehrmann, Arnau, Hugas and Vogel 1998, 685^{VP}
nig.rif.i.cans. L. adj. *niger* black; L. v. *facere* to make; M.L. n. *nigrificans* black-making.

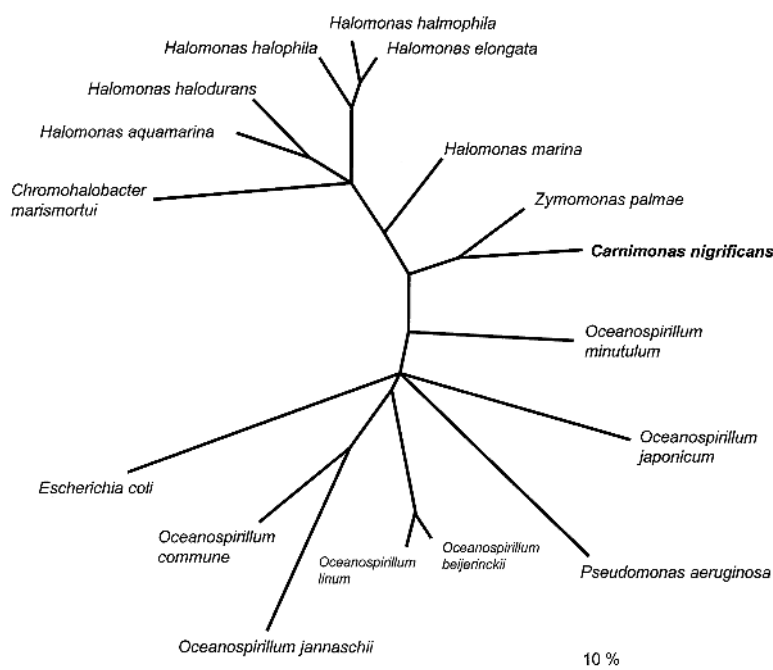
Characteristics are the same as for the genus. Colonies are nonpigmented, white, convex, shiny, and circular. Produce acid from glucose, fructose, maltose, xylose, melibiose,

and saccharose. Hydrolyze esculin and starch, but not gelatin, casein, or DNA. Voges–Proskauer negative. Arginine dihydrolase, urease, lecithinase, and phenylalanine deaminase negative. Do not produce indole.

The mol% G + C of the DNA is: 56 ± 0.3 (*T_m*).

Type strain: CTCBS1, CECT 4437, CIP 105703, NCIMB 13550.

GenBank accession number (16S rRNA): Y13299.

**FIGURE BXII.γ.118.** Phylogenetic relationships of *Carnimonas nigrificans* and selected representatives of the class *Gammaproteobacteria*.

Genus III. Chromohalobacter Ventosa, Gutierrez, García and Ruiz-Berraquero 1989, 384^{VP}
emend. Arahal, García, Ludwig, Schleifer and Ventosa 2001a, 1446)

ANTONIO VENTOSA

Chro.mo.ha'lo.bac'ter. Gr. n. *chroma* color; Gr. n. *halos* the sea, salt; M.L. n. *bacter* rod; M.L. masc.
n. *Chromohalobacter* colored salt rod.

Gram-negative, straight or sometimes slightly curved, rods (0.6–1.2 × 1.5–4.2 µm). Motile by polar or peritrichous flagella. Cells occur singly, in pairs, and in short chains. Spores are not formed. **Moderately halophilic.** Salt is required for growth. **The optimum salt concentration for growth is between 8 and 10%.** May grow at salt concentrations up to 30%. The broader ranges of temperature and pH observed for growth are 5–45°C (optimal 30–37°C) and pH 5.0–10.0 (optimal pH 7.5), respectively. **Aerobic.** Chemoorganotrophic. Catalase positive. **Oxidase negative. Most strains reduce nitrates.** Phenylalanine deaminase test is negative. Gelatin, starch, Tween 80, esculin, DNA, and tyrosine are not hydrolyzed. Acid is produced aerobically from D-glucose and other sugars. Carbohydrates, amino acids, and some polyols can serve as sole carbon sources. Colonies are cream to brown-yellow pigmented.

The mol% G + C of the DNA is: 62–66.

Type species: Chromohalobacter marismortui (ex Elazari-Volcani 1940) Ventosa, Gutierrez, García and Ruiz-Berraquero 1989, 384.

FURTHER DESCRIPTIVE INFORMATION

The genus *Chromohalobacter* was created by Ventosa et al. (1989) to accommodate seven moderately halophilic isolates obtained during the course of an extensive taxonomic study of bacteria isolated from salterns in the Mediterranean coast (Ventosa et al., 1982). These isolates showed phenotypic and chemotaxonomic features very similar to those of "*Chromobacterium marismortui*", a species originally described by Elazari-Volcani (1940) based on strains isolated from the Dead Sea. This species was described in the 7th edition of *Bergey's Manual of Determinative Bacteriology* (Breed et al., 1957), and in the 8th edition it was included in the genus *Chromobacterium* as a species *incertae sedis* since it did not produce violacein and lacked the typical flagellar arrangement of the genus *Chromobacterium* (Buchanan and Gibbons, 1974). It was not included in the Approved Lists of Bacterial Names (Skerman et al., 1980). Until recently, the genus *Chromohalobacter* was represented by a single species, *Chromohalobacter marismortui*, which was phylogenetically closely related to species of the genus *Halomonas* and included in the family *Halomonadaceae* (Mellado et al., 1995b). Very recent studies based on 16S rDNA gene sequence comparison and other phenotypic and molecular data supported the conclusion that two species previously described as members of the genus *Halomonas*, *Halomonas canadensis* and *Halomonas israelensis*, should be placed in the genus *Chromohalobacter*, as *Chromohalobacter canadensis* and *Chromohalobacter israelensis*, respectively (Arahal et al., 2001a). Finally, a study of two strains previously assigned to the species *Halomonas elongata*, strains DSM 3043 and ATCC 33174, showed that they represent a different species of the genus *Chromohalobacter*, and named them as *Chromohalobacter salexigens* (Arahal et al., 2001b).

All species currently described as members of the genus *Chromohalobacter* have a specific requirement for Na⁺. They grow optimally in media containing 8–10% salt and are defined as moderately halophilic microorganisms (Ventosa et al., 1998). Few studies have been carried out with the type species *C. marismortui*;

however, *C. canadensis* and *C. israelensis* have been extensively used for physiological studies. These species were formerly designated as strain Ba₁ (Rafaeli-Eshkol, 1968), isolated from the Dead Sea (*C. canadensis*) and strain NRCC 41277, isolated by Matheson et al. (1976) as a medium contaminant (*C. israelensis*). More detailed information about their physiological and biochemical features can be found in a recent review (Ventosa et al., 1998).

To cope with the high external salinity, halophilic bacteria need to balance their cytoplasm with the osmotic pressure exerted by the external medium. However, in *C. canadensis* and *C. israelensis*, as in many other moderate halophiles, the sum of the apparent intracellular concentration of Na⁺ and K⁺ ions is much lower than the external concentration (Matheson et al., 1976; Goldberg and Gilboa, 1978). They accumulate intracellular organic compounds, called compatible solutes, such as glycine betaine (Rafaeli-Eshkol and Avi-Dor, 1968; Shkedy-Vinkler and Avi-Dor, 1975). However, the most wide spread compatible solutes found in halophilic bacteria, and in *Chromohalobacter* in particular, are ectoine and its β-hydroxy derivative, hydroxyectoine (Ventosa et al., 1998). During recent years, several studies have focused on the osmoregulatory mechanisms of *C. salexigens* DSM 3043. It was selected as an excellent model organism because it displays one of the widest ranges of salt tolerance found in nature. In order to maintain its internal osmolarity and generate turgor in environments with high salinities, *C. salexigens* accumulates compounds such as glycine betaine, choline or choline-*O*-sulfate, when present externally. However, the main osmoadaptation mechanism is the *de novo* synthesis of ectoine and hydroxyectoine (Cánovas et al., 1996, 1997, 1998b, 1999; Nieto et al., 2000). Genes involved in the biosynthesis of ectoine and the oxidation of choline to betaine have been recently characterized (Cánovas et al., 1998a, 2000). The ectoine synthesis genes of *C. salexigens* and *Halomonas elongata* have lower sequence homology values than was expected from two closely related halophilic microorganisms (Nieto et al., 2000).

The genome size, determined by pulsed-field gel electrophoresis, of three strains of *C. marismortui* (including the type strain) ranged from 1770 to 2295 kb and that of *C. israelensis* was estimated to be 2490 kb (Mellado et al., 1998). Several plasmids have also been reported (Ventosa et al., 1994; Mellado et al., 1995a; Vargas et al., 1995a). The basic replicon of the narrow-host-range plasmid pCM1 from *C. marismortui* has been sequenced and characterized in detail (Mellado et al., 1995a). Cloning and shuttle vectors useful for the genetic manipulation of these bacteria have been reported (Mellado et al., 1995c; Vargas et al., 1995a). Conjugation is the only genetic transfer mechanism that has been described for this genus (Vargas et al., 1997).

ENRICHMENT AND ISOLATION PROCEDURES

Selective procedures for the enrichment and isolation of species of the genus *Chromohalobacter* have not been reported. They can be isolated from hypersaline environments by direct plating on complex media supplemented with a salt mixture and incubation at 30–37°C for 7–10 d. The isolation medium described by Ventosa et al. (1982) can be used.

MAINTENANCE PROCEDURES

For short-term storage, the species of the genus *Chromohalobacter* can be maintained on complex medium containing 10% salts. An appropriate medium is described by Ventosa et al. (1989). Long-term storage of members of this genus can be carried out by freeze-drying, storage at -80°C , or storage in liquid nitrogen.

DIFFERENTIATION OF THE GENUS *CHROMOHALOBACTER* FROM OTHER GENERA

The genus *Chromohalobacter* belongs to the family *Halomonadaceae* and is closely related to the genus *Halomonas* (Fig. BXII.γ.119), as well as to *Zymobacter* and *Carnimonas*. Since the genus *Halomonas* is taxonomically heterogeneous and contains greater than 20 species, it is very difficult to differentiate these two genera based on phenotypic or chemotaxonomic characteristics. Typical features of species of the genus *Chromohalobacter* are their optimal growth in media containing 8–10% salt and characteristic results for the biochemical tests reported in the genus description. Species of the genus *Chromohalobacter* can be phylogenetically differentiated from other related genera by comparison of 16S rRNA gene sequences.

Differential features of the four currently recognized species of the genus *Chromohalobacter* are given in Table BXII.γ.101.

ACKNOWLEDGMENTS

I am grateful to Drs. R.H. Vreeland and J.T. Staley for the critical reading of this article and to Dr. D.R. Arahal for help with the phylogenetic analysis.

FURTHER READING

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- Ventosa, A., J.J. Nieto and A. Oren. 1998. Biology of moderately halophilic aerobic bacteria. *Microbiol. Mol. Biol. Rev.* 62: 504–544.

List of species of the genus *Chromohalobacter*

1. ***Chromohalobacter marismortui*** (ex Elazari-Volcani 1940) Ventosa, Gutierrez, García and Ruiz-Berraquero 1989, 384^{VP} *ma.ris.mor'tu.i.* L. gen. n. *maris* of the sea; L. adj. *mortuus* dead; M.L. gen. n. *marismortui* of the Dead Sea.

See Table BXII.γ.101 and generic description for many features. Cells are Gram negative, rod shaped, and sometimes slightly curved, the length varies with the concentra-

tion of salt; at 10% salts, the cells are $0.6\text{--}1.0 \times 1.5\text{--}4.0$ μm ; at higher and lower salt concentrations, the cells are longer. Cells occur singly or in pairs. Motile by means of peritrichous flagella. Spores are not formed. On solid, complex media containing 10% salts, colonies are circular, convex, smooth, entire, and concentrically ringed with dark brown centers followed by bluish brown, grayish brown, and

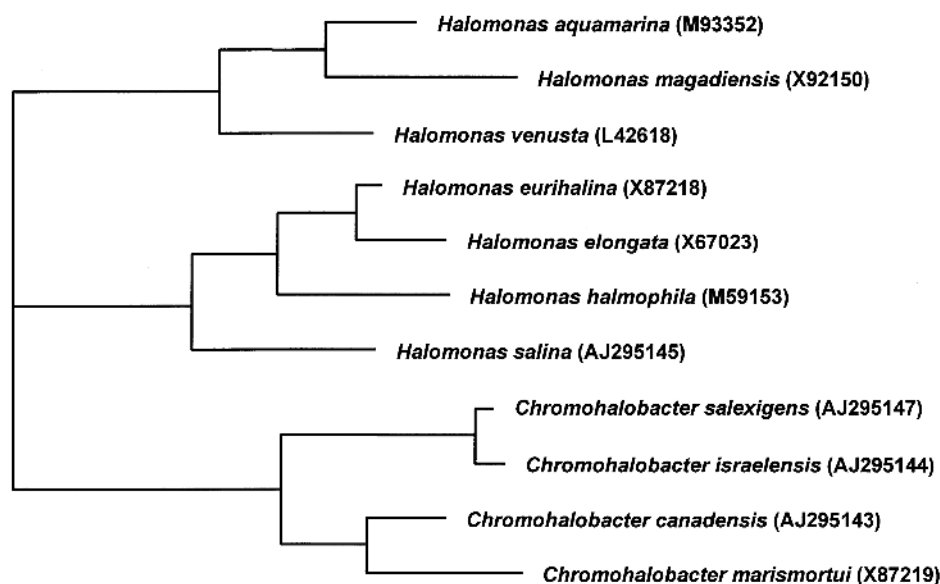


FIGURE BXII.γ.119. Phylogenetic tree derived from analysis of the 16S rRNA gene sequences of the type strain of species of the genus *Chromohalobacter* and some representative related species of the genus *Halomonas*. Figures in parentheses indicate the accession numbers of 16S rRNA gene sequence data. Bar = 5% sequence difference.

TABLE BXII.γ.101. Differential characteristics of species of the genus *Chromohalobacter*^a

Characteristic	1. <i>C. marismortui</i>	2. <i>C. canadensis</i>	3. <i>C. israelensis</i>	4. <i>C. salexigens</i>
Indole production	—	+	+	—
H ₂ S production	—	—	—	+
Simmons citrate	+	—	—	+
Hydrolysis of casein	—	—	—	+
Urease	—	—	—	+
Lysine decarboxylase	—	+	+	—
Ornithine decarboxylase	—	+	+	—
Acid production from:				
Arabinose	—	+	+	+
Maltose	+	—	+	+
Sucrose	+	—	+	+
D-Trehalose	+	—	—	—
Growth on:				
D-Cellobiose	—	+	+	—
Sucrose	+	—	+	+

^aFor symbols see standard definitions.

yellow rings. They produce a yellow pigment and a violet-blue pigment which is not violacein. Production of pigment is favored by suboptimal growth temperatures, glycerol, and a salt concentration of 10%. In liquid medium containing 10% salts, turbidity and a pellicle are produced; usually brown-to-yellow pigments are also produced.

Grows in media containing 1–30% salts, with an optimum at about 10% salts. Growth occurs at 5–45°C (optimum temperature is 37°C) and pH 5–10 (optimal growth at pH 7.5) on solid media containing 10% salts. Acid is produced without gas under aerobic conditions in marine oxidation–fermentation medium supplemented with 10% salts and D-glucose, D-galactose, maltose, lactose, D-arabinose, D-xylose, sucrose, trehalose, glycerol, and D-mannitol. Generally, nitrate is reduced to nitrite (except the type strain). Gelatin, casein, starch, esculin, tyrosine, Tween 80, and DNA are not hydrolyzed. Phosphatase, Voges–Proskauer, β-galactosidase, phenylalanine deaminase, and arginine dehydrolase tests are negative.

The following compounds are utilized as sole carbon and energy sources: dulcitol, D-fucose, D-galactose, D-glucuronate, D-glucose, glutamate, *meso*-inositol, maltose, D-mannitol, D-mannose, pyruvate, D-ribose, sucrose, D-sorbitol, and D-xylose. The following compounds are not utilized as sole carbon and energy sources: *N*-acetylglucosamine, amygdalin, DL-α-aminobutyrate, butyrate, cellobiose, citrate, esculin, *p*-hydroxybenzoate, hippurate, inulin, malonate, melibiose, oxalate, raffinose, salicylate, salicin, and D-tartrate. The following compounds are utilized as sole carbon, nitrogen, and energy sources: L-alanine, DL-arginine, L-glutamine, L-ornithine, L-proline, putrescine, and L-serine. The following compounds are not utilized as sole carbon, nitrogen, and energy sources: L-allantoin, betaine, creatine, ethionine, L-isoleucine, L-leucine, phenylalanine, sarcosine, L-threonine, and L-valine.

Isolated from the Dead Sea and marine salterns.

The mol% G + C of the DNA is: 62.1–64.9 (*T_m*).

Type strain: ATCC 17056, CCM 3518, DSM 6770, NCIMB 8731.

GenBank accession number (16S rRNA): X87219.

Additional Remarks: Other sequences: strain A-65 (X87220), strain A-100 (X87221), strain A-492 (X87222), and strain T1093 (U78719).

2. ***Chromohalobacter canadensis*** (Huval, Latta, Wallace, Kushner and Vreeland 1996) Arahal, García, Ludwig, Schleifer and Ventosa 2001a, 1447^{VP} (*Halomonas canadensis* Huval, Latta, Wallace, Kushner and Vreeland 1995, 1130.) *ca.na.den'sis*. M.L. n. *canadensis* pertaining to Canada, the country where the type strain was isolated.

See Table BXII.γ.101 and generic description for many features. Cells are Gram negative, straight or curved rods with rounded ends, 0.6–1.2 × 2.0–3.8 μm; cells occur singly, in pairs, or in short chains. Pleomorphic cells are not present. Both culture age and salt concentration affect chain length. Motile by a single polar flagellum. Spores are not formed. On solid complex media with 8% salts, colonies are translucent and convex with a smooth glistening surface, produce a nondiffusible white pigment, and have a diameter of 2–6 mm after 7 days of growth at 30°C.

Grows in media containing 3–25% NaCl from 15 to 30°C; grows in media with 8–32% NaCl at 45°C. Optimal growth at 7.5% NaCl. Growth occurs at 5–45°C (optimum temperature is 30°C) and from pH 5 to 9 on solid media containing 8% NaCl. Acid is produced from mannose, lactose, glycerol, and cellobiose but not from sucrose. Nitrate is reduced to nitrite. Phenylalanine deaminase and Voges–Proskauer tests are negative. Gelatin, starch, DNA, esculin, and agar are not hydrolyzed.

The following compounds are utilized as sole carbon and energy sources: acetate, citrate, DL-malate, succinate, lactate, D-glucose, xylose, L-arabinose, D-fructose, D-mannitol, *meso*-inositol, ethanol, mannose, esculin, glycerol, and gluconic acid. The following compounds are not utilized as sole carbon and energy sources: butyrate, isobutyrate, propionate, tartrate, benzoate, maltose, melibiose, L-sorbose, D-ribose, D-sorbitol, *n*-propanol, *n*-butanol, ethylene glycol, *n*-hexadecane, pyridine-1-oxide, formaldehyde, formamide, *N,N*-dimethylacetamide, and ethylenediamine. Isolated as a contaminant on Sehgal and Gibbons medium containing 25% NaCl.

The mol% G + C of the DNA is: 62 (*T_m*).

Type strain: ATCC 43984, DSM 6769, CECT 5385, CCM 4919, CIP 105571, NCIMB 13767, NRCC 41227.

GenBank accession number (16S rRNA): AJ295143.

3. ***Chromohalobacter israelensis*** (Huval, Latta, Wallace, Kushner and Vreeland 1996) Arahal, García, Ludwig, Schleifer

and Ventosa 2001a, 1447^{VP} (*Halomonas israelensis* Huval, Latta, Wallace, Kushner and Vreeland 1996, 1189.)
is.ra.el.en' sis. M.L. n. *israelensis* an inhabitant of Israel.

See Table BXII.γ.101 and generic description for many features. Gram negative. Cells are straight rods with rounded ends, $0.6\text{--}0.9 \times 1.5\text{--}4.2 \mu\text{m}$, occurring singly and occasionally in pairs or short chains. Pleomorphic cells are not present. On solid complex medium with 8% NaCl colonies are flat with entire margins, produce a nondiffusible cream pigment, and have 2–6 mm in diameter after 7 days.

Grows in media containing 3.5–20% NaCl (optimum at 8% NaCl). Growth occurs at 15–45°C (optimum temperature is 30°C) and from pH 5 to 9. Acid is produced without gas from mannose, lactose, glycerol, cellobiose, and sucrose. Nitrate is reduced to nitrite. Phenylalanine deaminase, Voges-Proskauer, and methyl red tests are negative. Gelatin, starch, esculin, and DNA are not hydrolyzed.

The following compounds are utilized as sole carbon and energy sources: cellobiose, lactose, mannose, esculin, glycerol, acetate, starch, gluconic acid, and sucrose.

The mol% G + C of the DNA is: 65 (T_m).

Type strain: ATCC 43985, DSM 6768, CECT 5287, CCM 4920, CIP 106853, NCIMB 13766.

GenBank accession number (16S rRNA): AJ295144.

4. **Chromohalobacter salexigens** Arahall, García, Vargas, Cánovas, Nieto and Ventosa 2001b, 1460^{VP}
sal.ex'i.gens. L. n. *sal* salt; L. v. *exigo* to demand; M.L. part. adj. *salessigens* salt-demanding.

See Table BXII.γ.101 and generic description for many features. Cells are Gram-negative rods, $0.7\text{--}1.0 \times 2.0\text{--}3.0 \mu\text{m}$; cells occur singly or in pairs. Motile. Spores are not formed. On solid, complex media containing 10% salts, colonies are cream, opaque, and circular and less than 2 mm in diameter; spreading may occur after extended incubation. In liquid medium containing 10% salts, a homogeneous turbidity is produced.

Genus IV. Zymobacter Okamoto, Taguchi, Nakamura, Ikenaga, Kuraishi and Yamasato 1995, 418^{VP} (Effective publication: Okamoto, Taguchi, Nakamura, Ikenaga, Kuraishi and Yamasato 1993, 336)

TOMOYUKI OKAMOTO, HIROSHI KURAISHI AND KAZUhide YAMASATO

Zy.mo.bac' ter. Gr. n. *zyme* leaven, ferment; M.L. n. *bacter* masc. equivalent of Gr. neut. n. *bakterion* rod ♂ sc. n. *Zymobacter* the fermenting rod.

Rod-shaped cells with rounded ends, $1.3\text{--}2.4 \times 0.7\text{--}0.9 \mu\text{m}$; usually single. **Motile by peritrichous flagella** that are nonsheathed. Gram negative. **Facultatively anaerobic**. Chemoorganotrophic. **Grow on and ferment 1 mol of glucose or the hexose moiety of maltose to produce approximately 2 mol each of ethanol and CO₂, with trace amounts of acids**. Ferment hexoses, α-linked di- and trisaccharides, and sugar alcohols. Growth occurs at pH 4.7–8.1. Catalase positive. **The predominant cellular fatty acids are oleic acid, cyclopropanoic acid of C_{19:0}, and palmitic acid. The hydroxylated acid is characteristically C_{12:0} 3OH. The quinone system is ubiquinone-9.**

The mol% G + C of the DNA is: 55.4–56.2 (HPLC).

Type species: **Zymobacter palmae** Okamoto, Taguchi, Nakamura, Ikenaga, Kuraishi and Yamasato 1995, 418 (Effective publication: Okamoto, Taguchi, Nakamura, Ikenaga, Kuraishi and Yamasato 1993, 336.)

Grows in media containing 0.9–25% salts, and optimal growth at 7.5–10% salts. Growth occurs at 15–45°C (optimum temperature is 37°C) and from pH 5 to 10 on liquid media containing 10% salts (optimal growth at pH 7.5). Acid is produced from L-arabinose, D-fructose, D-galactose, glycerol, D-glucose, lactose, maltose, D-mannose, sucrose, and D-xylose, but not from trehalose. Nitrate is reduced to nitrite but nitrite is not reduced. Gelatin, starch, esculin, DNA, and Tween 80 are not hydrolyzed. Methyl red positive. Indole and acetoin are not produced. Phenylalanine deaminase, lysine decarboxylase, and ornithine decarboxylase are not produced.

The following compounds are utilized as sole carbon and energy sources: L-arabinose, erythritol, D-fructose, D-galactose, D-glucose, maltose, D-mannitol, D-mannose, D-sorbitol, sucrose, D-trehalose, D-ribose, D-xylose, ethanol, glycerol, *meso*-inositol, dulcitol, acetate, α-aminovalerate, α-ketoglutarate, citrate, fumarate, DL-glycerate, glutamate, malate, malonate, propionate, D-saccharate, succinate, and D-tartrate. The following compounds are not utilized as sole carbon and energy sources: adonitol, cellobiose, L-fucose, α-lactose, D-melibiose, D-raffinose, L-rhamnose, galactosamine, gluconolactone, inulin, DL-α-aminobutyrate, butyrate, caprylate, lactate and oxalate. The following compounds are utilized as sole sources of carbon, nitrogen and energy: L-arginine, L-asparagine, betaine, glycine, L-glutamine, L-lysine, L-ornithine, L-proline, and L-serine. The following compounds are not utilized as sole sources of carbon, nitrogen and energy: L-alanine, creatine, L-methionine, putrescine, sarcosine, L-threonine, and L-valine.

Isolated from salterns.

The mol% G + C of the DNA is: 64.2–66.0 (T_m).

Type strain: ATCC BAA-138, CECT 5384, CCM 4921, CIP 106854, DSM 3043, NCIMB 13768.

GenBank accession number (16S rRNA): AJ295146.

Additional Remarks: Other sequence: AJ295147 (*C. salessigens* ATCC 33174).

FURTHER DESCRIPTIVE INFORMATION

The morphology of the cells is depicted in Fig. BXII.γ.120. The cells can possess as many as 20 peritrichous flagella.

When grown on MY agar¹, the colonies are round, entire, smooth, opaque, and milky white. Colonies of similar size develop under aerobic and anaerobic growth conditions. When the organism is grown in MY broth, the medium is slightly turbid, becoming cloudy with flocculent cells during the late log phase growth, then transparent due to sedimentation of cells during the stationary phase. No pellicle is formed. Growth is better in static cultures than in shaken cultures. *Zymobacter* requires nic-

1. MY agar (per liter): yeast extract (Difco), 10.0; maltose, 20.0; KH₂PO₄, 2.0; NaCl, 5.0; agar (Difco), 15.0; pH 6.0.

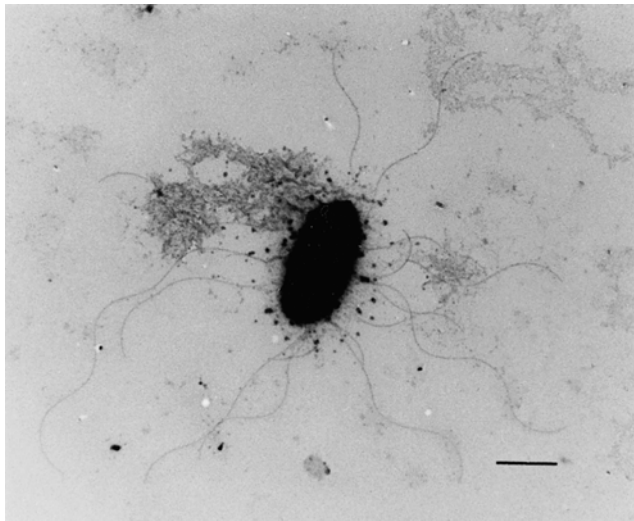


FIGURE BXII.γ.120. Electron micrograph of peritrichous flagella of *Zymobacter palmae* T109^T. Bar = 1 μm.

otinic acid for growth. Growth does not occur in the absence of a sugar or sugar alcohol.

Growth occurs at 15–37°C (optimum, 30°C) and pH 4.7–8.1 (optimum, 6.0). The organisms are neither halophilic nor halo-tolerant, but they are considerably tolerant to ethanol and pro-duce 5.8% ethanol from maltose after 6 d of fermentation.

The organism produces more than 0.5% (w/v) ethanol in the culture broth from glucose, fructose, maltose, sucrose, melibiose, raffinose, sorbitol, mannitol, and—depending on the strain—mannose and galactose. The specific growth rate on glucose is higher than that on maltose at sugar concentrations less than 7.5% (w/v). Growth is poor at glucose concentrations greater than 15% (w/v). The specific growth rate on maltose is substan-tially similar over a concentration range of 2.5–20% (w/v) (Table BXII.γ.102). Growth is initiated at 50% (w/v) maltose, but not at 25% (w/v) glucose; this divergence may be due to the differing osmotic pressures exerted by these sugars.

The genera *Zymomonas*, *Saccharobacter*, and *Zymobacter* are unique among bacteria in terms of their energy-yielding metab-olism, which produces ethanol, CO₂, and small amounts of other products. *Zymomonas* has been studied as a potential bacterium for the industrial production of ethanol, but its spectrum of fermentable sugars is limited to glucose, fructose, and sucrose (Ingram et al., 1998). *Zymobacter* (and *Saccharobacter*) may be pref-erable to *Zymomonas* in this regard, as it has a wider profile of fermentable sugars, especially maltose and raffinose.

Palmitic acid (C_{16:0}) comprises 46–54% of the whole cell fatty acids. Oleic acid (C_{18:1 ω9c}) C_{19:0 cyc} account for 30–42% (Table BXII.γ.103).

ENRICHMENT AND ISOLATION PROCEDURES

Okamoto et al. (1993) have used a two-step procedure to first isolate ethanol-tolerant bacteria and then select maltose-fer-menting, ethanol-producing isolates to obtain *Zymobacter* strains. The composition of the isolation medium is (g/l): yeast extract (Difco), 10.0; glucose, 20.0; KH₂PO₄, 2.0; Bacto agar (Difco), 15.0 g; pH 6.0. Ethanol is added to a concentration of 5% (v/v). To prevent multiplication of fungi and yeasts, the antibiotic Ka-bicidine (Wako Pure Chemical Industries, Osaka, Japan) is added to a concentration of 100 mg/l. Diluted samples are plated onto the medium and incubated at 30°C for 4 d. Isolates are incubated at 30°C for 4 d in MY medium, and those that produce concen-trations of ethanol in excess of 2% are selected. Four strains have been obtained from palm sap from among 500 samples of plant sap, fermented foods, and alcoholic beverages collected from tropical and subtropical regions of Brazil, Indonesia, Japan, Thai-land, and other countries.

MAINTENANCE PROCEDURES

Zymobacter cultures can be maintained by serial transfer every 2 weeks on MY agar and stored at 4°C. *Zymobacter* can be preserved for many years by lyophilization, L-drying, or freezing with 50% glycerol at –80°C.

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Characterization procedures for *Zymobacter* strains have been de-scribed by Okamoto et al. (1993). Conventional taxonomic fea-tures can be determined through standard methods. Test media that cannot support the growth must be supplemented with nic-otinic acid (1 mg/l) and/or yeast extract (10.0 g/l), as well as with glucose or maltose. Flagellation is observed when cells are grown at 20°C for 12 h in brain–heart infusion broth (Difco) supplemented with nicotinic acid (1 mg /l), maltose (20.0 g/l), and liver infusion (10% (v/v); pH 6.0. The liver infusion is pre-pared by gently boiling 100 g of sliced cattle liver in 250 ml water for 30 min and filtering through cloth. Cells can be shadowed with platinum and visualized by transmission electron micro-scopy. The basal liquid medium for testing growth-factor require-ment and fermentation of carbon compounds is composed of (g/l): K₂HPO₄, 7.0; KH₂PO₄, 2.0; MgSO₄·7H₂O, 0.1; (NH₄)₂SO₄, 1.0; pH 6.0. Maltose is added when assessing growth-factor re-quirements, and nicotinic acid is added when evaluating the fer-mentation of carbon compounds.

DIFFERENTIATION OF THE GENUS ZYMOBACTER FROM OTHER GENERA

Table BXII.γ.104 indicates the salient features that differentiate *Zymobacter* from other phylogenetically related genera. Char-acteristics that distinguish *Zymobacter* from *Halomonas* and *Chromo-halobacter* are ethanol and CO₂ production from glucose, growth in relationship to oxygen, halophilic property, and habitat.

TABLE BXII.γ.102. Specific growth rate of *Zymobacter palmae* T109^T as affected by the initial concentration of glucose and maltose

Sugar	Specific growth rate μ (h ⁻¹) at a sugar concentration of (%):							
	2.5	5	7.5	10	12.5	15	17.5	20
Glucose	0.47	0.43	0.38	0.26	0.19	0.08	0.07	0.05
Maltose	0.2	0.24	0.25	0.29	0.26	0.26	0.2	0.2

Phenotypically and ecologically, *Zymobacter* is similar to the ethanol-producing genera *Zymomonas* and *Saccharobacter*, both of which occur in sugar-containing niches, such as plant juices. The features that differentiate *Zymobacter* from these genera are highlighted in Table BXII.γ.105. *Zymobacter* is clearly differentiated from these two genera by flagellation, sugar fermentation, vitamin requirements for growth, osmotic tolerance, mol% G + C of DNA, and ubiquinone system.

TAXONOMIC COMMENTS

Zymobacter belongs to the family *Halomonadaceae*. The 16S rRNA gene of *Zymobacter palmae* T109^T (accession number D14555) is

TABLE BXII.γ.103. Cellular fatty acids of *Zymobacter palmae*^a

Fatty Acid	Range, %	Mean, %
C _{12:0}	5–6	5
C _{14:0}	3–4	4
C _{16:0}	46–54	51
C _{16:1}	<0.5	<0.5
C _{18:0}	4–6	5
C _{18:1 ω9c}	8–21	11
C _{19:0 cyc}	21–26	24
C _{10:0 3OH}	<0.5	<0.5
C _{12:0 3OH}	8–13	12

^aFrom an examination of four strains. The percentages of fatty acid compositions were calculated on the basis of total non-hydroxylated acids.

TABLE BXII.γ.104. Differential characteristics of the genus *Zymobacter* and other genera in the family *Halomonadaceae*^a

Characteristics	<i>Zymobacter</i>	<i>Halomonas</i>	<i>Chromohalobacter</i>
Motility	+	D	+
<i>Flagella:</i>			
Peritrichous	+	+	+
Polar	—	+	—
Mol% G + C of DNA	55.4–56.2	52–68	62.1–64.9
<i>Growth in relationship to oxygen:</i>			
Facultatively anaerobic	+	—	—
Aerobic	—	+ ^b	+
Glucose fermented to ethanol/CO ₂	+	—	—
Halophilic	—	+	+
<i>Habitat:</i>			
Palm sap	+	—	—
Saline environment	—	+	+

^aSymbols: see standard definitions.

^bSome strains can grow anaerobically in the presence of nitrate.

TABLE BXII.γ.105. Differential characteristics of *Zymobacter palmae* and other ethanol-producing bacteria^a

Characteristics	<i>Zymobacter palmae</i>	<i>Zymomonas mobilis</i>	<i>Saccharobacter fermentatus</i>
Cell diameter, μm	0.7–0.9	1.0–1.4	0.5–0.9
Cell length, μm	1.3–2.4	2.0–6.0	1.0–1.9
Motility	+	— (usually)	+
<i>Flagellar arrangement:</i>			
Peritrichous	+	—	+
Polar, 1–4 flagella	—	+	—
Utilization of citrate	—	—	+
Arginine dihydrolase	—	—	+
Phenylalanine deaminase	—	—	+
β-galactosidase	—	—	+
<i>Utilization of:</i>			
L-Arabinose	—	—	+
D-Xylose	—	—	+
Mannose	+	—	—
Maltose	+	—	+
Sucrose	+	d	+
Melibiose	+	—	+
L-Rhamnose	—	—	+
Trehalose	—	—	+
Lactose	+	—	+
Starch	—	—	+
<i>Growth factor requirement:</i>			
Nicotinic acid	+	—	—
Biotin	—	+	—
Pantothenate	—	+	—
<i>Osmotic tolerance to:</i>			
Glucose	<25%	40%	35%
Maltose	50%	—	—
Mol% G + C of DNA	55.4–56.2	47.5–49.5	63.3–63.8
Ubiquinone system	Ubiquinone-9	Ubiquinone-10	—
<i>Characteristic non-hydroxylated acids, %</i>			
C _{16:0}	51	12	—
C _{18:1} (oleic)	11	—	—
C _{18:1} (vaccenic)	—	69 ^b	—
C _{19:0 cyc}	24	—	—

^aSymbols: see standard definitions.

^bData from Carey and Ingram (1983).

TABLE BXII.γ.106. The signature characteristics of *Zymobacter palmae* and related bacteria

Position(s)	<i>Zymobacter palmae</i> ^a	<i>Halomonas</i> ^b	<i>Chromohalobacter marismortui</i> ^c	<i>Oceanospirillum linum</i>
76–93	6-bp stem	6-bp stem	6-bp stem	6-bp stem
484	A	A	A	G
486	C	C	C	U
640	G	G	G ^d	G
660 ^e	A	A	A	A
745	U	U	U	U
668 ^f	A	A	A	G
738	U	U	U	C
669 ^g	A	A	A	U
737	U	U	U	A
776	U	U	U	G
1124	U	U	G ^h	U
1297	U	U	U	G
1298	C	C	C	U
1423	A	A	A	G
1424	U	C	C	U
1439	C	U	U	C
1462	G	A	A ⁱ	G
1464	U	C	C ^j	C

^a*Zymobacter palmae* T109^T (D14555).^bData from Dobson and Franzmann, 1996.^c*Chromohalobacter marismortui* ATCC 17056^T (X87219).^dG for strains T1093 (U78719), A-65 (X87220), and A-492 (X87222).^eNucleotides at 660 and 745 form a base pair.^fNucleotides at 668 and 738 form a base pair.^gNucleotides at 669 and 737 form a base pair.^hG for strains ATCC 17056 (X87219), A-65 (X87220), A-100 (X87221), and A-492 (X87222); U for strain T1093 (U78719).ⁱStrains T1093 (U78719), A-65 (X87220), and A-492 (X87222).^jStrains T1093 (U78719), A-65 (X87220), and A-492 (X87222).

89.3–92.5% similar to that of *Halomonas* and 91.1% similar to that of *Chromohalobacter* (Dobson and Franzmann, 1996).

Zymobacter belongs to the family *Halomonadaceae*, which is in the class *Gammaproteobacteria*, and includes the marine genera *Halomonas* and *Chromohalobacter*. In contrast with these genera, *Zymobacter* is of terrestrial origin, is neither halophilic nor halotolerant, is facultatively anaerobic, and produces ethanol and CO₂. The nucleic acid similarity among the 16S rRNAs of these genera is about 90%, and, like *Halomonas* and *Chromohalobacter*, *Zymobacter* demonstrates the 15 signature sequences necessary to place it in the family *Halomonadaceae* (Dobson and Franzmann, 1996) (Table BXII.γ.106). *Zymobacter* (type strain of *Zymobacter palmae*) differs from *Chromohalobacter* and *Halomonas* at 16S rRNA positions 1424, 1439, 1462, and 1464 (*E. coli* numbering system), at which *Halomonas* has the sequences described as characteristic of its genera (Dobson and Franzmann, 1996) (Table BXII.γ.106). The 16S rRNA similarity of *Zymobacter* to another marine genus, *Oceanospirillum*, is 88.7–89.7%—similar to that between *Zymobacter* and *Halomonas* and *Chromohalobacter*. However, *Oceanospirillum* is phylogenetically distinct; it lacks the signature sequences of the *Halomonadaceae* (Dobson and Franzmann, 1996). Chemotaxonomically, the *Halomonadaceae* share the ubiquinone-9 respiratory system, whereas *Oceanospirillum* uses ubiquinone-8 (Sakane and Yokota, 1994). Because the genus *Zymobacter* and the species *Zymobacter palmae* were established in light of the taxonomic features of four strains isolated from a limited source (palm sap from the southern region of Japan), a precise circumscription of the attributes of this genus and species requires the isolation of strains from more varied sources.

The phenotypic features of *Zymobacter* are similar to those of the genera *Zymomonas* and *Saccharobacter*. All of these organisms are Gram negative, facultatively anaerobic, inhabit plant juice and the like, and produce ethanol and CO₂ from carbohydrates. Phylogenetically, *Zymobacter* is distinct from *Zymomonas*, which be-

TABLE BXII.γ.107. Other characteristics of *Zymobacter palmae*^a

Characteristics	Reaction or result
Catalase	+
Oxidase	—
Nicotinic acid required for growth	+
Methyl red test	+
Voges–Proskauer test	+
Utilization of citrate (Koser, Christensen)	—
Nitrate reduction	—
Indole production	—
Gelatin liquefaction	—
Hydrolysis of starch	—
Phenylalanine deaminase	—
Arginine dihydrolase	—
Lysine decarboxylase	—
Ornithine decarboxylase	—
α-Glucosidase	+
β-Galactosidase	—
Utilization of carbon sources:	
l-Arabinose	—
D-Xylose	—
D-Glucose	+
D-Fructose	+
D-Mannose	d
D-Galactose	d
Maltose	+
Sucrose	+
Melibiose	+
l-Rhamnose	—
Trehalose	—
Cellobiose	—
Lactose	—
Raffinose	+
Sorbitol	+
Mannitol	+
Dextrin	—
Inulin	—

^aSymbols: see standard definitions.

longs to the *Alphaproteobacteria*. Like *Zymobacter*, *Saccharobacter* is peritrichously flagellated, but it has a higher mol% G + C (63.5 versus 55.8). The generic description of *Saccharobacter* lacks a description of chemotaxonomic characteristics and phylogenetic

analysis. The phylogenetic relationship and exact differentiation of *Zymobacter* from *Saccharobacter* requires detailed comparative study.

List of species of the genus *Zymobacter*

1. ***Zymobacter palmae*** Okamoto, Taguchi, Nakamura, Ikenaga, Kuraishi and Yamasato 1995, 418^{VP} (Effective publication: Okamoto, Taguchi, Nakamura, Ikenaga, Kuraishi and Yamasato 1993, 336.)
pal' mae. L. gen. n. *palmae* of palm.

The characteristics are as given for the genus. See also Tables BXII.γ.104, BXII.γ.105, and BXII.γ.107. Isolated from palm sap in Okinawa Prefecture, Japan.

The mol% G + C of the DNA is: 55.4–56.2 (HPLC).

Type strain: T109, ATCC 51623, IAM 14233.

GenBank accession number (16S rRNA): D14555, AF211871.

Order IX. ***Pseudomonadales*** Orla-Jensen 1921, 270^{AL}

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

Pseu.do.mon.a.da'les. M.L. fem. n. *Pseudomonas* type genus of the order; -ales suffix to denote order; M.L. fem. n. *Pseudomonadales* the *Pseudomonas* order.

The order *Pseudomonadales* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rRNA sequences; the order contains the families *Pseudomonadaceae* and *Moraxellaceae*.

Aerobic chemoorganotrophs with respiratory metabolism. Most are motile by means of flagella. *Azomonas* spp. and *Azotobacter*

spp. fix nitrogen; *Azotobacter* spp. form cysts; *Moraxella* spp. inhabit the mucosa of animals and man.

Type genus: ***Pseudomonas*** Migula 1894, 237^{AL} (Nom. Cons., Opin. 5 of the Jud. Comm. 1952, 121.)

Family I. ***Pseudomonadaceae*** Winslow, Broadhurst, Buchanan, Krumwiede, Rogers and Smith 1917, 555^{AL}

GEORGE M. GARRITY, JULIA A. BELL AND TIMOTHY LILBURN

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The family *Pseudomonadaceae* was circumscribed for this volume on the basis of phylogenetic analysis of 16S rRNA sequences; the family contains the genera *Pseudomonas* (type genus), *Azomonas*, *Azotobacter*, *Cellvibrio*, *Mesophilobacter*, *Rhizobacter*, and *Rugamonas*. *Serpens* is also included.

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Genus I. ***Pseudomonas*** Migula 1894, 237^{AL} (Nom. Cons., Opin. 5 of the Jud. Comm. 1952, 121)*

NORBERTO J. PALLERONI

Pseu.do'mo.nas or *Pseu.do.mo'nas*. Gr. adj. *pseudes* false; Gr. n. *monas* a unit, monad; M.L. fem. n. *Pseudomonas* false monad.

Straight or slightly curved rods but not helical, 0.5–1.0 × 1.5–5.0 μm. **Most of the species do not accumulate granules of polyhydroxybutyrate**, but accumulation of polyhydroxyalkanoates of monomer lengths higher than C₄ may occur when growing on alkanes or gluconate. Do not produce prosthecae and are not surrounded by sheaths. No resting stages are known. Gram neg-

ative. **Motile by one or several polar flagella**; rarely nonmotile. In some species lateral flagella of short wavelength may also be formed. **Aerobic, having a strictly respiratory type of metabolism with oxygen as the terminal electron acceptor; in some cases nitrate can be used as an alternate electron acceptor**, allowing growth to occur anaerobically. **Xanthomonadins are not produced**. Most, if not all, species fail to grow under acid conditions (pH 4.5 or lower). Most species do not require organic growth factors. Oxidase positive or negative. Catalase positive. Chemoorganotrophic. **Strains of the species include in their composition the hydroxylated fatty acids C_{10:0} 3OH and C_{12:0}, and C_{12:0} 2OH, and ubiquinone Q-9**. Widely distributed in nature. Some species are pathogenic for humans, animals, or plants.

*Editorial Note: The literature search for the chapter on *Pseudomonas* was completed in January, 2000. During the course of unavoidable publication delays, a number of new species were described or reclassified after the chapter was completed. It was not possible to include these species in the text of to include their characteristics in the comparative tables. The reader is encouraged to consult the studies listed in the Further Reading section and the *International Journal of Systematic and Evolutionary Microbiology* (2000–2003).

longs to the *Alphaproteobacteria*. Like *Zymobacter*, *Saccharobacter* is peritrichously flagellated, but it has a higher mol% G + C (63.5 versus 55.8). The generic description of *Saccharobacter* lacks a description of chemotaxonomic characteristics and phylogenetic

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Note: The above-mentioned characteristics do not allow an absolute differentiation of this genus from other genera of aerobic pseudomonads¹ belonging to other ribosomal RNA groups. The following approaches add considerable solidity to the decision to assign newly isolated strains to *Pseudomonas*. One is the determination of sequence similarity in ribosomal RNA, which can be demonstrated either by hybridization techniques or, more practically, by nucleotide sequence determination. The second is determination of the fatty acid and ubiquinone composition. These two approaches will be discussed in separate sections of this chapter.

The mol% G + C of the DNA is: 58–69.

Type species: ***Pseudomonas aeruginosa*** (Schroeter 1872) Migula 1900, 884 (*Bacterium aeruginosum* Schroeter 1872, 126.)

FURTHER DESCRIPTIVE INFORMATION

General Comments The following sections provide further descriptive information on the properties of species assigned to the genus *Pseudomonas*, beyond the basic characteristics that have been highlighted in the above genus definition. Unfortunately, the comments that follow refer to only a few species, and the available information for these species is uneven. Thus, a literature survey shows that most references on *P. aeruginosa* are focused on its medical importance as an opportunistic pathogen; *P. putida* references are concentrated on its remarkable biochemical versatility; and those on *P. fluorescens* also highlight biochemical properties, some of which are directed to understanding its role in the promotion of plant growth. *P. stutzeri* and *P. mendocina* have attracted attention as denitrifiers, and, in recent years, an infamous side of their reputation is their potential activities as opportunistic pathogens.

Many of the species listed at the end of this chapter or in the tables are little known beyond the details published in the original species descriptions. In addition to these unavoidable limitations, the details that are given for the better known species of the genus represent, by necessity, a radical simplification of the mass of information found in the literature since the date of publication of the first volume of the *Manual* (Palleroni, 1984).

Cell morphology and fine structure The cells of *Pseudomonas* strains occasionally differ substantially in size and shape from the general definition. For instance, the original description of "*P. ovalis*" (now considered to be a synonym of *P. putida*) highlighted the oval shape of the cells, while the cells of other *P. putida* strains and of some plant pathogenic fluorescent species may far exceed the 5 µm indicated in the definition.

A morphological character that was used to differentiate members of the genus *Pseudomonas* from other aerobic pseudomonads was the inability of the former to accumulate endocellular granules of poly-β-hydroxybutyrate (PHB) when growing in media of low nitrogen content on various carbon sources. More recently, the fluorescent members of *Pseudomonas* were found to be able to accumulate polyhydroxyalkanoates (PHAs) composed of monomers of medium chain lengths (C₆ to C₁₂) when grown on carbon sources such as gluconate, alkanes, and alkenes (Huisman et al., 1989; Anderson and Dawes, 1990; Steinbüchel and Val-

entin, 1995). Further comments on this subject may be found in the section Procedures for Testing Special Characters, below.

Thin sections of cells of *Pseudomonas* show cell walls and membranes characteristic of Gram-negative bacteria. In freeze-etched preparations of *P. aeruginosa* cells, as many as nine layers can be defined, accounting for all the electron-dense and electron-transparent layers that can be observed in thin sections (Lickfield et al., 1972; Gilleland et al., 1973). Gilleland and collaborators have shown that the outer membrane of the cell wall could be split down the middle when the cells were freeze-etched in the presence of a cryoprotective agent (Gilleland et al., 1973), thus confirming an earlier interpretation of similar experiments. The outer layer thus separated from the outer membrane had in its inner face numerous spherical units, which are made up of protein and in some cases were aggregated in the shape of small rods. *P. aeruginosa* is extremely sensitive to the action of EDTA, which provokes cell lysis. The spherical elements were removed by EDTA treatment but could be restored by addition of magnesium ions, which also restored cell stability.

Flagella and pili (fimbriae) Typically, *Pseudomonas* cells have polar flagella. The flagellar insertion in some instances is not exactly polar but subpolar, and occasionally it may be difficult to differentiate the latter from the so-called degenerately peritrichous type observed in members of other genera. In addition to the polar flagella, lateral flagella of short wavelength may be produced by strains of some species (*P. stutzeri*, *P. mendocina*) and are shed much more easily than the polar flagella. Growth on solid media favors the formation of lateral flagella, which suggests that they may be involved in swarming of the population on solid surfaces (Shinoda and Okamoto, 1977).

The number of flagella has taxonomic importance. Based on the work performed at the University of California in Berkeley, the number of flagella (usually expressed as "one" or "more than one") was highly characteristic for all the strains of a species that have been examined. To get good results, it is advisable to follow well-controlled growth conditions and to express the results on a statistical basis, as indicated by Lautrop and Jessen (1964). Some practical recommendations and basic details of the technique described by these authors will be summarized in the section Procedures for Testing Special Characters. Unfortunately, they have not been followed in the description of most species. In Jessen's classical study of *P. aeruginosa*, 97% of the cells with stained flagella had only one flagellum, 3% two flagella at one pole, and only three cells (about 0.05% of the collection) had three flagella. According to the definition proposed by Lautrop and Jessen (1964), the group can be characterized as polar monotrichous. Only one strain deviated from the limit defined for monotrichous strains in having two flagella on 15% of the flagellated cells, instead of the accepted maximum of 10%.

The amino acid sequence of the flagella of a given species is not always homogeneous. An analysis of the virulence-associated locus *fliC* of *P. aeruginosa* revealed two types of flagellin genes, one variable and the other conserved. Differences in the flagellar sequences amount to about 35%, but the two proteins fold into similar structures during polymerization (Spangenberg et al., 1996). In two strains of *P. putida*, flagellins of apparent molecular masses of 81 kDa and 50 kDa, respectively, were identified. Within the chromosomal fragment encoding the large component, two genes homologous to two flagellin genes of *Salmonella typhimurium* and one gene homologous to a *P. aeruginosa* flagellin gene were identified. The deduced molecular mass of the product of

1. The word "pseudomonad" will be used quite often in this chapter. Although it may be clear to many, the precise meaning is given in the paper "The aerobic pseudomonads: a taxonomic study" by Stanier et al. (1966). It comprises all the species assigned to the genus *Pseudomonas* as classically defined. Ever since this publication, a fairly accurate definition has also been included in the editions of Webster Collegiate Dictionary.

the first flagellin gene was 68 kDa instead of 81 kDa, suggesting posttranslational modification. Both the N- and C-terminal sequences are conservative, while the middle section is variable (Winstanley et al., 1994). PCR amplification of flagellar genes of *P. aeruginosa* gave products that were analyzed for restriction fragment length polymorphism, and 13 groups could be defined, suggesting that this approach may provide a useful genetic marker for the study of genetic variation among closely related species (Winstanley et al., 1996).

Fimbriae (pili) of polar insertion have been reported for *P. aeruginosa* and *P. alcaligenes* in the early studies of Fuerst and Hayward (1969). No fimbriae have been observed in the strains of *P. fluorescens*, *P. chlororaphis*, and *P. putida* that have been examined.

The *P. aeruginosa* pili are about 6 nm wide, thinner than those of enteric bacteria. They act as receptors for various phages and are retractile (Bradley, 1972a, b), but not all the pili of this species have the property of withdrawing into the cell (Bradley, 1974). Fig. BXII.γ.121 shows the polar fimbriae of *P. aeruginosa*, whose genetic determinants are mobilized by the FP plasmids (Bradley, 1980b).

P. aeruginosa produces type-4 pili, which are found in a number of pathogenic bacteria and are involved in cell adhesion to epithelial cells (Woods et al., 1980; Doig et al., 1988). The adhesive region is located at the tip of the pilus, but this mode of attachment is not universal among fimbriae of different species (Smyth et al., 1996). In the C-terminal region of pilin there is a disulfide loop of 12–17 amino acids, and even though pili of different prototypes have little sequence similarity, the loops in all cases are structurally similar. However, the loops do not seem to be essential in maintaining the functionality of the binding domain. In addition to the fimbriae, exoenzyme S and flagella are additional adhesins of importance in the attachment process (Hahn, 1997).

Type-4 pili not only determine attachment to epithelial cells, but also cause a form of surface translocation called twitching motility (Henrichsen, 1975; Bradley, 1980a). Adhesion of *P. aeruginosa* cells to mucin is pilus-independent. A gene *fliO* complemented mutations in motility and adhesion to mucin (Simpson et al., 1995). A gene homologous to the *fliF* gene of other species has been identified in *P. aeruginosa*. It is involved in both motility and adhesion to mucin. Further characterization of the markers involved showed that the basal body structures of flagella also are important for adhesion (Arora et al., 1996).

Pilus formation is genetically very complex and depends on the expression of more than 30 genes distributed in at least six regions of the *P. aeruginosa* chromosome (Mattick et al., 1996; Tan et al., 1996; Watson et al., 1996a). The locations of a number of the pili genes have now been determined. Some of the genes encode proteins highly homologous to factors involved in protein secretion (Alm and Mattick, 1995, 1997; Alm et al., 1996b; Hahn, 1997). At least four different mechanisms are known to participate in the assembly of fimbriae from subunits (Vu-Thien et al., 1996).

Some years ago, it was found that the amino terminal sequence of *P. aeruginosa* pili had marked similarity with that of *Neisseria meningitidis* and *Moraxella nonliquefaciens* (Buchanan and Pearce, 1979). One of the gene clusters encodes proteins strikingly similar to chemotaxis components of enteric bacteria and the gliding *Myxococcus xanthus*. In *M. xanthus*, type-4 pili have a role in what has been called "social motility" (Darzins, 1994; Darzins and Russell, 1997; Wu et al., 1997b). Work done with

polyclonal antibodies raised against pilins expressed by *P. aeruginosa*, *Moraxella bovis*, *Neisseria gonorrhoeae*, *Dichelobacter nodosus*, and *Vibrio cholerae* showed that these polypeptides have conserved antigenic and, in most instances, immunogenic determinants, which are located in the highly similar amino-terminal domains (Patel et al., 1991).

AlgR is considered to be a regulator of alginate production in *P. aeruginosa*; in addition, it is required for twitching motility. Adjacent to the *algR* gene, a sensor gene (*fimS*) has been identified, and AlgR and FimS appear to represent another example of the two-component signal transduction systems that have been identified in bacteria. AlgU, an alternate sigma factor, also affects both alginate production and twitching motility, indicating that type-4 fimbriae and alginate, two important virulence determinants, are closely related (Whitchurch et al., 1996).

A gene region identified in nonfimbriated, phage-resistant transposon mutants of *P. syringae* pathovar *phaseolicola* (in this treatment described as a pathovar of *P. savastanoi*) has a high degree of similarity with a *P. aeruginosa* region required for pilus type-4 production (Alm and Mattick, 1995). Pili mutants that have lost the twitching motility characteristic of type-4 fimbriae, could be complemented in *trans* by homologous or heterologous subunits from the same strains (PAO or PAK) and from strains of other species such as *Dichelobacter nodosus* (Watson et al., 1996b).

Composition of cell envelope The cytoplasmic membrane separates the bacterial cytoplasm from other components of the cell envelopes, and its efficiency as a barrier is related to the lipid bilayer composition, which may undergo relatively rapid changes to adjust its fluidity to changes in the composition of the medium and the physical conditions of the environment.

In contrast to the cytoplasmic membrane, the structure of the outer membrane is asymmetrical. It is composed of a special lipid (lipopolysaccharide, LPS) with hydrocarbon chains giving low fluidity to the inner region of the LPS. Toward the outside region, chains of carbohydrates give to the whole cells a specific antigenic identity. The outer membrane is quite effective as a barrier against passage of molecules such as the antibiotics, which may have a deleterious effect on the cells, but at the same it must be permeable to nutrients. This discrimination is effected by the presence of porins.

The outer membrane proteins of pseudomonads carry the designation Opr followed by a letter or a letter and number (Hancock and Carey, 1980). One monomeric *Pseudomonas* protein, OprF, is similar to OmpA of *E. coli*; it is highly antigenic and not specific toward the solutes that enter the cells (Nikaido, 1992). In *Pseudomonas*, porins are composed of monomeric proteins, and not trimeric proteins as in *E. coli*. The outer membrane of *P. fluorescens* can change its permeability in direct relationship with growth temperature, even though no difference in composition could be found to correlate with this transition. The effect was attributed to a change in the structure of the porins that affects their function, since a decrease in the growth temperature induced a reduction in the conductance in the major component, OprF, suggesting a structural change (De et al., 1997).

Protein fractions representing the OprF of two *P. fluorescens* strains were purified and characterized. A sequence similarity of 94% was found among OprFs of strains from origins as different as milk and soil (De et al., 1995). OprF of *P. aeruginosa* is a major outer membrane component, and as such it is potentially inter-

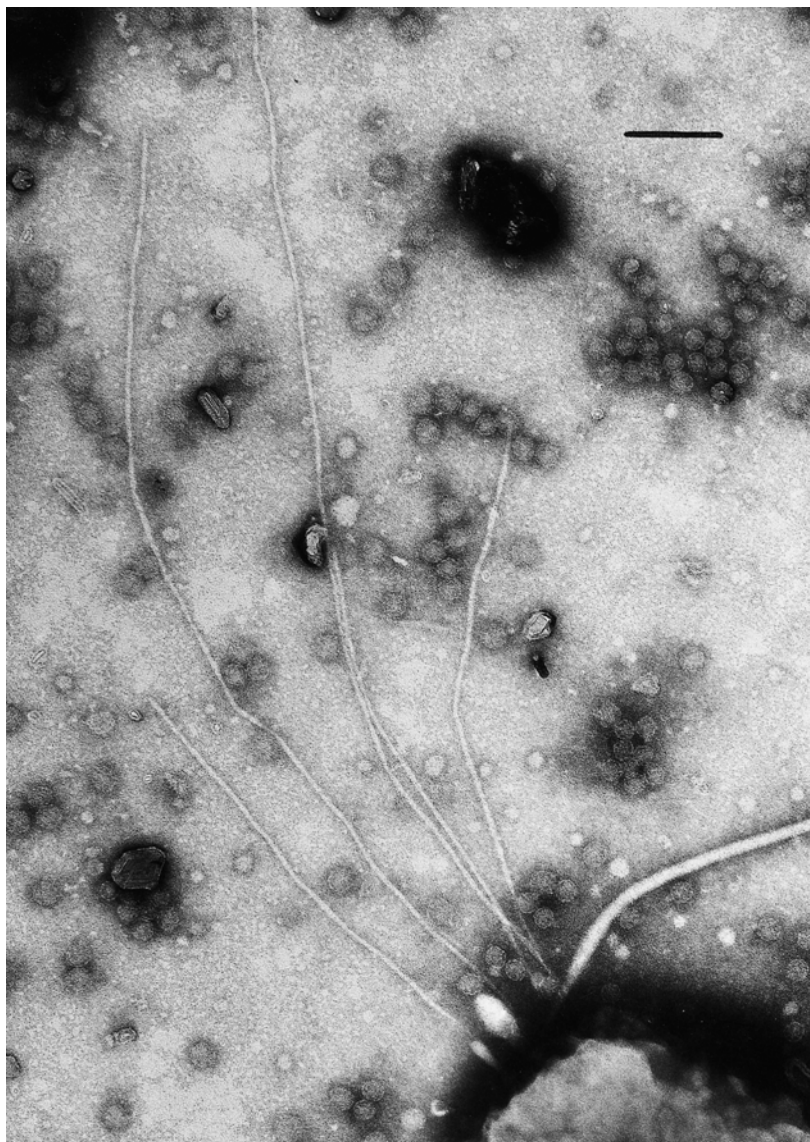


FIGURE BXII.γ.121. *Pseudomonas aeruginosa* strain PAC5 polar pili, which have been prevented from retracting by the adsorption of a pilus-specific bacteriophage; this can be seen scattered over the field. Bar = 0.1 μm. (Reproduced with permission from D.E. Bradley, Canadian Journal of Microbiology 26:155–160, 1980, ©National Research Council of Canada.)

esting for the preparation of vaccines (Rawling et al., 1995). There are differences in the location of OprFs between the two species. Thus, while in *P. fluorescens* the protein is embedded in the outer membrane in such a way that it offers a surface-exposed region rich in proline, this region is absent in *P. aeruginosa* and in *P. syringae*. In addition, in *P. aeruginosa*, the entire protein is embedded, without an exposed region to the outside (De Mot et al., 1994).

An outer membrane protein of *P. fluorescens* that was inducible under conditions of phosphate limitation was purified and characterized (Leopold et al., 1997). It did not show similarity to any of the known outer membrane proteins. Its distribution among different strains of the species was uneven, i.e., it was not present in all strains, which simply may reflect the internal heterogeneity of *P. fluorescens*. A method of immunofluorescence microscopy involving cell permeability was developed to visualize the specific

expression of the protein in cells exposed to limiting phosphate concentrations, in order to use the system to measure phosphate availability (Leopold et al., 1997). In *P. aeruginosa*, exposure to limiting phosphate provoked the expression of porin OprP. Only one of three lysine residues of this protein was required to form the specific phosphate binding site in OprP (Sukhan and Hancock, 1996).

Resistance to polymyxin B, gentamicin, and EDTA is associated with the outer membrane protein OprH in *P. aeruginosa*, which is expressed under magnesium-limiting conditions (Rehm and Hancock, 1996).

P. aeruginosa is able to simultaneously express two different LPS types, which migrate to different positions in electrophoresis and have been named bands A and B. The B band corresponds to the component that confers to the cell its O-antigen serological specificity and the A band to a common antigen. Two of the

eight genes that are part of the operon that controls band A biosynthesis and transport encode proteins that are highly similar to a number of proteins that are part of the ABC (acronym for "ATP-binding cassette") transport systems (Rocchetta and Lam, 1997). Of the two LPS O-polysaccharide species, one confers to the cells high hydrophobicity, and the other, hydrophilicity. The factors that provoke a change in the ratio between these two components affect both adhesion and survival (Makin and Beveridge, 1996).

Pigments and siderophores Early taxonomic treatments of the genus *Pseudomonas* included pigmentation as a generic character, but this is no longer valid. In fact, the colonies and other cell masses always display some colors due to normal cellular components, which, in some instances, become quite apparent. Thus, *P. stutzeri* is grouped with the nonpigmented species, even though the colonies of many strains become dark brown due to high concentration of cytochrome *c* in the cells.

In his comprehensive monograph on the taxonomy of *P. aeruginosa* and other fluorescent pseudomonads, Jessen (1965) listed six pigments produced by the type species: four phenazines (pyocyanin, pyorubin, chlororaphin, oxiphenazin), the *Pseudomonas* blue protein, and pyoverdine. Several phenazine pigments can be produced by a single strain (Chang and Blackwood, 1969). The best known and most characteristic of these pigments is the phenazine blue pigment pyocyanin, identified many years ago as the cause of the blue color of the pus of wounds infected with *P. aeruginosa*. Synthesis of pyocyanin by *P. aeruginosa* can be stimulated by growth in King A medium (King et al., 1954). It is inducible in a dose-dependent manner and its production is strongly enhanced by addition of L-N-(3-oxohexanoyl) homoserine lactone to the medium (Stead et al., 1996).

Phenazine pigments synthesized by other fluorescent pseudomonads are the green, almost insoluble, chlororaphin of *P. chlororaphis*, and the orange phenazine-monocarboxylic acid characteristic of *P. aureofaciens* (in the present treatment these taxa are considered to be subspecies of *P. chlororaphis*). The species "*P. lemonnieri*", which was considered by Stanier et al. (1966) as a biovar of *P. fluorescens*, produces an intracellular blue pigment that has been chemically characterized (Starr et al., 1960, 1967). Some strains of *P. aeruginosa* are able to produce melanin pigments (Mann, 1969).

Other important pigments from the physiological and taxonomic standpoints are the pyoverdines. They are the typical yellow-green pigments of the so-called fluorescent pseudomonads. Jessen (1965) found that his collection of strains of *P. aeruginosa* contained 328 fluorescent strains producing pyocyanin, 18 fluorescent strains not producing pyocyanin, five nonfluorescent strains with pyocyanin, and three nonpigmented strains. Strains of this species lacking one or more of the characteristic pigments, however, could be identified based on other phenotypic properties, and, in general, pigmentation is a most striking but not always dependable characteristic for species identification. In any case, identification of fluorescent *Pseudomonas* species by examination of the fluorescence profiles of pigments diffusing into the medium has been proposed by Shelly et al. (1980).

Pyoverdines are also physiologically important because they function as efficient siderophores (Meyer and Abdallah, 1978; Meyer and Hornsperger, 1978). Their production is enhanced under conditions of iron-starvation. All pyoverdines share a quinoleinic chromophore, which is linked to peptides of different compositions and sizes. This may vary from 6–12 D- and L-amino

acid residues. With minor exceptions, each fluorescent *Pseudomonas* type strain produces a pyoverdine with a specific amino acid composition (Budzikiewicz, 1993). However, within a single species there is a diversity of peptide structures.

The presence of D- and L-amino acid in pyoverdines as well as possible cyclic structures suggests a similarity the cyclic peptides of antibiotic properties produced by some Gram-positive bacteria and toxins synthesized by *Pseudomonas syringae* pathovars. These observations are indicative of a nonribosomal type of synthesis (Georges and Meyer, 1995; Merriman et al., 1995). In a search for components of such a system, high molecular weight cytoplasmic proteins have been identified. As expected, they are produced by cells of fluorescent pseudomonads in iron-deficient media, for which they have been named iron-repressed cytoplasmic proteins (IRCPs). They vary from M_r 180–600 kDa, and they give characteristic electrophoresis profiles for strains producing different pyoverdines. On the other hand, they are absent from mutants in pyoverdine synthesis and from cells of nonfluorescent pseudomonads (Georges and Meyer, 1995). As mentioned by Meyer et al. (1997), pyoverdines provide information of use in strain typing and epidemiological applications.

Using a system with a promoterless *lacZ* gene, 24 insertion mutants of *P. aeruginosa* unable to synthesize pyoverdine were isolated. All the mutations could be allocated within a 103-kb region ("pyoverdine region") at 47 min of the *P. aeruginosa* strain PAO genetic map, very near the catechol region (Tsuda et al., 1995). The genetic determinants of pyoverdine production occupy a region of at least 78 kb, but few genes have been characterized at present. *pvdE*, recently identified, codes for a protein that is a member of the ATP-binding cassette group of membrane transporters (McMorran et al., 1996). *pvdA* encodes an enzyme that catalyzes a key step in the synthesis. Three tightly iron-regulated regions are located in a fragment upstream of *pvdA*. Fur (ferric uptake repressor, a protein that controls expression of iron-repressible genes) indirectly controls *pvdA* transcription. When iron is not limiting, Fur blocks the *pvdA* promoter, thus inhibiting transcription of several pyoverdine genes (Leoni et al., 1996). Gene *pvdS* is also required for pyoverdine biosynthesis by *P. aeruginosa*. Under conditions of unlimited iron, Fur also acts here as a repressor, binding to the *pvdS* promoter and preventing expression (Cunliffe et al., 1995).

Because siderophores are iron-scavenging compounds, a large number of strains of *P. aeruginosa* have been analyzed by different methods for pyoverdine-mediated iron incorporation, and in all cases the collection could be subdivided into the same three groups, in spite of the inclusion of strains devoid of pyoverdine production in some of the groups (Meyer et al., 1997). The *P. aeruginosa* cells also are able to incorporate iron combined with siderophores of foreign origin. For instance, they can produce a receptor (PfeA) for the iron trapped by enterobactin, a siderophore synthesized by enteric bacteria. PfeA production depends on a regulator and a sensor, which are members of the family of two-component regulatory systems. The operon of regulator/sensor is probably regulated by iron, since the regulator gene in this system has a sequence similarity to the iron uptake regulator *fur*, the gene that codes for the above-mentioned Fur protein (Dean et al., 1996). Iron can be transported into *P. aeruginosa* not only by enterobactin but also by at least one of its breakdown products, 2,3-dihydroxybenzoyl-L-serine, in a process that is neither iron-repressible nor strongly energy-dependent (Spangenberg et al., 1995).

Attempts to complement a pyoverdine mutant from a *P. ae-*

ruginosa PAO cosmid bank resulted in the recovery of an apparent wild-type phenotype. Physical mapping indicated that the cloned fragment corresponded to a different region of the PAO chromosome and that the properties of the transconjugants were not the result of a true complementation. The yellow-green fluorescent compound was different in its properties from pyoverdine and it was named pseudoverdine. It lacks a peptide chain, but it resembles pyoverdine in its spectral properties (Stintzi et al., 1996).

Aside from pyoverdines, the fluorescent pseudomonads also produce other strain-specific, but chemically related, siderophores called pseudobactins. A method for isolation of pseudobactin from pseudomonads, and an assay for fluorescent siderophores based on reverse-phase HPLC, have been described (Nowak-Thompson and Gould, 1994). When grown in an iron-deficient medium, each of two strains of *P. putida* and *P. fluorescens* produced two different novel yellow-green fluorescent pseudobactins. All four compounds contained a dihydroxyquinoline-based chromophore. The receptor proteins of the two species are similar but not identical (Khalil-Rizvi et al., 1997). The composition of the corresponding peptides has been described, but the reason for the differences are not apparent, since the uptake system seems to consist of a single receptor in both organisms. Fragments of the peptide component of one pseudobactin have been synthesized. Antifungal peptides called pseudomycins have related structures, but the one mentioned here does not have activity against fungi (Koushik et al., 1997).

A *P. putida* strain WCS358 produces a fluorescent pseudobactin that has been described in detail (von der Hofstad et al., 1986). In spite of the results mentioned in the previous reference, the results here indicate that the ability of *P. putida* WCS358 to use different pseudobactins of various origins is related to the presence of multiple outer membrane receptor proteins (Koster et al., 1995).

In addition to the siderophores mentioned above, the fluorescent pseudomonads produce another one, pyochelin, which is not pigmented (that is, it does not have an absorption spectrum in the visible region), and appears to have low efficiency as an iron-scavenging compound (Liu and Shokrani, 1978). In the composition of pyochelin there is one molecule of salicylic acid (which is by itself a siderophore) condensed to two cysteinyl residues. In *P. aeruginosa*, the pathway of salicylate synthesis has been clarified: conversion of chorismate (which also acts as an intermediate of aromatic amino acid biosynthesis) to isochorismate, which is converted to salicylate plus pyruvate. These steps are catalyzed by two iron-repressible proteins, PchA and PchB (Serino et al., 1995).

The iron trapped by the siderophores is utilized by the producing organism presumably after reduction to ferrous ion, a step catalyzed by ferrisiderophore reductases. Ferripyoverdine reductase is ubiquitous among *Pseudomonas* species (Halle and Meyer, 1992), and since its activity is not under iron regulation, it is not part of the siderophore genetic system (J.-M. Meyer, personal communication).

There are earlier reports of other fluorescent compounds that have been isolated from fluorescent pseudomonads. These include four pteridine derivatives isolated from "*P. ovalis*" (presumably *P. putida*) (Suzuki and Goto, 1971); two fluorescent antibiotics isolated from *P. fluorescens*, which were named fluopsin C and fluopsin F (Shirahata et al., 1970); 6-hydroxymethylpteridine, an intermediate in folic acid synthesis (Viscontini and Fra-

ter-Schröder, 1968); and erithroneopterin of *P. putida* (Suzuki and Goto, 1972).

P. fragi, a nonpigmented member of the genus, does not produce siderophores in detectable amounts and it is very sensitive to iron-limiting conditions. Growth can be strongly stimulated by iron combined with siderophores of foreign origin. The siderophores include enterobactin, some pyoverdines, and siderophores of eucaryotic origin, such as transferrin, lactoferrin, and hemoglobin. Probably this property is related to the capacity of *P. fragi* to grow in milk, from which the first strain of this species was isolated many years ago (Gruber, 1905). Iron starvation in this species may induce the synthesis of the siderophore-mediated iron uptake system (Champomier-Vergès et al., 1996).

The following is a useful summary of siderophores produced by *Pseudomonas* species, kindly supplied by J.-M. Meyer from a manuscript submitted for publication. All strains of the fluorescent species *P. aeruginosa*, *P. fluorescens*, *P. chlororaphis*, and *P. putida* produce pyoverdines as main siderophores. Some strains also produce pyochelin and/or salicylic acid as secondary siderophores (Cox et al., 1981; Meyer et al., 1992; Visca et al., 1993). *P. stutzeri* ATCC 17588 produces desferrioxamines E and D2 (Meyer and Abdallah, 1980; Azelvandre, 1993); a different strain of *P. stutzeri* (RC7) gives a catechol-type siderophore (Chakraborty et al., 1990) and an unclassified *Pseudomonas* sp., aerobactin (Buyer et al., 1991). No siderophores have been detected in *P. stutzeri* YPL-1 (Lim et al., 1991), in *P. fragi* (Stintzi et al., 1996), and in *P. mendocina*. It is interesting to note here that the internal heterogeneity of the nonfluorescent species *P. stutzeri* is also reflected in the capacity for siderophore production.

P. alcaligenes, *P. mendocina*, and *P. flavescens* produce yellow to orange pigments that have not been chemically characterized. Strains of one of the biovars of *P. fluorescens*, which were originally assigned to the species "*P. lemonnieri*", produce an intracellular insoluble pigment of structure related to that of indigoidine, the purple pigment of "*Pseudomonas indigofera*". Both pigments are derivatives of 3,3'-bipyridyl (Kuhn et al., 1965). The pigment of "*P. lemonnieri*" has been reexamined, its chemical structure has been clarified, and it has received the name "lemonnieriin" (Ferguson et al., 1980; Jain and Whalley, 1980). Pigments that are not soluble in water and remain associated with the cell mass are found in many former *Pseudomonas* species now classified in other genera.

Nutrition and growth conditions Strains of *Pseudomonas* species can grow in minimal, chemically defined media with ammonium ions or nitrate as nitrogen source and a single organic compound as the sole carbon and energy source. Some of the species previously assigned to *Pseudomonas* have true growth factor requirements, but none of the species belonging to the rRNA group I (Palleroni, 1984) has an absolute dependence on these nutritional supplements. In media of minimal composition, strains of phytopathogenic *P. syringae* grow very slowly in comparison with strains of the main saprophytic species, and that growth is enhanced by addition of small amounts of complex organic materials (yeast extract, peptones). However, in most cases no true dependence on organic growth factors can be demonstrated. Pantothenate is required by strains of *P. syringae* pathovar *avellanae*, now considered as an independent species, *P. avellanae* (Janse et al., 1996) (see List of Species). In addition, organic growth factors are required by some species of uncertain phylogenetic position (*P. iners*, *P. lanceolata*, *P. spinosa*) (see List of Species). Occasionally, attempts have been made to improve

the poor growth of phytopathogenic *Pseudomonas* in chemically defined media. In one such proposal, a defined medium for "*P. tomato*" (*P. syringae* pathovar *tomato*) includes L-asparagine, L-glutamine, or L-threonine as a nitrogen source, and D-galactose as carbon source. The organism grows in this medium as well as in complex media (Bashan et al., 1982).

From time to time the capacity for nitrogen fixation has been claimed for some species that later on proved to be unable to perform this function under strictly controlled conditions. In the first edition of this *Manual*, it was assumed that none of the *Pseudomonas* species could be considered legitimate nitrogen fixers; however, this ability is reported to occur in strains of the nonfluorescent species *P. stutzeri* (Krotzky and Werner, 1987; Puente and Bashan, 1994), which is also a vigorous denitrifier.

Requirement for sodium ions has been observed for *Pseudomonas elongata* which, according to Anzai et al. (2000) should be placed in the genus *Microbulbifer* following further taxonomic studies, and for *P. halophila* (described in the section Other Species). This requirement, which is one of the characteristics typical of so-called marine eucaryotes, has not been determined in the majority of *Pseudomonas* species.

P. aeruginosa can utilize one of many different sources of sulfur in the medium. The list includes inorganic and organic compounds, from which the aromatic sulfur compounds are excluded. If sulfur sources other than the preferred ones (sulfate, cysteine, or thiocyanate) are the only ones available, a set of 10 sulfate starvation induced (SSI) proteins are upregulated (Hummerjohann et al., 1998). One of these proteins is periplasmic and has high affinity for sulfate. Even though no similarity of the other nine SSI proteins to other proteins of known function has been detected, they also may represent scavenging elements for the S-sources preferred by the cells. Studies on the genetics of the starvation response have indicated the role of cysteine biosynthetic intermediates and the possibility that at least two independent co-repressors are operative in *P. aeruginosa* (Hummerjohann et al., 1998).

The ability to grow in very simple mineral media at the expense of many organic compounds has served as the basis for extensive nutritional characterization of a large number of strains, providing a mass of phenotypic data ideally suited for taxonomic studies by numerical methods. Aside from their taxonomic implications, nutritional investigations on the utilization of certain groups of compounds have served as the bases for many interesting studies on metabolic pathways, their regulatory mechanisms, and their phylogenetic significance.

The best growth temperature for growth of most strains is approximately 28°C. Some species grow at a substantial rate at 4°C and thus can be considered psychrotrophic. For others the maximum temperature is about 45°C, and therefore they are not true thermophiles. None of the members of the genus tolerates acidic conditions and growth is invariably negative at pH 4.5.

Metabolism and metabolic pathways The metabolism of *Pseudomonas* is typically respiratory with oxygen as the terminal electron acceptor, but some species also can use nitrate as an alternate electron acceptor and can carry out oxygen-repressible denitrification (dissimilatory reduction of nitrate to N₂O or N₂). In most cases, denitrification is the property of all members of a given species, and only a few strains may be unable to denitrify. Denitrification is not a species characteristic in *P. fluorescens*; only some biovars are able to denitrify (Stanier et al., 1966).

Some cytochromes are involved in denitrification through the

participation of a special cytochrome oxidase that probably is a remnant of a very primitive mechanism dating from the pre-aerobic era of the planet (Yamanaka, 1964). An excellent review on the cellular and molecular aspects of denitrification has been published by Zumft (1997). *Pseudomonas* is rich in denitrifying species. The core structures of the denitrification apparatus in these and in denitrifiers belonging to other genera demand the participation of around 50 genes.

Assimilation of nitrate occurs through the reduction of nitrate to ammonia. Although there may be common intermediates in the assimilatory and dissimilatory routes of nitrate reduction (Hartingsveldt et al., 1971), the pathways are encoded by different sets of genes. Some mutations with pleiotrophic effects are due to alterations in a gene involved in molybdenum incorporation into the nitrate reductases (Sias et al., 1980). A comprehensive review on the diversity of enzymes involved in denitrification in the pseudomonads and the genetic organization is available (Zumft and Körner, 1997), as well as a description of the localization of the genes in the *P. aeruginosa* PAO map (Vollack et al., 1998). The genes for nitrate reduction (*nir*), nitric oxide reduction (*nor*), and nitrous oxide reduction (*nos*) have been located in a 30-kb gene cluster in the chromosome of *P. stutzeri* (Braun and Zumft, 1992), and a sequence analysis of a 9.72-kb internal segment that includes the genes is now available (Glockner and Zumft, 1996). Very useful accounts of the enzyme diversity and gene organization of the denitrification genes in denitrifying pseudomonads, as well as the regulatory elements of the denitrification system of *P. stutzeri*, have been published by Zumft and collaborators (Cuyper and Zumft, 1992; Zumft and Körner, 1997). As mentioned before, a review by Zumft (1997) includes much useful information on this subject.

The oxidative degradation of some substrates (particularly the aromatic compounds) or their intermediates by *Pseudomonas* occasionally involves the participation of oxygenases. Both mono- and dioxygenases coupled to a variety of electron donors are well represented in species of the genus. Oxygenases acting on aliphatic compounds such as alkanes may be part of complex systems, and this is also true for the oxidation of some compounds like camphor by *P. putida*, with steps involving oxygenases of considerable complexity that include the iron-sulfur protein putidaredoxin and cytochrome P-450_{CAM} (Gunsalus et al., 1971). Other systems include the iron protein rubredoxin (Lode and Coon, 1971) and cytochrome *o* (Peterson, 1970).

The classical reactions of the tricarboxylic acid cycle are found in all species of *Pseudomonas* that have been examined. A key reaction is the synthesis of citrate from oxaloacetate and acetyl-CoA, which is under a control system typical of absolute aerobic organisms (Weitzman and Jones, 1968). The control of peripheral catabolic enzymes (amidase, histidase, enzymes of aromatic compounds, and camphor metabolism) by intermediates of the tricarboxylic acid cycle are a manifestation of the central position that it occupies in metabolism. Those intermediates that are used for biosynthetic purposes can be replenished by carboxylation of pyruvate and by the action of the enzymes isocitrate lyase and malate synthase, which are part of the anaplerotic system known as the glyoxylate cycle (Kornberg and Madsen, 1958). A multienzyme complex of tricarboxylic acid cycle enzymes (fumarase, malate dehydrogenase, citrate synthase, aconitase, and isocitrate dehydrogenase) that catalyzes the reactions from fumarate to α -ketoglutarate has been identified in *P. aeruginosa* cells, from which it can be released by gentle osmotic lysis. The complex can be reconstituted from the individual enzymes provided that

one of the two citrate synthase isoenzymes is present (Mitchell, 1996).

A variety of macromolecules can be degraded by some strains by means of extracellular enzymes. Hydrolytic enzymes that have been studied in detail include the proteases of *P. aeruginosa* (Mori-hara, 1964; Mori-hara et al., 1965), which are important in infections caused by this organism. In addition to these early studies, others will be discussed in relation to pathogenesis to humans and animals.

One of the differences between *P. fluorescens* and *P. putida* is the ability of the former species to produce extracellular proteases, to which the property of gelatin liquefaction can be attributed. A metalloprotease of *P. fluorescens* that is inhibited by EDTA and has a trypsin-like activity has been found to be highly homologous to zinc metalloproteases of diverse origins (Kim et al., 1997a).

Carbohydrates The amylolytic activity of *P. stutzeri* is responsible for rapid starch hydrolysis, one of the characteristic phenotypic properties of the species. The enzymology of the exo-amylase, which is responsible for the formation of maltotetraose as end product, has been examined at the molecular level, and the enzyme has been cloned (Morishita et al., 1997). Phytopathogenic pseudomonads have been found to hydrolyze pectin (Hildebrand, 1971; Ohuchi and Tominaga, 1973, 1975; Wilkie et al., 1973), xylan (Maino et al., 1974), and glycosides (Hayward, 1977).

Common monosaccharides (glucose, fructose, galactose, L-arabinose) are used by strains of most species of the genus, but growth of some of the species (*P. stutzeri*, *P. mendocina*, *P. syringae*) may be slow. Most hexoses are degraded by the Entner-Doudoroff pathway, which was discovered in studies on *P. saccharophila* (Entner and Doudoroff, 1952).

The fluorescent pseudomonads have multiple peripheral pathways for glucose oxidation that converge for the synthesis of 6-phosphogluconate, which is further degraded by the Entner-Doudoroff pathway (Eisenberg et al., 1974). Of these routes, one involves direct oxidation of the sugar (oxidative pathway), and either gluconate or 2-ketogluconate can serve as a precursor of 6-phosphogluconate. However, in *P. putida*, 6-phosphogluconate is synthesized preferentially from 2-ketogluconate (Vicente and Cánovas, 1973). Induction of the oxidative pathway in *P. aeruginosa* can only occur in the presence of oxygen. Under denitrifying conditions, only the so-called phosphorylative pathway (starting with the phosphorylation of glucose) is operative in this organism (Hunt and Phibbs, 1981).

A rather peculiar situation is represented by the oxidative degradation of glucose by *P. putida*, which has been shown to oxidize the sugar with a dehydrogenase peripherally located in the cells. This results in gluconate accumulation in the surrounding medium, from which it is taken back into the cells after induction of a specific energy-dependent transport system. The properties of this system are different from a second one specific for glucose uptake by the same cells (Schleissner et al., 1997).

The metabolism of fructose by several species of fluorescent and nonfluorescent pseudomonads occurs by means of a phosphoenol-pyruvate (PEP) phosphotransferase system. The product is fructose-1-phosphate, which may be further phosphorylated and cleaved by an aldolase. An isomerization of free mannose to fructose can be demonstrated in cell-free extracts of strains of many species (Alicia Palleroni and N.J. Palleroni, unpublished), although the significance of this conversion is obscure at present.

An excellent review on alternative pathways of carbohydrate metabolism by pseudomonads is available (Lessie and Phibbs, 1984).

Polyhydroxyalkanoates At the time the main body of *Pseudomonas* taxonomy was being developed, the general ideas on reserve materials of the species of rRNA group I were vague. As mentioned before, although other groups of aerobic pseudomonads accumulate PHB in granules easily observable under phase microscopy, particularly when grown in media of low nitrogen and high carbon content, with few exceptions the species of rRNA similarity group I (Palleroni, 1984) now classified in the genus *Pseudomonas* did not seem to have the same capacity, and it was even assumed that the reserve material might be dispensable proteins, because of production of ammonia during respirometric studies on the endogenous metabolism of cells. Later it was discovered that the fluorescent pseudomonads (typical representative species of the genus *Pseudomonas*) were able to accumulate PHAs of medium chain length (C_6 to C_{12}) (de Smet et al., 1983; Lageveen et al., 1988). The studies confirmed the inability of these organisms to accumulate PHB, and this inability remains a reliable negative characteristic for differentiation from most other aerobic pseudomonads (Anderson and Dawes, 1990). It has been suggested that the ability to accumulate the medium chain length PHAs, which is not dependent on the presence of a plasmid, may be in itself of taxonomic value (Huisman et al., 1989). Additional comments on this subject will be found in the section Differentiation of the Genus *Pseudomonas* from Other Genera.

Hydrolysis of PHAs for the utilization of these carbon reserve materials can be catalyzed by extracellular depolymerases, of which one, produced by a strain of *P. fluorescens*, has been purified and characterized (Schirmer et al., 1993). The depolymerases are considered to belong to the family of serine hydrolases (Schirmer and Jendrosseck, 1994; Schirmer et al., 1995). All lipases and an esterase of *P. fluorescens* had hydrolytic action on triolein, but this property was absent from the PHA depolymerases tested, although these enzymes have the lipase consensus sequence in their structure. The finding confirms the differences in function between lipases and PHA depolymerases (Jaeger et al., 1995).

Biotechnological applications of some pseudomonad lipases have been reported (Tan et al., 1996).

Polyalcohols Genes involved in the conversion of 2,3-butylene glycol to central metabolites by *P. putida* have been characterized, and there is a high similarity between the genes that encode 2,3-butylene glycol dehydrogenase and those that encode alcohol dehydrogenases (Huang et al., 1994). Degradation of polyethylene glycol has been demonstrated with *P. stutzeri* (Obadadors and Aguilar, 1991). The mannitol dehydrogenase of strains of *P. fluorescens* has a broad specificity, since it is also capable of oxidation of other polyalcohols such as sorbitol and arabitol (Brunker et al., 1997).

Acetamide and biochemical evolution Acetamide is used for growth by several species of aerobic pseudomonads, among them *P. aeruginosa* (Stanier et al., 1966). It is hydrolyzed with liberation of ammonia, which is used then as a nitrogen source, but the enzyme can also catalyze acyl transfer reactions. Amides hydrolyzed by the wild-type enzyme include propionamide, acetamide, formamide, and butyramide, in order of decreasing rates of hydrolysis. Among the amides there are excellent inducers (lactam-

ide) as well as anti-inducers (butyramide). Clarke and collaborators found this system to be admirably suited for studies of experimental evolution, during which it became possible to evolve amidases with altered substrate specificities, and the work of this group, performed during many years, stands out as a beautiful example of experimentation on evolution achieved in the laboratory (Brown et al., 1969; Betz and Clarke, 1972; Brown and Clarke, 1972; Betz et al., 1974). Among the many papers that followed these, a general article by Clarke and Drew (1988) gives a good account of the experimental evolution of the system.

The genetic organization and complete sequence of the amidase operon is now known (Drew and Wilson, 1992), as well as details of its transcription antitermination regulation (Wilson et al., 1996).

Aromatic compounds For obvious reasons, as in the preceding sections, the short review that follows has been restricted to very few of the many valuable reports available in a well-populated literature on the degradation of aromatic compounds and some derivatives.

Members of the genus *Pseudomonas* are notorious for their capacity of aerobic degradation of a number of hydrocarbons, aromatic compounds, and their derivatives, of which there are natural compounds and final products or intermediates from industrial activities. A considerable number of these compounds are toxic and are found widely dispersed as environmental contaminants. Many of them can be used for growth by *Pseudomonas* species. Compounds as diverse as benzoate, *p*-hydroxybenzoate, mandelate, tryptophan, phthalate, salicylate, polycyclic compounds, and many derivatives may be metabolized by strains of the group following pathways that converge to a common intermediate, β -ketoacidipate. This intermediate is formed soon after the last aromatic intermediate is cleaved by a 1,2-dioxygenase, in a type of ring cleavage frequently referred to as an *ortho* cleavage. The attack of the aromatic rings and the mode of cleavage are properties of taxonomic importance that have been of help in the circumscription of the rRNA similarity group I, which comprises the *Pseudomonas* species. In recent times, much information has been added to our basic knowledge of this pathway, but few details will be mentioned here. A transporter and chemoreceptor protein from *P. putida* that is part of the β -ketoacidipate regulon, PcaK, was localized in the membrane when it was expressed in *E. coli*, adding to our understanding of active transport in aromatic compound metabolism (Nichols and Harwood, 1997).

The *ortho* ring cleavage, however, is not the only type of ring opening that can be catalyzed by these organisms, and the conditions adopted for the experimental protocols should be precisely stated. Thus, in the *P. putida* type strain, cleavage of the diphenolic intermediate catechol can be caused by either a 1,2-dioxygenase or a 2,3-dioxygenase (which causes a *meta* cleavage) depending on whether the substrate on which the cells had grown was benzoate or phenol, respectively (Feist and Hegeman, 1969). Often the genes encoding for *meta*-cleavage systems are located in plasmids (Austen and Dunn, 1980), as in the case of the well-known plasmid TOL, but this is not always so. Hewetson and collaborators have presented evidence of an *ortho* pathway for *p*-cresol degradation catalyzed by enzymes encoded by plasmid genes (Hewetson et al., 1978). The *ortho*-enzymes (1,2-dioxygenases, also known as pyrocatechases) in different *P. putida* strains have all a common ancestry (Nakai et al., 1995).

The pathways of aromatic compounds degradation are still being extensively analyzed. Regulatory mechanisms, genetic or-

ganization of the genes involved, immunological properties of the enzymes, and molecular studies as the bases for strain improvement in possible bioremediation applications are all subjects of great basic and practical importance. Detailed reports on the molecular aspects of the analysis of aromatic hydrocarbon degradation have been published (Zylstra and Gibson, 1991; Zylstra, 1994). Figs. BXII.γ.122 and BXII.γ.123 have been taken from one of these sources (Zylstra and Gibson, 1991). Fig. BXII.γ.122 is a graphical representation of the main differences among aerobic pseudomonads in the initial steps of pathways of toluene degradation, and illustrates differences among some members of rRNA groups I and II (represented by *Burkholderia cepacia* and *Ralstonia pickettii*). The ring fissions of the dihydroxy intermediates (catechol, protocatechuic acid, and 3-methyl-catechol) occur in different ways. For organisms carrying the TOL plasmid (A) and *P. putida* F1 (E), the cleavage of the catechols results from the action of a 2,3-dioxygenase (*meta* cleavage), while cleavage of the protocatechuate by *P. mendocina* is of the *ortho* type (1,2-dioxygenase). Further details of the initial steps of toluene degradation by *P. putida* F1 are shown in Fig. BXII.γ.123.

Tolerance to toluene can develop in cells that normally are killed by contact with solvents. A *P. putida* strain able to grow in the presence of high concentrations of aromatic hydrocarbons was isolated from enrichments. The tolerance was inducible in this strain. Electron micrographs showed that the cells had a periplasmic space wider than that in the nontolerant cells and

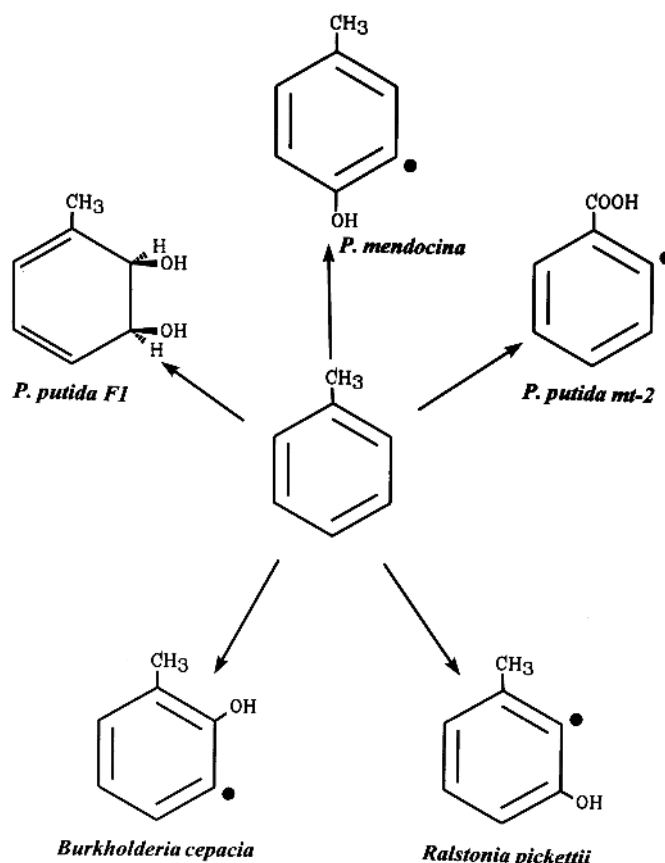


FIGURE BXII.γ.122. Pathways for the degradation of toluene by aerobic pseudomonads. The black dot indicates the position of the subsequent hydroxylation that precedes ring cleavage. For more details, see text and also Zylstra and Gibson (1991).

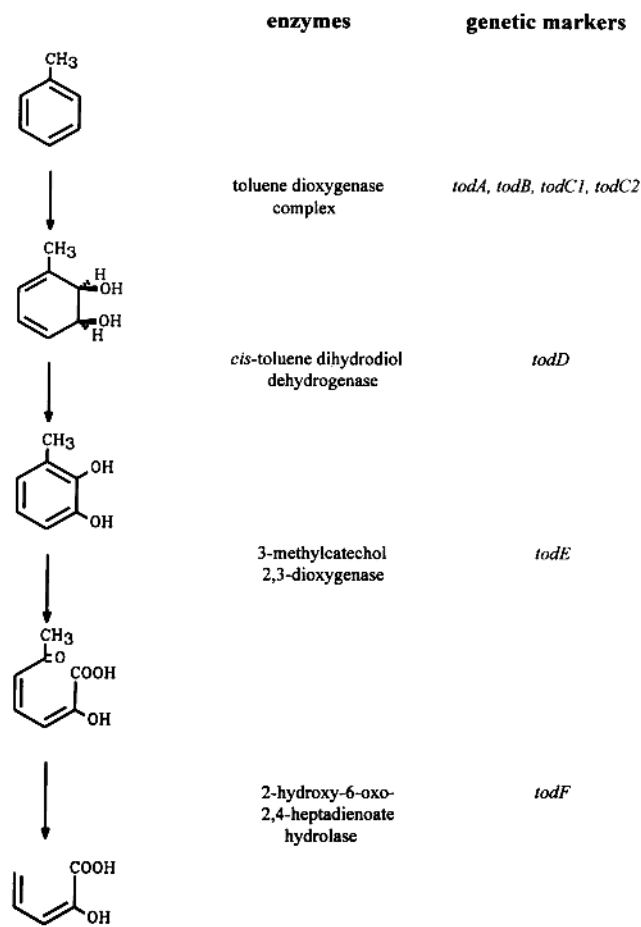


FIGURE BXII.γ.123. Initial steps in the pathway for the aerobic degradation of toluene by *Pseudomonas putida* strain F1. More details are found in Zylstra and Gibson (1991).

that the outer membrane integrity was preserved. Tolerance was also acquired to *m*- and *p*-xylene, and the strain gained the capacity of using these two compounds after receiving the plasmid TOL (Ramos et al., 1995). An energy-dependent efflux system may be part of the tolerance of this species to toluene (Isken and De Bont, 1996).

In a personal communication, J.A.M. de Bont has summarized as follows some of the mechanisms of tolerance to solvents: modification of the composition of the head groups of phospholipids (Ramos et al., 1997), of outer membrane proteins (Li et al., 1995; Ramos et al., 1997), of LPS (Pinkart et al., 1996), changes in the rate of turnover of membrane components (Pinkart and White, 1997), and most importantly, alterations in the composition of the fatty acids of phospholipids (Heipieper et al., 1994). This last is a very straightforward response, mostly due to the action of a *cis-trans* isomerase in *P. putida*. This enzyme has been cloned and expressed in *E. coli* (Holtwick et al., 1997). The isomerization is quite fast and does not demand an energy input. The presence of solvents triggers this activity, but other stress factors (heavy metals, high temperatures, low pH, and water stress due to salt) can act as inducers (Heipieper et al., 1996).

As expected, plasmid TOL, which was discovered in a strain now classified as *P. putida*, has continued receiving much attention. There are interesting relationships between the pathways encoded by TOL genes and those chromosomally located in the

host. Thus, when growing on unlimited amounts of toluene, a *P. putida* carrying TOL lost the capacity of using xylenes and toluates, but could grow on toluene by using initially the enzymes of the TOL upper pathway and the chromosomal system for benzoate metabolism (Brinkmann et al., 1994). Catabolite repression by succinate and glucose on the TOL enzymes has been examined (Duetz et al., 1994; Holtel et al., 1994).

The catabolism of toluene and *o*-xylene in *P. stutzeri* has been analyzed by gene cloning (Baggi et al., 1987; Bertoni et al., 1996).

In studies on the degradation of polycyclic aromatic hydrocarbons (PAHs), a mixed natural population in which *P. aeruginosa* and *P. putida* were represented was more efficient in the degradation of the poorly soluble components of the mixture than any of the pure cultures isolated from the population. The latter in turn were more active on the more water-soluble compounds (among them acenaphthene, fluorene, phenanthrene, and fluoranthene) (Trzesicka-Mlynarz and Ward, 1995).

Naphthalene degradation by *P. stutzeri* has been found to be catalyzed by enzymes encoded by highly homologous sets of catabolic genes, although the hosts had been allocated to four different genomovars (Rosselló-Mora et al., 1994c).

Many reports in the literature refer to the degradation by *Pseudomonas* species of a number of derivatives of simple aromatic compounds including *p*-cymene (Eaton, 1996, 1997), nitrophenols (Rani and Lalithakumari, 1994; Meulenbergh et al., 1996; Michan et al., 1997), and styrene (O'Connor et al., 1995; Marconi et al., 1996; Beltrametti et al., 1997).

Due to their widespread use in a number of applications based on highly desirable physical and chemical characteristics, the halogenated derivatives of hydrocarbons and aromatic compounds occupy an outstanding position in the studies on environmental microbiology. Many of the multitude of halogenated compounds are toxic and, due to their remarkable stability, they tend to persist in the environment, where they are slowly degraded by a number of organisms, under both aerobic and anaerobic conditions. In the aerobic degradation of these recalcitrant compounds, the pseudomonads are among the most active organisms, and some of them also are involved in the anaerobic degradation when nitrate is present as an alternate electron acceptor. The biodegradation of many halogenated organic compounds has been adequately summarized in reviews (Häggblom, 1992; Fetzner and Lingens, 1994).

In view of the fact that anaerobic bacteria are more active in the initial degradation of haloaromatics that are reductively dehalogenated, attempts have been made to combine this property with the ability of aerobic bacteria for active degradation of the aromatic ring. A proper combination of genes in a *P. putida* strain resulted in the metabolism of polyhalogenated compounds by sequential reductive and oxidative reactions (Wackett et al., 1994).

In *P. putida* chloroaromatic and methylaromatic compounds are degraded via *meta*-cleavage pathways. The chlorocatechol produced in the first case usually inactivates the 2,3-dioxygenase, but one strain was developed with a resistant dioxygenase and had the ability to degrade both toluene and chlorobenzene via the *meta*-cleavage route (Mars et al., 1997).

Phenol, monochlorophenols, monochlorobenzoate, dichlorophenols, 2,3,5-trichlorophenol, and several alkylbenzenes, were oxidized by a strain of *P. putida* by means of an inducible toluene dioxygenase. A study of this property revealed not only the remarkable versatility of the species, but also the influence

of other factors (e.g., growth rate) on the competence of the cells for use as bioremediation agents (Heald and Jenkins, 1996).

The degradative enzymes that convert these unusual compounds to intermediates of the central metabolism in distantly related hosts frequently show a high level of similarity. A rather extreme case is illustrated by a comparative study of the deduced amino acid sequences of components of the polychlorinated biphenyl degradation systems of the Gram-positive *Rhodococcus glob-erulus* and of Gram-negative bacteria, which showed a high similarity between the toluene dioxygenase of *P. putida* and the biphenyl dioxygenase of the Gram-positive organism, suggesting a possible transfer of genes crossing the Gram-positive/Gram-negative barrier (Asturias et al., 1995). This is certainly a reminder of the danger of drawing phylogenetic conclusions based on similarities found in metabolic systems involved in the degradation of these unusual organic substrates.

Amino acid catabolism One aspect that has attracted the attention of biochemists and bacteriologists for many years is the utilization of arginine by the pseudomonads. Among the degradative pathways, the arginine deiminase ("dihydrolase") system has been used by taxonomists since the early 1960s for differentiation of species. The reactions catalyzed by the system are the conversion of arginine to citrulline, and of citrulline to ornithine, with liberation of ammonia (Slade et al., 1954). Motility of the cells can be activated by air or arginine (Sherris et al., 1957). In the latter case the required energy is provided by the ATP formed in the citrulline to ornithine step. This system is constitutively produced by fluorescent pseudomonads but it is inhibited under aerobic conditions.

A rise of pH in the medium with arginine, because of ammonia liberation, indicates the presence of the dihydrolase system, and this is the basis of one of the determinative methods (Thornley, 1960). This method is still widely used because it is simpler (although less reliable) than methods based on arginine disappearance. Presence of arginine deiminase is not diagnostic for species of the genus *Pseudomonas* (for instance, *P. stutzeri* is negative, while species now assigned to other genera of aerobic pseudomonads are positive).

Several other pathways of arginine degradation have been described for species of *Pseudomonas*. They are characterized by the following key reactions: arginine oxidase in *P. putida* and arginine decarboxylase in fluorescent *Pseudomonas* species (Stalon and Mercenier, 1984), arginine dehydrogenase in *P. aeruginosa* (Jann et al., 1988), and arginine succinyl transferase in fluorescent *Pseudomonas* species and in *P. mendocina* (Vander Wauven and Stalon, 1985; Stalon et al., 1987). The arginine decarboxylase/agmatine deiminase pathway appears to be characteristic of *Pseudomonas*, and it is the source of polyamines (Stalon and Mercenier, 1984). A brief discussion on the importance of these compounds as taxonomic tools will be given below, in the section on taxonomic considerations.

Pseudomonas species use only one of the various agmatine catabolic pathways, in which there is participation of an inducible agmatine deiminase system. Evidence for the presence of arginase (which converts arginine to ornithine and urea) has not been found in any *Pseudomonas* species. Further studies on arginine metabolism by organisms of this genus have included a number of ureido and guanido compounds, with findings that suggest that the corresponding screening may be of taxonomic value (Tricot et al., 1990).

A study of the distribution and induction of guanidinoacetate amidinohydrolase (GAH), guanidinopropionate amidinohydrolase (GPH), and guanidinobutyrate amidinohydrolase (GBH) among some fluorescent species of *Pseudomonas* has shown the presence of GBH in most strains, with arginine as an inducer (Yorifuji et al., 1983). GPH was detected in *P. aeruginosa* only, and the results further indicate that this species lacks the *P. putida* enzyme(s) required for degradation of L-arginine to 4-guanidinobutyrate.

The information collected on the metabolism of arginine and related compounds suggests an interesting phylogenetic scenario. The arginine deiminase pathway may be a primitive remnant from the time when anaerobic conditions prevailed on our planet, and amino acids may have provided both building blocks and energy (Prieto et al., 1992). If so, the appearance of arginine oxidase may have occurred in more recent times.

In summary, in spite of the variety of arginine catabolic pathways among the aerobic pseudomonads, members of the genus *Pseudomonas* are characterized by the presence of the arginine decarboxylase/agmatine deiminase system. The finding of arginine dehydrogenase in *P. aeruginosa* (Jann et al., 1988) would require a survey of other species not yet tested in order to assess its phylogenetic significance.

Lysine catabolism can occur in *Pseudomonas* by at least three different pathways eventually converging to glutarate, which generates acetyl-CoA. For short, these pathways may be called the "oxygenase", the "pipecolate", and the "cadaverine" routes (Chang and Adams, 1971; Miller and Rodwell, 1971; Fothergill and Guest, 1977), and their distribution in species of *Pseudomonas* is indicated in Table BXII.γ.108.

Amino acid biosynthesis Catabolic pathways offer a larger variety of metabolic routes than the biosynthetic ones. A good example is the diversity of arginine degradative pathways discussed above. In contrast, similar anabolic routes can be used by widely different organisms. Biosynthetic pathways also are interconnected with other pathways and regulatory systems, and consequently there is a selective pressure favoring their conservation, thus preventing perturbations in pathways belonging to the same network. However, taken in combination with its regulatory mechanisms, a given biosynthetic pathway often can be used as a tool for the exploration of distant relationships among organisms.

The regulatory mechanisms of biosynthetic pathways of some aliphatic amino acids in the pseudomonads have interesting phylogenetic implications. The regulation of activity of aspartokinase and homoserine dehydrogenase is clearly different in organisms of the fluorescent group and species now assigned to other genera. Long ago it was found that *P. stutzeri* and *Burkholderia cepacia* (formerly *Pseudomonas cepacia*) resembled the fluorescent species (Cohen et al., 1969). In fact, this was one of the first findings suggesting a phylogenetic relationship between *P. stutzeri* and other species of *Pseudomonas*. However, *B. cepacia* also showed similarities with the fluorescent pseudomonads, in spite of its phylogenetic position in a different branch of *Proteobacteria*.

The multibranched pathway of biosynthesis of aromatic amino acids has offered not only more variations in biochemical details, but also a richer source of regulatory information. These are the reasons for which Jensen and his collaborators chose the interconnected pathways as a model for the study of phylogenetic relationships among the aerobic pseudomonads. The results of this work, which has direct taxonomic implications (Byng et al.,

TABLE BXII.γ.108. Distribution of lysine catabolic pathways in some fluorescent *Pseudomonas* species and *Burkholderia cepacia*^a

Lysing catabolic pathways	<i>P. aeruginosa</i>	<i>P. fluorescens</i>	<i>P. putida</i> biovar A	<i>P. putida</i> biovar B	<i>Burkholderia cepacia</i>
Oxygenase	—	+	+	+	—
Pipecolate	+	+	+	—	+
Cadaverine	+	+	—	—	—

^aFor symbols see standard definitions. Taken from Palleroni (1984); data from Miller and Rodwell (1971), Chang and Adams (1971), and Fothergill and Guest (1977).

1980, 1983; Whitaker et al., 1981b) were in agreement with the subdivision of the pseudomonads into five so-called rRNA homology groups (Palleroni et al., 1973). In fact, the enzymatic assays are simpler than the hybridization methods of ribosomal similarity studies, which makes the former good determinative tools indeed.

Glutamine synthetase, an important enzyme involved in nitrogen metabolism, has been subjected to immunological comparative studies that were expressed in combination with the results of nucleic acid hybridization experiments (Baumann and Baumann, 1978). These studies, aside from representing further confirmation of the internal subdivision of the pseudomonads, have useful determinative applications.

Genetics Some of the *Pseudomonas* species have attracted a great deal of attention from bacterial geneticists because of their widespread occurrence, their biological and medical importance, their nutritional and biochemical versatility, and the simplicity of conditions required for their cultivation in the laboratory.

Of them, *P. aeruginosa* is by far the best known from the genetic point of view. Strain PAO ("*Pseudomonas aeruginosa* One") is the one that has been most extensively studied. A detailed genomic map including more than 360 genes is now available (Holloway et al., 1994). This map represents the last of a number of reports from the Department of Genetics of Monash University, Clayton, Australia (see below), in combination with work performed elsewhere on the construction of physical maps. Recent additions to the genomic map available for strain PAO include the physical mapping of 32 genes by application of modern molecular techniques. The approach used will increase the number of genes included in databases, where now nearly 40% of the genes of the species are represented (Liao et al., 1996).

A second strain of *P. aeruginosa*, PAT ("*Pseudomonas aeruginosa* Two"), is also familiar to geneticists, and its map closely resembles that of PAO. The strains have different geographical origin, and their similarity is a confirmation of the homogeneity of the species already noticed by many taxonomists.

The three best-known systems of genetic exchange, namely, conjugation, transduction, and transformation, have been observed in strains of *P. aeruginosa*. Conjugation and transduction have been most important in genetic studies of this species. In contrast, transformation occurs in *P. stutzeri*, *P. mendocina*, and other nonfluorescent members of the genus under natural conditions. Conjugation, which results in transfer of substantial chromosomal segments, has been the most effective source of information for mapping purposes in *P. aeruginosa* genetic studies. The transfer depends on the chromosomal mobilizing ability (cma) of some plasmids, of which FP2 has been used extensively, and to a lesser extent, FP5, FP39, and FP110. Each of these fertility plasmids has one predominant attachment site, and this was the main reason that delayed the demonstration of circularity of the map. This was finally achieved based on 2- and 3-factor crosses with cma plasmids FP2, FP5, FP110, and R68.45 (Royle et al., 1981).

An approach recently introduced in the study of bacterial genomes combines the action of endonucleases that infrequently cut the chromosome, with pulsed-field agarose gel electrophoresis (PFGE), which makes possible the resolution of large DNA fragments. A limited number of cuts per genome is desirable for genomic restriction mapping (McClelland et al., 1987), with clear advantages over frequently cutting endonucleases for the production of fingerprints characteristic of different taxa (Mielenz et al., 1979; Dobritsa, 1985; Sorensen et al., 1985). Application of this principle has resulted in a practical method for constructing and analyzing macrorestriction patterns of 234 strains of different species of aerobic pseudomonads (Grothues and Tümmeler, 1991). The restriction nucleases *AsnI*, *DraI*, *SpeI*, *XbaI*, and *PacI* are among the most appropriate by being specific for AT-rich regions (which are less common than GC pairs in *Pseudomonas*), or for sites including the extremely uncommon tetranucleotide CTAG (McClelland et al., 1987).

The macrorestriction patterns are compared for the number and position of the bands, and use of appropriate equations often indicates a correlation between the estimated similarities of fingerprints and conventional taxonomic groupings. The results obtained by Grothues and Tümmeler (1991) in general confirm the classification of pseudomonads based on extensive numbers of phenotypic properties and nucleic acid hybridization studies.

Comparisons involving mol% G + C content of the chromosome, codon usage, and genome size can be used for the calculation of similarity coefficients useful for determinative purposes. The genome restriction patterns also have the practical advantage of helping in epidemiological studies. Differences in these parameters were found between strains susceptible or resistant to multiple antibiotics (Yamashita et al., 1997).

Differences in restriction fragment size distribution may result from various chromosomal rearrangements and/or mutations at the restriction sites, which suggests that the methodology based on restriction fingerprints should be supplemented with other approaches to determine the degree of similarity of the fragments. Large genomic rearrangements have been observed in *P. aeruginosa* strains isolated from clinical samples or from the environment (Schmidt et al., 1996c). The former came mostly from cystic fibrosis cases. A 95-kb plasmid was detected in environmental strains, and it was integrated in the chromosome in cystic fibrosis strains. Exchange of DNA blocks and large DNA inversions led to divergence of clones in this species. The presence of inversions only in cystic fibrosis strains suggests that this niche causes or tolerates substantial changes in the genome (Römling et al., 1997).

A study on pathovars of fluorescent plant pathogenic pseudomonads (Grothues and Rudolph, 1991) indicated that two strains belonging to the same pathovar, but of different origins, can give almost identical restriction fingerprints. However, this is not always the case. Pathogens of wide host range give more diverse restriction patterns than those of restricted host range. The methodology was also used for the examination of repre-

sentative strains of *P. stutzeri*, a species notorious for its heterogeneity (Stanier et al., 1966; Palleroni et al., 1970). The results show a marked correlation of genome structure with fatty acid composition, and with data of nucleic acid hybridization experiments (Rainey et al., 1994b). A high degree of heterogeneity in macrorestriction patterns was also observed by another group of workers, which did not correlate with the subdivision of the species in genomovars. The marked heterogeneity of this species is attributed, at least in part, to large chromosomal rearrangements (Ginard et al., 1997). The results of these two groups also indicate that the genome size of *P. stutzeri* ranges from 3.4–4.64 Mba for the strains subjected to their studies.

Pseudomonas species other than *P. aeruginosa* have been less rewarding subjects for chromosomal mapping. This is unfortunate in the case of a species like *P. putida*, which has been extensively investigated from the biochemical viewpoint. This species has presented serious problems for the development of a satisfactory chromosome transfer system, and, consequently, a less detailed chromosome map is available (Mylroie et al., 1977; Strom and Morgan, 1990). Circularity of the *P. putida* map strain PPN (related to strain ATCC 12633, which was the subject of many metabolic studies for many years) has been demonstrated (Dean and Morgan, 1983). A genetic map of *P. syringae* pathovar *syringae* is also available (Nordeen and Holloway, 1990). The situation is even more regrettable in the case of the very complex species *P. fluorescens*, with many strains extensively known for their phenotypic properties. A good starting point now is the report on a complete physical map of a strain of the species with a genome of 6.63 Mbp. A total of 139 restriction sites and 31 genes have been located in the map (Rainey and Bailey, 1996). Similarly, an efficient mutagenic procedure using electroporation for *P. fluorescens* and a transposon delivery vector has been described (Artiguenave et al., 1997). A *recA* mutant has been obtained, and sequence studies have shown that the chromosomal organization was very similar to that of *P. aeruginosa* and *Azotobacter vinelandii*. The regulatory region and the structural gene differed from those of *Burkholderia cepacia*. By insertion of a kanamycin cassette in the *recA* gene, the mutant obtained had an increased UV sensitivity and was much impaired in its recombinatorial activity (De Mot et al., 1993).

A *recA* mutant strain of *P. stutzeri* is also available (Vosman and Hellingwerf, 1991). It is completely deficient in natural transformation with chromosomal DNA and it is sensitive to UV and methyl-methane sulfonate. The wild-type gene complements an *E. coli recA* mutant (Bennasar et al., 1996).

Plasmids, phages, bacteriocins Plasmids are important components of the genetic makeup of *Pseudomonas*. Some of them act as fertility factors, some (R plasmids) may impart resistance to various agents, and others confer the capacity of degradation of unusual carbon sources, thus contributing to the nutritional versatility that is a striking feature of many members of the genus. A large number of *Pseudomonas* plasmids have been described in the past decades, and it is impossible to cover all references here. Some of the plasmids are mentioned in other sections of this chapter. Examples of the properties of *Pseudomonas* encoded by plasmid genes are:

1. Resistance characters to antibiotics and other antibacterial compounds, such as carbenicillin, chloramphenicol, gentamicin, kanamycin, streptomycin, tetracycline, tobramycin, sulfonamides;
2. Resistance to chemical and physical agents (borate, chromate, various metal ions, organomercurials, tellurite, ultraviolet radiation);
3. Resistance to bacteriophage propagation; interference with lysogenization by some temperate phages; DNA restriction and modification;
4. Resistance to bacteriocins.

Various other characteristics due to plasmid genetic determinants are the chromosome donor ability, donor-specific phage susceptibility, inhibition of bacteriocin production, fertility inhibition, incompatibility with other plasmids, and, most important for some species, utilization of various organic compounds not normally used for growth by species other than the aerobic pseudomonads. Occasionally plasmids are found that confer on the cells the capacity to use simple organic compounds. Thus, for instance, a *P. fluorescens* plasmid that carries genes of resistance to ampicillin, kanamycin, and streptomycin also allows the host to grow on malonate because of a plasmid gene encoding malonate decarboxylase. The plasmid (pPSF1) can be transferred to *E. coli*, where the ability to grow on malonate can also be expressed (Kim and Kim, 1994).

Plasmids can be classified most effectively by incompatibility, that is, by the incapacity of a given plasmid to coexist in the same cell with other plasmids of the same group. In *Pseudomonas*, at least 10 incompatibility groups have been defined (Jacoby, 1977; Korfhagen et al., 1978). Useful information on *P. aeruginosa* plasmids may be found in a chapter by Jacoby (Jacoby, 1979). In general it can be stated that most of the research with *Pseudomonas* plasmids has been focused on the fluorescent species, and the information on other species is fragmentary.

Plasmids of the various incompatibility groups have different host ranges. The widest range is that of the IncP-1 plasmids, while those of groups 2 and 5–9 are more specific. IncP-2 plasmids include R factors and plasmids carrying genes for the degradation of unusual carbon compounds. Some of them are among the largest of *Pseudomonas* plasmids, exceeding 300 MDa, while the majority range from 10–60 MDa. Outside of the IncP-2, degradative plasmids are also found in the P-9 group.

Degradative plasmids named CAM, OCT, SAL, NAH, TOL, and XYL are involved in the degradation of camphor, *n*-octanol, salicylate, naphthalene, toluates, and xylene, respectively, and have received much attention because they confer to the cells the capacity to degrade environmental pollutants. The naphthalene degradative enzymes of plasmids identified in *P. fluorescens* are also involved in the degradation of high molecular weight polyaromatic hydrocarbons other than naphthalene (Menn et al., 1993).

Of the above-mentioned plasmids, TOL, which was originally isolated from a strain now classified as *P. putida*, is the best known and it still receives a great deal of attention by bacteriologists. An excellent review of the properties of TOL and naturally occurring variants is available (Assinder and Williams, 1990). The upper and lower operons of this plasmid seem to have different origins. In support of this hypothesis, it has been found that their respective codon usages are different, but they are the same for the genes within each operon (Harayama, 1994). TOL can be transferred from cell to cell by conjugation, which occurs at a sufficiently high rate to maintain the plasmid in a dense microbial community without the help of selective pressures (Smets et al., 1993).

Comai and Kosuge (1980) have presented evidence for plas-

mid involvement in the oxidative degradation of tryptophan to the plant hormone indoleacetic acid (IAA) by *P. savastanoi*. Concentrations of IAA higher than those normally found in plant tissues result in gall formation (Gardan et al., 1992).

The range of antibiotic resistance determined by plasmids and the mechanisms by which the resistance is manifested are similar to those found in *E. coli*. Some plasmids are transferred from cell to cell by conjugation; others lack this capacity, but some of them can be mobilized by other conjugative plasmids.

Many different lytic and temperate bacteriophages have been identified in *Pseudomonas*, particularly in *P. aeruginosa*. Their morphological diversity is at least as great as for phages of other bacterial genera. Lysogeny is a very common phenomenon in *P. aeruginosa*, and transducing phages have been very useful in linkage studies. Most *Pseudomonas* phages contain double-stranded DNA, but some are RNA phages, including one with double-stranded RNA (Semancik et al., 1973; Vidaver et al., 1973). The sequence of *P. aeruginosa* single-stranded RNA phage PP7 has been determined and the results suggest that the phage is related to coliphages but branched off before the coliphages diverged into separate groups (Olsthoorn et al., 1995).

An example of a rather unusual application of a *P. stutzeri* bacteriophage has been its use for the selection of denitrification-negative mutants. The procedure takes advantage of the fact that the phage (ϕ PS5) adsorbs to the outer membrane protein NosA. Mutants defective in NosA production do not grow with N_2O and are resistant to the phage (Clark et al., 1989).

Even though host specificity is the rule, some bacteriophages attack different but related species, with interesting taxonomic implications (M.P. Starr, personal communication). This approach to bacterial classification was attempted many years ago by Billing, in an effort to improve the methodology for the differentiation of phytopathogenic pseudomonads (Billing, 1963, 1970a). The sensitivity tests were not very useful in themselves, but they were valuable as a complement of the biochemical tests of identification. Pathovars of *P. syringae* isolated from pear could be distinguished by phage sensitivity but not by serological tests from the strains isolated from cherry. On the other hand, pathogenic and saprophytic pseudomonads could not be distinguished by their lysotypes (phage types or phagovars) (Crosse and Garrett, 1963).

Phage sensitivity can also be used successfully as a method of typing *P. aeruginosa* strains, but the procedure is not considered sufficiently reliable to be used by itself as an epidemiological tool without comparison to other typing procedures (Brokopp and Farmer, 1979).

Some factors required for phage nucleic acid replication appear to have been conserved through evolution in some groups of Gram-negative bacteria. One of the elements required for *in vitro* replication of sex-specific single-stranded RNA coliphage Q β is the so-called host factor (HF). This is a heat-resistant RNA-binding protein of molecular weight 12,000, usually present in *E. coli* as an hexamer (Franze de Fernández et al., 1972). In *P. putida*, a polypeptide of molecular weight 11,000 gives an immunological cross-reaction with HF and allows Q β replication *in vitro* (DuBow and Blumenthal, 1975). Material cross-reacting with *E. coli* HF antiserum was found in extracts of *P. aeruginosa*, *P. fluorescens*, *P. putida*, and in other species of different rRNA similarity groups (Palleroni, 1984), but the material was heat stable only when it came from organisms of RNA similarity groups I and V (*Pseudomonas*, *Xanthomonas*, and *Stenotrophomonas*), which

are located in neighbor branches in the *Gamma*proteobacteria (DuBow and Ryan, 1977).

Bacteriocins are proteins produced by some bacterial strains, which have a lethal action on other strains of the same species. They have been frequently detected in *Pseudomonas*, and in *P. aeruginosa* they have been named pyocins (from the old name *P. pyocyanea*) and aeruginosins, although both names are improper. Pyocin can be mistaken for pyocyanin, the blue diffusible pigment characteristic of the species, and aeruginosin has been used for two red pigments produced by some strains of *P. aeruginosa* (aeruginosins A and B) (Holliman, 1957; Herbert and Holliman, 1964).

Different types of *P. aeruginosa* bacteriocins have been described. One resembles bacteriophage contractile tails, a second one has the appearance of slender, flexuous rods, and a third (S type) is amorphous and is sensitive to proteolytic attack (Govan, 1974a, b). A novel S-type pyocin has been characterized by molecular techniques (Duport et al., 1995).

Bacteriocin typing can be performed in two ways, which are based on production of these elements, or on sensitivity to them, respectively. The first approach is more commonly used, and it requires a collection of sensitive strains (Govan, 1978; Brokopp and Farmer, 1979). Bacteriocin production by species other than *P. aeruginosa* has been reported in a few instances (Vidaver et al., 1972; Smirnov et al., 1984). In a study on bacteriocin typing of unknown *P. syringae* strains, the specificity was uneven, and the highest level was shown for the case of the phaseolins of *P. syringae* pathovar *phaseolicola* (in this treatment described as a pathovar of *P. savastanoi*) (Vidaver et al., 1972). In an extension of these studies, the correlation found between bacteriocin type and host plant of origin of phytotoxin production was rather poor (Vidaver and Buckner, 1978). The approach is probably susceptible to further refinements, since 86% of the strains examined were able to produce bacteriocins.

Antigenic structure Again here, the literature is dominated by the number of papers dedicated to *P. aeruginosa*, reflecting in this case, as in others, the medical importance of this species.

Agglutination of intact *P. aeruginosa* cells can be caused by specific antibodies elicited in animals by specific components that are similar to the O-antigens of other Gram-negative bacteria. The heat-stable O-antigen, considered the most stable marker for *P. aeruginosa*, is represented by one of the components of the lipopolysaccharide (LPS). The specificity is related to the composition of the polysaccharide chains (the O antigens themselves) projecting to the outside of the cells. The LPS of *P. aeruginosa* has O-specific polysaccharide chains with unbranched oligosaccharides including amino sugars that are not acetylated (Wilkinson, 1983).

P. aeruginosa shows a marked serological diversity. The LPS composition determines at least 17 heat-stable O antigens, and antisera are adequate for serotype identification, in spite of variations in titer and specificity. In experiments using a panel of 48 monoclonal antibodies against eight of the serotypes, various degrees of activity were shown by some of the antibodies, one of which bound to all serotype strains and also to strains of *P. fluorescens* and *P. putida* (Gaston et al., 1986). A further refinement of this approach showed that the antigenic specificity of various parts of the LPS molecule could be clearly demonstrated by preparing monoclonal antibodies recognizing the inner core, outer core, and lipid A regions of LPS. Antibodies to the complex (lipid A + core + one repeat of the O-specific polysaccharide chain)

reacted with a lower number of serotypes than the antibodies recognizing the outer core without the oligosaccharide. The specificity was even lower for the antibodies that recognized the inner core, since they reacted not only with the largest number of serotypes tested, but with all the other Gram-negative species included in the experiment (de Kievit and Lam, 1994).

Mild heating of a cell suspension reduces the agglutinability by specific antisera, but the property can be fully recovered by a more intense treatment (100–120°C for 2–2.5 h). The alginate of mucous strains that can be selected under appropriate conditions (Govan and Fyfe, 1978) or isolated from cystic fibrosis cases (Doggett, 1969) does not interfere with the O-antigenicity.

In addition to the immune response against *P. aeruginosa* LPS, which in its early manifestation consists of IgM antibodies (Høiby, 1979), there is a humoral response to cross-reactive antigens that are present in other, mostly Gram-negative, bacteria (Høiby, 1975). A heat labile antigen common to a wide range of bacteria has been isolated from *P. aeruginosa* and shown to be an acidic protein composed of subunits of molecular weight 62,000, present in the cytoplasmic fraction of the cells (Sompolinsky et al., 1980a, b).

Various other antigens have been identified in *P. aeruginosa*. Heat-labile surface antigens are represented by flagella and fimbriae (Bradley and Pitt, 1975; Pitt and Bradley, 1975). An extracellular slime can elicit the production of an agglutinin, and exoenzymes such as phosphatases, proteases, and phospholipases can also act as antigens. The outer membrane protein OprF was subjected to epitope mapping, and, as mentioned before, the fact that it is a major component suggests that it may be a good candidate for use as a vaccine and as a target for monoclonal antibodies for immunotherapeutic applications and for diagnosis (Rawling et al., 1995).

Isolation and characterization of monoclonal antibodies for outer membrane antigens have been described (Hancock et al., 1982). As a practical identification tool for the identification of glucose oxidizing *Pseudomonas* species, Mutharia and Hancock (1985) have proposed the use of a monoclonal antibody (MA1-6), specific for a single antigenic epitope on the outer membrane lipoprotein H2 of *P. aeruginosa*. The epitope was detected in all 17 serotype strains of *P. aeruginosa*, in numerous clinical isolates of the same species, and in other *Pseudomonas* species but not in species of other rRNA similarity groups (Palleroni, 1984). Two strains of *P. aeruginosa* (out of a total of 52) failed to give the reaction. In a *P. aeruginosa* collection that included 30 environmental isolates, no major differences in outer membrane proteins (other than quantitative variations in lipoprotein H2 content) or lipopolysaccharide patterns were observed when compared with those of previously studied clinical isolates (Hancock and Chan, 1988).

Similar conclusions on the advantages of using monoclonal antibodies against outer membrane proteins, in addition to other tools useful for determinative purposes, have been reached from experiments on the detection of the lipoprotein I gene in species of *Pseudomonas* (De Vos et al., 1993), as an extension of the original observations by Saint-Onge et al. (1992).

Many schemes of serotyping have been proposed, but the system of Habs (1957) has gained wide acceptance and, with some modifications, is in general use. Habs defined 12 somatic groups that could be identified by agglutination tests. Different additional proposals and modifications include the addition of 5 O-groups. Standard strains and typing sera are commercially available.

In the opinion of Brokopp and Farmer (1979), serological typing of *P. aeruginosa* based on the O-antigens produces more reliable evidence for relatedness than can be obtained by other, less specific typing methods. Thermolabile surface antigens can also be used for serotyping, and various methods based on flagellar (H) antigens have been proposed (Verder and Evans, 1961; Lányi, 1970). These approaches have not been widely accepted, the main reason being the difficulties encountered in the preparation of specific flagellar antisera (Brokopp and Farmer, 1979).

For epidemiological purposes, serotyping is the most important typing method and, as usually practiced, consists of the reaction of a cell suspension toward a standard set of antibody preparations. Serotyping can be supplemented by other typing methods, such as production and sensitivity to bacteriocins and lytic phages, biotyping (or characterization of strains by their biochemical and physiological properties) and antibiograms, and sensitivity to antibiotics.

To these procedures we now have to add modern techniques based on molecular concepts. One of them is the arbitrary PCR fingerprinting of strains, which provides a simple and practical typing approach considered to be more discriminatory than the traditional serotyping scheme, although the maximum discriminatory power is achieved by a combination of both methodologies (Hernández et al., 1997). Comparative typing of *P. aeruginosa* also has been carried out by random amplification of polymorphic DNA or pulsed-field gel electrophoresis of DNA macrorestriction fragments (Renders et al., 1996), to which we can add the fingerprinting of whole cell proteins (Khan et al., 1996), and studies on comparative ribotyping and genome fingerprinting (Bennekov et al., 1996).

In the field of fluorescent plant pathogenic pseudomonads, the serological approach has had variable success, and has not helped substantially in the circumscription of nomenspecies. The lack of a solid taxonomic frame of reference has often made it very difficult to interpret the resolving power of the serological techniques. Otta and English (1971) were unable to define a precise serological differentiation of virulent strains from non-virulent ones. On the other hand, species-specific antigens could be identified in *P. syringae* pathovar *phaseolicola* (described here as a pathovar of *P. savastanoi*) (Guthrie, 1968) and in *P. syringae* pathovar *lachrymans* (Lucas and Grogan, 1969a, b). *P. syringae* strains isolated from different groups of host plants could be distinguished immunologically (Otta, 1977). In general, however, the reproducibility of serological data by different authors using different strains of the same nomenspecies appears to be rather poor.

Susceptibility to antibiotics Many aerobic pseudomonads are resistant to a number of antibacterial agents. This subject is of particular medical importance because members of this group are serious opportunistic human pathogens and are often isolated from patients and clinical materials. The capacity to resist antibiotics is included in some of the descriptions of new taxa.

As expected, most of the available information on *Pseudomonas* refers to *P. aeruginosa* because of its importance as a serious opportunistic human pathogen. The antibiotics that are most effective in the treatment of *P. aeruginosa* infections include some β -lactams, such as carbenicillin and ticarcillin, third-generation cephalosporins (cefsulodin, cephoperazone, ceftazidime), the synthetic monocyclic β -lactam aztreonam, carbapenems (among them the extremely broad-spectrum semisynthetic imipenem), the aminoglycosides (gentamicin, tobramycin), and the quino-

lones (ciprofloxacin). In practice, *P. aeruginosa* infections are treated with a mixture of tobramycin and some β -lactams active against *Pseudomonas*, such as piperacillin or azlocillin. Various quinolones are in use for the treatment of chronic or mild corneal or urinary infections.

P. aeruginosa and other fluorescent *Pseudomonas* species are in general resistant to β -lactams other than those indicated above. Carbenicillin is moderately effective and was extensively used in therapeutic applications either by itself or in mixtures with aminoglycosides. Carbenicillin produces cell enlargement and filament formation, but the sensitivity of the cells is not very high, and years ago the emergence of resistant *P. aeruginosa* mutants was reported (Lowbury et al., 1969; Gaman et al., 1976). The mucoid strains that are frequently found in infections in cystic fibrosis patients are somewhat more resistant to the antibiotic.

Resistance to carbenicillin may be due to β -lactamases coded for by genes carried by plasmids of the incompatibility group 1 (IncP-1) that can be transferred to *E. coli* (Lowbury et al., 1969), but similar genes may be found in host restricted IncP-2 plasmids (Bryan, 1979). Seven different kinds of β -lactamases have been identified in 24 *Pseudomonas* plasmids belonging to at least eight of the incompatibility groups (Jacoby and Matthew, 1979). β -lactamase genes are not invariably carried by plasmids and, in fact, there is in *P. aeruginosa* a constitutive penicillinase (Furth, 1975) and an inducible cephalosporinase (Sabath et al., 1965; Sykes and Matthew, 1976) whose genetic determinants are chromosomal.

Even though the third-generation cephalosporin ceftazidime is considered to be highly effective against many *P. putida* strains (Yang et al., 1996b), a metallo- β -lactamase gene *blaIMP* was detected by PCR in a number of Gram-negative organisms, among them resistant strains of *P. fluorescens* and *P. putida* (Senda et al., 1996). It is also interesting that the response to β -lactam antibiotics in *Pseudomonas* can result in their utilization as substrates for growth. A soil strain of *P. fluorescens* was able to grow at the expense of benzylpenicillin as carbon, nitrogen, and energy source (Johnsen, 1977). This property also is present in pseudomonads now assigned to different genera (for instance, *Burkholderia cepacia*) (Beckman and Lessie, 1979).

Aside from the production of enzymes such as the β -lactamases, the pseudomonads are able to resist many antibiotics by mechanisms such as a low cell wall permeability, the production of modifying enzymes, and efflux systems. The most abundant porin in the outer membrane, OprF, is probably very important in reducing the permeability to antibiotics, although other porins may be involved.

Enzymes capable of modification of aminoglycosides as a resistance mechanism have been described, and they correspond to the types already known for the enteric bacteria. These enzymes catalyze phosphorylation, adenylation, or acetylation of the antibiotics, although the resistance of *P. aeruginosa* to the aminoglycosides may be largely nonenzymatic.

A useful and comprehensive review on multidrug efflux systems present in both eucaryotes and bacteria has been published by Paulsen et al. (1996). Drug accumulation and efflux system studies suggest that *P. aeruginosa* has at least two different proton motive force-dependent efflux systems (Li et al., 1994, 1995). A chromosomal operon was discovered that confers multidrug resistance to this organism, with three genes, *mexA*, *mexB*, and *oprM*, encoding for components of a transport system originally shown to be able to correct a defect in iron metabolism (Poole et al.,

1993a, b; Ryley et al., 1995). Prior to this work, it was known that OprM was involved in conferring resistance to multiple drugs, and it was later confirmed as a component of the efflux system. The operon *mexAB/oprM* is inducible under iron-limiting conditions, and is co-regulated with components of the pyoverdinin-mediated iron transport. Mutants lacking *mexA* or *mexB* are unable to grow under conditions of iron limitation in the medium. All *P. aeruginosa* strains that have been examined have the operon, although its components seem to operate with unequal efficiencies according to the strain.

Mutations in the operon result in intracellular accumulation of chloramphenicol, tetracycline, norfloxacin, benzylpenicillin, carbenicillin, and quinolones. A second efflux system was later identified (*mexC-mexD-oprJ*) encoding proteins that prevent the cytoplasmic accumulation of quinolones, tetracycline, chloramphenicol, and the cepheims (Li et al., 1994, 1995; Liu et al., 1995b; Poole et al., 1996).

The most common mechanisms of gentamicin resistance in *P. aeruginosa* are enzymatic modifications of the antibiotic consisting of *N*-acetylation, *O*-adenylation, and, to a much lesser extent, phosphorylation. These properties are controlled by plasmid genes, mainly IncP-2 plasmids (Bryan et al., 1972, 1973, 1974; Jacoby, 1974a, b). Permeability factors may also be involved in the case of gentamicin and other aminoglycosides (Mathias et al., 1976; Bryan, 1979).

Polymyxins are very active against *P. aeruginosa* and other pseudomonads when tested *in vitro*, but their efficacy *in vivo* is limited. Resistance is unstable, and this condition seems to correlate with an increase in EDTA tolerance. Gilleland and Murray (1976) noted in these variant strains the disappearance of the particles characteristic of the concave cell wall layer separated by freeze-etching (see above in the section on fine structure), but the particles reappeared by growth in the absence of polymyxin, and acquisition of temporary resistance is accompanied by a decrease in the phosphorus content of the outer membrane.

Sensitivity to metals and metalloids Copper compounds are extensively used in agriculture, and since only minute amounts are needed for nutrition, the excess can affect higher organisms as well as procaryotes, among which the plant pathogens are of particular interest. In *P. syringae* strains that carry the operon *cop*, copper is excluded by combination with components of the periplasm and in outer membrane. A two-component sensory transduction mechanism similar to equivalent systems in other organisms operates in the regulation of expression of copper resistance, and it has strong similarity with a gene that seems to regulate the uptake or efflux of copper in "*Streptomyces lividans*" (Mills et al., 1993).

Inducible copper resistance in *P. aeruginosa* is encoded by chromosomal genes, although the strains that were examined had plasmids (Vargas et al., 1995b).

In *P. fluorescens* isolated from copper-contaminated soil, a chromosomal locus with determinants for copper resistance and competitive fitness was cloned, and genes responsible for conferring copper resistance and production of cytochrome *c* were identified (Yang et al., 1993a, 1996a). A *P. putida* strain isolated from electroplating effluent could accumulate cupric ions in a concentration as high as 6.5% of its dry weight. The capacity was highest when the culture was previously grown under sulfate-limiting conditions (Wong and So, 1993).

Acid-labile sulfide levels were found to be generally higher in silver-resistant *P. stutzeri* strains than in sensitive ones. The resis-

tance to the metal may be due to formation of silver sulfide complexes, since no complex formation with polyphosphate or metal-binding proteins has been found to be the cause (Slawson et al., 1992).

Three plasmids were found in a silver-resistant strain isolated from a silver mine. The largest plasmid ($MW\ 49.4 \times 10^6$), which specifies silver resistance, is nonconjugative, but it could be transferred to *P. putida* by mobilization with plasmid R68.45 (Haefeli et al., 1984).

P. fluorescens detoxifies aluminum by elaboration of a soluble, aluminum-complexing metabolite. When iron was present in the medium, the two trivalent metals were immobilized in a lipid-rich complex containing Al, Fe, and P, after an early stage in which aluminum was found to be associated with phosphatidylethanolamine (Appanna et al., 1995; Appanna and Hamel, 1996). Workers from the same group examined the adaptation of *P. fluorescens* to stress caused by excess of cesium (Appanna et al., 1996), and were also involved in determining the basis for resistance of the species to various metals (aluminum, iron, zinc, calcium, and gallium). These metals were supplied as complexes with citrate, which was completely oxidized. As in the case of the resistance to aluminum mentioned above, the metals appeared associated to phosphatidylethanolamine, which later was found in the lipid-rich complex where the metals were located (Appanna and St. Pierre, 1996). In a study on the mechanism of metal-citrate complexes by *P. fluorescens*, the biodegradation depended on the nature of the complex. Thus, the bidentate ferric, nickel, and zinc citrate complexes were readily degraded, while the tridentate cadmium and copper citrates were not. The latter, similarly to uranium citrate, were neither transported inside the cells nor metabolized by cell-free extracts (Joshi-Tope and Francis, 1995).

About one-half of the *P. aeruginosa* known plasmids confer resistance to mercuric ions (Jacoby and Shapiro, 1977). A plasmid (pPB) was found to confer mercury and organomercurial resistance to a *P. stutzeri* strain. The plasmid had two regions of functional and independently regulated *mer* genes, probably transcribed from different promoters (Reniero et al., 1995). Resistance to chromium, boron, and tellurium was also determined by some *Pseudomonas* plasmids (Summers et al., 1978). After growth in the presence of increasing concentrations of tellurite, strains of *P. aeruginosa* and *P. putida* harboring certain plasmids that determine tellurite resistance accumulated crystalline structures containing tellurium in their periplasmic space. From there, these materials were released into the medium in vesicles that are pinched off the outer membrane (Suzina et al., 1995).

Operons of genes coding for arsenic resistance are usually carried in plasmids. Arsenate is reduced to arsenite, which is eliminated by an export system. A chromosomal operon was identified and cloned in *E. coli* and was able to hybridize with chromosomal of other enteric bacteria and *P. aeruginosa*. This chromosomal operon may be the evolutionary precursor of the plasmid operons, with the advantage of a multicopy system as a means of natural amplification (Diorio et al., 1995).

Antibiotic production Plant root colonization by certain strains of the species *P. fluorescens* and *P. putida* may result in an enhancement of plant growth, an effect that is accompanied by a selective inhibition of other bacterial species and fungi. In part this action is due to the production of siderophores, which remove iron required by organisms that lack systems for utilization of the ferric siderophores. An example is pseudobactin 358 (pro-

duced by *P. putida* WBS358) whose structure, resembling that of other siderophores, has been elucidated (Van der Hofstad et al., 1986). The molecular aspects of iron assimilation by the pseudobactin-producing organisms have been analyzed (Leong et al., 1992).

Fragments of the peptide component of pseudobactin do not have fungicidal activity, although they resemble the antifungal compounds pseudomycins (Okonya et al., 1995). However, the fluorescent organisms do produce antifungal antibiotics. Thus, a strain of *P. fluorescens* produces the antibiotics pyrrolnitrin, pyoluteorin, and 2,4-diacetylphloroglucinol, which are able to suppress root diseases due to fungal pathogens. Of the three compounds, the latter represents an important factor in the control of plant diseases. A 6.5-kb chromosomal fragment has been isolated that contains the genes that encode the biosynthetic enzymes (Bangera and Thomashow, 1996). An examination of this and other molecular aspects has led to identifying a sigma factor encoded by gene *rpoS* that has an influence on the production of the antibiotics, on the biological control activity, and on the survival capacity of the strain on plant surfaces (Fedi et al., 1996). Pyrrolnitrin requires the participation of four genes, the only determinants that appear to be absolutely necessary for the complete biosynthesis (Hammer et al., 1997).

In a *P. fluorescens* strain, the production of the above antifungal compounds and of an extracellular protease can be abolished by mutations of an antibiotic production gene (*apdA*). Interestingly, the sequence of this gene is strikingly similar to that of genes that encode sensor kinases required for the pathogenicity of *P. syringae* pathovar *syringae* and *P. viridiflava*, suggesting that *apdA* encodes a putative sensor kinase component of a classical two-component regulatory system that, in this case, is required for the synthesis of a secondary metabolite (Corbell and Loper, 1995).

A different strain of *P. fluorescens* was able to produce three antibiotics: 2,4-diacetyl-phloroglucinol, pyoluteorin, and + (S)-dihydroaeruginosic acid. This last compound (detected for the first time in the reported investigations) inhibits phytopathogenic fungi—among them, *Septoria tritici* (Carmi et al., 1994). Antifungal compounds produced by *P. fluorescens* include chitinase, cyanide, and pyrrolnitrin, all of which have a protective action on a variety of seedlings against damping-off caused by *Rhizoctonia solani* (Gaffney et al., 1994). Brominated derivatives of pyrrolnitrin were synthesized by a strain of *P. aureofaciens* (in this treatment considered to be a subspecies of *P. chlororaphis*) (Van Pée et al., 1983). The characteristics of fluorescent species of the genus *Pseudomonas* that participate in the inhibition of plant root pathogens have been analyzed in a useful review (O'Sullivan and O'Gara, 1992).

Miscellaneous antibiotics produced by fluorescent pseudomonads include (1) karalycin, an antibiotic isolated from a strain in the *P. putida*/*P. fluorescens* cluster, with some inhibitory action on yeasts, and a weak antiviral activity against herpes simplex viruses (Lampis et al., 1996a, b); (2) fluviols, a group of antibiotics with antitumor activity that were isolated from *P. fluorescens* and described (Smirnov et al., 1997); (3) 2-alkyl-quinolones, which inhibit growth of *Helicobacter pylori*, were produced by *P. aeruginosa* strains (Lacey et al., 1995); (4) fluopsins C and F (Shirahata et al., 1970); and (5) obafluorin, a β -lactone produced by *P. fluorescens*, with activity against *Bacillus* species. In spite of its β -lactone structure, this compound is hydrolyzed by β -lactamases (Wells et al., 1984); fosfomycin (phosphonomycin), is produced by a strain of *P. syringae* (Shoji et al., 1986).

An interesting practical application has been found for alginate, which is notorious for causing complications in cystic fibrosis cases. The polysaccharide has been found to be a convenient carrier for seed inoculation with genetically modified *P. fluorescens* strains producing antifungal antibiotics (Russo et al., 1996).

Pathogenicity for plants and mushrooms Phytopathogenic pseudomonads are allocated to three of the five rRNA similarity groups (Palleroni, 1984). Only the ones in rRNA group I, the present genus *Pseudomonas*, will be discussed here. Various symptoms produced in plants by these organisms such as tumorous outgrowth, rot, blight or chlorosis, and necrosis are the consequence of alterations of the normal metabolism of plant cells by substances excreted by the pathogens. Among these excretions there are toxins, plant hormones, and enzymes capable of attacking various components of plant tissues.

Bibliographic sources of the original papers describing various hydrolytic enzymes produced by phytopathogens can be found in the first edition of the *Manual* (Palleroni, 1984). Various nomenclatures of plant-pathogenic bacteria described in this treatment as pathovars of *P. syringae* have been found to produce phytotoxins capable of producing disease symptoms in susceptible plants. Table BXII.γ.109, taken from published information (Durbin, 1992), presents a summary of well-known examples of toxins as well as their respective targets of mode of action. Several phytotoxins have an amino acid or peptide nature (Leisinger and Margraff, 1979), and one of them, tabtoxin, has a β -lactam structure, a rare example among secondary metabolites of *Pseudomonas* (Stewart, 1971; Durbin et al., 1978).

Toxins do not necessarily parallel the host specificity of the respective phytopathogen (Patil, 1974). Their importance as taxonomic tools is very limited but their practical significance is considerable. In certain instances their production correlates with the presence of plasmids, and this has provoked renewed interest in them (Leisinger and Margraff, 1979). Loss of a plasmid by treatment with acridine orange resulted in loss of syringomycin production (González and Vidaver, 1979), but no correlation was observed between phaseolotoxin production and the presence of plasmids in *P. syringae* pathovar *phaseolicola* (described

in this treatment as a pathovar of *P. savastanoi*) (Jamieson et al., 1981).

High concentrations of the plant hormone indole-3-acetic acid produced by *P. savastanoi* are thought to be the cause of tumors in plants of the family Oleaceae (Wilson and Magie, 1964).

The interpretations given to data on host-parasite interactions in plant pathology have been deeply influenced by experiments making use of the genetic variability of the host species and of the pathogenic organisms. The main ideas developed during the first three-quarters of the 20th century, and the consequent conceptual changes in the interpretation of the host-pathogen interactions have been reviewed (Ellingboe, 1981).

In recent times, work performed by H.H. Flor with a fungal pathogen of flax gave origin to the gene-for-gene hypothesis. According to this hypothesis, the development of disease or the production of a well-circumscribed hypersensitive reaction in plant tissues depends on matching genes of resistance in the host plants with avirulence genes in the pathogens. In genetic experiments, host resistance was found to be dominant over susceptibility, and the pathogen avirulence dominant over virulence. Matching host resistance with avirulence in the pathogen results in "incompatibility", manifested in a hypersensitive reaction in the host tissues. In the other three combinations (resistant host/virulent pathogen, susceptible host/avirulent pathogen, and susceptible host/virulent pathogen), a "compatible" association is established, which results in disease. These concepts are being subjected to scrutiny in the genetic study of a number of pathovars of the fluorescent species *P. syringae* (Vivian, 1992), which has opened new horizons in attempting to understand the role played by the avirulence genes in the gene-for-gene hypothesis.

Pathogenicity for humans and animals Some references on this subject have been cited on sections of this chapter dealing with outer membrane composition, genetics, pili, and flagella.

As stated by Mekalanos (1992) in an excellent review on environmental signals that control virulence in bacteria, an expanded view of virulence comprises not only the properties directly linked to the pathogenic ability of microorganisms (adherence to host tissues, production of toxins, invasion mechanisms, host defenses) but also other factors involved in "house-

TABLE BXII.γ.109. Some toxins produced by phytopathogenic *Pseudomonas* species and pathovars^a

Species	Toxins	Target or mechanism
<i>P. syringae</i>		
<i>P. syringae</i> pathovar <i>atropurpurea</i>	Coronatine	
<i>P. syringae</i> pathovar <i>coronafaciens</i>	Tabtoxinine- β -lactam	Glutamine synthetase
<i>P. syringae</i> pathovar <i>garcae</i>	Tabtoxinine- β -lactam	Glutamine synthetase
<i>P. syringae</i> pathovar <i>glycinea</i>	Coronatine, polysaccharide	
<i>P. syringae</i> pathovar <i>lachrymans</i>	Polysaccharides	
<i>P. syringae</i> pathovar <i>maculicola</i>	Coronatine	
<i>P. syringae</i> pathovar <i>morsprunorum</i>	Coronatine	
<i>P. syringae</i> pathovar <i>phaseolicola</i> ^b	Phaseolotoxin	Ornithine transcarbamylase
<i>P. syringae</i> pathovar <i>savastanoi</i> ^b	Indole acetate, cytokinins	
<i>P. syringae</i> pathovar <i>syringae</i>	Syringomycins	Plasma membrane
	Syringopeptins	
	Syringotoxins	Plasma membrane
<i>P. syringae</i> pathovar <i>tabaci</i>	Tabtoxinine- β -lactam	Glutamine synthetase
<i>P. syringae</i> pathovar <i>tagetis</i>	Tagetitoxin	Chloroplastic RNA polymerase
<i>P. syringae</i> pathovar <i>tomato</i>	Coronatine	
<i>P. tolaasii</i>	Tolaasin	Plasma membrane

^aFor symbols see standard definitions. Modified from Durbin (1992). The nomenclature used is the one preceding the proposal by Gardan et al. (1992).

^b*P. syringae* pathovar *savastanoi* is described in this treatment as an independent species, *P. savastanoi*, and pathovar *phaseolicola* is considered as a pathovar of *P. savastanoi*; see description of this species in the list of species).

keeping" functions. In regard to opportunistic pathogens such as the fluorescent pseudomonads and, in particular, *P. aeruginosa*, the functions in this "gray area" can lead to different interpretations. Thus, competition of *P. aeruginosa* for iron in the host tissues by production of a strong chelator such as pyoverdine is on the whole comparable to the reaction of this organism in an iron-deficient artificial medium. However, the fact that pyoverdine, the main iron-scavenging compound produced by *P. aeruginosa*, allows normal growth of pyoverdine-negative mutants in a medium containing human iron-transferrin complex justifies considering pyoverdine production as a component of *P. aeruginosa* virulence complex (Meyer et al., 1996). In addition, iron-bound pyoverdine acts as an efficient catalyst for hydroxyl radical ($\text{HO}\cdot$) formation, and it contributes to endothelial cell damage from exposure to the superoxide radical (O_2^-) and H_2O_2 (Brigitan et al., 1997).

P. aeruginosa seldom infects healthy individuals outside the hospital environment, and the condition of the host is essential in determining the clinical relevance of this opportunistic pathogen. Strains of *P. aeruginosa* can be isolated from a bewildering variety of sources, since perhaps it is one of the most, if not the most, common species in nature. Numbers are very low in human feces, which may be due to competition with other species, since the numbers increase significantly because of antibiotic treatments (Levison, 1977).

In chronically debilitated and immunocompromised patients, *P. aeruginosa* is capable of causing serious and even fatal infections. Individuals with extensive burns, or those who have been subjected to surgical procedures, catheterization, and treatment with broad-spectrum antibiotics, are particularly vulnerable targets. Factors involved in the transmission of *P. aeruginosa* in hospitals have been reviewed and analyzed, and preventive measures are being recommended (Doring et al., 1996).

Iron and osmolarity are mentioned by Mekalanos as environmental signals inducing in *P. aeruginosa* the expression of dissimilar virulence determinants. Control of these and other factors provoked by environmental signals usually act at the transcription level (Mekalanos, 1992). *P. aeruginosa* controls the expression of multiple genes coding for virulence factors by means of LasR, a transcriptional activator, a key component that acts in a cell-density manner ("quorum sensing"), with participation of a *Pseudomonas* autoinducer and an *N*-acylhomoserine lactone. Production of virulence factors (exotoxin A, elastase, alkaline protease, alginate, phospholipases, and rhamnolipids acting as extracellular surfactants) can be stimulated in mutants by gene transfer, or by addition of the synthetic *N*-acylhomoserine lactones (Ochsner and Reiser, 1995). Characteristically, quorum sensing becomes maximal at the time the culture is entering the stationary phase of growth (Rust et al., 1996; Albus et al., 1997).

A global activator, GacA, is responsible for the production of exoenzymes and secondary metabolites in *P. aeruginosa*. Its inactivation resulted in reduced formation of the cell density signal and of LasR. Amplification of the *gacA* gene carried on a multicopy plasmid causes early and enhanced production of LasR and of the lactone. GacA is important in the regulation of the synthesis of virulence factors (Reimann et al., 1997).

P. aeruginosa exotoxin A is a virulence factor resembling diphtheria toxin in its mode of action. It is able to transfer an ADP-ribosyl group to elongation factor 2, causing translation termination, inhibition of protein synthesis, and cell death (Vasil et al., 1977). The toxin is a protein of 613 amino acid residues, of which residues 60–120 are of importance for excretion and

contain information for interaction with eucaryotic cells (Lu and Lory, 1996). The toxin is excreted when the cells grow under iron-limiting conditions, and its production is repressed when iron is not limiting, which suggests that the ferric uptake regulator Fur is involved, as in pyoverdine synthesis. However, the situation is more complicated, and accessory iron regulatory systems may be involved (Barton et al., 1996; Ochsner et al., 1996). The genetic picture is quite complicated. A regulatory gene (*ptxR*) is involved, and when introduced in a multicopy plasmid, it increases toxin production four- to fivefold, while other virulence factors remain unchanged.

P. aeruginosa is able to produce several proteases. In fact, the organism can cause serious corneal infections and conditions that complicate diseases of the respiratory tract and burn cases, and in all these instances proteases are responsible for the tissues alterations. The most notorious of the proteases as a virulence factor is elastase, with activity on elastin, which lines the blood vessels. The protease is produced in the cell as a precursor of 54 kDa, but it is excreted as a smaller molecule (39.5 kDa) (Moriyama and Homma, 1985). Various other proteases produced by the same organism are the cause of lesions in different human and animal tissues. Their importance in pathogenesis is not easy to determine, but mutants defective in protease formation are known to be less virulent.

Alginate is an exopolysaccharide produced by several species of *Pseudomonas*. Various aspects of this extracellular product already have been mentioned, one of which refers to the relationship to fimbriae, which, like alginate, are involved in adherence to tissues (Baker, 1990). A recent review covers important aspects of alginate biosynthesis by bacteria (Gacesa, 1998). The effect that environmental sensory signals have on alginate production as a virulence factor has been examined by DeVault et al. (1989). Specific factors that have been examined include osmolarity, which affects alginate formation by various fluorescent pseudomonads (Singh et al., 1992), and metals (Kidambi et al., 1995).

The *P. aeruginosa* alginate has been the one most thoroughly studied, but production of the polysaccharide also has been detected and characterized in *P. mendocina* (Govan et al., 1981; Hacking et al., 1983; Anderson et al., 1987; Sengha et al., 1989), *P. putida* (Govan et al., 1981), *P. fluorescens* (Govan et al., 1981; Conti et al., 1994; Smit et al., 1996), and even *P. syringae* (Kidambi et al., 1995). Sequences of alginate genes have been detected in other alginate producers and nonproducers belonging to rRNA similarity group I and in species of other rRNA similarity groups (Palleroni, 1984; Fialho et al., 1990; Fett et al., 1992). Similarity of the genetic determinants of alginate in *P. aeruginosa* and *Azotobacter vinelandii* (Ertesvag et al., 1995; Rehm et al., 1996) suggests either a transfer of these determinants among related members of *Proteobacteria*, or else a relic of common ancestry.

Interest in the *P. aeruginosa* alginate, which was first identified in *Pseudomonas* more than three decades ago (Linker and Jones, 1966), derives mostly from its importance as a virulence factor causing respiratory tract infections that aggravate the condition of many cystic fibrosis patients.

Alginate is composed of alternating units of mannuronic and L-guluronic acids joined by a β (1,4) linkage, where part of the carboxyl groups of mannuronic acid are acetylated. The polymer does not show the repetition of a certain type of unit, but instead there are regions where one or the other uronic acid predominates.

The steps of alginate biosynthesis are formulated starting with fructose-6-phosphate, going through mannose-6-phosphate,

mannose-1-phosphate, GDP-mannose, and GDP-mannuronic acid (steps catalyzed by enzymes encoded by the genes *algA*, *C*, *A*, and *D*, respectively), followed by polymerization, acetylation, epimerization, and final export of the alginate polymer (Chakrabarty, 1998). The genetic aspect of the process is quite complicated because it includes participation of many biosynthetic and regulatory proteins encoded by many more genes than the ones above indicated. The genes have been mapped in three regions of the *P. aeruginosa* chromosome (Goldberg, 1992).

The polymer is not used as a reserve material, and its synthesis demands energy that is not recovered by the cells. It is hard to think of a role other than protection of the pathogen against the host defenses. Secretion of alginate represents a detoxification mechanism to protect the bacterial cells against accumulation of toxic energy-rich nucleoside triphosphates under conditions that prevent active multiplication (Chakrabarty, 1998).

P. stutzeri is frequently isolated from clinical materials but rarely causes disease. In the few cases in which it was the cause of infections, the patients often had another serious underlying disease, and they responded to antibiotics (aminoglycosides, some of the β -lactams, or cephalosporins) (Noble and Overman, 1994).

Ecology Numerous natural materials are good sources for the isolation of strains of *Pseudomonas*, as well as of strains of species of other rRNA groups now assigned to different genera. It is unfortunate that few procedures can be recommended for the selective isolation of *Pseudomonas* species with exclusion of these other organisms. In some of the habitats, strains of *Pseudomonas* species may represent a minority of the microbial flora; however, certain conditions—a pH close to neutrality, organic matter in solution, a temperature in the mesophilic range, a good supply of dissolved oxygen—together with a capacity for rapid growth in the absence of complex organic factors can favor their predominance. Even in media of extremely low nutrient content, pseudomonads occasionally multiply to a considerable extent. Thus, *P. aeruginosa* has been found capable of growth at the expense of minor impurities present in hospital distilled water (Favero et al., 1971).

Strains of many species are ubiquitous, and isolation data often throw little light on their ecology. When dealing with organisms of such versatility, ecological conclusions are particularly difficult to draw. Of the 57 strains of fluorescent *Pseudomonas* species isolated by den Dooren de Jong, 23 had their origin in soil, and with one exception all were classified as *P. putida*, while all other strains were gelatin-liquefiers isolated from water, and were assigned to *P. fluorescens* (den Dooren de Jong, 1926). However, the conclusion that *P. putida* is a soil organism whereas *P. fluorescens* predominates in water could not be supported by an analysis of isolation data of many other strains from the collections of the Department of Bacteriology at Berkeley, California, and of the Seruminstitut in Copenhagen (unpublished observations). Obviously, a decision on this point is not easy in view of the possibility of cross-contamination of materials from the two habitats.

In recent years, more precise data regarding habitats have become available. An early observation suggests that the predominant fluorescent pseudomonads in wheat rhizosphere comprise one of the biotypes of *P. fluorescens* (Sands and Rovira, 1971). Many other reports have confirmed the fact that these organisms proliferate in general in plant rhizospheres, where they seem to have a stimulating effect on plant growth. In part, this may be

due to inhibition of plant pathogenic organisms by iron starvation (Kloepper et al., 1980; Sarniguet et al., 1995) and to the utilization of the iron-siderophore complex by the plants (Powell et al., 1980). An additional beneficial effect is suggested by studies on the rhizosphere of plants growing in the Canadian Arctic, in which *P. putida* can grow and stimulate root elongation of plants during spring and winter at 5°C, as shown by using a strain that can survive exposure to -20°C to -50°C. Following growth at 5°C, the strain secreted a protein with antifreeze activity (Sun et al., 1995). A numerical taxonomic analysis of strains of fluorescent pseudomonads associated with the roots of tomato plants has been published (Stenström et al., 1990).

A recent survey indicates that of the species of bacteria isolated from the rhizosphere of wheat, 40% stimulated plant growth, 40% were inhibitory, and 20% had no effect (Lugtenberg and De Weger, 1992). The properties of fluorescent *Pseudomonas* species that are involved in the suppression of pathogenic species in the root system of plants have been reviewed (O'Sullivan and O' Gara, 1992).

A gnotobiotic system for the study of colonization of plant rhizospheres by *Pseudomonas*, based on seed inoculation and subsequent estimation of different parameters, has shown that many strains were outcompeted by a chosen *P. fluorescens* strain. Slow-growing and auxotrophic mutants of this strain were at a disadvantage, indicating that growth rate and the capacity for multiplication in the absence of exogenous growth factors were important for the persistence of a strain in the rhizosphere community (Simons et al., 1996).

Many species of *Pseudomonas* have been isolated from *Lolium* leaves (Stout, 1960). *P. fluorescens* has been found frequently, and *Burkholderia cepacia* and *Stenotrophomonas maltophilia* less so (Austin et al., 1978). In their review of the literature, these authors mention isolation of *P. fluorescens* from leaves of *Phaseolus vulgaris*, *Fagus*, and *Pinus* by various workers, and consider that this species is indigenous to leaf surfaces. Whether the strains can become opportunistic plant pathogens is an open question, but some of the epiphytic pseudomonads have definite pathogenic potentialities, as was suggested by Billing (1970b) for *P. viridiflava*.

Plant pathogenic pseudomonads are normally isolated from lesions in plant hosts, which are natural enrichment cultures. Because of their ecological niches, these bacteria may offer some of the most interesting materials for the study of bacterial speciation (Palleroni and Doudoroff, 1972). In general, animals are not as good sources of *Pseudomonas* species, unless these are involved in infections as opportunistic pathogens.

In different areas, speciation of plant pathogens may proceed at different rates according to the selective forces imposed by various environmental factors. Some pathovars of pathogenic species may develop as distinctive ecotypes according to the selective pressures predominating in a given area (Garrett et al., 1966).

Continued association with living hosts is important to the survival of members of the *P. syringae* group (Schroth et al., 1981). The association does not necessarily have to involve lesions, since the pathogen may be able to survive as an epiphyte on hosts as well as on nonhosts (Ercolani et al., 1974). Survival of the *P. syringae* pathovars in soil may not be long, in contrast with other fluorescent species (Schroth et al., 1981).

A brief discussion of a few points related to modern biotechnological applications of *Pseudomonas* strains may be pertinent to the present discussion. It has been reassuring to observe that the introduction of a plant growth-promoting strain of *P. fluorescens* and a genetically modified derivative did not appreciably disturb

the indigenous microflora when introduced to the soil (Mahaffee and Kloepper, 1997), and that carbon-starved *P. fluorescens* had a good survival capacity and resistance to stress in this environment (Van Overbeek et al., 1995).

The attractive possibility of using genetic means to improve the metabolic versatility of *Pseudomonas* species for the degradation of environmental pollutants has raised questions about the convenience of keeping the lateral spread of genes in the environment under control. These questions have stimulated the development of ingenious methodologies to quantify the degree of horizontal transfer of genes (Jaenecke et al., 1996), to prevent such transfer (Diaz et al., 1994), and to contain the spread of inoculated strains (Jensen et al., 1993b). Modern biotechnological developments also have resulted in improvements for observation of cells in natural environments. These improvements include the combined use of an appropriate probe and scanning confocal laser microscopy (Moller et al., 1996b), as well as in the construction of a biosensor for monitoring metabolic activities *in situ* with a bioluminescent reporter *P. fluorescens* strain (Heitzer et al., 1994).

Pseudomonas species are important members of natural microbial communities. The capacity of these organisms to react to environmental changes is in part related to their capacity to exchange genetic material. An evaluation of this process may be attempted using a strategy that tries to avoid selecting for the characters acquired by the recipients during exchange. The core of the procedure is the use in donor cells of an engineered *lazZ* reporter gene whose expression is shut down by chromosomal repressors at the levels of transcription and translation. The gene will be freely expressed when it can escape repression in the new host, which, consequently, can be identified (Jaenecke et al., 1996).

The consequences of the high surface/volume ratio in prokaryotes are manifested, among other things, in concentrating on their surface components of the liquid medium. Fine-grained minerals precipitate around the cell clusters as consequence of their activities when the cells are either freely suspended ("planktonic"), or attached to surfaces with formation of a biofilm. LPS is important in the attachment, which results in a biofilm. Membrane vesicles containing degradative enzymes may bleb off the cell surface and have a predatory action against cells of surrounding populations, with liberation of nutrients that help the growth of the biofilm, as suggested by results obtained with a *P. aeruginosa* model system (Beveridge et al., 1997).

ENRICHMENT AND ISOLATION PROCEDURES

Direct isolation of *Pseudomonas* species often can be achieved from many natural materials, especially soil and water. From these materials, direct isolation in solid complex or minimal media is feasible. When heavy contamination with fungi is expected, these media can be supplemented with antifungal compounds such as cycloheximide (Actidione), which is added to the medium after autoclaving (final concentration, 20–50 µg/ml), from a concentrated solution that does not need to be sterilized.

A medium that is frequently used for direct isolation especially by plant pathologists is medium B² of King et al. (1954), which enhances pyoverdinin production by fluorescent organisms but is also satisfactory for nonpigmented strains as a general purpose medium. Some selective solid media have been developed based

on medium B. An example is a medium that contains penicillin G, novobiocin, and cycloheximide (Sands and Rovira, 1970). These compounds do not inhibit the fluorescent pseudomonads, and their colonies can be identified on the plates by the characteristic diffusible, fluorescent pigment. A modification of this medium later was published for the isolation of *Pseudomonas* species with pectolytic properties (Sands et al., 1972). The medium contains sodium polypectate, and the pectolytic activity is detected by flooding the plates after growth with a solution of hexadecyltrimethylammonium bromide, which precipitates intact pectin. Once identified, the colonies are isolated as soon as possible since the compound is toxic to the cells.

Two selective media containing the detergent lauroyl sarcosine and the antibiotic trimethoprim, and differing in having either Casamino acids (medium S1) or L-asparagine (medium S2), gave good selectivity and detection of fluorescence on initial plating. Of the two media, S1 gave the highest recovery, although S2 was more selective (Gould et al., 1985).

Selective media for the oxidase-negative fluorescent phytopathogens have been described (Sands et al., 1980).

Some methods for the isolation of certain species may go through an initial enrichment step. The lesions produced by plant pathogenic pseudomonads on various plant organs represent a natural enrichment, from which direct isolation can be attempted. The lesions selected should not be too old, when other organisms may have gained access to the diseased tissues. Isolation of denitrifying pseudomonads (*P. aeruginosa*, *P. stutzeri*, *P. mendocina*, some biovars of *P. fluorescens*, etc.) after enrichment is often successful. The denitrification experiment can be performed using particular carbon sources as electron donors, and the incubation can be done at a temperature that favors multiplication of the desired organism. However, due to our ignorance of the proper conditions, or unpredicted transformations by other organisms in the microflora, these experiments may fail to give the expected results.

It is convenient to keep in mind that that species of aerobic pseudomonads other than the ones mentioned above may predominate in the enrichment cultures. In the chapter on *Burkholderia* in this *Manual*, a table has been included that summarizes the properties of *Pseudomonas* species that are able to denitrify, together with selected species of denitrifiers of other rRNA groups.

Methods of enrichment and isolation are dispersed in a number of publications, and some of them are very specific for species originally assigned to *Pseudomonas* but now allocated to newly created genera and phylogenetically belonging to other rRNA groups (Palleroni, 1984) than group I. Additional details may be found in the second edition of the treatise *The Prokaryotes* (Palleroni, 1992a), and in the first edition of this *Manual* (Palleroni, 1984).

MAINTENANCE PROCEDURES

Most *Pseudomonas* strains can be maintained on slants of common bacteriological media (nutrient agar or other standard complex media, or various chemically defined media with the addition of 0.5% yeast extract and 0.1% lactate or glycerol), with transfers every 1–2 months. Slants can be stored at 4–8°C. The collection examined by Stanier et al. (1966) included a number of strains of fluorescent and nonfluorescent *Pseudomonas* species that had been kept on slants of ordinary media with periodic transfers since the 1920s, and their phenotypic properties appeared to have remained essentially unchanged. In contrast, some strains

2. Medium B (g/l): protease-peptone (Difco), 20.0; Bacto-agar (Difco), 15.0; glycerol, 10.0; K₂PO₄, 1.5; and MgSO₄·7H₂O, 1.5. The pH is adjusted to 7.2.

of *Pseudomonas* species as well as members of other rRNA groups (Palleroni, 1984) are not so easy to maintain for long periods in slants in the refrigerator.

Lyophilization is generally reliable for long-term preservation, particularly when proteinaceous material (e.g., skim milk) is used as an ingredient of the suspending fluid. There are many variations of lyophilization procedures traditionally followed by various laboratories, with good results. A substantial proportion of the cell mass of cultures of *Pseudomonas* species does not survive even the most careful lyophilization procedure, and lyophilization has been replaced in many laboratories by freezing cultures at a temperature of -20°C or lower (usually from about -70° to -80°C in mechanical freezers or at lower temperatures in liquid nitrogen) after addition of a cryoprotective agent to the liquid before freezing (for instance, 5–10% glycerol, final concentration).

PROCEDURES FOR TESTING SPECIAL CHARACTERS

Flagellar number and insertion The observation of these characteristics is best done on smears prepared by the method described by Jessen (1965), combined with results expressed on a statistical basis, as mentioned earlier (Lautrop and Jessen, 1964). On clean microscope slides, thin smears are prepared from dilute suspensions in distilled water of bacteria that have grown in broth agar cultures at 23°C or 30°C for about 18 h. Best results are obtained using freshly poured solid media whose surface has been dried in an incubator for no more than 1 h. The number of cells with flagella in the smears often varies with the degree of dryness of the agar medium in which the strains have been growing. The smears are stained for flagella by the method of Leifson (1960)³. Only slides with well-dispersed cells and well stained for flagella should be used. The number of flagella on each of the first 50–100 flagellated cells is noted. Cells with flagella at both ends are disregarded, since they probably represent cells in an early stage of their division. Some results of this technique have been mentioned in the early section on flagella and pili (fimbriae).

Screening of nutritional and other physiological properties The medium described by Stanier et al. (1966) has been used extensively for nutritional screenings and for testing various physiological properties of cultures of pseudomonads. One disadvantage of this medium is that not all cultures tolerate well the heavily chelated composition of the medium. More satisfactory results are obtained with a formulation that is both simpler and safer, such as the medium recommended by Palleroni and Doudoroff (1972)⁴, which is most satisfactory for autotrophic and heterotrophic enrichments and as a general medium for cultivation and short-term preservation. Minimal media of similar

composition have been described (Schlegel and Lafferty, 1971; Zavarzin and Nozhevnikova, 1977). These last authors recommend supplementation of the medium with minor elements for best results with fastidious organisms, but such additions are unnecessary when dealing with *Pseudomonas* strains.

Good heterotrophic growth in all these minimal media can be obtained by addition of a single organic compound as a carbon and energy source (usually 0.1%, final concentration). Experimental details of the nutritional analysis have been extensively discussed elsewhere (Stanier et al., 1966; Palleroni and Doudoroff, 1972). Ideally, the analysis should be performed under conditions that prevent cross-feeding or competition among the various strains when the medium is solidified with agar and several strains are patched on a single plate. Organic impurities in the agar may also be a serious problem. For critical tests, liquid media may be preferable, although the amount of work involved and the facilities required may be considerably greater. Carbon compounds that are to be tested in different concentrations or that require special treatments for their use in nutritional screenings have been discussed by Palleroni and Doudoroff (1972). Description of many species of aerobic pseudomonads performed at Berkeley included nutritional studies using many organic compounds as substrates. The studies were extended to the plant pathogenic pseudomonads in two laboratories (Misaghi and Grogan, 1969; Sands et al., 1970).

At present, the studies on nutritional properties of various prokaryotic groups are more often performed using commercial kits designed to reduce the labor involved in traditional procedures. Some of them (API, Biotype-100 strips, Biolog GN Microplate System, Hayward, CA) have been used in studies on *Pseudomonas*, and the results will be briefly discussed below in the section on taxonomic comments.

Biosynthesis of medium-chain-length polyhydroxyalkanoic acids (mcl-PHAs) The following observations on the biosynthesis and detection of mcl-PHAs have been provided by Birgit Kessler, a member of B. Witholt's group at the Swiss Federal Institute of Technology in Zürich.

1. As with poly- β -hydroxybutyrate (PHB), accumulation of mcl-PHAs can be elicited in a medium of high C/N ratio, and the polymer can be detected after staining with Sudan Black or by direct observation under phase contrast microscopy. However, granules of the two polymers cannot be differentiated by either procedure, and the identification procedure may involve either gas chromatographic analysis (Timm and Steinbüchel, 1990) or a solubility test. The mcl-PHAs are soluble in acetone, while PHB is not.
2. Assimilation of medium- and long-chain-length fatty acids (saturated, unsaturated, or certain branched and substituted acids) enables the cells to accumulate mcl-PHAs. Some *Pseudomonas* strains (*P. aeruginosa* PAO and *P. putida* KT2442) accumulate mcl-PHAs when grown on carbohydrates. In contrast, *P. oleovorans* lacks this capacity, but it accumulates mcl-PHAs when grown on alkanes. Growth on butyrate and β -hydroxybutyrate induces production of small amounts of PHAs, and the granules can be seen under phase contrast directly or after staining with Sudan Black.
3. Several species of aerobic pseudomonads have been tested for their capacity of PHB and/or PHA accumulation (Timm and Steinbüchel, 1990). Production of both types of polymers by a single strain did not receive special attention, but recently a strain of an unidentified *Pseudomonas* sp. was found

3. Leifson's reagent for flagella staining (Leifson, 1960): 1.2% solution of basic fuchsin in 96% ethanol, 3% solution of tannic acid in water, and 1.5% solution of NaCl in water. Equal parts of the three reagents are mixed before use. The mixture can be left overnight in a refrigerator, where a precipitate may form. The clear stain is applied over the smear, and after about 2–3 min a fine precipitate will form. At this point, the staining is interrupted by washing the slides with water at room temperature. The final point may have to be ascertained by trial and error before optimal conditions are defined. Absolute cleanliness of the slides is essential for good results.

4. Medium of Palleroni and Doudoroff (1972) (g/l of 0.33 M Na-K phosphate buffer, pH 6.8): NH_4Cl , 1.0; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 0.5; ferric ammonium citrate, 0.05; and CaCl_2 , 0.005. The first two ingredients are added to the buffer and sterilized by autoclaving. The ferric ammonium citrate and calcium chloride are added aseptically from a single stock solution that has been sterilized by filtration.

with such capacity when grown on fatty acids or carbohydrates.

Pigment production Perhaps the most widely used medium for pyoverdine production is the medium B of King et al. (1954). Paton (1959) has described a medium whose preparation involves extensive treatment of the solution and the agar with iron-chelating agents, but the removal of iron is so complete that it often results in inhibition of bacterial growth (Garibaldi, 1967). Garibaldi recommends instead a medium that is based on the well-known iron-chelating capacity of conalbumin, a component of egg-white.⁵ Luisetti and collaborators (1972) recommend a medium⁶ for the enhancement of fluorescent pigment production by plant pathogenic and other pseudomonads that fail to fluoresce in the medium B of King and collaborators.

A point frequently incorrectly stated or disregarded altogether in descriptions of procedures for the detection of pyoverdins by fluorescent pseudomonads is the type of ultraviolet lamp to be used for observing fluorescence. Only the "true" fluorescent pigment will fluoresce under a source of short wavelength ultraviolet light (around 254 nm). Some species belonging to rRNA group II (Palleroni, 1984) (*Burkholderia cepacia*, *B. gladioli*, *B. caryophylli*) produce diffusible yellow-green pigments that are sometimes mistaken for the fluorescent siderophores.

Medium A of King et al. (1954) is generally recommended for the production of phenazine pigments to which pyocyanin belongs. In our experience, this medium is not always very effective, but unfortunately no alternative can be recommended. Phenazine pigment production, therefore, appears erratic, particularly with cultures that have been kept for a long time under ordinary laboratory conditions.

The blue pigment of *P. fluorescens* biovar IV ("*P. lemonnieri*") can be produced abundantly by fresh cultures in the potato medium⁷ familiar to mycologists. As mentioned earlier, the pigment is related to indigo iodine (Starr et al., 1967). Interestingly, this same medium induces the production of indigo iodine by an unrelated organism, *Corynebacterium insidiosum* (Starr, 1958).

Ring fission mechanisms The *ortho* or *meta* cleavage of two central intermediates in the metabolism of aromatic compounds (catechol and protocatechuate) can be tested by the method originally suggested by K. Hosokawa (Stanier et al., 1966). After growth in a chemically defined medium with an aromatic substrate, the cells are suspended in 0.02 M Tris buffer (pH 8). To each 2-ml portion of suspension are added a drop of toluene and 0.2 ml of either 0.1 M catechol or 0.1 M protocatechuate. A bright yellow color appearing in a short time (usually within

a few seconds) indicates a *meta* cleavage. The tubes are shaken for 1 h at 30°C, and then tested for the appearance of β -ketoadipate (indicative of an *ortho* cleavage) by the Rothera reaction, as follows. Solid ammonium sulfate is added to saturation and the pH is brought up to about 10 by the addition of two drops of 5 N ammonium hydroxide. One drop of freshly prepared 25% sodium nitroprusside in water is added. A deep purple color indicates a positive reaction.

Arginine dihydrolase reaction The method giving the most unequivocal results is the direct one, based on an estimation of arginine disappearance under anaerobic conditions. A bacterial suspension (200 Klett units, as measured with green filter 54) is incubated for 2 h at 30°C in the presence of arginine (2.5×10^{-4} M) under anaerobic conditions. The tubes are immersed in a boiling water bath for 15 min and arginine is estimated by the method of Rosenberg in a sample of the supernatant after removing the cells. A control, without addition of arginine, is included in the experiment (Rosenberg et al., 1956).

The Rosenberg procedure is cumbersome and time-consuming, and other methods appear to be much simpler and sufficiently reliable. Among them, the method of Thornley (1960) is perhaps the most convenient. As recommended by Lelliott et al. (1966), Thornley's medium⁸ can be conveniently dispensed in 3-ml portions into 5-ml screw cap vials. After autoclaving, the vials are inoculated by stabbing and sealed with melted paraffin. Anaerobic formation of ammonia from arginine can be detected by a change of color of the indicator within 3 d.

Acid production from carbohydrates As pointed out by Palleroni and Doudoroff (1972), the reaction of acid production from carbohydrates is not necessarily correlated with nutritional data, since the acid produced by oxidation of a sugar may not always be a suitable carbon source for growth. However, this should not diminish the taxonomic value of the test. More serious, though, is the objection of redundancy, in that a single enzyme may affect the oxidation of several different sugars (Weimberg, 1962).

Acid production from carbohydrates has been used very extensively for descriptive purposes since the last century, and it is still being used in various laboratories. It is described here among the recommended procedures, provided that the results are interpreted with proper reservations. The method of choice is that described by Hugh and Leifson (1953), which gives clear and reproducible results. The main advantage of this method is the use of a medium containing a low concentration of peptone⁹. If the concentration is high, the deamination can neutralize a positive acid reaction from the sugar.

Hydrolysis of Tween 80 This reaction is indicative of lipolytic activity, and is carried out in the medium of Sierra (1957)¹⁰. The medium is usually dispensed in Petri dishes and is spot-inoculated with the cultures. During the incubation, the plates are observed daily for opacity around the patches, due to formation of insoluble calcium soaps (Sierra, 1957). If the reaction has been negative for 10 d, the bacterial mass may be scraped off to observe

5. Garibaldi's medium: sterile egg white is drawn aseptically from eggs that have been sterilized externally by immersion in 70% ethanol for 5 min, drained and flamed to remove the residual alcohol. The egg white is warmed to 45°C and added aseptically to give a final concentration of 10% (v/v) to any commercial complex solid medium that has been autoclaved and cooled to 45°C. After mixing, the medium is dispensed into Petri dishes. A solution of the protein conalbumin (available commercially) that has been sterilized by filtration can be used instead of egg white at a final concentration of 1.7 mg/ml of medium.

6. Medium of Luisetti et al. (1972) (g/l): vitamin-free Casamino acids, 10.0; K_2HPO_4 , 1.0; $MgSO_4 \cdot 7H_2O$, 1.0; sucrose, 10.0; gelatin, 30.0; and agar, 20.0; pH 7.0.

7. A formula recommended by D.C. Hildebrand (personal communication), who has isolated a number of strains of these organisms, is as follows. Sliced potatoes (250 g) are steeped in 1 liter of water at 65°C for 1 h. The preparation is filtered through three layers of cheesecloth, and 2.0 g of glucose and 20.0 g of agar are added to the solution. The medium is sterilized at 20 psi for 30 min.

8. Thornley's medium (g/l): peptone, 1.0; NaCl, 5.0; K_2HPO_4 , 0.3; agar, 3.0; phenol red, 0.01; and L-arginine-HCl, 10.0; pH 7.2.

9. Hugh-Leifson medium (g/l): peptone, 2.0; NaCl, 5.0; K_2HPO_4 , 0.3; agar, 3.0; bromothymol blue, 0.03; and carbohydrate, 10.0; pH 7.1.

10. Sierra's medium (g/l): Bacto-peptone (Difco), 10.0; NaCl, 5.0; $CaCl_2 \cdot H_2O$, 0.1; and agar, 17.0. Tween 80 is sterilized separately and added to the medium after autoclaving to give a final concentration of 10.0 g/l.

the medium under the patch. The presence of a precipitate, however, may be due to release of endocellular lipases by lysis of part of the population, and therefore, for practical purposes, the reaction is considered negative.

Nitrate reduction and denitrification Reduction of nitrate to nitrite can be tested according to Lelliott et al. (1966), and denitrification according to Stanier et al. (1966). Some of the problems encountered in the denitrification test have been discussed by Palleroni and Doudoroff (1972).

Miscellaneous tests Additional tests that are performed in the identification of *Pseudomonas* species include the following, with references to bibliographic sources for the methods: oxidase test (Kovács, 1956; Stanier et al., 1966); levan formation from sucrose (Lelliott et al., 1966; Stanier et al., 1966); gelatinase (Skerman, 1967); and the egg yolk reaction (Lelliott et al., 1966; Stanier et al., 1966). In testing for catalase, the usual procedure of placing a drop of hydrogen peroxide solution on top of a colony and observing the formation of bubbles is usually satisfactory, but occasionally the reaction may be so weak as to require observation under a dissection microscope, or sensitive detection methods involving the use of an oxygen electrode (Auling et al., 1978).

A number of additional techniques have been added as taxonomic tools to supplement the phenotypic analysis of the aerobic pseudomonads. The results of many of these methods will be summarized below under general taxonomic comments, and the papers cited are sources of information on the methodologies followed in each case. The methods described in these bibliographic references cover various aspects of nucleic and protein studies, the analysis of the fatty acid composition, as well as the description of numerical procedures for data processing.

DIFFERENTIATION OF THE GENUS *PSEUDOMONAS* FROM OTHER GENERA

The first edition of the *Manual* included a discussion of the properties that distinguish the aerobic pseudomonads of the genus *Pseudomonas sensu lato* from other phenotypically similar Gram-negative, nonsporulating aerobic bacteria. The reader is referred to this source for comments on this subject in its broader sense (Palleroni, 1984).

Over a quarter of a century ago, strains of *Pseudomonas* species were subjected to experiments of ribosomal RNA–DNA hybridization. Based on the conservative nature of the ribosomal RNA cistrons that had been demonstrated in species of the genus *Bacillus* (Doi and Igarashi, 1965; Dubnau et al., 1965), these experiments attempted to disentangle the phylogenetic complexity that was suspected in the unrelated genus *Pseudomonas*. The results were indeed very striking. The rRNA–DNA hybridization experiments clearly suggested an internal subdivision of *Pseudomonas* into five rRNA groups that seemed “to be very distantly related to each other phylogenetically . . . [deserving] at least independent genus (and possibly family or order) assignment” (Palleroni et al., 1973). One of these groups, which included the type species, *P. aeruginosa* (rRNA group I), was to retain the genus name. It showed a closer relationship to *Xanthomonas* and to *E. coli* than to other groups of pseudomonads.

It was soon realized by other workers that extension of this approach to comparative studies of other bacterial taxa could yield results of use “as an index of general relatedness [to] provide a sound base on which to construct taxonomic or maybe identification schemes” (Johnson and Francis, 1975). This pro-

phetic statement found ample corroboration in the years that followed.

Pseudomonas species were examined again at Gent (Belgium) on a much broader basis, but using essentially the same experimental approach, nucleic acid hybridization, and the results, published 10 years later (De Vos and De Ley, 1983), confirmed the conclusions of the original observations. In view of the phylogenetic diversity demonstrated in the genus *Pseudomonas* as classically defined, one of the striking consequences from the taxonomic point of view was that the set of basic properties that had been considered characteristic for this group of organisms—i.e., rod shaped, Gram negative, nonsporeforming, motile by means of polar flagella, and aerobic—was inadequate for practical diagnosis of the genus *Pseudomonas*. The circumscription of *Pseudomonas* is now restricted to the members of rRNA similarity group I of Palleroni et al. (1973), and it is unfortunate that the similarities to members of other groups (which, in the first place, was the basic reason for their original assignment to the genus *Pseudomonas*) make it difficult to define discrete sets of differential phenotypic characters.

The hybridization methods originally designed for the evaluation of rRNA similarities (Palleroni et al., 1973; De Vos and De Ley, 1983) were eventually replaced by other procedures (oligonucleotide cataloging, rDNA nucleotide sequences). Their use, as expected, confirmed the results of the earlier hybridization experiments. Sequencing the 16S rDNA gene after amplification by PCR is now universally used as the basis for assignment of new species to the genus *Pseudomonas*. As shall be mentioned below, fatty acid analysis can also be effective in differentiating *Pseudomonas* from the other similarity groups.

Pseudomonas sensu stricto (rRNA similarity group I) is still a group of considerable heterogeneity. Internal subgroups can be defined based on plant pathogenicity or pigment production. Among the pigments, the fluorescent siderophores are characteristic of the so-called fluorescent pseudomonads, and production of these compounds is sufficient *per se* to allocate a strain in the genus *Pseudomonas*.

However, the genus also includes nonfluorescent species. It is interesting to note that during the early phenotypic studies carried out at Berkeley, in spite of the differences in pigmentation, a phylogenetic relationship was suspected between some pigmented and nonpigmented species. Obviously, the extensive phenotypic study to which the strains had been subjected implied the expression of a considerable proportion of genes, and it was observed that “the nutritional spectrum of the [alcaligenes group, that is represented by strains that are the least nutritionally versatile of all aerobic pseudomonads] recall[ed] in many respects that of the fluorescent pseudomonads, but it [was] much narrower, notably in the areas of sugars and sugar-acids, and of aromatic compounds” (Stanier et al., 1966). In agreement with the above inference, the numerical taxonomic studies of Sneath et al. (1981), based on part of the published data from the Berkeley group, resulted in a three-dimensional graphical representation, where the so-called fluorescent complex is placed on the same side of the horizontal plane as the nonfluorescent species *P. stutzeri* and *P. alcaligenes*. Although discrete sets of phenotypic properties appear to be inadequate for an absolute differentiation of *Pseudomonas* from other genera of aerobic pseudomonads, the wealth of information contained in extensive nutritional screenings can provide, after appropriate processing, a generally satisfactory graphical representation of the relationships between *Pseudomonas* and other rRNA similarity groups (Sneath et al.,

1981). The position of nonfluorescent species is uncertain in graphical representations of values obtained using the Gower or the pattern coefficients, but association of *P. stutzeri* and *P. alcaligenes* is more distinct when using three-dimensional representation by principal coordinate analysis of Euclidean distances (Sneath et al., 1981). Moreover, the phylogenetic representation of Fig. BXII.γ.124 shows that the nonfluorescent species *P. alcaligenes* and *P. pseudoalcaligenes*, are placed in the neighborhood of the versatile fluorescent species *Pseudomonas aeruginosa*.

The original descriptions of the various phenotypic groups of *Pseudomonas* species included the inability of species of group I to accumulate PHB granules as carbon reserve material, with the exception of some *P. pseudoalcaligenes* strains that appeared to accumulate small amounts of the polymer. The diagnostic value of this negative characteristic appeared to be rather limited: on the one hand, the species of group V (*Xanthomonas* and *Stenotrophomonas*) are also negative, and on the other, the present treatment includes some species (*P. corrugata*, *P. amygdali*, *P. ficusereectae*) that have been reported in the literature to be able to accumulate PHB. However, *P. corrugata* and *P. ficusereectae* actually do not synthesize PHB but instead accumulate PHAs of different monomer composition than PHB, thus buttressing the taxonomic value of inability to accumulate PHB as a generic character (Kessler and Palleroni, 2000). The lack of ability to form PHB is shared by species of group V (*Xanthomonas* and *Stenotrophomonas*), which is interesting in view of the relatively close relationship of this group with group I (*Pseudomonas*). (See also the chapter on *Stenotrophomonas* in this *Manual*.)

The main body of the present treatment of the genus *Pseudomonas* comprises the species listed in group I of the first edition of the *Manual*. In addition, there are species that were not included in the early experiments, although they were mentioned as probable members of group I, and were later assigned to *Pseudomonas* mainly from additional screenings performed at the University of Gent, Belgium (De Vos and De Ley, 1983; De Vos et al., 1985, 1989). An excellent source of information on the genuine *Pseudomonas* species as well as the present position of many species assigned to *Pseudomonas* but now transferred to rRNA groups other than group I (Palleroni, 1984) is the review article by Kersters et al. (1996). The list of species below will refer with special detail to species whose assignment to *Pseudomonas* has been decided based on phylogenetic criteria.

TAXONOMIC COMMENTS

The proposed division of the aerobic pseudomonads into five RNA similarity groups has received confirmation from numerous contributions by workers in many different laboratories. The investigations have followed various approaches, including investigations on metabolic pathways and their regulatory mechanisms, nucleic acid similarity studies, amino acid sequences of selected proteins, immunological studies, and cell wall composition. The results obtained by application of these approaches were extensively discussed in previous reviews (Palleroni, 1975, 1986, 1992a, b, 1993), and determinative keys have been proposed (Palleroni, 1977; Bergan, 1981; Stolp and Gadkari, 1981). The data add considerable precision to the circumscription of the genus *Pseudomonas*, and some of the methodologies can be of help for determinative purposes.

Interest in the taxonomic intricacies of this genus has not subsided, and of particular interest is a compilation of papers published by members of a consortium of European research groups in vol. 19, pp. 465–568 of *Systematic and Applied Microbi-*

ology. These reports represent to a great extent further explorations into the formidable phenotypic and genomic diversity of the genus *Pseudomonas* as presently circumscribed using a number of experimental approaches.

In addition to the techniques for the determination of special properties already described, this section includes a discussion of other methods or variations of previous methodologies that have been particularly informative in taxonomic analysis of *Pseudomonas* species. Details of the experimental procedures may be found in the respective bibliographic sources.

Nutritional and metabolic studies Even though the formidable body of phenotypic information collected at Berkeley constituted a very appropriate subject for numerical taxonomic studies, this type of data treatment was reported only for a fraction of the available strains. A considerable amount of work was done in this area, but it remained unpublished, except for a few selected instances (Sands et al., 1970; Palleroni et al., 1972; Champion et al., 1980). Numerical treatment of phenotypic data also is available in original Ph.D. theses from the Department of Bacteriology (Ballard, 1970; Barrett Ralston, 1972).

A more extensive numerical analysis of published nutritional data of the Berkeley collection has given results in very good agreement with the proposed groupings, as indicated by Sneath et al. (1981). Their paper includes an illustrative three-dimensional representation using the system of principal coordinate analysis. A different system of data processing not requiring a complete matrix with all pairwise comparisons also results in useful three-dimensional representations, and it has been applied to published data of the pseudomonads (Hildebrand et al., 1984).

In addition to the extensive phenotypic studies performed at Berkeley on many strains of aerobic pseudomonads following conventional bacteriological techniques, in recent times additional observations have been performed using available commercial kits. Many strains of *Pseudomonas* species were subjected to nutritional studies using API Biotype-100 strips (BioMérieux, La Balme les Grottes, France) for carbon assimilation and the Biolog GN MicroPlate System (Biolog Inc.), which is based on substrate oxidation as detected by the change of color of an indicator (Grimont et al., 1996). Each of these systems included a list of nearly 100 organic compounds, the data obtained were analyzed by computerized systems, and the results were represented graphically in dendrogram form. *P. aeruginosa*, *P. tolaasii*, *P. mendocina*, *P. cichorii*, *P. viridiflava*, *P. fragi*, *P. stutzeri*, *P. agarici*, *P. alcaligenes*, and *P. pseudoalcaligenes* appeared as homogeneous phenons. In contrast, some of the fluorescent species (*P. putida*, *P. fluorescens*, *P. marginalis*) did not form discrete clusters but were found distributed in several phenons, confirming the heterogeneity previously reported elsewhere.

In general, according to Grimont et al. (1996), there was a remarkable correlation between the Biotype-100 data and the results obtained over 30 years ago using a more conventional methodology (Stanier et al., 1966; Ogawa et al., 1992). This is in agreement with the basic idea behind the two approaches, which rely on the utilization of substrates for growth. The work of Stanier et al. (1966) did not include testing for the production of acid from carbohydrates, a technique widely used at the time. As discussed elsewhere (Palleroni and Doudoroff, 1972) and in a previous section of this chapter, in spite of their reproducibility the results of the oxidation tests may be redundant; i.e., one enzyme may produce acid from more than one substrate. On



FIGURE BXII.γ.124. Phylogenetic tree of species of the genus *Pseudomonas* derived from the 16S rDNA sequences. Bootstrap values of 80% or higher are indicated at the branch points. *Escherichia coli* (V00348) was used as the root organism. (Reproduced with permission from Y. Anzai et al., International Journal of Systematic and Evolutionary Bacteriology 50: 1563–1589, 2000, ©International Union of Microbiological Societies.)

the other hand, acid production does not necessarily correlate with utilization of a given substrate for growth. However, acid production from carbohydrates in some instances is useful and provides an additional metabolic test.

A study of strains of *P. putida* and other fluorescent pseudomonads was performed by a determination of the assimilation of carbon compounds of three chemical families (carbohydrates, acids, and amino acids); this was done using the API system (BioMérieux, La Balme les Grottes, France) and following a methodology described elsewhere for a numerical taxonomic study on nonfluorescent species of *Pseudomonas* (Gavini et al., 1989a). This determination was supplemented by an extensive enzymatic study using kits from the same manufacturer for testing 59 peptidase, 10 esterase, and 20 oxidase activities (Elomari et al., 1994). Five main phenotypic clusters were obtained by application of the unweighted pair group average linked method and the Dice similarity coefficient. Of these, one group (II) was identified as *P. putida* biotype A, and could be subdivided into four subclusters, IIa–IIId. Strains of cluster II were characterized by ribotyping (see below) using the restriction enzyme *PvuII*, and seven ribotype subclusters were obtained. Two phenotypic subclusters correlated completely with two ribotype clusters. Based on the results of these studies, the authors urged a revision of *P. putida* and its biovars (of which A represents the “genuine” *P. putida*). The studies indicated the convenience of creating a third biovar, in confirmation of a similar suggestion that emerged from the studies of Barrett et al. (1986), and they serve to illustrate once again the enormous complexity of the species of the fluorescent group. In future studies, the *P. putida* biovars may be circumscribed more precisely as different species (Elomari et al., 1994).

The phenetic taxonomy of a large number of fluorescent pseudomonads isolated from tomato roots was performed using Jaccard similarity coefficients (Stenström et al., 1990). The field strains belong to the *P. fluorescens*/*P. putida* complex, and a suggestion was made to convert the biovars of the first species to the rank of individual species. Once again, the *P. putida* strains included a group that differed from the two known biovars (Stenström et al., 1990).

The usefulness of 36 determinative tests was estimated for 32 pathovars of the phytopathogenic fluorescent species *P. syringae*. The results allowed differentiation of pathovars in some cases, but in others, no distinctive clusters could be detected (Young and Triggs, 1994).

A numerical taxonomic analysis of the nonfluorescent species of RNA similarity group I (*Pseudomonas*), including a large number of strains, suggested that there may be future subdivision in a larger number of species than those recognized at present (Gavini et al., 1989a). Since the analysis was restricted to phenotypic properties, further work using molecular approaches will be necessary to substantiate such suggestion. Other examples illustrate the power of the numerical analysis of extensive phenotypic data for defining clusters that, upon further analysis, may be segregated as biovars or even described as independent species within a population of related strains. One case in point is the circumscription of *P. lundensis*, which, as mentioned in the species description (see below in list of species), had been previously identified as a well-defined cluster within a collection of strains isolated from meat (Molin and Ternström, 1982, 1986). It was similarly defined independently by an analysis of a collection of fluorescent strains from the Berkeley collection (Barrett et al., 1986). The studies resulted in defining four clusters of strains, of which two were identified as *P. fragi*. The other two

clusters, of fluorescent pseudomonads, could not be identified as any of the described biovars of *P. fluorescens* or *P. putida*. The authors emphasized the value of substrate utilization tests for elucidating the relationships among *Pseudomonas* strains (Shaw and Latty, 1982).

The numerical analysis of a collection of strains isolated from tomato pith necrosis demonstrated that the strains of *P. corrugata* constituted a single phenon. This study served as the basis for an emended description of this species (Sutra et al., 1997). Numerical taxonomic studies of strains isolated from natural mineral waters led to the discovery of the new species *P. rhodesiae* (Coroler et al., 1996) and *P. veronii* (Elomari et al., 1996).

Mol% G + C studies The original phenotypic studies by Stanier and collaborators in 1966 were supplemented with base composition determinations of the DNA of the strains (Mandel, 1966), which were later extended for inclusion in the description of new species under study in the same laboratory. In modern species descriptions, mol% G + C values of the DNAs are routinely included, although they are still missing for some of the species assigned to the genus.

DNA–DNA hybridization studies The phenotypic studies on the aerobic pseudomonads were followed by DNA–DNA hybridization studies, and, by virtue of the wide range of similarity values obtained, the results provided the first clues of the profound genomic differences among the species of this group of organisms. The results eventually suggested the need for similarity studies involving molecules of a more conservative nature. Within the genus *Pseudomonas*, the DNA–DNA hybridization studies included experiments with strains of the *P. stutzeri* group (Wendt-Potthoff et al., 1992), with fluorescent organisms (Palleroni et al., 1972), and with miscellaneous members of other groups. In all these experiments, the hybridizations were done by the competition technique of Johnson and Ordal (1968), but in later experiments the S1 nuclease procedure described by Johnson (1994a) was chosen.

The most important taxonomic application of DNA–DNA hybridization techniques involving the whole genome is in defining relationships at the species level. When applied to strains of aerobic pseudomonads, the hybridization experiments confirmed the relationships among species of the same phenotypic groups; at the same time, as mentioned above, the values were very low or negligible for species of different groups, even among strains of species that unequivocally could be assigned to the same genus. In spite of these observations, in some instances the results of DNA hybridization experiments occasionally have been used to attempt the precise circumscription of genera. One example was the proposal of two new generic names (*Chryseomonas* and *Flavimonas*) (Holmes et al., 1987) for two species previously assigned to the genus *Pseudomonas* (Kodama et al., 1985); the low DNA similarity values obtained in hybridization with DNA of *Pseudomonas* species were taken as the basis for such a proposal. Later studies suggested that the newly created generic names should in fact be considered as junior subjective synonyms of *Pseudomonas* (Anzai et al., 1997), and they are described as such in the present treatment.

Restriction analysis of the whole genome A useful approach adopted for the characterization of bacterial genomes combines the action of endonucleases that infrequently cut the chromosome with pulsed-field agarose gel electrophoresis for the separation of large DNA fragments. A limited number of cuts per